



**GREEN
CLIMATE
FUND**

Meeting of the Board

15 – 18 July 2024

Songdo, Incheon, Republic of Korea

Provisional agenda item 10

GCF/B.39/02/Add.07

24 June 2024

Consideration of funding proposals - Addendum VII

Funding proposal package for FP232

Summary

This addendum contains the following seven parts:

- a) A funding proposal titled “Jordan Integrated Landscape Management Initiative (JILMI)”;
- b) No-objection letter issued by the national designated authority(ies) or focal point(s);
- c) Environmental and social report(s) disclosure;
- d) Secretariat’s assessment;
- e) Independent Technical Advisory Panel’s assessment;
- f) Response from the accredited entity to the independent Technical Advisory Panel’s assessment; and
- g) Gender documentation.

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Funding Proposal

Project/Programme title:	<u>Jordan Integrated Landscape Management Initiative (JILMI)</u>
Country(ies):	<u>Jordan</u>
Accredited Entity:	<u>UNEP</u>
Date of first submission:	<u>[2021/03/12]</u>
Date of current submission	<u>[2024/03/13]</u>
Version number	<u>[V.006]</u>



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NOTE TO ACCREDITED ENTITIES ON THE USE OF THE FUNDING PROPOSAL TEMPLATE

- Accredited Entities should provide summary information in the proposal with cross-reference to annexes such as feasibility studies, gender action plan, term sheet, etc.
- Accredited Entities should ensure that annexes provided are consistent with the details provided in the funding proposal. Updates to the funding proposal and/or annexes must be reflected in all relevant documents.
- The total number of pages for the funding proposal (excluding annexes) **should not exceed 60**. Proposals exceeding the prescribed length will not be assessed within the usual service standard time.
- The recommended font is Arial, size 11.
- Under the [GCF Information Disclosure Policy](#), project and programme funding proposals will be disclosed on the GCF website, simultaneous with the submission to the Board, subject to the redaction of any information that may not be disclosed pursuant to the IDP. Accredited Entities are asked to fill out information on disclosure in section G.4.

Please submit the completed proposal to:

fundingproposal@gcfund.org

Please use the following name convention for the file name:

“FP- [Accredited Entity Short Name]-[Country/Region]-[YYYY/MM/DD]”

A. PROJECT/PROGRAMME SUMMARY

A.1. Project or programme	Project	A.2. Public or private sector	Public				
A.3. Request for Proposals (RFP)	If the funding proposal is being submitted in response to a specific GCF Request for Proposals, indicate which RFP it is targeted for. Please note that there is a separate template for the Simplified Approval Process and REDD+. Not applicable						
A.4. Result area(s)	Check the applicable GCF result area(s) that the <u>overall</u> proposed project/programme targets below. For each checked result area(s), indicate the estimated percentage of GCF and Co-financers' contribution devoted to it. The total of the percentages when summed should be 100% for GCF and Co-financers' contribution respectively.						
		GCF contribution	Co-financers' contribution¹				
	Mitigation total	Enter number %	Enter number %				
	☐ Energy access and power generation:	Enter number %	Enter number %				
	☐ Low-emission transport:	Enter number %	Enter number %				
	☐ Buildings, cities, industries and appliances:	Enter number %	Enter number %				
	☐ Forestry and land use:	Enter number %	Enter number %				
	Adaptation total	Enter number %	Enter number %				
	☒ Most vulnerable people, communities and regions:	30 %	30 %				
	☒ Health and well-being, and food and water security:	40 %	40 %				
	☐ Infrastructure and built environment	Enter number %	Enter number %				
	☒ Ecosystem and ecosystem services	30 %	30 %				
A.5. Expected mitigation outcome (Core indicator 1: GHG emissions reduced, avoided or removed / sequestered)	Indicate greenhouse gas (GHG) emission reductions or removals in tCO ₂ e over total lifespan of the project/programme ²	A.6. Expected adaptation outcome (Core indicator 2: direct and indirect beneficiaries reached)	Indicate total number of direct and indirect beneficiaries <table border="1"> <tr> <td>246,942 direct</td> <td>756,000 indirect</td> </tr> <tr> <td>2.2% of total pop.</td> <td>7.2% of total pop.</td> </tr> </table>	246,942 direct	756,000 indirect	2.2% of total pop.	7.2% of total pop.
246,942 direct	756,000 indirect						
2.2% of total pop.	7.2% of total pop.						
A.7. Total financing (GCF + co-finance³)	60,509,951 USD	A.9. Project size	Medium (Upto USD 250 million)				
A.8. Total GCF funding requested	44,904,203 USD <i>For multi-country proposals, please fill out annex 17.</i>						
A.10. Financial instrument(s) requested for the GCF funding	Mark all that apply and provide total amounts. The sum of all total amounts should be consistent with A.8. Grant <u>44,904,203 USD</u> ☐ Equity <u>Enter number</u> ☐ Loan <u>Enter number</u> ☐ Results-based payment <u>Enter number</u> ☐ Guarantee <u>Enter number</u>						

¹ Co-financer's contribution means the financial resources required, whether Public Finance or Private Finance, in addition to the GCF contribution (i.e. GCF financial resources requested by the Accredited Entity) to implement the project or programme described in the funding proposal.

² The total lifespan of the project/programme is defined as the maximum number of years over which the outcomes of the investment are expected to be effective. This is different from the project/programme implementation period.

³ Refer to the [Policy of Co-financing](#) of the GCF.

A.11. Implementation period	8 years	A.12. Total lifespan	25 years
A.13. Expected date of AE internal approval	<i>This is the date that the Accredited Entity obtained/will obtain its own approval to implement the project/programme, if available.</i> 11/30/2023	A.14. ESS category	<i>Refer to the AE's safeguard policy and GCF ESS Standards to assess your FP category.</i> B
A.15. Has this FP been submitted as a CN before?	Yes ☒ No ☐	A.16. Has Readiness or PPF support been used to prepare this FP?	Yes ☒ No ☐ PPF support has been accessed and used to prepare this Funding Proposal. There is no request for further PPF resources.
A.17. Is this FP included in the entity work programme?	Yes ☒ No ☐	A.18. Is this FP included in the country programme?	Yes ☒ No ☐
A.19. Complementarity and coherence	<i>Does the project/programme complement other climate finance funding (e.g. GEF, AF, CIF, etc.)? If yes, please elaborate in section B.1.</i> Yes ☒ No ☐		
A.20. Executing Entity information	Government of Jordan represented by the Ministry of Environment (MoEnv); Lead Executing Entity: IUCN		
A.21. Executive summary (max. 750 words, approximately 1.5 pages)			
1. Historically, Jordan's climate has been arid, with a high degree of rainfall variability and temperatures in summer (June–August) that often exceed 40°C. Approximately 94% of Jordan's annual rainfall is lost through evaporation, limiting the amount of freshwater available in the country. These dry conditions have resulted in Jordan's current ranking as the fifth most water-stressed country in the world, with only 125 cubic metres per year (CM/yr) of water available per capita⁴. 2. This baseline water stress is exacerbated by climate change, with climate projections under RCP4.5 and RCP8.5 showing increasingly dry conditions in the country. This drying effect is a result of an increase in maximum temperatures and a reduction in annual average rainfall, which is causing extended dry periods and a shorter rainfall season, with an increased frequency of high-intensity rainfall events. The increasing deviation from natural cycles of wet and dry periods in the country associated with climate change is also resulting in an observed increase in the frequency of severe droughts over the past 50 years. These extreme drought periods are expected to continue becoming increasingly frequent under future climate change scenarios. Such shifts in rainfall, combined with the effects of an increase in the average temperature by 2–4°C and the resulting increase in evaporation rates will cause a reduction in safe yield water supplies beyond Jordan's existing deficit of 416 MCM/yr. Aquifer recharge rates are also projected to decline, resulting in a reduction in the available amount of groundwater. This will cause reduced baseflow in rivers and consequently also a decrease in surface-water availability. This problem will be further compounded by the increasing demand for freshwater for irrigation and domestic consumption linked to rising temperatures which will result in increases in groundwater abstraction rates. 3. Notably, the impacts of climate change will result in a considerable reduction in Jordan's agricultural production. For example, barley production is expected to decline by up to 50% under future climate conditions, which will severely affect local food supplies and consequently also food security in the country. The potential severity of this climate impact was demonstrated during a severe drought between 1998 and 2012 — worst drought recorded in Jordan over the past 900 years. During this period, rainfall decreased by up to 70% compared with long-term averages, resulting in severe agricultural, socio-economic and environmental impacts⁵. The drought caused a reduction in cereal harvests to only 0.6% of the country's total needs, diminished non-cereal crop production by 23% and reduced meat and dairy production by			

⁴ WRI. 2019. Aqueduct baseline water stress rankings. Available at: <https://www.wri.org/aqueduct/>.

⁵ Drought Conditions and Management Strategies in Jordan. Prepared for the Initiative on 'Capacity Development to Support National Drought Management' Policies Workshop. 2014.

40%⁶. Such impacts will also affect livestock production through disruptions in the supply of fodder. Already, shortages of up to 77% in the amount of fodder available for livestock have been observed, and are partly attributable to climate change over the past decades. Therefore, there is a need to not only strengthen the supply of water resources in Jordan, but also transition food systems from those reliant on climate-vulnerable crops like barely to those which are more diverse and resilient to climate change, such as quinoa and pearl millet, which contribute to food security under climate change conditions.

4. Climate change is also strongly affecting the functioning of ecosystems in Jordan and consequently the services they provide — particularly maintenance of the hydrological cycle. Ecosystems in the country are already being degraded by numerous non-climatic drivers, including: i) land loss or degradation from urbanisation and agriculture; ii) overgrazing; iii) unregulated fuelwood harvesting; iv) water over-extraction; v) species over-exploitation; vi) the introduction of invasive plant and animal species; vii) mining; and viii) pollution. Climate change is accelerating this degradation — with numerous adverse impacts, including increased soil erosion and sedimentation of dams and waterways, decreased quality and quantity of water resources, increased forest dieback, changes in species composition, desertification, habitat degradation and species loss — and threatening the climate-resilience of communities reliant on these ecosystems' services.
5. In the absence of urgent climate change adaptation interventions at the landscape level, the projected drying trend across Jordan will have increasingly severe impacts on Jordan's water supplies, extending the water supply deficit to 687 MCM/yr by 2040⁷. Combined with the compounding impacts on ecosystems, this water shortfall will have significant impacts on the livelihoods and well-being of local communities across Jordan.
6. The Government of the Hashemite Kingdom of Jordan (GoJ) is requesting GCF grant finance to respond to the above-mentioned climate threats and impacts. GCF finance will be combined with in-kind government (MoENV) and donor (Consortium of NGOs) co-financing to initiate a paradigm shift in the approach to land and water resources management in Jordan that will result in climate-resilient sustainable development nationwide.
7. The objective of the proposed project is to reduce the negative impacts of climate change on water resources in Jordan through improved aquifer recharge in a landscape-level approach. The proposed approach includes: i) improved ecosystem management through land restoration, improved agricultural practices and managed aquifer recharge; ii) rainwater harvesting; iii) evaporation reduction from the King Talal Dam; and vi) strengthened governance. A Water Fund will be established to scale up good landscape management practices and to promote water conservation in the agriculture sector which currently takes the largest share of water demand: agriculture. The Water Fund will leverage financing from private (industry and commercial agriculture) and public sector stakeholders (municipalities) — downstream water users — for aquifer replenishment services from the restoration and management in upstream watersheds. The project will be implemented at the Yarmouk, King Talal Dam and Kafraïn watersheds in the Jordan Valley.
8. Under the proposed project, on-the-ground, climate-resilient ILWRM interventions will be implemented across three sites within the Jordan Valley. These interventions will include 8,000 ha of restored ecosystems, 12 rockfill dams for managed aquifer recharge, 3,000 rooftop rainwater harvesting tanks and 1 MW of floating solar panels for evaporation reduction — leading to an ultimate increase in the country's annual water supply of ~8.9 MCM after eight years (project duration) and a potential cumulative increase of ~154.5 MCM after and 25 years. Upscaling of all four on-the-ground interventions by others is expected to cumulatively yield an increase in water resources of 34 MCM under a conservative scaling-up scenario and 344 MCM for a best-case scenario, both after 25 years. Furthermore, the project will demonstrate to decision-makers that the implementation of climate-resilient ILWRM as a national response to climate change has the potential to enable Jordan to balance its water demand with water supply in the long term despite climate change impacts.
9. Project interventions will directly benefit a total of 246,942 people (~2.2% of the Jordanian population) — 135,736 men (55%) and 111,206 women (45%) — in the Jordan Valley by reducing their climate vulnerability. Of this total: i) 62,446 (30,607 women and 31,839 men) are from the Yarmouk site — Khaled

⁶ FAO. 1999. Worst drought in decades decimates cereal crops in Jordan. Available at: <http://www.fao.org/NEWS/GLOBAL/GW9907-e.htm>.

⁷ Government of Jordan. 2014. Jordan's Third National Communication on Climate Change.

Bin Al Waleed and Al Seru municipalities; ii) 105,841 (45,179 women and 60,662 men) are from the King Talal Dam site — New Ardah, Deir Alla and M'addi municipalities; and iii) 78,655 (35,420 women and 43,235 men) are from the Kafraïn site — Shouna Alwasta and Mahes municipalities. Direct beneficiaries include all people who will benefit directly from a project activity or the result of an activity. For example, under Output 2.1, 1,200 smallholder farming households (~6,000 people — an average of 5 people per smallholder household) across the three sites will benefit from on-farm interventions, while 2,800 households across the sites (~14,000 people) will benefit directly from rainwater harvesting infrastructure implemented on houses. Direct beneficiaries also include all vulnerable peoples in the project's three target areas, as they are all expected to benefit from the augmentation of water resources under Outputs 2 and 3, either via improvements in water supplies for domestic and/or agricultural usage. Under Output 3, all 172,179 people living in the municipalities of the three target sites (and related municipalities) listed above will benefit from the improved protection, supply and use of water for domestic purposes under climate change conditions. Specifically, they will directly benefit from enhancements in the quality and quantity of ground and surface water through Activity 3.1. This will be achieved through the construction of rockfill dams to support managed aquifer recharge and the rehabilitation of active freshwater springs.

10. As integrated land and water resources management interventions (ILWRM) are scaled up via the Water Fund (Activity 2.4), it is expected that an additional 4,000 ha of land will be placed under improved management for water resources benefits, with at least one additional (adjacent) municipality per watershed site benefitting directly. The 4,000 ha is related to total water attenuation potential for the three watersheds, in addition to the 8,000 ha to be restored directly by the project (see Table 27 and Figures 116-118 in Annex 2). The rainfall attenuation potential represents the slowing of overland water flows which reduces the amount of water running off the land and increases the amount of time available for water infiltration, and hence groundwater recharge. The Water Fund will scale the set of restoration and land management interventions in Component 2 to target the entire 12,000 ha with potential for restoration to support water attenuation. Consequently, an additional 74,763 people (41,604 men and 33,159 women)⁸ are expected to benefit directly from the project.
11. The indirect beneficiaries of improved watershed management across the northern Jordan Valley will include all the region's remaining municipalities, amounting to 756,000 people (355,320 women and 400,680 men). Indirect benefits will include: i) the enhancement of the quality and quantity of water resources across the northern Jordan Valley area; ii) strengthened local governance to implement climate-resilient ILWRM; iii) reduced rate at which water is lost from the King Abdullah Canal through evaporation; and iv) improved awareness on the impacts of climate changes and the adaptation options to address them. Interventions at the regulatory and institutional level will indirectly benefit the Jordanian population as a whole, amounting to ~10 million people.
12. Co-benefits will accrue through the proposed project: GHG emission reductions and women's empowerment. Other sustainable development potential includes reduced losses and damages from flood attenuation potential associated with re-vegetated landscapes; improved health outcomes from cleaner water and enhanced food security; and enhanced biodiversity. In terms of gender-responsive development impacts, the proposed project will support a transformation in the water sector to one in which women are included in the decision-making for and management of water resources at the community level, including being involved in monitoring of water resource use and maintenance of community-level water infrastructure. This will allow for their specific water-related climate-change adaptation needs to be addressed. The project's community-level gender-responsive approach also includes promoting gender-inclusive climate-resilient livelihoods that will benefit female-headed households (can be implemented at home and allow for the strengthening of adaptive capacity). Tailored awareness-raising and outreach programmes will also allow women to become leaders in the promotion of climate change adaptation in their communities.
13. Ultimately, the proposed project will confer climate change adaptation benefits to the country's entire population. These benefits will be achieved by reducing the adverse impacts of climate change on Jordan's water supplies and by providing the country's main economic sectors and ecosystems with mechanisms and interventions to build long-term climate resilience. Moreover, the proposed GCF project will enable Jordan to serve as an example of how a transformative approach to climate change adaptation can combat the increasingly arid conditions resulting from climate change in the Middle East and Mediterranean landscapes. The proposed adaptation solution contributes directly to the GCF's Updated Strategic Plan (2024-2027). This includes references to programming priorities: i) Mitigation and Adaptation: Supporting

⁸ https://dosweb.dos.gov.jo/DataBank/Population/Population_Estimares/Municipalities.pdf

paradigm shifts across sectors, including to climate-resilient ecosystems and food systems; ii) Adaptation: Addressing urgent and immediate adaptation and resilience needs, for particularly vulnerable countries. Relevant GCF Targeted Results for 2024-2027 include: 4. Food: Support for developing countries that results in 190 to 280 million beneficiaries adopting low-emission climate-resilient agricultural and fisheries practices, securing livelihoods while reconfiguring food systems; 5. Ecosystems: Support for developing countries that results in 120 million to 190 million hectares of terrestrial and marine areas conserved, restored or brought under sustainable management; and 9. 40-70 approved proposals for adaptation projects, including for locally-led adaptation action.

B. PROJECT/PROGRAMME INFORMATION

B.1. Climate rationale and context (max. 1000 words, approximately 2 pages)

Context

14. The Hashemite Kingdom of Jordan (hereafter referred to as Jordan) is a Middle Eastern country situated on the eastern bank of the Jordan River. The land area is divided into three main topographic regions, namely the: i) north-western central highlands with Mediterranean forest and the Irano-Turanian steppe; ii) western low-lying subtropical Jordan Valley; and iii) eastern arid desert (Badia), which covers 80% of the country⁹. Most of Jordan's population of 10.5 million people (51% male and 49% female)¹⁰ are concentrated in the north-western region of the country (predominantly in the Jordan Rift Valley; 1,194 people/km²), which is attributable to the area's fertile soils, higher rainfall and consequently greater agricultural productivity and prospects, relative to other parts of the country. Over the past six decades, the population of Jordan has increased 12-fold, driven by inter alia a historically high rate of endogenous population growth and an influx of refugees and immigrants as a result of crises in the region, such as the Syrian conflict. This rapid increase in Jordan's population has contributed to several developmental challenges for the country, including: i) shortages in food, water and other natural resources; ii) reduced groundwater recharge and increased erosion resulting from overexploitation and degradation of ecosystems linked to agricultural and urban expansion; iii) limited housing and employment opportunities; and iv) overburdening of the education system, health services, and urban infrastructure^{11,12}. Of these challenges, water insecurity is a primary challenge. Jordan is currently ranked as the fifth most water-stressed country in the world¹³, with only 125 CM/yr of water available per person¹⁴. This water scarcity is a function of: i) naturally low rainfall levels, as well as high temperatures and evaporation rates; ii) limited availability of shared transboundary surface water resources; iii) an increasing population and influx of refugees from neighbouring countries exacerbating water demand; and iv) an increase in unsustainable agricultural and industrial development. The current water usage in Jordan already exceeds safe yields¹⁵ by 416 MCM/yr¹⁶, leading to the over-extraction of renewable and non-renewable groundwater reserves.

15. The country's economy faces considerable challenges to accommodate the increasing demand for resources from its growing population. Although Jordan is an upper-middle-income country and is classified as an emerging economy, it is among the smallest in the Middle East and North Africa (MENA) region, with a Gross Domestic Product (GDP) of ~US\$40 billion. The largest contributors to Jordan's GDP are the: i) services (~67%); ii) industry (~29%); and iii) agriculture (~4%) sectors. Because of a slowing economy and influx of refugees, Jordan has high unemployment levels. Labour force participation — characterised by large gender differences — is 64% for men and 14% for women¹⁷, while unemployment rates are 17% for men and 27% for women.

Climate profile

16. Jordan is characterised by an arid climate with hot and dry summers contrasted with short, cool winters. Average temperatures range from 9–22°C in the winter months (from December to March) to 26–39°C in the summer months (from April until November)¹⁸. The dry Badia in the east receives less than 200 mm of rainfall annually, while the central highlands in the northwest receive the highest levels of rainfall at 350–500 mm, mostly through intense winter storm events¹⁹. Increasing rainfall variability in Jordan has resulted in more frequent drought and flash flood events, as well as greater levels of erosion in degraded landscapes.

Climate change in Jordan²⁰

⁹ Palmer C. 2013. Atlas of Jordan. Presses de l'Ifpo, Institut français du Proche-Orient. Beirut, Lebanon.

¹⁰ Department of Statistics (DOS). 2018. Available at: <http://dosweb.dos.gov.jo/>.

¹¹ Nsour F, Saleh K, Aranki I, Obaid B, Sweidan M, Abu-Haidar DA, Mowafi A, Khasawneh G, Mfarej EB (eds). 2012. *Jordan Population and Family Health Survey 2012*. Department of Statistics. Amman.

¹² Alshoubaki We, Harris M. 2019. The Impact of Syrian refugees on a receiving state's public expenditure: Evidence from Jordan. *International Journal of Business and Social Science* 10.

¹³ WRI. 2019. Aqueduct baseline water stress rankings. Available at: <https://www.wri.org/aqueduct/>.

¹⁴ Hammouri N, Adbulla F, Qdais HA & Freiwan M. 2015. Assessing the impacts of climate change on water resources of Jordan. *World Water Congress XV*. International Water Resources Association (IWRA), Edinburgh.

¹⁵ The amount of water that can be withdrawn from available sources without producing an undesired effect.

¹⁶ Ministry of Water and Irrigation. 2017. Jordan water sector — Facts and figures. MWI, Amman.

¹⁷ The World Bank, 2019. Gender Data Portal. Available at: <http://datatopics.worldbank.org/gender/country/jordan>

¹⁸ USAID. 2017. Climate Change Risk Profile in Jordan: Climate Risk Profile.

¹⁹ Ibid.

²⁰ Climatic data presented in this sub-section was sourced from the dynamic downscaling analyses of simulations of multi-model ensembles through the Coordinated Regional Climate Downscaling Experiment (CORDEX). This includes nine different GCMs coupled with two RCMs for two RCPs (4.5 and 8.5), which were used to assess future projections compared as compared to historic reference data (1980–2010). Hydrological data presented in this sub-section, including the impacts of climate change conditions, is derived from sources using an ensemble of RCMs.

17. Detailed primary and secondary climate analyses for the proposed project are presented in Annex 2: Feasibility Study, in Section 2 and Appendix 6, respectively. The information presented below is a summary of what is presented in the more detailed analyses, which should be consulted as necessary.

18. Observations of historical climate trends (1950–2015) in Jordan show that the country has already experienced significant changes in both temperature and rainfall. Since 1950, minimum and maximum temperatures have increased by 0.4–2.8°C and 0.3–1.8°C, respectively²¹ (Figure 1). In terms of rainfall, average annual values have decreased by 5–20% since 1950, with the average number of rainy days decreasing by 3–10% during the same period²² (Figure 2). This decrease is marked by extended drought periods, as well as periods of increased rainfall intensity as a result of a shorter rainfall season. This rainfall decline, combined with extreme temperatures, has resulted in prolonged drought periods in the country. Droughts are projected to increase in frequency and severity, with the number of years affected by meteorological drought in a 30-year period expected to increase from seven (1981–2010) to 28 (2071–2100)²³.

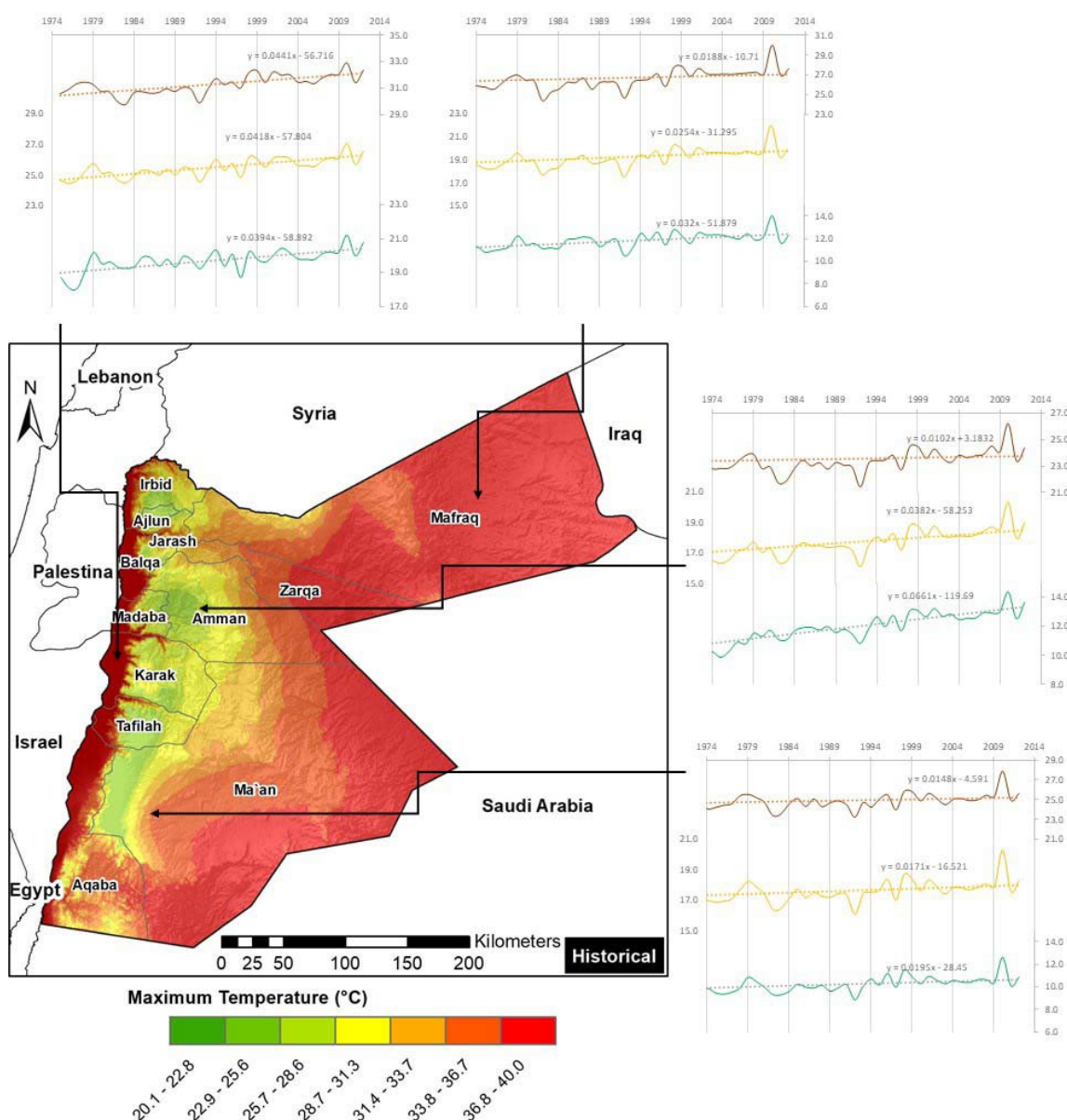


Figure 1. Observed maximum temperature trends in Jordan.

²¹ The Hashemite Kingdom of Jordan. 2014. Jordan's Third National Communication on Climate Change.

²² USAID. 2017. Climate Change Risk Profile in Jordan: Climate Risk Profile.

²³ Ibid.

²³ Rajsekhar D and Gorelick S, 2017. Increasing drought in Jordan, Climate change and cascading Syrian land use impacts on reducing transboundary flow. *Science Advances* 2017; 3:e1700581.

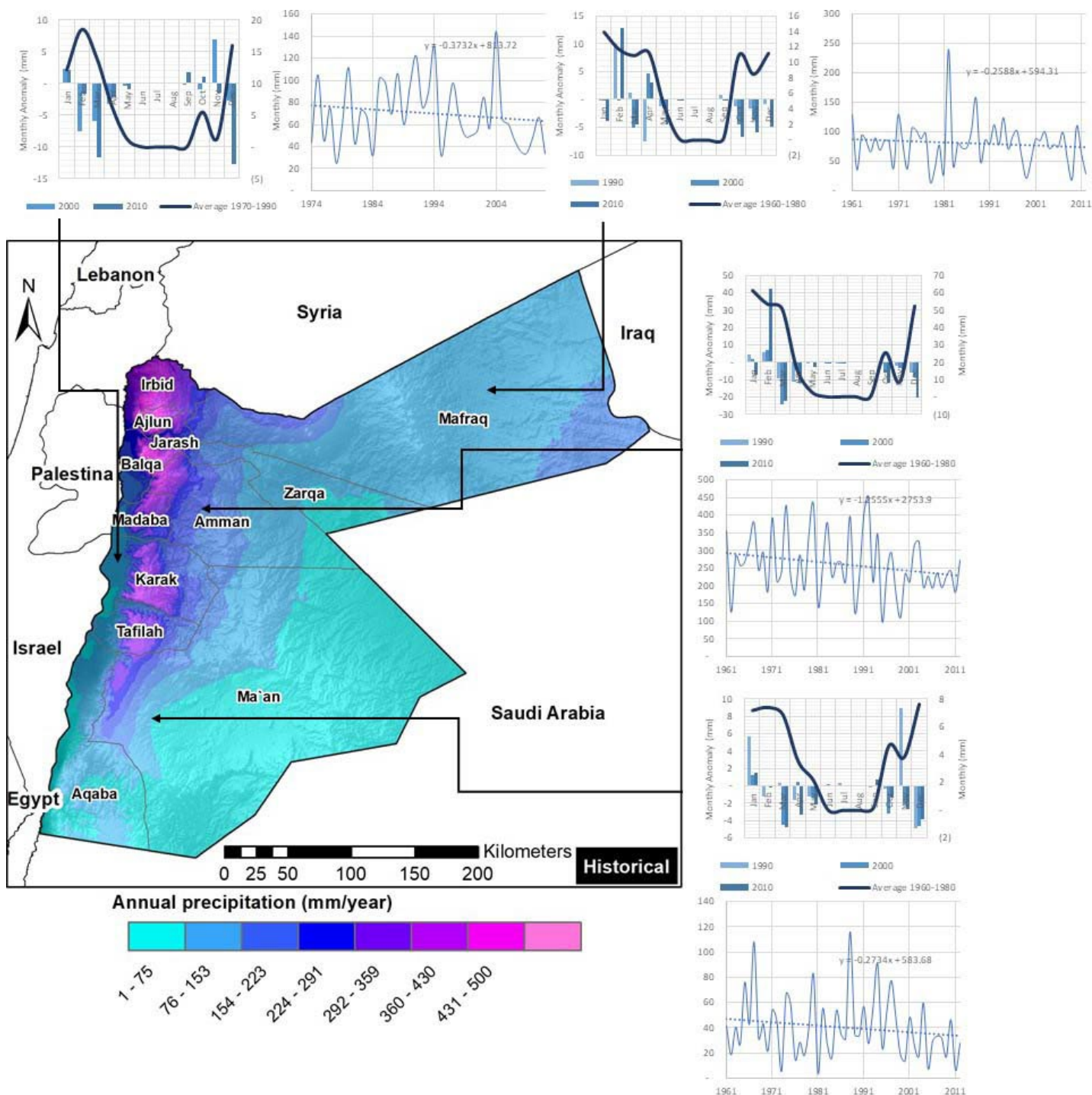


Figure 2. Observed rainfall trends in Jordan.

19. Changes in the standard precipitation evapotranspiration index (SPEI) were compared over time to determine if climate change has already impacted water availability in Jordan²⁴. The SPEI is a good measure of drought severity, where a value of +3 indicates exceptionally wet months and -3 indicates exceptionally dry months. Figure 3 shows changes in SPEI in Jordan from 1954–2019. A 48-month running average is used to determine long-term trends instead of short-term meteorological anomalies. Although there are natural oscillations over time, from the late 1990s there is a strong shift towards the prevalence of drier months, with the gradual decrease in SPEI values indicating more extreme drought conditions in Jordan. The current SPEI trends differ from natural cyclic phases and are likely a consequence of climate change. This means that Jordan is currently experiencing the impacts of

climate change in water scarcity both from hydrological and agricultural droughts that result from these meteorological droughts.

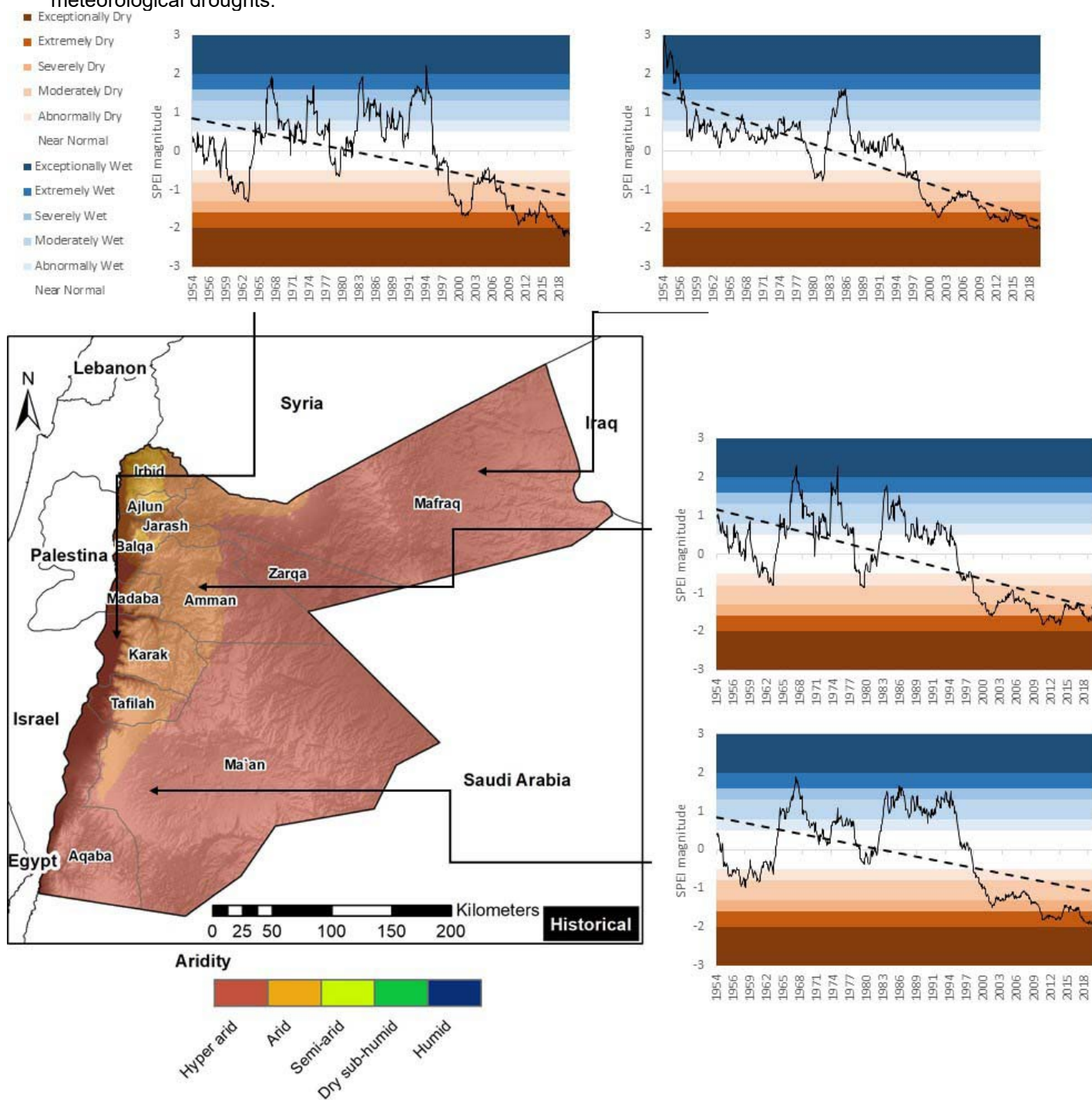


Figure 3. Observed drought trends in Jordan.

20. As Jordan is already impacted by extreme baseline climate conditions — resulting in ecosystems and communities existing on the threshold of survivability — even small changes in temperature or rainfall are expected to have considerable impacts. By the end of the 21st century, projections indicate that Jordan will experience: i) a warmer climate — where, between 2070–2100, the average temperature is expected to increase by between 2.1°C (RCP4.5) and 4°C (RCP8.5); ii) more heatwaves — where average maximum summer temperatures for the whole country are expected to exceed 42°C (RCP4.5) and 44°C (RCP8.5) (Figure 4); iii) a drier climate — where, between 2070–2100, total annual rainfall is expected to decrease by between 15% (RCP4.5) and 21% (RCP8.5); iv) more droughts — where from 2040 to 2070 the average number of consecutive dry days is likely to increase by a maximum of 30% under both RCP4.5 and RCP8.5²⁵; and v) an increase in extreme rainfall events within a

²⁵ Government of Jordan. 2014. Jordan's Third National Communication on Climate Change.

shortened rainy season. The above temperature and rainfall projections for 2071–2100 agree with those from the GCF World Meteorological Organisation (WMO) platform, which predicts an increase in temperature of 1.5–3.2 (with all 18 CORDEX models projecting rising temperatures) and 16% decline in precipitation under RCP4.5 (projected by 16 out of the 18 CORDEX models).

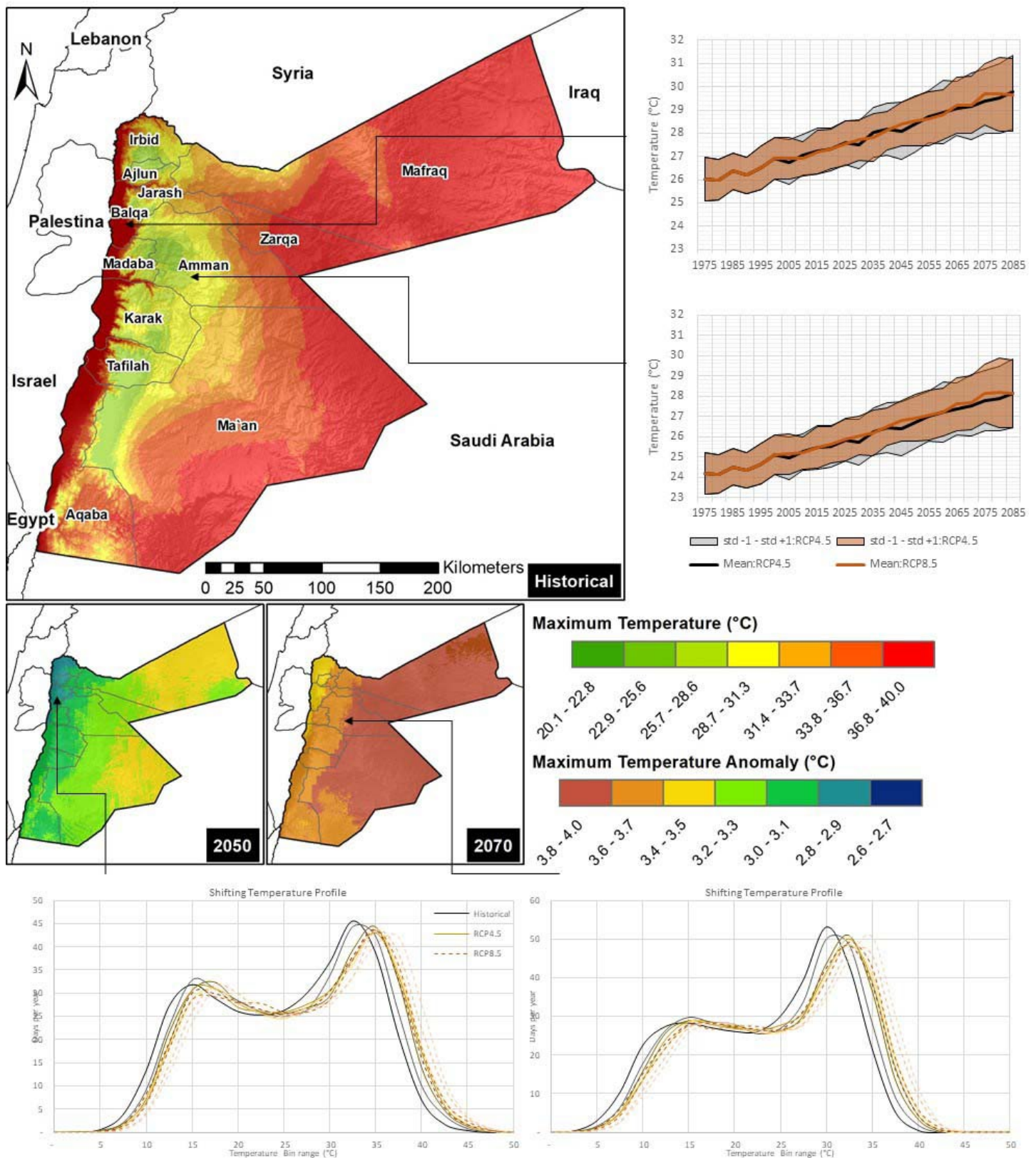


Figure 4. Projected peak maximum temperature change and temperature profile shifts for Jordan.

21. Current annual rainfall has decreased in recent decades and this trend is projected to increase into the future with some estimates showing an overall decrease of between 5–60% between now and the end of the century (Figure 5). This is expected to lead to enhanced drought frequency and severity, as well as intensified aridity. The current trends and continuation of decreases in the projected rainfall are expected to only exacerbate water scarcity in the future. There are projected increases in the number of dry days but also in the rainfall intensity in some months. The mid-season rainfall is projected to be lower each week relative to the historical period (1980-2010).

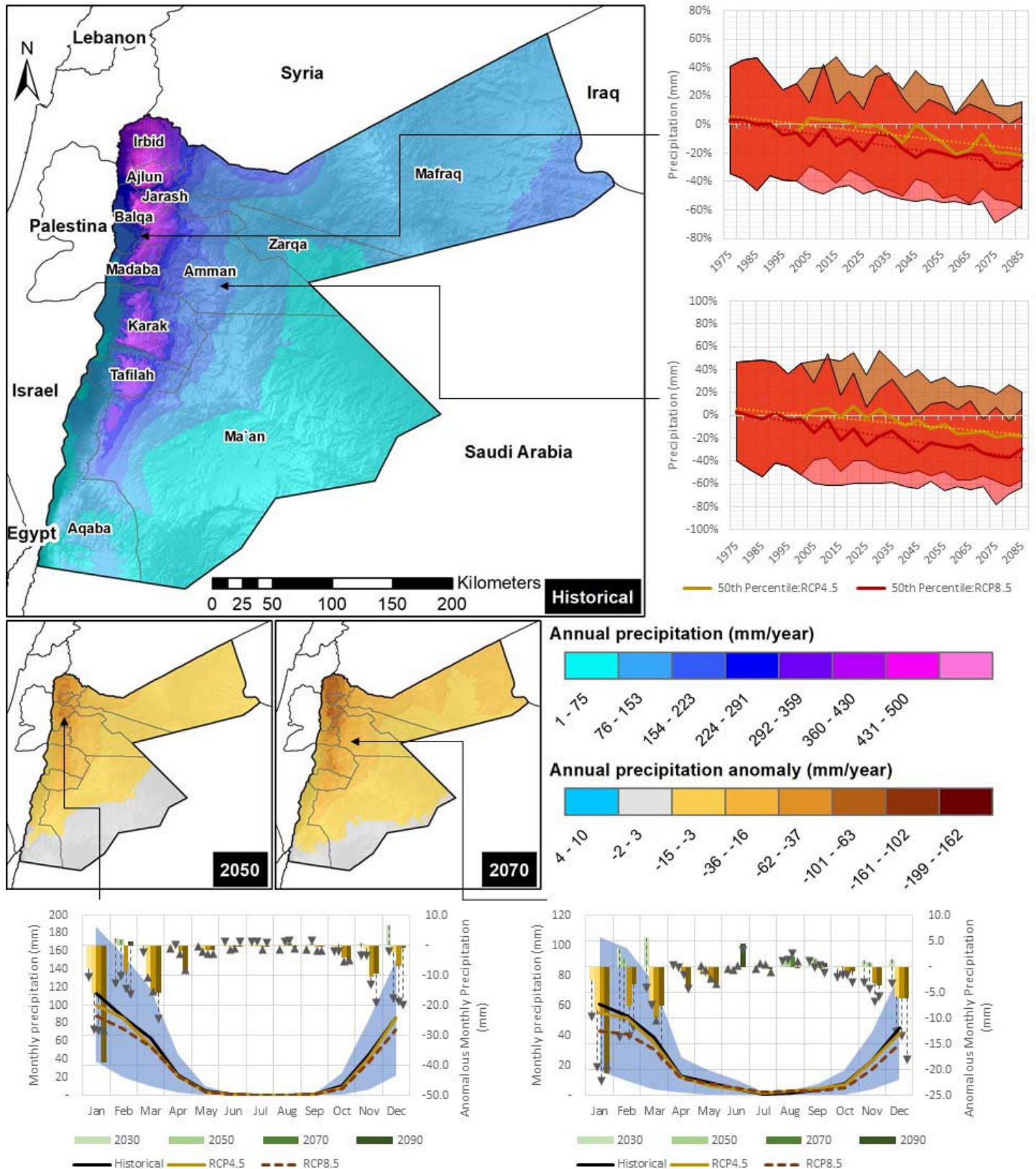


Figure 5. Long-term precipitation changes and monthly rainfall anomaly.

22. Jordan will experience an average increase in ETp of about 70–100 mm by the year 2050, and 150 mm by the year 2100²⁶. An increase in ETp, along with a decreasing trend in Standard Precipitation Indices (SPI) — above zero during the baseline period 1981–2010 (wet years); to below zero by 2085 (dry years) — combined with the projected

changes in rainfall, is likely to increase the frequency, duration and severity of drought events²⁷ (Figure 6). The long-term (1990-2020) SPEI average of Jordan is between Near Normal Dry to Moderately/Severely Dry (-0.29 to -1.21). Drought conditions in Jordan are also expected to be intensified by reductions in soil moisture and streamflow. These variables are likely to decrease by 28% and 58% respectively, by 2100 under RCP 8.5²⁸. Under RCP8.5, the probability of years having simultaneous drought as well as average annual temperature increases of greater than 1 SD above the baseline is projected to increase from less than 10% to more than 90% by 2100²⁹.

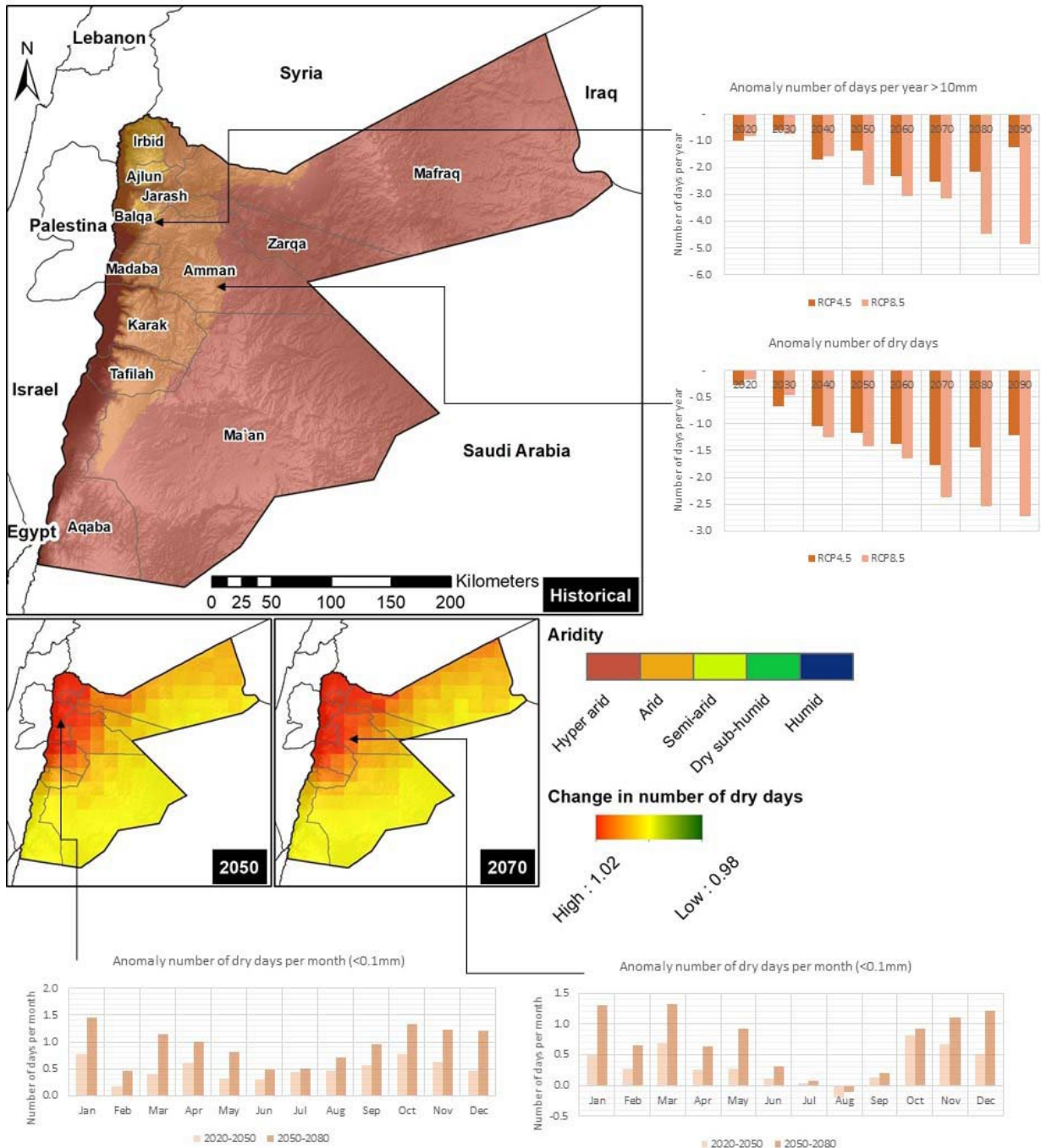


Figure 6. Aridity and long-term change in dry days in Jordan.

Climate change impacts and vulnerabilities

23. Current and predicted climate change are presently and will continue to have multiple negative impacts on Jordan, particularly with regards to water resources, ecosystems and the agriculture sector. These impacts are summarised below and provided in more detail in Annex 2: Feasibility Study.

Water resources

24. Jordan's worsening climatic conditions and the above-mentioned compounding factors have resulted in increased water stress in the country over the past decade, with water availability decreasing by 145 CM/yr per person between 2007–2017³⁰. Extreme temperatures and the ongoing drying trend in Jordan are the primary causes of severe water scarcity in the country, with the resultant evaporation, for example, accounting for the loss of 7,636 MCM of water in 2017³¹, which is 94% of the annual rainfall Jordan received that year. Between 2007 and 2017, the impacts of climate change contributed to water availability decreasing by 145 CM/yr per capita. Jordan consequently ranks as the fifth most water-stressed country in the world with only 125 CM/yr currently available per capita — well below the international water poverty line of 500 CM/yr³².
25. Most of Jordan's water supply (59%) is extracted from 10 renewable and two non-renewable groundwater basins, which are currently being over-extracted by ~169% and ~124% of safe yield, respectively³³. The current rate of abstraction is ~190 MCM/yr higher than the estimated safe yield of ~275 MCM/yr for all renewable groundwater combined. For non-renewable groundwater, the rate of abstraction is ~314 MCM/yr higher than the estimated safe yield of ~143 MCM/yr³⁴. Currently, water-use quotas and water-saving policies are implemented by the Ministry of Water and Irrigation (MWI) and its subsidiaries (such as the Jordan Valley Authority) to limit over-extraction. Groundwater is accessed either through springs that reach the surface or through wells. Of the country's 861 springs, 130 have run dry as of 2013³⁵. Concurrently, the current number of wells in Jordan exceeds 3,200, a number that has increased by 10% over the last decade³⁶. The safe yield of all renewable basins combined equates to 275 MCM/yr — which is 190 MCM/yr less than the current abstraction — which in turn is only a fraction of Jordan's current water needs (1,077 MCM/yr). On average, the water deficit from all non-renewable groundwater basins is 124% of the safe yield, while renewable groundwater basins show a deficit of 169%³⁷.
26. Reduced rainfall and increased evaporation from higher temperatures related to climate change have already resulted in observed impacts on Jordan's surface water sources. For example, the direct input of rainfall through runoff into the Upper Jordan River — the primary surface water source for most of the population — has decreased by 10% over 50 years, from 2,138 MCM/yr in 1950 to 1,930 MCM/yr in 2000³⁸. This reduction in runoff in the Upper Jordan River watershed is projected to decrease by a further 23%, from an average of 375 mm/yr⁻¹ in the 1961–1990 period to 289 mm/yr⁻¹ in the 2070–2099 period³⁹. Overall, a 51% reduction in streamflow as a result of the combined effect of reduced runoff and increased evaporation under climate change conditions is projected to occur in Jordan⁴⁰.
27. As surface water resources become less reliable under conditions of climate change, and wet seasons become shorter than before, there will be a greater dependence of communities on groundwater. With ~60% of Jordan's total water supply — and 73% of the municipal supply⁴¹ — already coming from groundwater, any reduction in groundwater recharge will have severe implications for water security in the country⁴². A climate change-induced

³¹ Ministry of Water and Irrigation. 2017. Jordan water sector — Facts and figures. MWI, Amman.

³² Hammouri N, Adbulla F, Qdais HA & Freiwan M. 2015. Assessing the impacts of climate change on water resources of Jordan. World Water Congress XV. International Water Resources Association (IWRA), Edinburgh.

³³ Ministry of Water and Irrigation. 2017. Jordan water sector — Facts and figures. MWI, Amman.

³⁴ FAO. 2008. Jordan country profile: Water resources. Available at: http://www.fao.org/nr/water/aquastat/countries_regions/Profile_segments/JOR-WR_eng.stm.

³⁵ Ministry of Water and Irrigation. 2018. Water Yearbook 2016/2017. MWI, Amman.

³⁶ Ministry of Water and Irrigation. 2017. Jordan water sector — Facts and figures. MWI, Amman.

³⁷ Ministry of Water and Irrigation. 2017. Jordan water sector — Facts and figures. MWI, Amman.

³⁸ Molle F & Wester P. 2009. River Basin Trajectories: Societies, Environments and Development. CABI International, United Kingdom.

³⁹ Kunstmann H, Suppan P, Heckl A & Rimmer A. 2007. Regional climate change in the Middle East and impact on hydrology in the Upper Jordan catchment. *IAHS Publ.* 313:141–149.

⁴⁰ Rajsekhar D & Gorelick SM. 2017. Increasing drought in Jordan: Climate change and cascading Syrian land use impacts on reducing transboundary flow. *Science Advances*, 8:1–16.

⁴¹ Ministry of Water and Irrigation. 2017. Jordan water sector — Facts and figures. MWI, Amman.

⁴² Wade AJ, Black E, Brayshaw DJ, El-Bastawesy M, Holmes PAC, Butterfield D, Nuimet S & Jamjoum K. 2010. A model-based assessment of the effects of projected climate change on the water resources of Jordan. *Philosophical Transactions of the Royal Society*, 368: 5151–5172.

decline in annual rainfall by 12% is predicted to reduce groundwater recharge by more than 25% by 2099^{43,44}. This is likely to result in a reduction of renewable water supplies by 416 MCM/yr (996 to 580 MCM/yr)⁴⁵.

28. Given the combination of impacts from climate change, by 2022 the country's water resources⁴⁶ (both renewable and non-renewable) are expected to be limited to ~1,132 MCM/yr. By this time, demand is predicted to increase to ~1,635 MCM/yr, resulting in a deficit of ~503 MCM/yr from a current baseline of 416 MCM/yr (increase of 87 MCM)⁴⁷. Without urgent climate change adaptation interventions, Jordan's water deficit will increase by an additional 184 MCM to 687 MCM/yr by 2040⁴⁸.

Ecosystems and ecosystem services

29. Climate change is strongly affecting the functioning of ecosystems and their services in Jordan, which are already being impacted by numerous non-climatic threats, including: i) land loss or degradation from urbanisation and agriculture; ii) overgrazing; iii) unregulated fuelwood harvesting; iv) water over-extraction; v) species over-exploitation; vi) the introduction of invasive plant and animal species; vii) mining; and viii) pollution⁴⁹. These non-climatic drivers are expected to be exacerbated by climate change, which is predicted to impact ecosystems through increased forest dieback, changes in community composition, desertification, habitat degradation and species loss⁵⁰.

30. Most notable in the context of Jordan is the effect climate change-induced vegetation loss is expected to have on water balances. While intact rangelands in Jordan experience runoff losses of 9–21 mm/yr, degraded vegetation increase these losses to ~20–28 mm/yr by decreasing infiltration and promoting runoff⁵¹. Forest loss and landscape degradation also decrease surface roughness, resulting in increased albedo and temperatures, as well as lower evapotranspiration rates, altering moisture and heat fluxes between terrestrial surfaces and the atmosphere⁵². Indeed, by changing these processes, large-scale forest dieback or desertification linked to climate change is expected to reduce rainfall in some regions by as much as 30%⁵³, further limiting water availability. The loss of vegetation cover will also reduce groundwater infiltration and soil stability. Desertification and ecosystem deterioration related to climate change and land-use changes result in soil degradation, leading to reduced soil organic carbon and impoverished soil structure⁵⁴. As a result, infiltration and water retention capacity of the soil will be reduced, which will, in turn, lead to a decline in aquifer recharge. In addition, bare soils resulting from ecosystem degradation increase the risk of flooding, particularly during extreme rainfall events, which are expected to increase in frequency and intensity under future climate conditions. This is because the loss of vegetation results in soil hardening or compaction, thereby reducing infiltration and increasing runoff⁵⁵.

Agriculture

31. Agriculture accounts for 65–75% of Jordan's total water consumption⁵⁶. As a result, water scarcity poses a risk to current food systems and food security in the country as the diminishing water supply results in lower agricultural yields and an increase in the need to import food. Rainfed agriculture — including locally valuable crops such as barley and wheat — is the most climate-vulnerable form of cultivation in Jordan⁵⁷. Under future climate change, a

⁴³ Al-Qaisi BM. 2010. Climate change effects on water resources in Amman Zarqa Basin — Jordan. Individual Project Report: Climate Change – Mitigation and Adaptation. Ministry of Water and Irrigation, Amman.

⁴⁴ Kunstmann H, Suppan P, Heckl A & Rimmer A. 2007. Regional climate change in the Middle East and impact on hydrology in the Upper Jordan catchment. *IAHS Publ.* 313: 141–149.

⁴⁵ Margane A, Borgstedt A & Subah A. 2008. Water resources protection efforts in Jordan and their contribution to a sustainable water resources management. *Climatic Changes and Water Resources in the Middle East and North Africa*. 325–345.

⁴⁶ Renewable, non-renewable groundwater and unconventional water sources such as wastewater treatment and desalination

⁴⁷ Hammouri N, Al-Qinna M, Salahat M, Adamowski J & Prasher SO. 2015. Community based adaptation options for climate change impacts on water resources: The case for Jordan. *Journal of Water and Land Development*, 26: 3–17.

⁴⁸ Government of Jordan. 2014. Jordan's Third National Communication on Climate Change.

⁴⁹ The General Corporation for the Environment Protection. 2001. Conservation and sustainable use of biological diversity in Jordan: First National Report. Amman, Jordan.

⁵⁰ Government of Jordan. 2014. Jordan's Third National Communication on Climate Change.

⁵¹ Haddad M. 2019. Exploring Jordan's rangeland transition: Merging restoration experiment with Modelling — a case study from Al Majdiyya Village.

⁵² Ellison D, Morris CE, Locatelli B, Sheil D, Cohen J, Muriyarso D, et al. 2017. Trees, forests and water: Cool insights for a hot world. *Global Environmental Change*, 43: 51–61.

⁵³ Lawrence D & Vandecar K. 2015. Effects of tropical deforestation on climate and agriculture. *Nature Climate Change*, 5: 27–36.

⁵⁴ Nyberg G, Tobella AB, Kinyangi A, Ilstedt U. 2012. Soil property changes over a 120-year chronosequence from forest to agriculture in western Kenya. *Hydrological Earth System Science*, 16: 2085–2094.

⁵⁵ Ellison D, Morris CE, Locatelli B, Sheil D, Cohen J, Muriyarso D, et al. 2017. Trees, forests and water: Cool insights for a hot world. *Global Environmental Change*, 43: 51–61.

⁵⁶ Vasquez O, Khraishy M (eds). 2016. Market overview and guide to Jordanian market requirements. USDA Foreign Agricultural Service. Amman.

⁵⁷ Al-Bakri J, Suleiman A, Abdulla F & Ayad J. 2011. Potential impact of climate change on rainfed agriculture of a semi-arid basin in Jordan. *Physics and chemistry of the Earth*, 35: 125–134.

reduction in precipitation of 20% (by 2055 under RCP4.5) will result in a decline in expected yield for rainfed barley by 35%, and for wheat by 21%⁵⁸. The total yield for irrigated agriculture is expected to decrease by 5% by 2050⁵⁹. In addition to the impacts on crop agriculture, climate change will have a considerable impact on livestock production because of reduced cultivated fodder and rangeland vegetation. Declines in barley yields and the fertility of pasturelands as a result of climate change have already led to a 77% shortage in livestock feed⁶⁰. During the severe drought of 1998–2012, the 70% reduction in rainfall resulted in a decline in cereal yields to only 0.6% of the country's needs and non-cereal crop production decreased by 23%, while meat and dairy production declined by 40%⁶¹. Economic losses from inadequate water supply are estimated to cost Jordan ~0.34% of its GDP each year⁶². Elevated temperatures and reduced rainfall will also increase the demand for water for irrigation of crops by 14% and 28% for 2030 and 2050 respectively, placing additional stress on Jordan's water supply⁶³.

32. Under future climate change conditions, increasingly variable and decreasing rainfall, as well as more severe droughts, will not only limit the optimal time available for planting crops, but also the types of crops that can be grown. For example, a reduction in precipitation of 20% (by 2055 under RCP4.5) will result in a decline in expected yield for rainfed barley by 35%, and for wheat by 21%⁶⁴. By 2050, the combined effect of a 20% reduction in rainfall and an increase in temperature by 4°C is predicted to decrease barley production by 50%⁶⁵. An increase in temperature is also predicted to lower the yields of irrigated crops. At 1°C above current values, irrigated vegetable yields in Jordan are expected to decrease by 5%. This decrease is predicted to double to 10% by 2050 with an increase in temperature of 2°C. Elevated temperatures and reduced rainfall will also increase the demand for water for irrigation of crops by 14% and 28% for 2030 and 2050 respectively, placing additional stress on Jordan's water supply⁶⁶. This stress will be compounded by reductions in overall water availability. Therefore, there is a need to not only strengthen the supply of water resources in Jordan, but also transition food systems from those reliant on climate-vulnerable crops like barely to those which are more diverse and resilient to climate change, such as quinoa and pearl millet, which contribute to food security under climate change conditions.

Vulnerability of local populations

33. Vulnerability in Jordan is largely driven by income levels and poverty rates. Unemployment levels⁶⁷ in Jordan are high, ranking 17th highest in the world in 2019 with ~14.9% of the total labour force unemployed⁶⁸. High unemployment is attributable to a slowing economy and the refugee influx, particularly into urban centres⁶⁹. Approximately 90% of registered Syrian refugees in Jordan that live outside of refugee camps have settled in the governorates of Amman, Mafrqa, Irbid, and Zarqa⁷⁰. Since the beginning of the Syrian conflicts in 2011, these governorates — as well as the small northern governorate of Jerash — have experienced the highest growth in the unemployment rate nationally⁷¹. The rate in these areas has increased by ~3% since 2011, compared with a ~1% decrease elsewhere in Jordan, indicating how Syrian refugees compete with Jordanians in the labour market⁷².
34. Women and youths have the highest respective unemployment rates, with unemployed women particularly reflecting the greatest proportion of unemployed people in Jordan⁷³. Employment amongst women is extremely low

⁵⁸ Ibid.

⁵⁹ Al-Bakri JT, Salahat M, Suleiman A, Suifan M, Hamdan MR, Khresat S & Kandakji T. 2013. Impact of climate and land use changes on water and food security in Jordan: Implications for transcending 'The Tragedy of the Commons'. *Sustainability*, 5: 724–748.

⁶⁰ USAID. 2016. Climate change risk profile: Jordan.

⁶¹ FAO. 1999. Worst drought in decades decimates cereal crops in Jordan. Available at: <http://www.fao.org/NEWS/GLOBAL/GW9907-e.htm>.

⁶² World Bank (ed). 2018. Beyond scarcity. Water security in the Middle East and North Africa. The World Bank Washington.

⁶³ Ibid.

⁶⁴ Al-Bakri J, Suleiman A, Abdulla F & Ayad J. 2011. Potential impact of climate change on rainfed agriculture of a semi-arid basin in Jordan. *Physics and chemistry of the Earth*, 35: 125–134.

⁶⁵ Al-Bakri JT, Salahat M, Suleiman A, Suifan M, Hamdan MR, Khresat S & Kandakji T. 2013. Impact of climate and land use changes on water and food security in Jordan: Implications for transcending 'The Tragedy of the Commons'. *Sustainability*, 5: 724–748.

⁶⁶ Ibid.

⁶⁷ Using the narrow definition, unemployment level refers to the share of a country's labour force that is without work but is available for and/or seeking employment.

⁶⁸ World Bank. 2019. Unemployment, total (% of total labour force).

⁶⁹ Amjad R, Aslan J, Borgnäs E, Chandran D, Clark E, Ferreira dos Passos A, Joo J, Mohajer O (eds). 2017. *Examining barriers to workforce inclusion of Syrian refugees in Jordan*. International Labour Office. Geneva.

⁷⁰ Appendino M, Cerisola M, Martin E, Salman F, Tiffin A, Morra P, Sobrinho N, Wagner H (eds). 2017. *Jordan. Selected issues*. International Monetary Fund. Washington, DC.

⁷¹ Ibid.

⁷² Ibid.

⁷³ World Bank, 2017. Jordan's Economic Outlook – April 2018. Available at: <https://www.worldbank.org/en/country/jordan/publication/economic-outlook-april-2018>.

— only ~13% of women over 15 years of age are included in the paid labour force⁷⁴. In the Global Gender Gap Index of 2015, which provides information on labour market participation, Jordan ranked 140th out of 145 analysed countries⁷⁵. Youth unemployment is also high in Jordan, particularly among those holding university degrees⁷⁶. The male-female divide in labour market participation is further evident in women's low levels of employment, despite elevated levels of education⁷⁷. While more than 75% of unemployed men in Jordan hold a high school diploma at most, the equivalent proportion of unemployed women hold at least a high school diploma, and many also hold tertiary qualifications⁷⁸. Unemployment and insufficient dependable income intersect and contribute to poverty in Jordan, where the proportion of the population below the poverty line⁷⁹ was ~14% in 2010, half of whom were 'working poor'^{80,81,82}. Like unemployment, poverty in Jordan is concentrated in the governorates that have historically hosted the largest numbers of refugees — with the highest number of poor households occurring in the urban areas of the governorates of Amman, Irbid, Zarqa and Mafrq⁸³. Across the project's sites of Yarmouk, King Talal Dam and Kafraïn, poverty rates are 15%, 18% and 13%, respectively. Approximately 13% of the total population lives in rural areas, where poverty is more prevalent than in urban areas and living conditions are difficult. As an arid country with less than 5% of land available for farming and little to no rainfall, prospects of self-sustainability through farming are limited. The most vulnerable groups include large rural households with low literacy and education levels, women-headed households, households 12% with sick or elderly people, and households that do not own land or have very little land.

35. The project target area is located within the Jordan Valley — the primary agricultural region of the country. The majority of local communities in the region depend on agriculture for their livelihoods, 25% of whom live below the poverty line⁸⁴. This dependence on agriculture combined with local poverty rates and low adaptive capacity, make these communities and the food systems (barley- and wheat-based) they rely on for food security highly vulnerable to the impacts of climate change, particularly reductions in water resources. As agricultural activities are vulnerable to temperature fluctuations, increased exposure to droughts and greater precipitation variability, crop productivity in the Jordan Valley is directly affected by climate change. Under future climate conditions, decreased agricultural yields will continue to reduce food security and the income potential for local communities who already face high poverty rates⁸⁵ without a transition to climate-resilient food systems.
36. Future climate change will additionally present multiple health risks to Jordan's most vulnerable inhabitants, including children, women and refugees. Climate change impacts on health include an increase in the incidence of heat stress-related illnesses and exposure to diseases from reduced water quality⁸⁶. Also, the health of the population is adversely impacted by reduced food security related to lower agricultural yields⁸⁷. As yields decline, the total quantity of food available in the country decreases, causing food prices to fluctuate. Consequently, higher prices lead to a reduction in the diversity of food that households can afford, causing malnutrition⁸⁸.

Institutional context

37. Within Jordan, institutions of relevance to integrated landscape and water resources management include *inter alia* the: i) Ministry of Environment (MoEnv) which is responsible for the country's legal and policy framework relating to environmental issues; and ii) Ministry of Water and Irrigation (MWI), which is responsible for overseeing the water sector, water supply and wastewater system. The MWI's mandate includes the formulation of national water

⁷⁴ Josua M (ed). 2016. *Jordan at a Glance: History, Politics, Economy*. Bundeszentrale für politische Bildung. Bonn.

⁷⁵ Ibid.

⁷⁶ Ibid.

⁷⁷ The Hashemite Kingdom of Jordan (ed). 2011. *Jordan's National Employment Strategy 2011-2020*. Ministry of Planning and International Cooperation. Amman.

⁷⁸ Ibid.

⁷⁹ In Jordan, the poverty line was JD.814 (US\$1,147) per year in 2010.

⁸⁰ The working poor are working people whose incomes fall below a given poverty line as a result of insufficient work hours and/or low wages.

⁸¹ World Bank, 2018. The World Bank in Jordan – Data. Available at: <https://data.worldbank.org/country/jordan>.

⁸² The Hashemite Kingdom of Jordan (ed). 2013b. *Needs Assessment Review of the Impact of the Syrian Crisis on Jordan*. Ministry of Planning and International Cooperation. Amman.

⁸³ Ibid.

⁸⁴ Sidahmed, A.E., Rabboh, W.A., Khresat, S., and Karablieh, E., 2012. Assessment of the Agricultural Sector in Jordan.

⁸⁵ Poverty rate is the percentage of people whose income falls below the poverty line, which is taken as half the median household income of the total population. A higher poverty rate indicates a population with a higher proportion of individuals below the poverty line.

⁸⁶ USAID. 2016. Climate change risk profile: Jordan.

⁸⁷ Approximately 25% of those living below the poverty line in Jordan rely on agriculture for subsistence and as a source of income. Although food is currently still readily available in Jordan, the slowing economy and declining agricultural productivity are increasingly threatening food security for many households. The average Jordanian family already spends ~40% of its income on food.

⁸⁸ Bawadi HA, Tayyem RF, Dwairy AN & Al-Akour N. 2012. Prevalence of food insecurity among women in northern Jordan. *Journal of Health, Population, and Nutrition*, 30: 49.

strategies and policies, research and development, information system management as well as the provision, standardisation and consolidation of centralised water-related data⁸⁹. The MWI operates through two independent entities under its purview, namely the: i) Water Authority of Jordan (WAJ) — which is responsible for the operational functions of all Jordan's water and sewage systems for both municipal and industrial users; and ii) Jordan Valley Authority (JVA) — which is responsible for the socio-economic development of the Jordan Rift Valley, including matters relating to water and irrigation⁹⁰. Additional details on these institutions are presented in Annex 2: Feasibility Study.

Organisational challenges

38. For the water sector and MWI, there is a growing need for a more consolidated, harmonized and coordinated water management organization, backed by a strong legal/regulatory framework especially when the IWRM approach, supported by key management instruments, is being adopted and reinforced to optimize water resource use. A harmonized national structure needs to be empowered to implement operational plans for water resourcing, distribution and delivery systems; mobilize public/private capital for expansion and improvement of infrastructure; sustainably recover operating and maintenance costs; protect the quality of water resources and water-dependent ecosystems; and protect the rights of consumers. The national water sector would require a reworked governance strategy and institutional framework that rationalizes, consolidates and reorganizes the core governance functions for: (i) national policy and planning; (ii) management of operations and quality assurance; (iii) delivery of sustainable sector services; (iv) regulatory and normative functions; and (v) sector coordination and client interface. This in turn requires a revised organizational structure and regulatory framework that factors in the new realities facing the sector⁹¹.

Related projects/interventions

39. Considerable baseline investments are being made through public expenditure and donor-funded initiatives to increase the resilience of Jordan's vulnerable communities to the impacts of climate change. Lessons learned and best practices from these investments have been incorporated into the design of the proposed project to replicate successful techniques and ensure that there is complementarity between the project and existing actions.

40. During the development of JILMI's Funding Proposal package, the Funding Proposal for another project in the Hashemite Kingdom of Jordan was approved for funding by the GCF board. The project, titled "Building resilience to cope with climate change through improving water use efficiency in the agriculture sector (BRCCJ)" or FP155⁹², will be executed jointly by the Food and Agriculture Organisation (FAO), United Nations Development Programme (UNDP), Ministry of Environment (MoEnv) and Ministry of Water and Irrigation (MWI). BRCCJ's goal is to increase the climate resilience of water management systems and the farming community in four governorates located in the Dead Sea Basin; Karak, Madaba, Talifah and Ma'an. To achieve this the project has been designed to address the lack of infrastructure, limited capacity of rural farming households and weak institutional capacity through three components. These include: Component 1 – Climate resilient water systems for enhanced water security; Component 2 – Climate change resilience for enhanced livelihoods and food security; and Component 3 – Scaling up climate adaptation into policy and across actors (institutions, private sector and civil society). While the BRCCJ project focusses on addressing climate change impacts in the agricultural sector through improved water use efficiency at the smallholder level, the proposed project will use a novel broader, integrated approach at the landscape level to address the climate change impacts of rising temperatures, droughts and floods not only on agriculture but on communities as a whole. Table 1 provides a summary of the key similarities and differences, while Table 2 provides a summary of complementarities between the two projects. A more detailed comparison is provided in Annex 29: Analysis of similarities, differences and complementarities between JILMI and BRCCJ (FP155). One approved, JILMI's Accredited Entity (AE) and Executing Entities (EEs) will coordinate with their BRCCJ counterparts to ensure that the two projects work closely together during implementation to foster alignment and a collective contribution towards increasing climate-resilient sustainable development in Jordan.

Table 1. Summary of key similarities and differences between JILMI and BRCCJ.

Project Aspect	Jordan Integrated Landscape Management initiative (JILMI)	Building resilience to cope with climate change through improving water use efficiency in the agriculture sector (BRCCJ) – FP155
Theme	Integrated land and water resources management (ILWRM)	Water use efficiency in the agriculture sector

⁸⁹ The Hashemite Kingdom of Jordan. Nd. Ministry of Water and Irrigation Profile. Available:

<https://jordan.gov.jo/wps/portal/Home/GovernmentEntities/Ministries/Ministry/Ministry%20of%20Water%20and%20Irrigation?na meEntity=Ministry%20of%20Water%20and%20Irrigation&entityType=ministry>

⁹⁰ Ibid.

⁹¹ Source: https://thewaternetwork.com/_/water-in-the-middle-east/article-FfV/a-summary-of-jordan-national-water-strategy-2016-2025-NtvJ5wUAZVqd1oPOqsorg

⁹² GCF. 2021. Building resilience to cope with climate change in Jordan through improving water use efficiency in the agriculture sector. Available at: <https://www.greenclimate.fund/project/fp155#details>

Project Aspect	Jordan Integrated Landscape Management initiative (JILMI)	Building resilience to cope with climate change through improving water use efficiency in the agriculture sector (BRCCJ) – FP155
Target area ⁹³	Governates of Irbid and Balqa in the northern Jordan Valley	Governates of Karak, Madaba, Talifa and Maan in the Dead Sea Basin
Goal	IF investments are made in sustainable land management, water storage and the protection of surface and groundwater resources and associated policies, regulations and capacities strengthened, THEN water security will be enhanced, BECAUSE a strengthened enabling environment and technology adoption reduces exposure and vulnerability to temperature increases and droughts	IF investments are made in water efficient infrastructure, farmer's knowledge of water saving and conserving technologies and awareness of climate adaptive technologies and practices and Government capacity in planning, information dissemination and policy development THEN vulnerable communities will be able to increase their resilience and enhance their livelihoods and improve their health, well-being and food security BECAUSE of their adoption of new practices and government
Approach	The objective is to reduce the negative impacts of climate change on water resources in Jordan through a landscape-level approach. The proposed approach includes: i) improved ecosystem management; ii) managed aquifer recharge; iii) rainwater harvesting; iv) evaporation reduction; v) improved agricultural practices; and vi) strengthened governance. The approach will be implemented through four components: <u>Component 1:</u> Climate change adaptation governance in Jordan <u>Component 2:</u> ILWRM under conditions of climate change <u>Component 3:</u> Improved protection, supply and use of water resources under conditions of climate change <u>Component 4:</u> Knowledge management and awareness on climate change adaptation	The objective is to increase the resilience to climate change of water management systems and the farming community. The project is designed to deal with a lack of infrastructure, limited capacity of rural farming households and weak institutional capacity through three components: <u>Component 1:</u> Climate resilient water systems for enhanced water security <u>Component 2:</u> Climate change resilience for enhanced livelihoods and food security <u>Component 3:</u> Scaling-up climate adaptation into policy and across actors
Outcomes / results	<u>Outcome 1:</u> Increased public budgets for adaptation <u>Outcome 2:</u> Climate-resilient ILWRM practiced in three Jordan Valley watersheds <u>Outcome 3:</u> Enhanced water security for households in target areas <u>Outcome 4:</u> Household behavioural change achieved through improved information and outreach	<u>Expected result 1:</u> Enhanced water availability to address climate change risks <u>Expected result 2:</u> Enhanced capacity of people to deal with climate change risks <u>Expected result 3:</u> Gender resilience tools and practices to adapt to climate change are mainstreamed into national policy, educational, administrative and social frameworks.
Results areas/Fund-level impacts	A1.0 Increased resilience and enhanced livelihoods of the most vulnerable people, communities and regions A2.0 Increased resilience of health and well-being, and food and water security A4.0 Improved resilience of ecosystems and ecosystem services	A1.0 Increased resilience and enhanced livelihoods of the most vulnerable people, communities and regions A2.0 Increased resilience of health and well-being, and food and water security

Table 2. Component-level analysis of similarities differences and complementarities between JILMI and BRCCJ.

BRCCJ Components	Similarities	Differences	Complementarities
Component 1: Climate resilient water systems for enhanced water security	- Component 3 of JILMI focuses on improving the protection supply and use of water resources under conditions of climate change to strengthen water security. - Both projects include rooftop rainwater harvesting as an activity.	- Different target sites. Component 1 of BRCCJ focuses on improving water availability at the farming community level in the Dead Sea Basin, while JILMI's interventions under its Component 3 focus on the northern Jordan Valley and will benefit multiple stakeholder groups. - While Component 1 of BRCCJ focuses on increasing the water availability for specific target farming communities, JILMI's Component 3 is focused on attenuating water resources at a landscape level benefitting multiple stakeholder groups. - BRCCJ includes the use of water-saving domestic devices and reclaimed water, while JILMI includes managed aquifer recharge (MAR), the rehabilitation of freshwater springs and floating solar panels for evaporation reduction. - JILMI includes, under Component 2, ecosystem restoration to increase water supplies at the landscape level. This is complemented by the implementation of climate-resilient watershed management plans in three watersheds of the northern Jordan Valley.	- As the two projects are being implemented in two different target areas their various interventions aimed at enhancing water security under climate change conditions are complementary, as together, their combined impact on climate resilience is greater, i.e., they have an additive impact on reducing Jordan's current and future water deficit. - With BRCCJ's interventions targeted at increasing water security focused primarily on the agriculture sector in Jordan, it provides a demonstration of interventions and impacts which could be included as part of a climate-resilient ILWRM approach being promoted by JILMI. - As the two projects are largely using different approaches to improve water security, they provide an opportunity for demonstrating their effectiveness under different conditions (different sites) in Jordan, providing a greater suite of potential adaptation solutions that can be replicated and upscaled in the future.

⁹³ A map showing the location of the two projects is presented in Annex 29.

BRCCJ Components	Similarities	Differences	Complementarities
		JILMI will be implementing in-field water conservation and harvesting measures through Component 2 to improve production on smallholder farms.	
Component 2: Climate change resilience for enhanced livelihoods and food security	Through JILMI's Component 2, climate-resilient additional livelihoods and agricultural practices will be introduced to reduce the dependency of beneficiary communities on climate-vulnerable livelihoods that are heavily reliant on water resources and responsible for ecosystem degradation.	- Different target sites. BRCCJ is targeting the Dead Sea Basin area (Madaba, Karak, Tafilah and Ma'an governorates), while JILMI is targeting the northern Jordan Valley (Balqa and Irbid governorates). - While intended to enhance the adaptive capacity of households under conditions of climate change, the climate-resilient livelihoods introduced through JILMI are also intended to mitigate baseline drivers of ecosystem degradation, thereby supporting restoration interventions and provide immediate tangible benefits for target communities while they wait for those related to restoration to accrue. While the activities will contribute to local food security, this is a co-benefit of the livelihoods. - Livelihood interventions under BRCCJ are only targeting farming households, while those related to JILMI will target all households around ecosystem restoration sites (including farming households and those dependent on other livelihood activities).	- As the projects are being implemented at different sites their livelihood-related activities are complementary as together, they are increasing the climate resilience of a broader group of beneficiaries. - With the contexts and conditions at BRCCJ's sites in the Dead Sea Basin and JILMI's sites in the northern Jordan Valley being different, the two projects will provide examples of different approaches to improving the climate-resilience of livelihoods, informing replication and upscaling across the rest of Jordan. - JILMI's investments into climate-resilient agriculture on smallholder farms can draw from the capacity, best practices and lessons learned through BRCCJ's climate-resilient agricultural interventions.
Component 3: Scaling-up climate adaptation into policy and across actors (institutions, private sector, civil society).	- Both projects have identified the lack of integration of climate change adaptation across sectors as a barrier in Jordan. - Component 1 of JILMI focuses on strengthening climate change adaptation governance in Jordan through the strengthening of the capacity of relevant actors and integrating climate-resilient ILWRM into the country's relevant policy, planning and legal frameworks as a climate change adaptation approach.	- BRCCJ's Component 3 focuses on mainstreaming adaptive tools and practices to adapt to water scarcity into national policy and educational frameworks, as well as the administrative, economic and social frameworks of the Dead Sea Basin area. - JILMI's Component 1 focuses primarily on strengthening governance for climate-resilient ILWRM at the national level, as well as in the Jordan Valley.	With BRCCJ planning to mainstream adaptive tools and practices for water security into national policy frameworks, JILMI complements this by focusing on the strengthening of adaptation governance in Jordan through capacity building and the integration of climate-resilient ILWRM into policy.

41. In addition to BRCCJ, other relevant baseline investments will be complemented by JILMI. The most relevant of these projects are presented in Table 3 below, with further projects and information provided in Annex 2: Feasibility Study.

Table 3. Summary of the most relevant past and ongoing projects and investments that will be complemented by the proposed project.

Project name, date, fund and amount	Description and complementarities
<i>Increasing the resilience of both displaced persons and host communities to climate change-related water challenges in Jordan and Lebanon (2021–2025; AF; US\$13,968,139)</i>	This project aims to increase the response to water-related climate change vulnerability in Jordan and Lebanon within the context of the ongoing Syrian crisis. The project specifically focusses on addressing the needs of both displaced peoples (DPs) from Syria and host communities in the target countries, especially women and the youth. For Jordan, the project is targeted within the governorates of Irbid and Mafraq; the former which also includes the target area for the JILMI project. The AF project is complementary to the GCF JILMI project in several ways. Urban master plans to increase institutional capacity to manage climate change and water scarcity under the AF project aligns with Outcome 1 of the JILMI project that aims to mainstream ILWRM at the national and sub-national level. Unlike the JILMI project, this capacity building is focussed on urban settings and are at a municipal level only. Awareness raising for DPs and host communities on climate change adaptation measures will complement the proposed project's adaptation outreach programme (Activity 4.1). Awareness raising in the AF project will specifically focus on capacitating DPs and host communities on the water availability adaptation options introduced by the project, while the JILMI project's outreach programme includes a focus on overall education on climate change and adaptation. Activities 2.3 and 3.3 of the JILMI project will be complemented by the expansion of similar unconventional water harvesting and supply techniques through

Project name, date, fund and amount	Description and complementarities
	the AF projects, including rooftop rainwater harvesting, water efficient irrigation and permaculture. Although there is overlap between these aspects of the projects, the target sites are geographically removed within Irbid, with the proposed project interventions focussing on the Jordan River Valley and the AF project prioritising the eastern areas of the governorate.
<i>Healthy Ecosystems for Rangeland Development (HERD): Sustainable Rangeland Management Strategies and Practices</i> (2017–2021; GEF, US\$15,879,968) ⁹⁴	The HERD project was developed with the objective of strengthening the restoration and sustainable management of pastoral rangelands in Egypt and Jordan to enable the continued provision of ecosystem services and protection of biodiversity. The proposed project will complement HERD activities within Jordan through: i) strengthened governance and capacity for integrated landscape and management (ILWRM) under Activity 1.1; and ii) outreach programmes to raise awareness within local communities of ILWRM under Output 4. The proposed project will additionally adopt best practices for rangeland restoration and community-based management implemented by HERD. As HERD is implemented by UNEP (GCF AE) and executed by IUCN (GCF EE), potential for complementarity and avoidance of duplication has been maximised during design and will be a key focus during implementation.
<i>Jordan Water Management Initiative</i> (WMI; 2016–2021; USAID, US\$33,247,544) ⁹⁵	The objective of the WMI project is to address Jordan's growing water scarcity by increasing the accountability and sustainability of water sector management and governance. This will be accomplished by supporting and strengthening the Jordanian government's policy development and implementation, as well as its capacity-building efforts. The proposed project's strengthened governance and capacity for climate-resilient ILWRM (Outcome 1) will complement the WMI project's focus on water sector management and governance.
<i>Increasing the resilience of poor and vulnerable communities to climate change</i> (2016–2020; Adaptation Fund (AF), ~US\$9,226,000) ⁹⁶	Implemented by Jordan's Ministry of Planning and International Cooperation (MOPIC), the overall objective of this project is to increase the adaptive capacity of Jordan's agricultural sector to climate change-induced water shortages and food security stresses. The proposed project will build on existing activities aimed at reinforcing the concept of a participatory water and agricultural development and management approach that involves users, planners and policymakers at all levels (Output 1.1). Lessons learned regarding the use of non-conventional water sources have been used to inform the design of the proposed project, with some technologies being replicated. For example, Output 2.1 of the JILMI project will build on the AF project's use of water harvesting technologies through improved watershed management. The proposed project will additionally replicate and assist in expanding the use of permaculture techniques which address challenges relating to soil and irrigation water quality (Activity 2.3). An important gap in the AF project is that ecosystem-based adaptation (EbA) interventions were not implemented for the improvement of water resources and increasing climate resilience. The proposed project's climate-resilient ILWRM approach will fill this gap, therefore providing an alternative to conventional adaptation solutions.
<i>Jordan Watershed Restoration Project</i> (2015–2020; USDA, ~US\$180,000) ⁹⁷	Implemented by the International Centre for Agricultural Research in the Dry Areas (ICARDA), the project objective is to create vegetation islands within the Badia by planting seedlings in Vallerani ⁹⁸ structures. This will reduce the amount of rainwater lost as overland runoff and enable a 50% increase in groundwater recharge. Moreover, the vegetation islands are expected to enhance natural revegetation, improve ecosystem viability and increase the productivity of rangelands for livestock fodder. The proposed project will replicate the successful interventions of this ongoing project in its target watersheds (Output 2.1). Additionally, the proposed project will build on the work of the Jordan Watershed Restoration Project through the use of watershed restoration and management as an adaptation approach.
<i>Sustainable use of Ecosystem Services in Jordan — Energy and Climate Fund</i> (EKF-ESS; 2014–2019; GIZ, ~US\$5,420,000) ⁹⁹	The project's objective is to promote national policy-making processes that account for biodiversity and ecosystem services as well as the importance of climate change adaptation and ensure they become a permanent focus within Jordanian policies. The proposed project will complement and build on the work of the EKF-ESS by promoting climate-resilient ILWRM mainstreaming into relevant national and local policies (Activity 1.2), including a focus on climate change. Specifically, the 'Ecosystem Service Opportunities' Framework will be used and expanded upon to find appropriate instruments for ecosystem service protection based on location-specific opportunities. In addition, the proposed project will highlight the link between ecosystem services and adaptation through its climate-resilient ILWRM approach, implementing on-the-ground activities that will complement policy changes made by EKF-ESS (Activity 2.2).
<i>Managing Water Resources in Jordan</i> (2006–2018; GIZ) ¹⁰⁰	The project's objective was to enable national water institutions to better respond to the mounting pressure of the country's limited water resources by providing strategic, technical and expert support. The project has led to an improvement in the management capabilities of the MWI, the establishment of decentralised, semi-autonomous water utilities with improved capability to provide a reliable source of water, Water User Associations within the agricultural sector and a water resource modelling unit. The proposed project will build on this project by promoting integrated watershed management and establishing Watershed Management Committees (WMCs) that will contribute to providing alternative sources of water to local communities.

⁹⁴ GEF. 2019. Healthy Ecosystems for Rangeland Development (HERD): Sustainable Rangeland Management Strategies and Practices. Available at: <https://www.thegef.org/project/healthy-ecosystems-rangeland-development-herd-sustainable-rangeland-management-strategies>

⁹⁵ Melikyan L, Al Deen Al Shbool M, Matar F, Zoubi R, Fityan R & Landis J. 2018. USAID/Jordan water management initiative (WMI). INTEGRATED. <https://www.globalwaters.org/HowWeWork/Activities/water-management-initiative>.

⁹⁶ Adaptation Fund Board Project and Programme Review Committee. 2015. Proposal for Jordan. Available at: <https://www.adaptation-fund.org/project/increasing-the-resilience-of-poor-and-vulnerable-communities-to-climate-change-impacts-in-jordan-through-implementing-innovative-projects-in-water-and-agriculture-in-support-of-adaptation-to-climate-4/>.

⁹⁷ Available at: <https://mel.cgiar.org/projects/jordan-watershed-restoration#about>.

⁹⁸ Vallerani structures are furrows created by a Vallerani plow which follow contour lines and are angled to more effectively capture runoff as water flows downhill.

⁹⁹ GIZ. 2019. Sustainable use of ecosystem services in Jordan. Available at: <https://www.giz.de/en/worldwide/28658.html>.

¹⁰⁰ GIZ. 2019. Managing water resources in Jordan. Available at: <https://www.giz.de/en/worldwide/17213.html>.

Project name, date, fund and amount	Description and complementarities
<i>MENARID: Mainstreaming Sustainable Land and Water Management Practices</i> (2009–2015; GEF, ~US\$29,740,000) ¹⁰¹	This project formed part of the Middle East and North Africa Regional Program for Promoting Integrated Sustainable Land Development (MENARID). The main objective of the project was to reduce the causes and negative impacts of land degradation and promote the incorporation of an ecosystem-based approach into public-supported productive and poverty-reduction activities. The proposed project will draw on this project's information management systems, which were developed to monitor land resources and link land changes to natural processes, human interventions and project interventions when developing restoration protocols. Specifically, these information management systems will assist the JILMI project by providing a decision-support tool on SLM for governmental stakeholders (Activity 1.1). Best practices from Project Demonstration Agro-Ecosystems will be used to inform the promotion of climate-resilient livelihood options under Activity 2.3 of the JILMI project. The implementation of the proposed project's participatory integrated watershed management plans will complement and draw on this project's participatory action plans that focus on sustainable land and water management. In addition, the proposed project will build on the institutional structures and capacities enhanced through this project to promote the awareness of sustainable land and water management practices within local communities. With MoEnv (GCF EE) being one of the executing agencies of MENARID, potential for complementarity and avoidance of duplication has been maximised during design and will be a key focus during implementation.
<i>Securing Rights and Restoring Lands for Improved Livelihoods</i> (2011–2013; EU) ¹⁰²	The overall objective of this global project was to reduce poverty and promote the sustainable management and restoration of ecosystems in drylands and rangelands. Specifically, the objective of the project was the conservation, restoration and sustainable management of ecosystem services as the basis for improved livelihoods. The three-year Jordan component of the project built the capacity of local communities to manage and restore their ecosystems and improve livelihoods. The proposed project will draw on best practices and lessons learned from this project's ecosystem services valuation and rural community income generation options to inform participatory climate-resilient ILM plans as well as climate-resilient livelihood and agricultural practices (Activity 2.3). Land-management rights, including women's land rights, were negotiated by participating communities over state-held lands. The project also promoted the revival of customary institutions such as Hima practices for rangeland governance and management. Moreover, the Jordanian National Rangeland Strategy was revised to emphasise community-based approaches. The proposed project will build on these activities by drafting further amendments to Jordan's land-management policy, planning and legal frameworks (Activity 1.2).
<i>Integrated Ecosystem and Natural Resource Management in the Jordan Rift Valley</i> (2007–2013; GEF, ~US\$12.7 million) ¹⁰³	The objective of this project was to secure the ecological integrity of the Jordan Rift Valley as a globally important ecological corridor and migratory pathway for birds. Interventions included: i) implementing generally accepted principles of integrated ecosystem management relating to land use in the Jordan Valley; and ii) establishing a network of well-managed protected areas (PAs) and special conservation areas (SCAs) that meet ecological, social and economic needs. The proposed project will draw from best practices and lessons learned about integrated ecosystem management and the related socio-economic benefits during implementation. In addition, the proposed project will complement the efforts of the GEF project by using an ILWRM approach to enhance the conservation of ecosystems and their services to local communities to support adaptation to climate change. Specific complementarities include: i) increase the mandate for landscape management of major development agencies, including the JVA; ii) increasing awareness amongst beneficiary communities on sustainable ecosystem management and resilient livelihood practices; and iii) expanding options for alternative and resilient livelihoods for vulnerable communities. Although there is some potential for overlap between the two projects, the JILMI project will differ in that it has a more specific focus on climate change and adaptation. Socio-economic activities and livelihoods introduced by the JILMI project will also be different but complementary to this GEF project. Additionally, shared implementing and technical partners — including RSCN and JVA — will ensure there is minimal duplication and maximum complementarity between the projects.
<i>Community-based Optimisation of the Management of Scarce Water Resources in Agriculture in Central-West Asia and North Africa Project</i> (CWANA; 2003–2013; IFAD, AFESD, OPEC Fund for International Development and IDRC) ¹⁰⁴	This regional project consisted of two phases, with the first phase taking place from 2003–2008 and the second phase from 2010–2013. The short-term objective of this project was to develop and test water management options to increase water productivity and optimise water use within three main agro-systems in the target regions, namely irrigated, rainfed and Badia systems. Outputs 2 and 3 of the proposed project will build on best practices regarding rainwater harvesting and rangeland rehabilitation developed by CWANA's two-phased on-the-ground interventions.

¹⁰¹ The GEF Trust Fund. 2008. Mainstreaming Sustainable Land and Water Management Practices: Project Identification Form. Available at: <https://www.thegef.org/project/menarid-mainstreaming-sustainable-land-and-water-management-practices>.

¹⁰² IUCN. 2019. Securing rights and restoring lands for improved livelihoods. Available at: <https://www.iucn.org/regions/west-asia/projects/completed-projects/securing-rights-and-restoring-lands-improved-livelihoods>.

¹⁰³ World Bank. 2014. Integrated Ecosystem Management in the Jordan Rift Valley: Implementation Completion and Results Report. Available at: <http://projects.worldbank.org/P075534/integrated-ecosystem-management-jordan-rift-valley-gef?lang=en>.

¹⁰⁴ ICARDA. 2011. Water benchmarks of CWANA: Community-based Optimisation of the Management of Scarce Water Resources in Agriculture in CWANA. Second Phase 1st Annual Progress Report 2010–2011.

B.2. (a). Theory of change narrative and diagram (max. 1500 words, approximately 3 pages plus diagram)

Problem statement

42. Jordan is currently experiencing extreme climatic conditions, which contributes to its ranking as the fifth most water-stressed country in the world. Climate change is causing an increase in temperatures and a decrease in rainfall, compounding the ongoing drying trend across the country. These changes are already having and will continue to have a considerable impact on local communities, particularly through several pronounced effects on water resources. First, the drying trend resulting from climate change is reducing the country's naturally limited and over-exploited groundwater and surface water sources. The reduction in water resources under future climate conditions will exacerbate existing challenges related to water quantity and quality. Second, by increasing the severity of already extreme climatic conditions, climate change will exacerbate the degradation and desertification of Jordan's ecosystems. Many of the country's ecosystems are already threatened and fragmented by inter alia: i) the unregulated collection of fuelwood and non-timber forest products (NTFPs); ii) overgrazing; iii) pollution; iv) invasive animal and plant species; iv) illegal hunting; and v) urban and agricultural expansion. Under future climate conditions, these pressures will increase and continue to inhibit the supply of numerous ecosystem services (e.g., regulation of water flow and quantity) on which many communities rely for livelihoods, as well as for buffering the impacts of climate change. Third, a reduction in water resources will decrease the productivity of agriculture and other water-dependent livelihoods, which are already pressured by extreme baseline temperatures and water scarcity. This includes pastoralism as the extent of rangelands shrink and become less able to supply fodder for the rearing of livestock. Despite several past and ongoing interventions being implemented in Jordan to address the impacts of climate change on water resources, these have lacked an integrated approach, limiting their effectiveness at the landscape level. Furthermore, limited focus has been placed on the potential contribution of EbA to improve water resources under climate change conditions, and how, as part of an integrated approach, it can result in multiple co-benefits, which will further strengthen the climate-resilience of vulnerable communities. Problem tree and impact pathway analysis diagrams for the proposed project are presented in Section 5 of Annex 2: Feasibility Study. These diagrams map out the relationships between the causes (climatic and non-climatic) and impacts relevant to the project's preferred solution.

Target communities

43. The proposed project will target local communities and small-scale farmers, including women and female-headed households, who are particularly vulnerable to the impacts of climate change on water resources, which continue to reduce water, food and livelihood security in these communities. This increased vulnerability will be considerable in the Jordan Valley, where agriculture is the primary source of employment. Additionally, their resilience to these impacts is limited because of their inflexibility in cropping and irrigation systems. Specifically, mixed farm systems are common in the project area and comprise the least financially secure category of smallholder tenure as well as being under rental or sharecropping agreements. These systems are allocated a water quota of 550 CM/ha/yr, with drip-irrigation being the most commonly used method of water use for crops. The average: i) revenue is US\$1,050/ha/yr; ii) farm size is 2 ha; and iii) number of family workers is 7 per farm.

Preferred adaptation solution

44. For Jordan to adapt to the impacts of climate change and strengthen the climate resilience of vulnerable communities, it is necessary to secure the provision of water resources. This is achieved by shifting from a compartmentalised, sector-specific approach to landscape and water resource management to one that is integrated, participatory and adaptation-focused, and that contributes to gender-responsive climate-resilient sustainable development in the country. An integrated landscape and water resources management (ILWRM) approach in Jordan should include a strong focus on a combination of: i) improved ecosystem management; ii) interventions to improve groundwater recharge; iii) rainwater harvesting; iv) evaporation reduction; v) strengthened governance on water resources in the Jordan Valley; and vi) considers gender-related water needs and the unique challenges faced by women in terms of water use and access¹⁰⁵. The ILWRM approach to be implemented under the proposed project will incorporate these practices to enhance the climate resilience of water resources. Consequently, the productivity of agricultural and natural resource-based livelihoods (including those implemented predominantly by women) under climate change conditions — as well as domestic water supplies and ultimately the resilience of vulnerable communities — will be enhanced. The adoption of an ILWRM approach is most effective when supported by developing an enabling environment for climate-resilient ILWRM at both the governmental and non-governmental levels. The proposed adaptation solution contributes directly to the GCF's Updated Strategic Plan (2024-2027). This includes references to programming priorities: i) Mitigation and Adaptation: Supporting paradigm shifts across sectors, including to climate-resilient ecosystems and food systems; ii) Adaptation: Addressing urgent and immediate adaptation and resilience needs, for particularly vulnerable countries. Relevant GCF Targeted Results for 2024-2027 include: 4. Food: Support for developing countries that results in 190 to 280

¹⁰⁵ As identified and confirmed through consultations with key actors and communities (see Annex 6).

million beneficiaries adopting low-emission climate-resilient agricultural and fisheries practices, securing livelihoods while reconfiguring food systems; 5. Ecosystems: Support for developing countries that results in 120 million to 190 million hectares of terrestrial and marine areas conserved, restored or brought under sustainable management; and 9. 40-70 approved proposals for adaptation projects, including for locally-led adaptation action. The preferred adaptation solution is expanded on in Section 6 of Annex 2: Feasibility Study, in which further details on the causal pathways and links to the project's objectives are provided.

Barriers to the adaptation solution

45. Barrier 1: Insufficient institutional and technical coordination capacity, as well as limited mandates of state and non-state actors for climate change adaptation, including climate-resilient ILWRM¹⁰⁶. This is particularly relevant for the MoEnv and its subsidiaries — including the JVA — as well as the MWI and Ministry of Local Administration (MLA). Technical and institutional capacity shortfalls are the result of *inter alia*: i) limited coordination between and within ministries and subsidiaries, leading to conflicting goals or duplication of efforts; ii) the existence of a culture of competition for financial resources — particularly a limited national budget — between ministries; and iii) insufficient clarification of responsibility for project completion. Limitations in state and non-state actors' mandates are the result of insufficient integration of, and gaps in, relevant policy and legislature focusing on climate change adaptation. While the need for climate change adaptation has been acknowledged by the Jordanian government, institutional and technical capacity within government institutions for adaptation planning, implementation and enforcement remains insufficient.
46. In the water sector, legislation surrounding the management of water resources and associated regulations provide little support in facilitating adaptation action related to groundwater management¹⁰⁷. For example, the mandate of the JVA¹⁰⁸ to effectively implement watershed conservation is currently limited, as most of the factors influencing water supply in the valley — including surface runoff and soil erosion — occur in areas outside of the jurisdiction of the JVA¹⁰⁹. The JVA also faces a wide range of other challenges, including disregard for the water-saving policies of the Authority's water supply system by farmers. For example, water use quotas to prevent the over-exploitation of limited water supplies are assigned but largely ignored¹¹⁰. Other examples of misuse and abuse of the JVA's water supply system include *inter alia*: i) deliberate damage to water meters; ii) unauthorised adjustment of valves, flow-limiting devices and pressure regulators; and iii) improper or unlawful conduct such as providing unmerited benefits to relatives and associates, or acceptance of monetary donations for personal or political gain¹¹¹. As a result of these practices, several adverse socioeconomic impacts occur, including increased maintenance costs, loss of public funds, low agricultural production, as well as additional social costs as a result of social instability in farming communities¹¹². The loss of income from and expenditure required to address these impacts makes it difficult for the JVA to effectively implement adaptation interventions in the Jordan Valley, further increasing the vulnerability of local communities.
47. Cross-sectoral coordination and dialogue related to adaptation and watershed management are also insufficient, with limited emphasis on the interdependence between sectors such as water, energy and food. The lack of unified guiding principles at policy-level is resulting in fragmented implementation of projects relating to climate change, reducing their potential impact. Cross-sectoral coordination and dialogue are also inhibited by informal activities which are outside of the control of state actors, such as the drilling of illegal wells and the unsustainable use of water for agricultural purposes. The economic benefits, rewards and privileges related to these activities have contributed to differing opinions across sectors in terms of the most appropriate policy decisions and the allocation of resources, in addition to interference with national reforms related to adaptation. As a result, improved management and technical solutions related to the adaptation of water and other sectors are alone not enough for a coordinated response to adaptation in Jordan — with reforms towards an inclusive and coordinated approach to adaptation in Jordan therefore also required¹¹³.

¹⁰⁶ While identified as a barrier, this also constitutes a risk to project activities that require institutional effectiveness (such as Activity 1.2).

¹⁰⁷ Bany Yasin H. 2018. The Case of The Hashemite Kingdom of Jordan. In ACT Alliance (Ed.), Enhanced Climate Action in Response to 1.5°C of Global Warming. Scaling Up Nationally Determined Contributions. Available at: https://www.preventionweb.net/files/62199_actalliancereport1.5c.pdf

¹⁰⁸ The institution charged with the economic and social development of the Jordan Valley.

¹⁰⁹ Salman A. Al-Karablieh E, Regner HJ, Wolff HP & Haddadin M. 2008. Participatory irrigation water management in the Jordan Valley. *Water Policy*, 10: 305–322.

¹¹⁰ Ibid.

¹¹¹ Ibid.

¹¹² Ibid.

¹¹³ Hussein H. 2018. Tomatoes, tribes, bananas, and businessmen: An analysis of the shadow state and of the politics of water in Jordan. *Environmental Science & Policy*, 84: 170–176.

48. Barrier 2: Limited availability of and access to climate change adaptation information and data. The constrained access of decision-makers to relevant tools, knowledge and advice to make particularly tough decisions related to climate change is one of the most considerable barriers to adaptation in Jordan. While there have been advancements in recent years, such as the establishment of a Drought Management Unit by MWI and NARC to provide information on drought management, including adaptation interventions, availability of and access to climate change adaptation information and data remains constrained. Increasing the resilience of communities in the country is currently inhibited by the insufficient availability of technological information on adaptation and sustainable development in general¹¹⁴. The current gaps in information impede the ability of decision-makers to effectively integrate climate change adaptation into development strategies and planning, particularly in the water and agriculture sectors.
49. Barrier 3: Insufficient awareness by politicians, community leaders, decision-makers and the public of the impacts of climate change and appropriate adaptation responses. This relates to the limited awareness of policymakers, main institutions and the public of the biophysical, humanitarian, economic and social costs of climate change, as well as the benefits of adaptation actions. While awareness of climate change of politicians within Jordan's government is increasing, the perception that poverty and unemployment are the primary challenges to meeting the country's Sustainable Development Goals (SDGs) remains¹¹⁵. Although recent steps, such as the establishment of a national body in response to recent floods, have been taken, these remain reactive and political leadership towards the improved management of water resources under climate change conditions is lacking. At the public level, awareness on water scarcity, drought and climate change adaptation is moderate and limited particularly in the water sector¹¹⁶.
50. Barrier 4: Insufficient financial capacity for climate change adaptation. No formal mechanism for the integration of financial requirements for climate change adaptation into national budgets is current available. A considerable funding gap exists between the country's adaptation goals and its public finances. Effective climate change adaptation in Jordan is challenging and remains highly conditional on the availability of financing¹¹⁷. While the country has acknowledged the need to commit funds to climate change adaptation and mitigation, there is no formal mechanism for the integration of these requirements into the national budget, which along with the current deficit in domestic funding impede the implementation of necessary activities^{118,119}. This is compounded by a lack of access to finance for rural communities and local groups to support climate change adaptation at the local and household levels¹²⁰.
51. Barrier 5: Limited awareness amongst decision-makers (including MWI and MoA) and farmers of the financial and economic effectiveness of dryland-adapted water management, including the management of green water, in Jordan as opposed to unsustainable abstraction of groundwater. While considerable work has been done towards the improvement, adaptation and integration of dryland-adapted water management (such as the management of green water, rainwater harvesting and other agro-ecological approaches for conserving soil moisture) into landscape management practices — particularly within the agricultural sector — its implementation in Jordan has been limited¹²¹. This is compounded by a lack of knowledge on the relationship between land degradation and dryland-adapted water management. Currently, local actors (such as WUAs in the agriculture sector, Development Advisory Committees and agricultural cooperatives) are only experimenting with dryland-adapted water management approaches such as rainwater harvesting through grants provided by international organisations and non-government organisations (NGOs)¹²². As a result, access to dryland-adapted water management technologies is restricted, limiting their uptake¹²³. Systemic challenges are also preventing the expansion of the dryland-adapted

¹¹⁴ DHI Hungary. 2017. Water and Climate Change Practice Framework Assessment Report, Jordan. Available at: http://watersum.rec.org/uploads/library/Books%20and%20Documents-%20Draft/Assessment%20Report_CCH_JORDAN.pdf

¹¹⁵ GoJ. 2014. Jordan's Third National Communication on Climate Change.

¹¹⁶ Bany Yasin H. 2018. The Case of The Hashemite Kingdom of Jordan. In ACT Alliance (Ed.), Enhanced Climate Action in Response to 1.5°C of Global Warming. Scaling Up Nationally Determined Contributions. Available at: https://www.preventionweb.net/files/62199_actalliancereport1.5c.pdf

¹¹⁷ Ibid.

¹¹⁸ Ibid.

¹¹⁹ Ministry of Foreign Affairs of the Netherlands, 2018. Climate Change Profile. Jordan.

¹²⁰ <https://carnegieendowment.org/sada/89628>

¹²¹ Sixt GN, Klerkx L & Griffin TS. 2018. Transitions in water harvesting practices in Jordan's rainfed agricultural systems: Systemic problems and blocking mechanisms in an emerging technological innovation system. *Environmental Science & Policy*, 84: 235–249.

¹²² Lahn G, Grafham O & Elsayed Sparr A. 2016. Refugees and Energy Resilience in Jordan. Available at: <https://www.chathamhouse.org/sites/default/files/publications/research/2016-08-03-refugees-energy-jordan-lahn-grafham-sparr.pdf>

¹²³ Bany Yasin H. 2018. The Case of The Hashemite Kingdom of Jordan. In ACT Alliance (Ed.), Enhanced Climate Action in Response to 1.5°C of Global Warming. Scaling Up Nationally Determined Contributions. Available at: https://www.preventionweb.net/files/62199_actalliancereport1.5c.pdf

water management sector in Jordan. For example, subsidies for groundwater abstraction are causing overexploitation of the resource because of the perception that short-term benefits of this practice are greater than the long-term costs¹²⁴.

52. Barrier 6: Gender inequality. The socio-economic situation for women in Jordan is categorized by high unemployment rates, economic inequality, political marginalization of youth and women, and a mismatch between educational skills and skills sought by the labor market. Inequalities remain in participation in economic and political spheres, as well as in legal rights related to marriage, inheritance, divorce and employment. There are indications that, within society, boys are preferred to girls and decision-making within the home is largely in the hands of husbands and male family members. Structural and systemic gender inequality is a major obstacle impeding women's empowerment in Jordan, preventing women from exercising their right to lead a full life in both public and private spheres. Men control access to resources in areas where they own the key productive assets such as land, livestock and medium to large businesses. The capacity of women to participate in and implement climate change adaptation activities is also reduced by a lack of access to finance, along with their high illiteracy, unemployment and poverty rates in Jordan.

Project approach to overcoming barriers to adaptation

53. The proposed project will overcome the abovementioned barriers and consequently enhance the climate resilience of Jordan's water sector through four interconnected outputs, described in detail in Section B.3. **Error! Reference source not found.** articulates which project activities and sub-activities have been designed specifically to address the described barriers. The project's theory of change diagram is presented in Figure 7 below.
54. Barrier 1 will be addressed by increasing the institutional and technical capacity of relevant actors (for example, MoEnv, MoA, JVA, and MLA, and other major institutions or bodies) in a gender-responsive manner to plan and implement climate-resilient ILWRM. Capacity-building interventions will include developing and implementing a gender-inclusive training programme for staff members and assisting them to develop gender-responsive participatory climate-resilient watershed management plans for the target watersheds in the Jordan Valley. This will be complemented by building institutional capacity at the local level. The mandate of these institutions for climate-resilient ILWRM in the Jordan Valley will be strengthened via the integration of ILWRM into Jordan's land policy, planning and legal frameworks. This will include defining roles, responsibilities and jurisdictions at the institutional level, as well as mainstreaming adaptation into national budget lines, which additionally contributes addressing Barrier 4.
55. Barrier 1 will additionally be overcome through the establishment of gender-inclusive WMCs within the Jordan Valley to ensure collaboration and dialogue between all stakeholders (including local communities) involved in watershed management. Members of these WMCs will receive gender-responsive technical training, enabling them to collaborate directly with the local governmental institutions and municipalities. This will be supported by the development and implementation of participatory climate-resilient integrated watershed management (IWM) plans. The plans will define the roles and responsibilities of stakeholders and provide a coordinated direction on all future interventions within the watersheds. These plans will be complemented by the establishment of a Water Fund to strengthen and maintain the restoration of upstream ecosystems to ensure the sustained supply of services under climate change conditions — most notably the supply of water to downstream users (more information is presented under the description of Activity 2.1.5 in Section B.3. and Section 8.5 of the Feasibility Study). The Water Fund will serve as an additional platform through which the collaborative management of watersheds and water resources takes place, ensuring that all stakeholders play a role and benefit. Moreover, the Water Fund will be a mechanism through which finances are channelled for the maintenance and upscaling of upstream ecosystem management interventions to attenuate water resource supplies, as part of an EbA approach, which will contribute to addressing Barrier 4.
56. Capacity-building activities will be supported by the development of knowledge-management tools and the establishment of a national climate change adaptation knowledge repository. As a result, access to best practices and lessons learned on adaptation will be increased, thereby addressing Barrier 2. This will facilitate both the planning and implementation of adaptation solutions, including EbA and ILWRM.
57. To improve the awareness of decision-makers and the public of the impacts of climate change and appropriate adaptation responses (Barrier 3), a gender-inclusive adaptation outreach programme will be implemented under the proposed project. This will include both national and site-specific awareness-raising activities and will be

¹²⁴ Sixt GN, Klerkx L & Griffin TS. 2018. Transitions in water harvesting practices in Jordan's rainfed agricultural systems: Systemic problems and blocking mechanisms in an emerging technological innovation system. *Environmental Science & Policy*, 84: 235–249.

championed by a Jordanian public figure to stimulate interest and support from women, men and youth. At the policymaking and planning level, the project will generate and share evidence of the economic benefits of ILWRM to project beneficiaries. Within the Jordan Valley, community centres — including mosques and churches — will be equipped in a gender-responsive manner to serve as climate change vulnerability and adaptation awareness-raising hubs, while capacity-building activities targeting decision-makers will also be used to enhance their awareness. In addition, local media platforms within project target communities — such as television and radio — will be used to further promote awareness among local people.

58. While traditional methods of the harvesting of rainwater have historically been widely practised in Jordan, the use of effective, modern techniques is currently limited. To increase the uptake of rainwater harvesting technologies in Jordan (Barrier 5), local communities at project sites, including smallholder farmers, will be trained on the installation, use and maintenance of systems for the harvesting of rainwater. This training, along with the installation of appropriate technologies at community centres and local institutions, will build awareness for and provide a practical demonstration of the benefits of rainwater harvesting for local communities. Complementing this will be activities aimed at generating economic evidence on the adaptation benefits of climate-resilient ILWRM for use by policy- and decision-makers.
59. Gender inequality (Barrier 6) will be addressed by raising awareness on the different vulnerabilities to and impacts of climate change on men and women. This will also be addressed through the livelihoods- and water supply and management-based activities, where the specific needs of female beneficiaries related to vulnerabilities and climate change adaptation are emphasised during implementation. The project will support the establishment of Watershed Management Committees (WMCs), local decision-making bodies focused on the climate-resilient management of watersheds and their resources. This is an opportunity to improve the representation and participation of women in the management of their local watersheds and associated resources to support climate change adaptation in their communities. In terms of access to finance, the on-granting facility and the Water Fund's grant mechanism will provide an opportunity for women to access grant finance to the support the implementation of gender-responsive ILWRM initiatives, empowering them to contribute to climate change adaptation in their own communities that directly benefits local women. All proposals submitted for consideration for funding by the on-granting facility and the Water Fund grant mechanism will need to be gender responsive, promoting the inclusion of women, considering their roles in society and have activities specifically tailored for women and align with the project's gender action plan (GAP). This will be complemented by the introduction of gender-responsive climate-resilient livelihoods that will specifically target female-headed households. In addition, local communities, including women, will be empowered to monitor and manage local water supplies more effectively. This will include developing gender-responsive community-based water use monitoring and management plans for project beneficiary communities; implementing a gender-inclusive outreach campaign to train community members to monitor and manage the use of clean water (100% female representation); and promoting additional livelihoods related to the monitoring and management of clean water (including at least 50% female beneficiaries). Please in additional information in Annex 8: Gender Assessment and Action Plan.
60. Together, the abovementioned interventions will overcome the barriers to climate change adaptation in Jordan and thereby catalyse a paradigm shift towards climate-resilient sustainable development through the adopting of an integrated approach to land and water resources management. This shift will ultimately reduce the impact of climate change on local communities by improving the provision and management of ecosystem services, with a focus on water resources, and strengthening the adaptive capacity of vulnerable communities.

Table 4. Project activities and sub-activities mapped against the barriers they are designed to address.

Barrier	Project activity	Project sub-activity
Insufficient institutional and technical coordination capacity, as well as limited mandates of state and non-state actors for climate change adaptation, including climate-resilient ILWRM	Activity 1.1.1 Strengthen the mandate and capacity of main government institutions for climate-resilient ILWRM in Jordan	Sub-activity 1.1.1.1 Develop land use guidelines, standards and protocols for climate-resilient ILWRM in the Jordan Valley Sub-activity 1.1.1.2 Train staff from relevant national and subnational institutions to use climate data, and the results of climate analyses and risk assessments in policy- and decision-making processes, as well as planning and disaster risk management
	Activity 1.1.2. Draft amendments to Jordan's policy, planning and legal frameworks for agriculture, land, water and natural resources, to mainstream climate-resilient ILWRM as a climate change adaptation approach	Sub-activity 1.1.2.1. Train policy- and decision-makers on the integration of climate-resilient ILWRM practices into policies, plans and strategies Sub-activity 1.1.2.2. Train relevant national and sub-national planners on climate-resilient ILWRM planning
	Activity 1.1.4. Strengthen the mandate and capacity of non-state actors for climate-resilient ILWRM in the Jordan Valley	Sub-activity 1.1.4.1. Establish WMCs, under the leadership of the JVA and local governors, at each of the three target watersheds to facilitate dialogue and participatory watershed management between upstream and downstream stakeholders Sub-activity 1.1.4.2. Develop a training programme on climate-resilient integrated watershed management, including EbA, targeting the WMC members
		Sub-activity 1.1.4.3. Train relevant watershed stakeholders (WMC members) on climate-resilient ILWRM and its economic benefits based on the programme developed under Sub-activity 1.4.2 Sub-activity 1.1.4.4. Train staff from relevant sub-national institutions to use climate data, and the results of climate analyses and risk assessments in policy- and decision-making processes, as well as planning and disaster risk management Sub-activity 1.1.4.5. Develop participatory climate-resilient integrated watershed management plans for the three target watersheds
Limited availability of and access to climate change adaptation information and data	Activity 4.1.2. Improve knowledge management for climate change adaptation in Jordan to support national and sub-national decision-making and build communities of practice	Sub-activity 4.1.2.1. Knowledge-management tools developed to facilitate sectoral and cross-sectoral learning and mainstreaming of climate change adaptation approaches (including EbA) by policymakers and planners Sub-activity 4.1.2.2. Collect lessons and best practices across Jordan (including on the project's ILWRM investments) and between arid states in the Mediterranean region and Middle East, as well as those generated via the project on an annual basis using the knowledge management tools. Sub-activity 4.1.2.5. User groups trained to use knowledge management tools to increase their access to knowledge/information of climate-resilient ILWRM approaches
Insufficient awareness by politicians, community leaders, decision-makers and the public of the impacts of climate change and appropriate adaptation responses	Activity 1.1.3. Generate evidence of the economic benefits of ILWRM to project beneficiaries for use by policymakers and planners	All sub-activities under Activity 1.1.3 (1.1.3.1 – 1.1.3.5)
	Activity 4.1.1 Implement a gender-responsive climate change adaptation outreach programme across the Jordan Valley	All sub-activities under Activity 4.1.1 (4.1.1.1 – 4.1.1.6)

Barrier	Project activity	Project sub-activity
Insufficient financial capacity for climate change adaptation	Activity 1.1.2. Draft amendments to Jordan's policy, planning and legal frameworks for agriculture, land, water and natural resources, to mainstream climate-resilient ILWRM as a climate change adaptation approach	Sub-activity 1.1.2.4. Prepare policy briefs with recommendations for the integration of climate-resilient ILWRM practices, and the results of climate analyses and risk assessments into relevant national policies, plans and strategies
	Activity 2.1.4. Establish a Water Fund in the Jordan Valley to maintain landscape investment for enhanced water resources	All sub-activities under Activity 2.1.4 (2.1.4.1 – 2.1.4.9)
Limited awareness amongst decision-makers (including MWI and MoA) and farmers of the financial and economic effectiveness of dryland-adapted water management, including the management of green water, in Jordan as opposed to unsustainable abstraction of groundwater	Activity 1.1.3. Generate evidence of the economic benefits of ILWRM to project beneficiaries for use by policymakers and planners	All sub-activities under Activity 1.1.3 (1.1.3.1 – 1.1.3.5)
	Activity 3.1.3. Introduce small-scale rainwater harvesting interventions in the project's three target areas to improve access to clean water under climate change conditions in a gender-responsive manner	Sub-activity 3.1.3.1. Install small-scale water harvesting interventions at community centres and local institutions at target sites in the Jordan Valley Sub-activity 3.1.3.2. Install small-scale water harvesting interventions at homes and schools at target sites in the Jordan Valley (100% of women trained in their operation and maintenance)
	Activity 4.1.1. Implement a gender-responsive climate change adaptation outreach programme across the Jordan Valley	All sub-activities under Activity 4.1.1 (4.1.1.1 – 4.1.1.6)
Gender inequality	Activity 4.1.1 Implement a gender-responsive climate change adaption outreach programme across the Jordan Valley	Relevant actions as per the GAP (Annex 8) - Needs of men and women considered in the design of the programme, and that outreach activities are gender responsive. - Awareness-raising on the impacts of climate change focuses on how men and women are affected differently and what their vulnerabilities and adaptation needs are. - Women, women's groups and institutions working with women are targeted with specific awareness-raising activities. - Schedule outreach activities at times when women can attend. This may include organizing activities outside of lunch hours to avoid conflicts with women's household responsibilities and limiting the length of activities so that women could attend and still see to other responsibilities. - Organize childcare services, or in spaces where children may be accommodated during outreach activities. - Identification of communication channels and media that are appropriate for women. - Providing transport for women to outreach activities as necessary. - Inclusion of key outreach activities for women at the household level — considerate of women's daily schedules and availability, and preferred engagement approaches.
	Activity 2.1.3. Introduce gender-responsive climate-resilient additional livelihoods and agricultural practices in the project's three intervention areas	- Working with WMCs and Technical Partners to develop selection criteria that are targeted at vulnerable and marginalised groups, including women. Female headed households will be specifically targeted, particularly as the alternative livelihood activities are considered to have a relative low workload. - Provide targeted training for women at their homes or provide transport to training sessions. - Schedule training at times when women can attend. This may include organizing training outside of lunch hours to avoid conflicts with women's household responsibilities and limiting the length of training so that women could attend and still see to other activities. - Organize childcare services, or in spaces where children may be accommodated during training.

Barrier	Project activity	Project sub-activity
		- The implementation of livelihood activities will consider the needs, aspirations, capacities and limitations of women. - Ensure that women farmers and their role in the agricultural sector are better understood before devising an engagement strategy which aims towards empowering them to co-lead the transformative approach for the farming sector in the target areas; - Take a flexible approach in forming the groups for coordination and capacity building, considering both mixed-sex groups and single – sex groups; - If deemed beneficial, organise women-only training sessions and where possible invite female extension workers to allow participants to feel comfortable to express their ideas and needs; - Schedule training at times when women can attend. This may include organizing training outside of lunch hours to avoid conflicts with women's household responsibilities and limiting the length of training so that women could attend and still see to other activities; - Include training modules for women and men training participants on GBV and access to community feedback mechanisms, complaints boxes, and referral pathways to obtain support for survivors of GBV; - Provide transportation if travel distances would hamper their participation; - Organize childcare services, or in spaces where children may be accommodated during training; and - Ensure the location of the demo plots allows women to easily access these without spending too much time travelling also considering any protection/GBV risks that may arise from travelling to and from the demo plots. - Ensuring women are fully able to participate in the saving associations; and - Ensuring women are equally benefitting from the distribution of agriculture inputs, including vulnerable women who may not be part of a farmers' cooperative.
	Activity 3.1.3. Introduce small-scale rainwater harvesting interventions in the project's three target areas to improve access to clean water under climate change conditions in a gender-responsive manner	- An exercise to determine the beneficiary profile and select the technologies to ensure these are appropriate to address specific needs of women. This may include having separate meetings with women to determine their needs and discuss various technology options and how they would benefit them. - Develop and support the implementation of an assessment (e.g. through focus group discussions or interviews with key informants) to determine the impact of water harvesting and on water availability and women's access to clean water.
	Activity 1.1.4. Strengthen the mandate and capacity of non-state actors for climate-resilient ILWRM in the Jordan Valley	- WMC capacity development activities specifically targeting women to ensure that all participants have the same knowledge to work together more effectively. Include training modules for women and men, training participants on GBV and access to community feedback mechanisms, complaints boxes, and referral pathways to obtain support for survivors of GBV. - Schedule training/meetings at times when women can attend. This may include organizing training outside of lunch hours to avoid conflicts with women's household responsibilities and limiting the length of training so that women could attend and still see to other activities. - Provide transportation if travel distances would hamper the participation of women. - Organize childcare services, or in spaces where children may be accommodated during training or meetings. - Schedule meetings at times when women can attend. This may include organizing training outside of lunch hours to avoid conflicts with women's household responsibilities and limiting the length of training so that women could attend and still see to other activities. - Provide transportation if travel distances would hamper the participation of women.

Barrier	Project activity	Project sub-activity
		- Organize childcare services, or in spaces where children may be accommodated during training or meetings. - Carrying out an in-depth needs assessment of women in each target watershed to inform the planning process; - Hosting separate meetings to capture women's views during the planning process and have representatives of women's groups publicly represent their voices during decision-making processes; - Organize meetings at a time of day and location conducive to their schedules and sensitive to women's domestic responsibilities; and - Promoting awareness among women and men on incidence of and avoiding gender-based violence as well as access to community feedback mechanisms, complaints boxes, and referral pathways to obtain support. - The Gender Officer will screen the capacity-building programme for the IASC GAM score (using the IASC online tool) and will provide inputs and advise to ensure they reach the agreed rating of 3.
	Activity 2.1.4. Establish a Water Fund in the Jordan Valley to maintain landscape investment for enhanced water resources	- Hosting separate meetings to capture women's views during the planning process and have representatives of women's groups publicly represent their voices during decision-making processes; and - Organize meetings at a time of day and location conducive to their schedules and sensitive to women's domestic responsibilities. - Ensure that all proposals submitted under the on-granting facility and the Water Fund's grant mechanism are gender-responsive, promoting the inclusion of women, considering their roles in society and have activities specifically tailored for women and align with the GAP as a whole.
	Activity 3.1.4. Empower communities, including women, to monitor and manage clean water supplies more effectively	- Hosting separate meetings to capture women's views during the planning process and have representatives of women's groups publicly represent their voices during decision-making processes; - Organize meetings at a time of day and location conducive to their schedules and sensitive to women's domestic responsibilities; and - Promoting awareness among women and men on incidence of and avoiding gender-based violence as well as access to community feedback mechanisms, complaints boxes, and referral pathways to obtain support. - Designing awareness raising activities that are gender-responsive and speak to the needs of women, - Organize awareness-raising activities at a time of day and location conducive to their schedules and sensitive to women's domestic responsibilities, including home visits; and - Arrange transport for women to attend awareness-raising activities if necessary. - Training activities specifically targeting women to ensure that all participants have the same knowledge to work together more effectively. Include training modules for women, training participants on GBV and access to community feedback mechanisms, complaints boxes, and referral pathways to obtain support for survivors of GBV. - Schedule training/meetings at times when women can attend. This may include organizing training outside of lunch hours to avoid conflicts with women's household responsibilities and limiting the length of training so that women could attend and still see to other activities. - Provide transportation if travel distances would hamper the participation of women. - Organize childcare services, or in spaces where children may be accommodated during training or meetings.

Barrier	Project activity	Project sub-activity
		• Select women (including female farmers) to be waterwise ambassadors for the monitoring mechanisms. • Working with WMCs and Technical Partners to develop selection criteria that are targeted at vulnerable and marginalised groups, including women. Women will be specifically targeted. • Provide targeted training for women at their homes or provide transport to training sessions. • Schedule training at times when women can attend. This may include organizing training outside of lunch hours to avoid conflicts with women's household responsibilities and limiting the length of training so that women could attend and still see to other activities. • Organize childcare services, or in spaces where children may be accommodated during training. • The implementation of livelihood activities will consider the needs, aspirations, capacities and limitations of women.

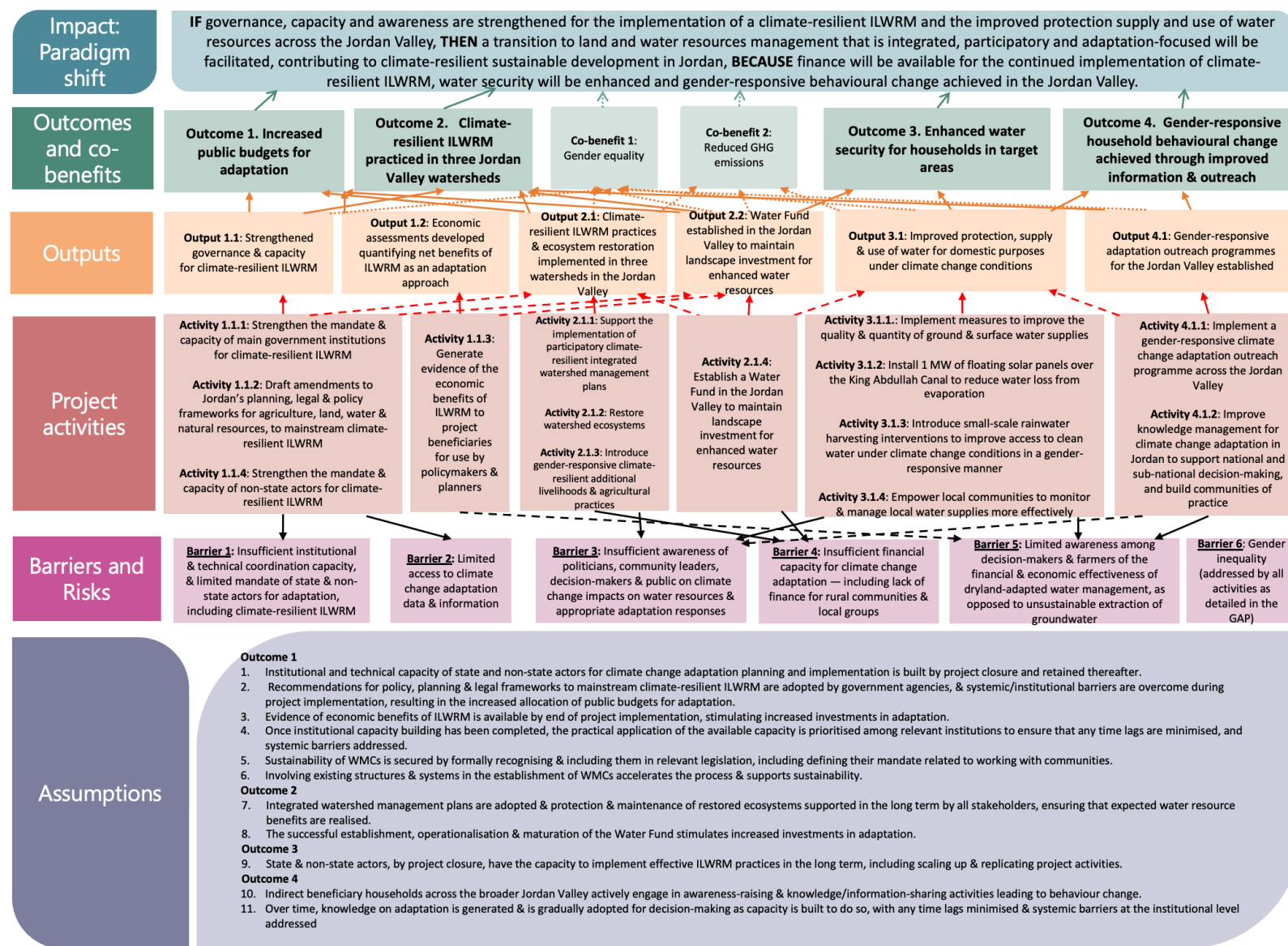


Figure 7. Theory of change diagram.

B.2. (b). Outcome mapping to GCF results areas and co-benefit categorization

Project outcomes:

- Outcome 1. Increased public budgets for adaptation;
- Outcome 2. Climate-resilient ILWRM practiced in three Jordan Valley watersheds;
- Outcome 3. Enhanced water security for households in target areas; and
- Outcome 4. Household behavioural change achieved through improved information and outreach.

Outcome number	GCF Mitigation Results Area (MRA 1-4)				GCF Adaptation Results Area (ARA 1-4)			
	MRA 1 Energy generation and access	MRA 2 Low-emission transport	MRA 3 Building, cities, industries, appliances	MRA 4 Forestry and land use	ARA 1 Most vulnerable people and communities	ARA 2 Health, well-being, food and water security	ARA 3 Infrastructure and built environment	ARA 4 Ecosystems and ecosystem services
Outcome 1	☐	☐	☐	☐	☒	☒	☐	☒
Outcome 2	☐	☐	☐	☐	☐	☒	☐	☒
Outcome 3	☐	☐	☐	☐	☒	☒	☐	☐
Outcome 4	☐	☐	☐	☐	☒	☒	☐	☒

Project co-benefits:

- Co-benefit 1: Gender equality — to be measured via the GAP (Annex 8); and
- Co-benefit 2: Reduced GHG emissions

Co-benefit number	Co-benefit					
	Environmental	Social	Economic	Gender	Adaptation	Mitigation
Co-benefit 1	☐	☐	☐	☒	☐	☐
Co-benefit 2	☐	☐	☐	☐	☐	☒

B.3. Project/programme description (max. 2500 words, approximately 5 pages)

61. The objective of the proposed project is to reduce the negative impacts of climate change on water resources in Jordan through a landscape-level approach — specifically Integrated Landscape and Water Resources Management (ILWRM) — including improved ecosystem management, enhanced protection, supply and use of water, behavioural change and strengthened governance. This objective is strongly aligned with the GCF's paradigm shift objective of increased climate-resilient sustainable development. Through the project, the management of water resources at a landscape level in Jordan will shift away from a business-as-usual model that is uncoordinated and compartmentalised to one that is integrated across economic sectors, government ministries and departments, municipalities, non-government organisations and civil society, as well as into water resources and land-management policy. This proactive and transformative approach will include direct investments into the restoration of ecosystems in watersheds of the northern Jordan Valley to strengthen their resilience to climate change, as well as increase their capacity to supply ecosystem services such as increased water resources and flood attenuation. These services will buffer the impacts of climate change on the lives and livelihoods of local people, while also responding to gender-specific needs. Increased water supply under climate change conditions through watershed restoration in the Jordan Valley will be supplemented by implementing: i) rockfill dams to promote managed aquifer recharge; ii) floating solar panels over the King Abdullah Canal to reduce evaporation and supplement local electricity supplies; iii) small-scale rooftop water harvesting interventions in vulnerable communities; iv) water harvesting and conservation practices on smallholder farms; and v) women-led community-based water monitoring and management. This project's focus on land, agriculture and water is aligned with — and seeks to build on and complement — major GoJ development and adaptation planning priorities.
62. The maintenance and upscaling of ecosystem management and restoration activities will be fostered by a Water Fund, which will leverage GCF resources to crowd-in private and public investment into ecosystems-based adaptation (EbA),

thereby overcoming the key barrier of insufficient finance for the long-term sustainability of adaptation interventions. This will be done via the Fund's long-term financing strategy, which will include securing agreements with major water users (private sector) who are willing to pay for ongoing ecosystem management in return for the water resources generated — a PES model. Payers for ongoing ecosystem management would include major water users in downstream areas of the Jordan Valley who would benefit from and rely on increased water resources generated via the Fund's interventions in upstream ecosystems. In addition to catalysing financing for adaptation, participatory climate-resilient ILWRM plans will be implemented in the target watersheds, namely the Jordan Valley, Yarmouk, Amman Zarqa and Jordan Rift Valley Side Wadis Basins. This will be complemented by facilitating the mainstreaming of climate-resilient ILWRM into Jordan's policy, planning and legal frameworks in agriculture, land, water and natural resource management to promote the replication and upscaling of project interventions across the country. Institutional and technical capacity for this mainstreaming will be built within government — for example, MoEnv and JVA — and non-governmental institutions responsible for land management in Jordan. Finally, the improvement of climate change adaptation knowledge management under the project will contribute to capacity building activities and promote informed decision-making by state and non-state actors involved in ILWRM.

63. An overview of the design of the proposed project's four components is provided below. Specific detail on the technical design of the activities under each component is presented in Section 10 of Annex 2: Feasibility Study.

Component 1: Climate change adaptation governance in Jordan

Outcome 1: Increased public budgets for adaptation

Output 1.1 Strengthened governance and capacity for climate-resilient integrated landscape and water resources management (ILWRM) and Output 1.2: Economic assessments developed quantifying net benefits of ILWRM as an adaptation approach

64. Outputs 1.1 and 1.2 will contribute to GCF Outcome A5.0 — Strengthened institutional and regulatory systems for climate-responsive planning and development — and A8.0 — Strengthened awareness of climate threats and risk reduction processes — by enhancing governance and capacity for climate-resilient ILWRM in Jordan. To achieve this, the mandates and capacities of governmental (such as MoEnv, JVA, MoA, WAJ and MWI¹²⁵) and non-governmental (such as NGOs) stakeholders for landscape management across the Jordan Valley will be built through several interventions, including the development of land-use guidelines, standards and protocols for climate-resilient ILWRM in the Jordan Valley. This will include clarifying the roles and responsibilities of all relevant actors.
65. The mandates of the institutions involved in climate-resilient ILWRM will be strengthened through its integration into relevant policies, plans and legislature, including, *inter alia*, Jordan's Decentralisation Programme and National Development Plan. To ensure that this is done effectively, several capacity-building actions targeting national- and local-level decision-makers and planners will be implemented. These include training: i) policy and decision-makers on the mainstreaming of climate-resilient ILWRM; ii) planners on climate-resilient ILWRM planning; and iii) staff from relevant national and sub-national institutions to use climate data, the results of climate analyses and risk assessments in decision-making processes, as well as planning and disaster risk management related to ILWRM. The capacity built through these trainings¹²⁶ will facilitate the preparation of policy briefs with recommendations for the integration of climate-resilient ILWRM practices into relevant policies, plans, strategies and legislature.
66. Currently, an unsustainable model of sectoralised water resource management exists in Jordan that does not prioritise landscape-level ecosystem resilience. However, adaptation of the country's population through conservation and adaptive measures in land and water resources management can be achieved through an integrated approach to conserving Jordan's water resources under climate change conditions. Baseline gaps identified in the National Water Strategy (2016–2025) include that it does not reflect the initiated decline of available freshwater, presuming that alternative water supplies would increase in availability from 2025 (which is at odds with preceding and proceeding National Communications to the UNFCCC). Addressing this is part of the Jordan Water Management Initiative's (2016–2021) scope, by increasing the accountability and sustainability of water sector management and governance. In addition, the National Water Strategy highlights gaps in how the GoJ manages water demand in the agricultural sector, as well as specific interventions proposed to address these gaps. These interventions and broader themes, along with specific options have been detailed under Section 1.3.2 'Relevant national policies, strategies and frameworks' of Annex 2: Feasibility Study. The development of a legislative framework on climate change was identified as a gap through Vision 2025 and the EDP — which emphasised the need for a legal framework to regulate national climate adaptation activities, in addition to enhancing the capacities of national institutions and experts to address climate change impacts. Addressing these gaps has been supported by the AF-funded project entitled '*Increasing the*

¹²⁵ Please see Section 1.3.1 Legal, institutional and policy context of the Feasibility Study for further information on the relevant governmental stakeholders.

¹²⁶ Delivered with guidance from the policy and institutional expert to be housed within the Project Management Unit (PMU).

resilience of poor and vulnerable communities to climate change', which makes specific reference to Vision 2025 as a priority policy to be reviewed and amended. In addition, the Climate Change Policy for a Resilient Water Sector aims to bridge the gap between priority areas to support policy coherence by building upon other policies and plans in the water sector. For the proposed project to catalyse a paradigm shift in land and water resource management in Jordan, policy changes are necessary to facilitate a move away from the current business-as-usual approach.

67. To ensure a participatory and cross-sectoral approach to landscape management, WMCs will be established at each of the project's three target areas. Staff of WMC member institutions, will be trained on climate-resilient watershed management to enhance technical capacity at both the local and national level. Training will differ according to the role of the stakeholder as defined in the land-use guidelines, standards and protocols, which will be used to develop a training programme for climate-resilient watershed management.
68. Outputs 1.1 and 1.2 will see the establishment of an enabling environment for the implementation of climate-resilient ILWRM at scale across Jordan through a combination of activities. These include strengthening the mandate and capacity of main government institutions and non-state actors for ILWRM; mainstreaming ILWRM as an adaptation approach into regulatory frameworks for land, water, agriculture and natural resources; and generating evidence of economic benefits of ILWRM (as an adaptation approach) for use by policy makers and planners. Through its policy-related activities, the project will support budget reforms, increasing the allocation of public budgets to adaptation (ILWRM in particular), which, along with the Water Fund (Activity 2.5), would support scaling up of the proposed project's interventions in the medium to long term (see Component 2 and Section D.2.). This will support a long-term balance in the country's water resources in the face of climate change.
69. Under the project's delegated execution structure, MoEnv will act as Executing Entity for Component 1.

Activity 1.1.1. Strengthen the mandate and capacity of main government institutions — including MoEnv, MWI and their subsidiaries¹²⁷ — for climate-resilient ILWRM in Jordan.

70. Activity 1.1.1 includes two sub-activities aimed at strengthening the mandate and capacity of Jordan's government institutions involved in water resources management to incorporate climate-resilient ILWRM into national and sub-national policy- and decision-making processes, supporting the scaling up and replication of the project's interventions. These sub-activities include developing land use guidelines, standards and protocols for climate-resilient ILWRM and training national and sub-national policy- and decision-makers on the use of climate-related data. Institutional strengthening under Activity 1.1 includes extending the mandate of the Jordan Valley Authority (JVA) — whose current mandate also includes oversight of water and irrigation policy implementation — to facilitate the uptake of ILWRM principles in the decision-making process for sustainable socioeconomic development.

Sub-activity 1.1.1.1 Develop land use guidelines, standards and protocols for climate-resilient ILWRM in the Jordan Valley.

71. Guidelines, standards and protocols will be developed in year 1 to facilitate a standardised and coordinated approach to climate-resilient ILWRM across the Jordan Valley. These guidelines, standards and protocols are necessary because knowledge, technical and institutional capacity to implement activities related to climate change adaptation remains limited. The materials will be developed with, and for, the state and non-state actors responsible for landscape and water resources management including the MoEnv, NGOs, MLI, MLA, DLS, AGM, ASEZA, JVA, MoA, WAJ, MWI and WMCs. This participatory approach will ensure vertical and horizontal coordination in ILWRM across the Jordan Valley, working within existing coordination and planning structures. It will also target both supply- and demand-side factors of vulnerability and prevent maladaptation.
72. The protocols, standards and guidelines will: i) define roles and responsibilities for the relevant actors; ii) promote gender-inclusive community engagement and participation in landscape management; iii) prioritise supply- and demand-side landscape-management interventions across the watersheds of the Jordan Valley in consideration of and building on MWI's water-use quotas and water-saving policies, with a focus on enhancing the climate resilience of local communities; iv) describe how fiscal and/or market incentives (e.g., via tax reductions) can be used to encourage poor rural communities to invest in rainwater harvesting infrastructure (such as rooftop water tanks) or other water-saving, -efficiency and -storage options; and v) ensure that all development activities planned for and implemented in the relevant watersheds are part of a coordinated and standardised ILWRM approach to climate change adaptation, by linking them to the watershed management plans developed under Sub-activity 1.4.5. These protocols, guidelines and standards will seek to build upon existing planning and development processes, to coordinate and integrate activities, bridge gaps and address priorities previously identified (see Section 10 of Annex 2: Feasibility

¹²⁷ Institutions through which the ministries operate at the local level, such as the Jordan Valley Authority (JVA).

Study). Their development will consider the results of the latest climate analyses (including long-term climate projections), risk assessments and hydrological modelling¹²⁸ (under future climate scenarios).

Sub-activity 1.1.1.2. Train staff from relevant national and subnational institutions to use climate data, and the results of climate analyses and risk assessments in policy- and decision-making processes, as well as planning and disaster risk management.

73. Staff from relevant **national-level** institutions will be trained to interpret and apply climate data, as well as the results of climate analyses and risk assessments to improve the mainstreaming of climate-resilient ILWRM into Jordan's policy-making and planning processes. The gender-inclusive training for national-level staff, which will include an annual three-day workshop for 30 people (at least 50% female representation), will also focus on the use of this data for disaster risk management related to ILWRM. National-level staff to be trained will include representatives from MoEnv (including the Climate Change Directorate), MWI, MoA, and members of the National Committee on Climate Change¹²⁹. In total, 24 training days are allocated across the project's 8-year implementation period.
74. To enhance the technical capacity of **sub-national-level** stakeholders for climate-resilient ILWRM, training for staff will be conducted on the use of climate data, the results of climate analyses and risk assessments in gender-responsive policy- and decision-making processes, as well as planning and disaster risk management related to ILWRM. The gender-inclusive training will target 34 local-level-staff members (at least 50% female representation) during an annual three-day workshop in the northern Jordan Valley. Local-level staff to be trained will comprise: i) 8 staff from the Irbid Governate, and 3 staff each from its relevant districts, including At-Taibah, Al-Wastiyah, Qasabah Irbid and Al-Kurah (20 staff in total); and ii) 8 staff from the Balqa Governate, and 3 staff each from its relevant districts, including Dir Alla and Ash-Shunah al-Janubiyah (14 staff in total). In total, 24 training days are allocated across the project's 8-year implementation period.

Activity 1.1.2. Draft amendments to Jordan's policy, planning and legal frameworks for agriculture, land, water and natural resources, to mainstream climate-resilient ILWRM as a climate change adaptation approach.

75. Activity 1.1.2 will follow the development of ILWRM guidelines, standards and protocols (Sub-activity 1.1.1.1), as well as the training undertaken in Sub-activity 1.1.1.2 to mainstream climate-resilient ILWRM into policy, planning and legal frameworks relating to agriculture, land, water and natural resources. Sub-activities will involve: i) training of national and sub-national planning as well as policy- and decision-makers; ii) reviewing of policies, plans and legislation; iii) preparing of policy briefs; and iv) convening workshops to validate the policy briefs. The mainstreaming of ILWRM into agriculture, land, water and natural resources regulatory frameworks, contributes to the establishment of an enabling environment required to support the scaling up of the project's on-the-ground interventions aimed at improving the water resource balance in Jordan (through improved budget allocations, for example).

Sub-activity 1.1.2.1. Train policy- and decision-makers on the integration of climate-resilient ILWRM practices into policies, plans and strategies.

76. Sub-activity 1.2.1 will enhance the capacity of policy- and decision-makers from institutions including but not limited to MoEnv, JVA, MoA, WAJ and MWI to integrate climate-resilient ILWRM practices into policies, plans and strategies through annual (from year 2 to 8) two-day workshops presented by a local policy expert for at least 25 people, including a minimum of 50% women.

Sub-activity 1.1.2.2. Train relevant national and sub-national planners on climate-resilient ILWRM planning.

77. Training under Sub-activity 1.1.2.1 will be complemented by the training of national and sub-national development and land-use planners on gender-responsive climate-resilient ILWRM planning through one two-day national-level workshop for at least 20 people (year 2), and two two-day sub-national workshops for at least 20 people (year 2), each including at least 50% women. The 40 individuals trained will comprise: i) 10 staff from the Irbid Governate, and 4 staff each from its relevant districts, including At-Taibah, Al-Wastiyah, Qasabah Irbid and Al-Kurah (22 staff in total); and ii) 10 staff from the Balqa Governate, and 4 staff each from its relevant districts, including Dir Alla and Ash-Shunah al-Janubiyah (18 staff in total).

Sub-activity 1.1.2.3. Review Jordan's national land-management policies, plans and legislation to identify gaps and opportunities for the consideration of climate-resilient ILWRM (including EbA, climate analyses and risk assessments) as an adaptation approach.

¹²⁸ An international consultant will work with a local hydrological modelling expert to undertake long-term hydrological modelling for the Jordan Valley under long-term climate scenarios and develop a risk assessment to inform climate-resilient ILWRM planning.

¹²⁹ Including the Ministry of Planning and International Cooperation, Ministry of Social Development and the Jordan Meteorology Department.

78. Once the training under Sub-activities 1.1.2.1 and 1.1.2.2 has been completed, trained individuals will, in year 3, review Jordan's national and local policies, plans and legislation related to water resources and landscape management to identify gaps and opportunities for the consideration of gender-responsive climate-resilient ILWRM as an adaptation approach — such as the Agriculture Law No. 44 of 2002, Jordan's Rangeland Strategy, the Climate Change Policy for a Resilient Water Sector, as well as MWI's water-use quota's and water-saving policies. The review process will be supported by a policy and planning expert (procured by the project) and project Technical Partners. Also involved in the reviews will be staff trained through Sub-Activity 1.1.1.2, who will identify gaps for the integration of climate-related data into policies, plans and disaster risk management for ILWRM, consider the need for revisions to be made to water-use quotas and water saving policies, as well as identify fiscal and/or market incentives (e.g., via tax reductions) for poor rural communities (including female-headed households) to invest in rainwater harvesting infrastructure (such as rooftop water tanks) or other water-saving, -efficiency and -storage options. These tasks will be informed by climate data, as well as results from the latest long-term climate analyses and risk assessments (including hydrological modelling of water resources in the Jordan Valley under future climate scenarios to inform ILWRM).

79. Sub-activity 1.1.2.3 dependencies:

- Training under Sub-Activity 1.1.1.2 — ongoing throughout project implementation, starting from year 1.
- Training under sub-activities 1.1.2.1 and 1.1.2.2 — completed in year 2.

Sub-activity 1.1.2.4. Prepare policy briefs with recommendations for the integration of climate-resilient ILWRM practices, and the results of climate analyses and risk assessments into relevant national policies, plans and strategies.

80. Based on the reviews undertaken in Sub-activity 1.1.2.3, policy briefs — to inform ongoing and anticipated policy-development processes — will be developed in the fourth year of project implementation. These will include recommendations for the mainstreaming of gender-responsive climate-resilient ILWRM and the results of long-term climate analyses and risk assessments (including assessments based on hydrological modelling of the Jordan Valley's water resources under future climate conditions) into the relevant policies, legislation and plans, as well as recommendations for the use of fiscal and/or market incentives (e.g., via tax reductions) for poor rural communities and vulnerable groups (such as female-headed households) to invest in rainwater harvesting infrastructure (such as rooftop water tanks) or other water-saving, -efficiency and -storage options — at both the national and local levels (such as the Agriculture Law No. 44 of 2002, Jordan's Rangeland Strategy and the Climate Change Policy for a Resilient Water Sector). To ensure sustainability, recommendations presented in the policy briefs will also support the commitment of policy and budget allocations to support the ILWRM elements introduced by the project, as well as to the improved management of water resources and water infrastructure.

81. Sub-activity 1.1.2.4 dependencies:

- Policy reviews under Sub-activity 1.1.2.3 — completed in year 3.

Sub-activity 1.1.2.5. Convene a workshop to present the policy briefs prepared under 1.1.2.4 to representatives from relevant ministries and departments (with at least 50% female participation) for validation.

82. Once policy briefs have been developed under Sub-activity 1.1.2.4, a multi-sectoral workshop (with at least 50% female participation) will be hosted by the NCCC in the second half of year 4 to validate the policy briefs, including ensuring that they are gender responsive. Moreover, the workshop will enhance sectoral coordination in terms of ILWRM in Jordan. Once validated, the policy briefs will be submitted to the GoJ via the NCCC.

83. Sub-activity 1.1.2.5 dependencies:

- Policy briefs developed under Sub-activity 1.1.2.4 — completed in the first half of year 4.

Activity 1.1.3. Generate evidence of the economic benefits of ILWRM to project beneficiaries for use by policymakers and planners.

84. To prioritise an ILWRM approach and ensure that it is implemented at scale, decision-makers are required to understand and appreciate the relevant interlinked ecological processes and how they directly produce economic benefits. The relevant information and evidence will be generated by the monitoring and analysis of ILWRM interventions implemented under the project. This analysis will consider: i) avoided costs — impacts or losses that would otherwise occur in the absence of ecosystem services, such as reduced water resource quantity and quality; and ii) replacement costs — the implication of implementing 'grey' artificial systems (as opposed to 'green' nature-based solutions) that would be required to deliver the same services as ecosystems. Where feasible, monitoring of the following sustainable development co-benefits will be included: flood attenuation, enhanced biodiversity, decreased pollution of water resources, improved access to clean water, strengthened food security, increased climate resilience of pastoral livelihoods and energy generation from floating solar.

85. The monitoring of the climate-resilient ILWRM investments will be guided by scientists (including encouraging the involvement of female scientists), but conducted wherever feasible by community members via WMCs. The long-term data sets created will be used for adaptive management of project interventions throughout the duration of the project and to inform future climate-resilient ILWRM investments across Jordan, and a full impact study of the project will be conducted in the project's final two years. The full socioeconomic and environmental benefits of restoration investments are often maximised a decade or more after implementation. For this reason, the long-term data sets from the project will be archived within Jordanian research institutions (including relevant NGOs and project Technical Partners) and the long-term monitoring of the GCF project's investments will also be institutionalised. Similarly, detailed, long-term Operations and Maintenance (O&M) plans will be developed for all ecological and physical infrastructure constructed by the project. Government co-financing for O&M will include: i) financing of ongoing restoration, conservation and management of those ecosystems restored by the project; and ii) maintaining the mechanisms and tools established by the project.
86. The GoJ will include the monitoring and analysis in their ongoing operations to contribute to the capacity building at the policy- and decision-making level and policy reform process under Activity 1.1.2 to integrate climate-resilient ILWRM into Jordan's planning, policy-making and budgeting process. This will facilitate the embedding of coordination and operation of these mechanisms and structures in budgets and mandates beyond the life of the project. These plans will also include long-term funding commitments for enhancing the ecological infrastructure and maintaining the physical infrastructure associated with livelihood interventions. By recognising ILWRM's positive environmental, economic and social co-benefits to communities, cross-sectoral cooperation and integrated planning to conserve and augment water resources will be fostered.

Sub-activity 1.1.3.1. Gather primary and secondary data from the project's ILWRM interventions.

87. To establish the necessary evidence base for ILWRM in Jordan, primary and secondary data will be generated during the monitoring and analysis of ILWRM interventions implemented under the project. This will include collecting data through a 1-day gender-inclusive stakeholder engagement workshop at each of the project's three sites during years 6 and 7. The data gathered will be focussed on *inter alia* flood levels, biodiversity, the levels of pollution of water resources, the availability of and access to clean water (including by women), food security, and energy generation from floating solar.
88. Sub-activity 1.1.3.1 dependencies:
- Component 2's ILWRM ecosystem restoration activities — ongoing throughout project implementation.
 - Component 3's activities to improve the protection, supply and use of water for domestic purposes — ongoing throughout project implementation.

Sub-activity 1.1.3.2. Analyse monitoring data from ILWRM interventions implemented under the project to identify benefit streams for valuing ILWRM benefits.

89. Monitoring data from ILWRM interventions collected under Sub-activity 1.1.3.1 will be analysed in year 7 to identify benefit streams for valuing ILWRM benefits under different climate scenarios. This analysis will include a focus on ecosystem services generated by EbA-related project interventions — particularly the regulation of the hydrological system and production of water. The process of collecting these data as part of the monitoring process will be guided by scientists, but wherever feasible will be undertaken by community members via WMCs.

90. Sub-activity 1.1.3.2 dependencies:

- Collection of data under Sub-activity 1.1.3.1 — completed between years 6 and 7.

Sub-activity 1.1.3.3. Identify data streams and modelling methodology for valuing ILWRM benefits.

91. Based on the analysis of monitoring data from ILWRM interventions under Sub-activity 1.1.3.2, data streams and a modelling methodology for valuing ILWRM benefits will be identified by the specialists (scientists) procured by the project. Data streams will include *inter alia* long-term data sets from the project — archived within Jordanian research institutions — as well as the results of climate analyses and risk assessments which will be stored in a publicly accessible online database (i.e., a national climate change adaptation knowledge repository developed under Activity 4.2).
92. Sub-activity 1.1.3.3 dependencies:
- Analysis of ILWRM monitoring data under Sub-activity 1.1.3.2 — completed in year 7.

Sub-activity 1.1.3.4. Carry out valuation modelling exercises of benefit streams identified under 1.1.3.1.

93. Projections of the values generated by the benefit streams for climate-resilient ILWRM will be modelled by the specialists (scientists) procured by the project in year 8.

94. Sub-activity 1.1.3.4 dependencies:

- Identification of data streams and modelling methodology under Sub-activity 1.1.3.4 — completed in year 7.

Sub-activity 1.1.3.5. Prepare full and summary reports of the analysis and results, including briefing notes for decision-makers highlighting evidence of ILWRM as an adaptation approach and specific policy recommendations.

95. To enable policy- and decision-makers to effectively use the data generated under Activity 1.1.3, full and summary reports of the analysis and results will be prepared in year 8 of the project. This will include briefing notes highlighting the evidence for and benefits of ILWRM as an adaptation approach, as well as specific policy recommendations to address the particular needs highlighted in the reports.

96. Sub-activity 1.1.3.5 dependencies:

- Results of valuation exercises carried out under Sub-activity 1.1.3.4 — completed in year 7.

Activity 1.1.4. Strengthen the mandate and capacity of non-state actors for climate-resilient ILWRM in the Jordan Valley.

97. Activity 1.1.4 complements the state actor capacity building under Activities 1.1.1 and 1.1.2 by strengthening the mandate and capacity of non-state actors for climate-resilient ILWRM in a gender-inclusive manner. This will be done through the establishment and training of Watershed Management Committees (WMCs) that promote the inclusion of women in decision-making, training staff in relevant sub-national institutions, developing integrated watershed management plans and implementing ILWRM capacity building for NGOs. Inclusion of at least 50% women in WMCs will be prioritised. By improving local-level mandates and capacities for ILWRM, scaling up and replication of the project's interventions will be supported, contributing to an improvement in water resource supplies both locally and nationally under conditions of climate change.

Sub-activity 1.1.4.1. Establish WMCs, under the leadership of the JVA and local governors, at each of the three target watersheds to facilitate dialogue and participatory watershed management between upstream and downstream stakeholders. This will include working with MoEnv for the formal recognition of WMCs in relevant legislature — including defining their responsibility towards working with communities.

98. To promote the representation and participation of all relevant stakeholders — including government, NGOs, CSOs, local communities, farmers and the private sector — in gender-inclusive watershed management in the Jordan Valley, WMCs (with at least 50% female and 20% youth representation) will be established within each of the project's three target sites (Yarmouk, King Talal Dam and Kafraïn)¹³⁰. JVA and local governors¹³¹ will co-chair the WMCs, convening quarterly meetings to discuss ongoing activities, as well as any plans for future activities related to watershed management. Members will include representatives from all stakeholders involved in landscape management, including governates (Irbid and Balqa) and districts¹³², local municipalities, local communities (including women's groups), as well as project Technical Partners. Once established, the project will work closely with the MoEnv to formally recognise and include WMCs in relevant legislation. This will not only reinforce their contribution to the project and strengthen the participation of local stakeholders (communities in particular) but contribute to a paradigm shift in community participation in development projects. Fostering the formal recognition of WMCs in the relevant environmental legislature will ensure their recognition and authority, contributing to the representation of communities (including women and youth) and building their trust in the project. Moreover, the local experience and footprint of the consortium of NGOs involved in project execution will contribute to successful engagement with local-level stakeholders, helping to mitigate the potential risk of the WMCs' course being redefined post project. Furthermore, by recognising them in legislation and formally defining their mandate, the medium- to long-term sustainability of the WMCs will be enhanced, particular from the perspective of community involvement. As a result, the communities will no longer be marginalised (including vulnerable groups such as women and youth), but have key roles to play in the implementation and decision-making processes.

99. The presence of WMCs will ensure that target communities' needs and concerns are considered in the development and implementation of water monitoring and management plans. This will include specifically focusing on the needs of vulnerable groups such as women and youth. Representatives of target communities will be members of the WMCs, providing them with direct inputs into decision-making processes and ensuring that relevant information is always made

¹³⁰ Draft ToR for the role and function of WMCs are presented in Annex 28. These will be further developed during project implementation by the project Technical Partners.

¹³¹ Yarmouk site – Khaled Bin Al Waleed municipality; King Talal Dam site – New Ardah and Deir Alla municipalities; Kafraïn site – Shouna Alwasta municipality.

¹³² Relevant districts include: i) Irbid Governate – At-Taibah, Al-Wastiyah, Qasabah Irbid and Al-Kurah Districts; and ii) Balqa – Dir Alla and Ash-Shuna al Janubiyah Districts.

available to them. These community representatives will be members of community-based committees already present on the ground in the three project areas, such as Development Advisory Committees, Stakeholders Advisory Groups, agricultural cooperatives, women's cooperatives, youth groups and Water User Associations. By making use of existing community structures and systems will accelerate the time required to establish the WMCs and support their sustainability. The approval and implementation of watershed management activities via the WMCs will be informed by the guidelines, standards and protocols developed through Activity 1.1.

100. Several activities and sub-activities are dependent on the formal establishment of the WMCs. These include 1.1.4.2, 1.1.4.4, 1.1.4.5, 2.1.3, 2.2.2, 2.3.2, 2.3.1, 2.3.3, 2.3.5, 2.4.1, 2.4.3, 2.4.4, 2.4.5, 2.4.6 and 3.4.1 — dependencies have been mapped across the implementation timeline in further detail in Annex 5. Focus will therefore be placed on ensuring that their inclusion in legislature, constitution and operationalisation is prioritised and finalised in year 1 of project implementation.

Sub-activity 1.1.4.2. Develop a training programme on climate-resilient integrated watershed management, including EbA, targeting the WMC members.

101. The technical capacity of WMC members to perform their roles will be strengthened through a tailored gender-responsive training programme developed under the project by international and local experts and supported by project Technical Partners during year 1 of the project. This training will be designed to focus on climate-resilient ILWRM and will specifically target WMC members including JVA, LDCs, local leaders, decentralisation committees, local councils and other relevant stakeholders.

Sub-activity 1.1.4.3. Train relevant watershed stakeholders (WMC members) on climate-resilient ILWRM and its economic benefits based on the programme developed under Sub-activity 1.1.4.2.

102. Based on the programme developed under Sub-activity 1.1.4.2, training of stakeholders in the project's target watersheds (WMC members) will consist of one five-day workshop convened for each of the project's three target areas (in year 2), i.e., three workshops in total, with at least 30 people trained per workshop, of which at least 50% will be women. The gender-responsive training will be implemented in coordination with the JVA, local NGOs and WMCs and facilitated by the international and local experts who designed the programme. Follow-up trainings will take place biennially (years 4, 6 and 8).

103. Sub-activity 1.1.4.3 dependencies:

- Establishment of WMCs under Sub-activity 1.1.4.1 — completed in year 1.
- Development of training programme via Sub-activity 1.1.4.2 — completed in year 1.
- Economic benefits of climate-resilient ILWRM reported under Sub-activity 1.1.3.5 — throughout project implementation.

Sub-activity 1.1.4.4. Train staff from relevant sub-national institutions to use climate data, and the results of climate analyses and risk assessments in policy- and decision-making processes, as well as planning and disaster risk management.

104. To enhance the technical capacity of relevant sub-national institutions represented on WMCs for climate-resilient ILWRM, gender-inclusive training for staff (including planners) will be conducted on using the results of long-term climate analyses and risk assessments (including assessments based on hydrological modelling of the Jordan Valley's water resources under future climate conditions developed under Sub-activity 1.1.1.1) in gender-responsive planning and disaster risk management related to ILWRM. The training will target 30 staff members (including 50% female representation) during a five-day workshop at a central venue in the northern Jordan Valley during year 2.

105. Sub-activity 1.1.4.4 dependencies:

- Hydrological modelling of the Jordan Valley's water resources under future climate conditions developed under Sub-activity 1.1.1.1 — completed in year 1.
- Economic benefits of climate-resilient ILWRM reported under Sub-activity 1.1.3.5 — throughout project implementation.

Sub-activity 1.1.4.5. Develop participatory climate-resilient integrated watershed management plans for the three target watersheds.

106. The skills and expertise gained from the training under Sub-activity 1.1.4.4 will contribute to the development of participatory climate-resilient integrated watershed management plans that promote gender equity by WMCs for their respective watersheds, with assistance from land-use planners and adaptation experts during year 2. The plans will draw on the ILWRM guidelines, standards and protocols developed under Sub-activity 1.1.1.1. Once developed, validation workshops will be held with national- and local-level government (including the MoEnv, MWI, MoA, JVA and NCCC), WMCs and sectoral stakeholders. The validation workshops will be used to ensure that the plans are aligned

with government policies, are gender-responsive, as well as to confirm financial support from government and donor organisations for their implementation.

107. Sub-activity 1.1.4.5 dependencies:

- ILWRM guidelines, standards and protocols developed under Sub-activity 1.1.1.1 — completed in year 1.
- Establishment of WMCs under Sub-activity 1.1.4.1 — completed in year 1.
- Training of relevant staff conducted under Sub-activity 1.1.4.4 — completed in year 2.

Sub-activity 1.1.4.6. Develop and implement an ILWRM capacity-building programme for relevant NGOs.

108. The sustainability of a climate-resilient ILWRM approach across the Jordan Valley will be further strengthened by the design and implementation of a gender-inclusive capacity-building programme for NGOs working in the area in years 4, 5, 6, 7 and 8 of the project. NGOs targeted for climate-resilient ILWRM capacity building include IUCN, RSCN, JREDS, PAF, HGD, JOHUD, RBG and FPEC¹³³. The capacity-building programme will be based on the specific needs, expertise and area of influence of each NGO, which will be determined through a consultative process. Focus areas of the programme will include inter alia: i) technical training; ii) the integration of climate-resilient ILWRM into ongoing programmes; and iii) the identification of and capacity building for new initiatives that NGOs can contribute to in the future, building on the activities of the proposed project.

Component 2: Integrated landscape and water resources management (ILWRM) under conditions of climate change

Outcome 2. Climate-resilient ILWRM practiced in three Jordan Valley watersheds

Output 2.1 Climate-resilient ILWRM practices and ecosystem restoration implemented in three watersheds in the Jordan Valley

109. Without the implementation of adaptation measures such as active and passive restoration of forest ecosystems, the supply of ecosystem services will be compromised under climate change conditions through continued degradation aggravating the vulnerability of local communities, particularly groups such as women and youth. Vegetation loss from desertification and ecosystem deterioration related to climate change has the potential to disrupt other important water-related ecosystem services in Jordan such as the reduction of runoff and prevention of soil erosion during heavy rainfall events. This is because the loss of vegetation results in soil hardening or compaction, thereby reducing infiltration, increasing runoff and leading to a higher likelihood of flooding¹³⁴. A positive feedback loop results whereby the deterioration in soil quality caused by increased erosion prevents vegetation from establishing on bare or degraded areas, further increasing the impact of flooding events linked to climate change. A loss of 11% of rangeland in northern Jordan, for example, could result in an increase in soil loss from erosion of ~1 tonne/ha over 17 years¹³⁵. Current watershed restoration activities in Jordan include both passive and active approaches that provide best practices for the proposed project, which will build on these activities. The removal or exclusion of sources of ecosystem degradation through passive restoration, for example, has been successfully implemented in areas such as the Zarqa and Yarmouk River basins and Swaimeh located north of the Dead Sea through the Hima system, with considerable economic benefits (Section D.6. of this Funding Proposal and Section 8.2.3 of Annex 2: Feasibility Study). Projects that specifically incorporate restoration and ILWRM in Jordan include inter alia the Sustainable use of Ecosystem Services in Jordan — Energy and Climate Fund, Building Resilience to Cope with Climate Change in Jordan Through Improving Water Use Efficiency in the Agriculture Sector (BRCCJ), Healthy Ecosystems for Rangeland Development (HERD), National Programme for Rangeland Rehabilitation and Development, and the Jordan Watershed Restoration Project. Further detail on these projects, as well as additional projects are provided in Section B.1 of this Funding Proposal and Section 7 of the Feasibility Study (Annex 2).

110. The project will also introduce gender-responsive climate-resilient livelihood options, including beekeeping and vegetable gardening (via hydroponics and permaculture). Numerous initiatives have been developed to improve the beekeeping sector in Jordan. The National Center for Agricultural Research and Extension (NCARE) established the Bee Research Department, which is mandated to conserve the indigenous honeybees, *Apis syriaca*, for research purposes¹³⁶. Support to Jordan's beekeeping industry has also been provided by the USAID's Jordan Local Enterprise

¹³³ i) Royal Botanical Garden (RBG); ii) Future Pioneers Empowering Communities (FPEC); iii) Horizons for Green Development (HGD); iv) International Union for Conservation of Nature and Natural Resources (IUCN), Regional Office for West Asia (ROWA); v) Jordan Hashemite Fund for Human Development (JOHUD); vi) Princess Alia Foundation (PAF); vii) Jordanian Royal Society for the Conservation of Nature (RSCN); and viii) The Royal Marine Conservation Society of Jordan (JREDS).

¹³⁴ Ellison D, Morris CE, Locatelli B, Sheil D, Cohen J, Muriyarso D, et al. 2017. Trees, forests and water: Cool insights for a hot world. *Global Environmental Change*, 43: 51–61.

¹³⁵ Alkharabsheh MM, Alexandridis TK, Bilas G, Misopolinos N & Silleos N. 2013. Impact of land cover change on soil erosion hazard in northern Jordan using remote sensing and GIS. *Procedia Environmental Sciences* 19:912–921.

¹³⁶ The Fifth National Report on the Implementation of the Convention on Biological Diversity. 2015. Available at: <https://www.cbd.int/doc/world/jo/jo-nr-05-en.pdf>

Support Project (LENS), an ongoing project that aims to encourage sustainable economic growth and develop the potential of vulnerable communities by combining local economic development efforts with private sector initiatives¹³⁷. Vegetable gardening has previously been promoted by projects such as the Greening the Desert Project¹³⁸, which ongoing work includes: i) supplying permaculture gardens for local schools; ii) empowering women through permaculture sustainable livelihoods; and iii) donating wicking beds¹³⁹ to vulnerable families (including female-headed households) for vegetable gardening.

111. Output 2.1 will contribute to GCF Outcome A7.0 — strengthened adaptive capacity and reduced exposure to climate risks — by implementing an ILWRM approach and restoring ecosystems in a climate-resilient manner in the Yarmouk, King Talal Dam and Kafraïn areas of the Jordan Valley (Figure 8**Error! Reference source not found.** This will benefit all community groups in these areas, including the most vulnerable (women and youth). The three biogeographical zones and associated ecosystems, including unique species, plant assemblages and non-climatic threats, will be taken into account when developing site- and ecosystem-specific restoration protocols. Included in the protocols will be recommendations for climate-resilient plant species, water harvesting and conservation measures¹⁴⁰ as well as maintenance guidelines. Restoration protocols will be used to inform the restoration of ecosystems within watershed areas of the intervention sites and will subsequently allow for upscaling after project completion. Restoration activities will include: i) passive restoration through the traditional Hima system; and ii) active restoration through re-establishing vegetative cover in degraded areas and afforestation. The restoration of the target ecosystems will strengthen the supply of ecosystem services, such as water, fodder and NTFPs — such as fuelwood and edible and medicinal plants, to buffer the impacts of climate change on the lives and livelihoods of local communities. Current baseline demand for water in the project target areas is ~292 MCM/yr. Both contracted labourers and local communities (men and women) surrounding the restoration sites, will be involved in the implementation of restoration activities under Output 2.1.
112. Active restoration activities will be led by the project's Technical Partners who will oversee local labourers (predominantly underprivileged community members including local women). Restoration labourers will be trained on the relevant site- and ecosystem-specific protocols to ensure effective implementation in a gender-responsive manner. Selection of restoration labourers will occur via WMCs with oversight from relevant Technical Partners, with a focus on gender equity. The benefits of the project's restoration activities will be enhanced by the implementation of participatory gender-responsive climate-resilient integrated watershed management plans in the three target areas. This will include the implementation of forest, rangeland, wadi and riparian management interventions to enhance the climate resilience and conservation of these ecosystems in the target watersheds. The plans will include options for additional natural resource-based livelihoods that are climate resilient, gender responsive and appropriate for the local context, in addition to being complementary to the watershed restoration and management activities. Furthermore, climate-resilient agricultural techniques¹⁴¹, including crop diversification, drought-tolerant crop varieties and permaculture, will be introduced on smallholder farms (headed by male and female farmers) at project sites — supported by National Agricultural Research Centre (NARC).
113. Gender-responsive climate-resilient agricultural techniques will be implemented on smallholder farms to increase productivity under climate change conditions, providing immediate benefits for local communities, while at the same time reducing the pressure on degraded ecosystems undergoing restoration. Such techniques will include the installation of in-field rainwater harvesting systems. These interventions have been designed to build upon and scale up existing practices and technologies that have proven to be effective in Jordan, including those which directly focus on the development of female farmers. For example, rainwater harvesting being successfully used in Jordan, in-field rainwater harvesting structures have been used to improve soil structure and decrease soil erosion rates by increasing soil organic matter and intercepting runoff water at relatively short-distance intervals^{142,143}. These structures have also successfully supported the re-establishment of drought-tolerant forage shrubs in degraded rangelands. An economic analysis undertaken for the Badia Benchmark Project, for example, demonstrated that planting of shrubs with harvested rainwater produced an economic internal rate of return (EIRR) estimated at ~17%, with the addition of numerous environmental benefits.

¹³⁷ USAID Jordan Local Enterprise Support Project (LENS). N.d. Available at: <https://www.usaid.gov/jordan/fact-sheets/usaid-jordan-local-enterprise-support-network>

¹³⁸ Greening the Desert Project. 2019. Available at: <https://greeningthedesertproject.org/>.

¹³⁹ Wicking beds are agricultural systems used in water scarce countries that rely on underground water reservoirs filled with decomposing organic matter.

¹⁴⁰ Such as: contour bunds for trees; semi-circular bunds; triangular bunds; planting pits; negarim; vallerani micro basins; eyebrow terraces; Meskat runoff basins.

¹⁴¹ Drawing from local and regional best practices.

¹⁴² Critchley W, Siegert K, Chapman C, Fink M. 1991. *Water harvesting: A manual for the design and construction of water harvesting schemes for plant production*. Rome: Food and Agriculture Organization of the United Nations (FAO).

¹⁴³ Sprong C. 2019. Soil water dynamics affected by micro rainwater harvesting structures in Jordanian Badia restoration context. MSc thesis, Universiteit Utrecht, Utrecht.

Under the project's delegated execution structure, IUCN will act as Executing Entity for Component 2.

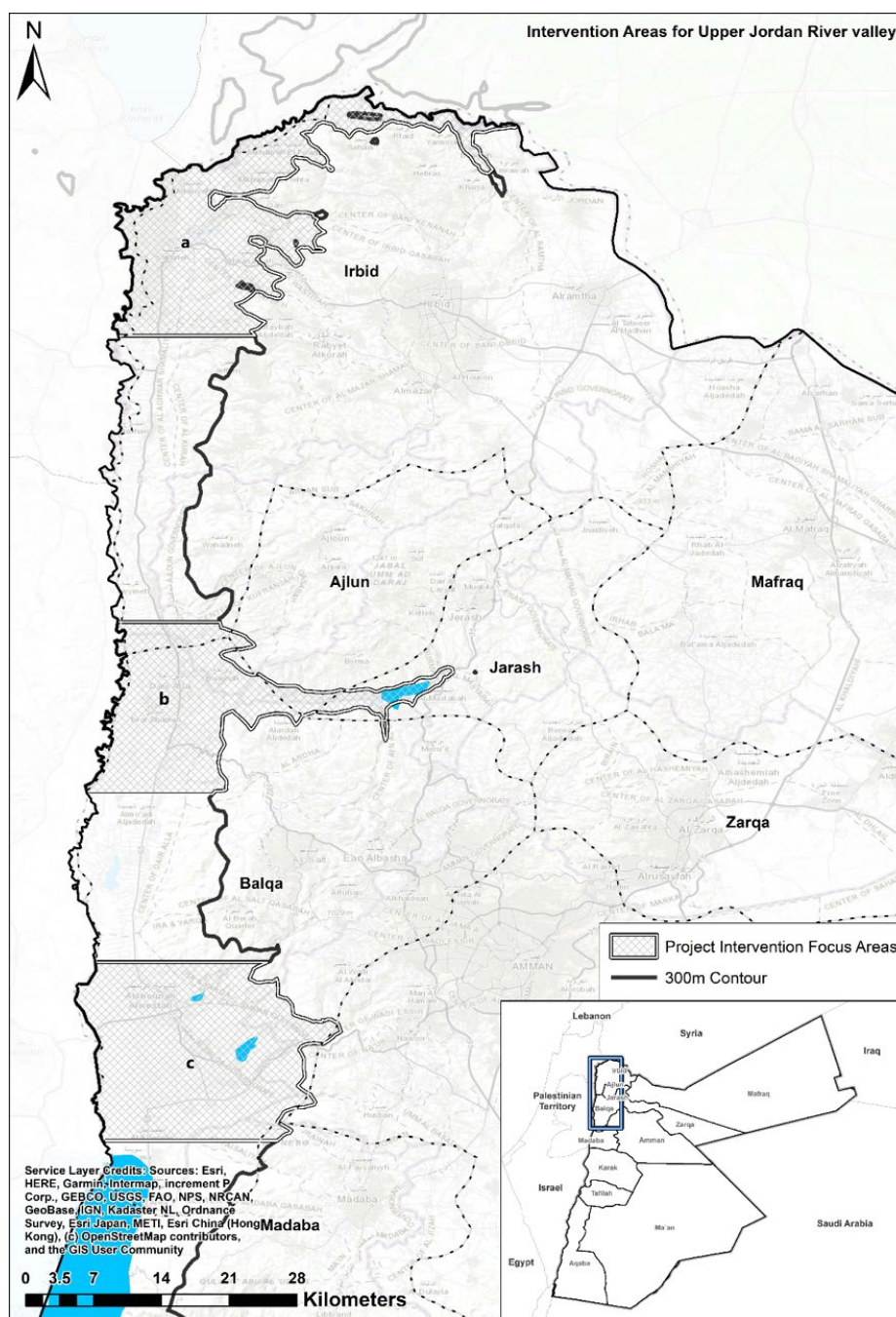


Figure 8. Project intervention area in the Jordan Valley: a) Yarmouk Forest and downstream areas; b) King Talal Dam and downstream areas; and c) Kafraïn and downstream areas.

Activity 2.1.1 Support the implementation of participatory climate-resilient integrated watershed management plans (developed under Activity 1.1.4) in the Yarmouk, King Talal Dam and Kafraïn areas of the Jordan Valley.

114. Under Activity 2.1.1, the gender-responsive climate-resilient integrated watershed management plans developed for each of the project intervention sites will be implemented by project Technical Partners (TPs) with support from WMCs. This will afford local communities (including women) management rights over their local ecosystems, incentivising the enforcement of land-management laws, resulting in reduced human-driven ecosystem degradation and increased support for restoration activities. To facilitate this activity, best practices and lessons learned including those related to gender equity, will be drawn from previous and ongoing projects with a focus on community-based landscape management and restoration. These projects include: i) HERD; and ii) the Community-based Optimisation of the Management of Scarce Water Resources in Agriculture in the CWANA Project (Section 7.2 of Annex 2: Feasibility Study, provides further detail). The implementation of the plans will be divided across Sub-

activities 2.1.1.1 and 2.1.1.2, which will be supported by Sub-activity 2.1.1.3 which involves the establishment of community-based mechanisms for monitoring and evaluation of the implemented measures.

Sub-activity 2.1.1.1. Implement climate-resilient watershed management (WSM) measures at relevant project sites.

115. Under sub-activity 2.1.1.1, climate-resilient watershed management interventions presented in the plans developed for each of the project's intervention sites (Sub-activity 1.1.4.5) will be implemented in collaboration with local communities via WMCs. By including WMCs, the participation and input of women and youth in the implementation of WSM measures will be promoted. Implementation of the measures will be overseen by the project's TPs from year 3 to 8 of project implementation.

116. Sub-activity 2.1.1.1 dependencies:

- Development of participatory climate-resilient integrated watershed management plans under Sub-activity 1.1.4.5 — completed in year 2.

Sub-activity 2.1.1.2. Establish mechanisms required for the implementation of gender-responsive climate-resilient additional livelihoods and agricultural practices under Activity 2.1.3.

117. Under Sub-activity 2.1.1.2, the mechanisms for implementing climate-resilient natural resource-based additional livelihoods — including vegetable gardening (hydroponics and permaculture) and beekeeping — and agricultural practices on smallholder farms — including selection of climate-resilient crop varieties and permaculture — will be designed and established with support from the National Agricultural Research Council (NARC), WMCs and project TPs in year 2 of the project. The mechanisms will ensure complementarity between the climate-resilient additional livelihoods and agricultural practices (to be implemented via Activity 2.1.3), as well as watershed management interventions implemented through this activity (Sub-activity 2.1.1.1), including ensuring that the selection and design thereof for each site is undertaken to ensure that they fit within the water management system implemented by the project, the impacts of climate change, market conditions (to improve income generation) and speak to the needs of various beneficiary groups (including women and youth). See additional details in Sections 8.8, 8.9 and 8.10 of Annex 2.

Sub-activity 2.1.1.3. Establish and implement community-based participatory mechanisms for monitoring and evaluating change in community climate resilience and behavioural change, including community fora.

118. To monitor the effectiveness of the climate-resilient watershed management plans, community-based participatory mechanisms (including fora) for monitoring change in community climate resilience, including behavioural changes related to awareness-raising activities under Output 4, as well as groundwater levels through taking reading through piezometers following the installation of 12 rockfill dams (year 3, Activity 3.1.1) within the project target sites will be established. This will include establishing community fora specifically for women and youth to ensure that their feedback is captured via a setting where they do not feel intimidated by the presence of men. These mechanisms will inform and draw on the project's knowledge-management interventions implemented under Activity 4.1.2 and will be overseen by WMCs and Local Development Committees (LDCs) with guidance provided by project TPs.

119. Sub-activity 2.1.1.3 dependencies:

- Development of participatory climate-resilient integrated watershed management plans under Sub-activity 1.1.4.5 — completed in year 2.
- Implementation of climate-resilient WSM measures under Sub-activity 2.1.1.1 — ongoing from year 3.
- Establishment of WMCs under Sub-activity 1.1.4.1 — completed in year 1.

Activity 2.1.2. Restore ecosystems in the watersheds of the Yarmouk, King Talal Dam and Kafraïn areas of the Jordan Valley.

120. Activity 2.1.2 consists of four sub-activities aimed at supporting and implementing climate-resilient ecosystem restoration, including the development of protocols, establishment of nurseries and the implementation of small-scale rainwater harvesting interventions at the restoration sites. The activity will result in the climate-resilient restoration of forest, rangeland, riparian and wadi ecosystems across the project's three intervention areas (Figure 3). The extent of restoration measures will include: i) 3,178 ha within the Yarmouk site; ii) 2,309 ha within the King Talal Dam site; and iii) 2,512 ha within the Kafraïn site. The ecosystem restoration interventions will prioritise increasing water resource supplies across an arid landscape under climate change conditions. Co-benefits will include the increased supply of other ecosystem services such as flood attenuation of the surrounding areas, as well as increased water and fodder supplies, to support the predominantly agricultural livelihoods of local communities.

Sub-activity 2.1.2.1. Develop protocols for the climate-resilient restoration approaches identified for each project site, based on baselines assessments.

121. Sub-activity 2.1.2.1 includes the development of detailed protocols for climate-resilient ecosystem restoration interventions at each site, led by the Royal Botanical Garden (RBG). Protocols will be developed for activities including: i) passive restoration, via the traditional Hima system for example; and ii) active restoration through re-establishing vegetative cover in degraded areas and afforestation. Included in the protocols will be inter alia lists of recommended plant species, restoration techniques, planting schedules, maintenance guidelines and recommendations for small-scale in-field rainwater harvesting interventions (to be implemented under Sub-activity 2.1.2.4). The protocols — based on short- and medium-term climate change projections — will build on lessons learned and best practices from other arid region restoration initiatives in Jordan and globally (Sections 7, 8.2 and 8.3 of Annex 2: Feasibility Study, provide additional details). In addition, the protocols will be informed by baseline assessments of the target sites' ecosystems, which will be overseen by the project's Technical Partners, as well as the results of long-term climate analyses and risk assessments, and suitability mapping using appropriate software such as the Open Foris platform (open source)¹⁴⁴. The baseline assessments will include consultations with local resource users to ensure a participatory approach. The development of restoration protocols will be led by the project's TPs in collaboration with relevant stakeholders including government, academics, practitioners and technicians of ongoing initiatives. Local resource users will also be consulted during the development of restoration protocols to ensure that traditional knowledge is considered.

Sub-activity 2.1.2.2. Implement climate-resilient restoration of degraded watersheds at project sites, 3,178 ha within the Yarmouk site, 2,309 ha within the King Talal Site, and 2,512 ha within the Kafraïn site.

122. Under Sub-activity 2.1.2.2, the restoration protocols developed via Sub-activity 2.2.1 will be implemented starting in year 2 of project implementation. Multi-use and climate-resilient plant species that can tolerate drought conditions and extreme temperatures, will be used for the restoration interventions, as detailed in the protocols. Depending on the site, species that are adapted to inter alia increased temperatures, more extreme rainfall events and more severe droughts will be selected. The seedlings to be used for the restoration interventions will be supplied by a central nursery located at the RBG and community-managed nurseries (primarily managed by women — at least 75% representation) at each intervention site (Sub-Activity 2.1.2.3).

123. The restoration will use as many indigenous species as possible because species diversity will improve resilience/stability of the ecosystem during extreme climatic conditions. While most of the restoration will occur by planting indigenous species, natural regeneration will be used where appropriate. However, many of the sites are likely to be too dry and/or too degraded for natural regeneration to be successful. The exact mix of planting versus natural regeneration will be determined and detailed in the restoration protocols to be developed by RBG. Using climate-resilient and multi-use species that supply a diverse range of ecosystem goods and services, including NTFPs, will enhance the livelihoods and food systems of local communities despite climate change conditions. Some of the benefits will accrue in the short-term, using fast-growing plant species, while other benefits will accrue over decades (for example, timber from slow-growing, hardwood species). Climate change will take place over the ensuing decades, and the planning for long-term benefits (which contribute to resilience) is part of the adaptation strategy. Planting schedules for each selected plant species will also be included in the protocols developed under Sub-activity 2.1.2.1. Incorporating climate change projections in the design of restoration protocols will climate-proof business-as-usual restoration interventions.

124. Implementation of restoration activities will be led by the project TPs, drawing from their experience and knowledge from previous restoration initiatives across Jordan. TPs will be supported by local WMCs (who will ensure the input and buy-in from local institutions), CSOs and CBOs. Local community members, including women (at least 30%), will be hired through a process overseen the project's TPs and WMCs on a temporary basis to assist with restoration activities, with roles assigned based on gender. This will follow and build on a similar model to the GIZ's Cash for Work project (more information available in Section 7.2 of Annex 2), which temporarily employs disadvantaged individuals to carry out work to reduce erosion in dam catchment areas¹⁴⁵.

125. Sub-activity 2.1.2.2 dependencies:

- Establishment of WMCs under Sub-activity 1.1.4.1 — completed in year 1.
- Development of restoration protocols under Sub-activity 2.1.2.1 — completed in year 1.
- Establishment of nurseries via Sub-activity 2.1.2.3 — completed in year 2.

Sub-activity 2.1.2.3. Establish a central nursery at the RBG and community-managed nurseries (including the provision of seed and seedlings of species identified under 2.1.2.1) at each intervention site to grow the seedlings required for climate-resilient restoration.

¹⁴⁴ <https://www.openforis.org/>

¹⁴⁵ GIZ. 2019. Cash for work: Protecting reservoirs in Jordan. Available at: <https://www.giz.de/en/worldwide/60465.html>.

126. Female-headed community-managed nurseries will be established — supported by project Technical Partners — at project intervention sites to provide seedlings for restoration activities. A central nursery will be established at the RBG to provide seed and seedling stock for the community-managed nurseries. Plant species to be propagated at the nurseries will be defined in the restoration protocols developed under Sub-activity 2.1.2.1. The use of community nurseries for enhancing biodiversity and ecosystem services have only gained traction in the past decade in Jordan¹⁴⁶. The establishment of community-managed nurseries under the proposed project will draw from the best practices of the USAID's sustainable Economic and Environmental Development (SEED) project (see Section 7 of Annex 2 for more details) and will ensure that at least 75% of nursery workers are female.

127. Sub-activity 2.1.2.3 dependencies:

- Plant species lists in restoration protocols developed under Sub-activity 2.1.2.1 — completed in year 1.

Sub-activity 2.1.2.4. Implement small-scale rainwater harvesting interventions at restoration sites to enhance the success of restoration interventions.

128. Active restoration will be complemented by the implementation of small-scale, in-field rainwater harvesting interventions (e.g., bunds, planting pits, Negarim and Vallerani basins, and runoff strips) and irrigation (e.g., porous hose and deep pipe) within the restoration sites, improving soil moisture and accordingly plant establishment (see Section 8.3 of Annex 2). The selection of in-field rainwater harvesting interventions for the restoration sites will be detailed in the restoration protocols (Sub-activity 2.1.2.1).

129. Sub-activity 2.1.2.4 dependencies:

- Recommendation on small-scale rainwater harvesting interventions presented in restoration protocols developed under Sub-activity 2.1.2.1 — completed in year 1.

Activity 2.1.3. Introduce gender-responsive climate-resilient additional livelihoods and agricultural practices in the project's three intervention areas.

130. Under Activity 2.1.3, climate-resilient gender-responsive additional livelihoods and agricultural practices will be introduced through five sub-activities to reduce the dependency of beneficiary communities on climate-vulnerable livelihoods that are heavily reliant on water resources and responsible for ecosystem degradation. This approach will complement ecosystem restoration under Activity 2.1.2 by both reducing unsustainable degradative practices and promoting livelihoods that are reliant on healthy ecosystems. As a result, the conservation as well as ongoing restoration of the ecosystems in the target watersheds will be promoted. This activity will provide immediate tangible benefits to the target communities who will only benefit from restored ecosystems in the long term. Additionally, strong support for the introduction of climate-resilient additional livelihoods was received from target communities during consultations, including those targeting women, as current livelihoods are being negatively impacted by current climate changes (see Annex 7). Consequently, it will foster their continued support for the restoration activities, which is essential for their success. In addition, the introduced livelihoods will prevent the augmentation of water resources achieved via the project from being nullified by local communities who may interpret this as an opportunity to continue with practices that result in an over abstraction of water. The additional climate-resilient livelihoods will include those that can be implemented by either men or women, such as vegetable gardening and beekeeping, as well as a transition to climate-resilient agricultural practices and post-harvest practices (that will reduce food loss and waste) will be implemented under the guidance of project Technical Partners — with technical support provided by NARC. This will support a transition to climate-resilient food systems, which support food security under climate change conditions — in alignment with the GCF's Updated Strategic Plan's (2024-2027) fourth target results area¹⁴⁷.

Sub-activity 2.1.3.1. Technical partner site visits to promote and train local communities on beekeeping as a climate-resilient additional livelihood to incentivize ecosystem restoration.

131. Beekeeping will be introduced at 150 households in total. Broken down across sites according to percentage of total direct beneficiaries¹⁸⁰, this includes 29 households from Yarmouk, 73 households from King Talal Dam and 48 households from Kafra. Each household will receive 10–20 modern beehives and equipment. Each beehive is expected to produce ~10 kg of honey per year from year three of project implementation, which can be sold at ~US\$14 per kg. Major adaptation strategies to increase climate resilience of beekeeping will include the provision of gender-inclusive training on good beekeeping practices that include knowledge of climate change and their impacts, enabling beneficiaries to select the right location, the appropriate maintenance and harvesting methods. Training will include one annual two-day training workshop (years 2 to 8) at each of the three project intervention areas for the beekeeping

¹⁴⁶ Al-Satari Y, Alokour A, Al-Zoubi E & Worekat A. 2018. Effectiveness of nursery planting dates of *Atriplex halimus* and *Salsola vermiculata* on decreasing seedlings production cost in Jordan. *Trans Stellar*. 3(5):13-18.

¹⁴⁷ <https://www.google.com/url?sa=t&rct=j&q=&esrc=s&source=web&cd=&cad=rja&uact=8&ved=2ahUKEwi-0smtg86CaxUWxQIHHelCD8UQFnoECAGQAQ&url=https%3A%2F%2Fwww.greenclimate.fund%2Fabout%2Fstrategic-plan&usg=AOvVaw1WW6ljo745gOag5pyF5By&opi=89978449>

expert (with support from Technical Partners and WMCs). Beneficiaries (at least 50% female-headed households) will be selected via consultations between project Technical Partners, WMCs and a local beekeeping expert who will be hired to assist with the technical delivery of and training for this livelihood. See Section 8.8 of Annex 2 for more details on beekeeping in Jordan. If deemed feasible during the design and implementation of the Water Fund (Activity 2.4), beneficiaries may qualify for payment for pollination services via the Funds PES mechanism, which could allow them to further scale up this livelihood.

132. Sub-activity 2.1.3.1 dependencies:

- Establishment of WMCs under Sub-activity 1.1.4.1 — completed in year 1.
- Establishment of mechanisms required for the implementation of climate-resilient additional livelihoods under Sub-activity 2.1.1.2 — completed in year 2.

Sub-activity 2.1.3.2. Technical partner site visits to promote and train local communities (at least 75% women) on vegetable gardening through hydroponics and permaculture (wicking beds) as an additional livelihood.

133. Vegetable gardening in the form of hydroponics and/or permaculture will be introduced at 600 households in total (at least 75% female headed). Broken down across sites according to percentage of total direct beneficiaries¹⁸⁰, this includes 117 households from Yarmouk, 293 households from King Talal Dam and 190 households from Kafraïn. Depending on the situation and needs at each of the beneficiary households, either hydroponics, permaculture, or a combination of these will be introduced¹⁴⁸. Hydroponic production has a low-water/high-energy input requirement, making it best suited to dry regions with abundant renewable energy¹⁴⁹. Sub-activity 2.1.3.2 will therefore also include the introduction of photovoltaic (PV) systems at the 600 households to power the hydroponics systems, combined with the training of community beneficiaries on these use and maintenance from year 4 to 8 of the project. The introduction of and training on the use of renewable energy and energy-efficient technologies will be overseen by project TPs, including identifying and selecting beneficiaries through consultations with WMCs. To avoid conflict related to the selection of beneficiaries to be supported, an important selection criterion will include willingness to pay part of the cost of the PV systems as well as to cover maintenance costs over time.

134. Once established, yields from hydroponic production systems can be up to 11 times greater than with traditional agriculture¹⁵⁰ — additional details on the benefits of hydroponics are presented in Sections 7 and 8.9 of Annex 2. Under the proposed project, tomatoes, cucumbers, peppers and eggplants will be grown under the hydroponic systems. The primary input required to set up a hydroponic system for vegetable growing is a drip irrigation system, which is easy to install and manage, and is relatively low cost. Other inputs will include tunnels, hydroponic drainage channels, planting pots or bags, and planting media¹⁵¹.

135. Permaculture will be introduced at beneficiary households in the form of wicking beds. Predominantly used for the growing of vegetables at the household level, wicking beds are self-irrigating, providing water to plants for several weeks, depending on the climate, season and location. Wicking beds consume up to 80% less water than conventional vegetable gardening methods. The beds require limited maintenance and the proliferation of weeds amongst vegetable plants is reduced because of the dry soil surface. Wicking beds of 1.5 m x 1.5 m will be constructed with beneficiary households and are expected to produce up to 100 kg of vegetables — as selected by the beneficiary household¹⁵² — per year. Each beneficiary household will be equipped with 10 wicking beds. Inputs for the establishment of a wicking bed include: i) plastic, wood or brick to construct the basin; ii) plastic lining; iii) PVC piping; iv) compost; v) soil; and vi) seedlings.

136. A hydroponics and permaculture expert will be procured to oversee the technical design and delivery of the vegetable gardening interventions with support provided by the TPs and WMCs. Additional details on permaculture are presented in Section 8.10 of Annex 2. Training of communities will include four 2-day training workshops at each of the project sites in years 2 and 5 (24 in total).

¹⁴⁸ It is expected that hydroponics systems will be introduced at 50 households per site, while the remaining 150 households will benefit from the establishment of permaculture systems.

¹⁴⁹ Barbosa, G., Gadelha, F., Kublik, N., Proctor, A., Reichelm, L., Weissinger, E., Wohlleb, G. and Halden, R., 2015. Comparison of Land, Water, and Energy Requirements of Lettuce Grown Using Hydroponic vs. Conventional Agricultural Methods. *International Journal of Environmental Research and Public Health*, 12(6), pp.6879-6891.

¹⁵⁰ Number is for annual lettuce production per kg per m². Barbosa, G. et al. 2015. Comparison of Land, Water, and Energy Requirements of Lettuce Grown Using Hydroponic vs. Conventional Agricultural Methods. *International Journal of Environmental Research and Public Health*, 12(6), pp.6879-6891.

¹⁵¹ USAID-Jordan & ECOConsult. Hydroponic System Costs and Suppliers. <http://hgfi-jo.org/pdf/Hydroponic-Kit-Supplies-Jun15.pdf>

¹⁵² It is recommended that garlic and onions are grown as plant disease control measures.

137. Sub-activity 2.1.3.2 dependencies:

- Establishment of WMCs under Sub-activity 1.1.4.1 — completed in year 1.
- Establishment of mechanisms required for the implementation of climate-resilient additional livelihoods under Sub-activity 2.1.1.2 — completed in year 2.

Sub-activity 2.1.3.3. Technical partner site visits to promote the uptake of climate-resilient agricultural practices on 1,200 smallholder farms across the three intervention areas to improve productivity under climate change conditions.

138. Climate-resilient agricultural techniques promoted on 1,200 smallholder farms through the project will include permaculture, agroforestry, conservation agriculture, organic farming, the planting of appropriate drought-resilient crop varieties and post-harvest processing to reduce food loss and waste from year 2 to 8. Broken down across sites according to percentage of total direct beneficiaries¹⁸⁰, this includes 234 smallholder farms from Yarmouk, 586 smallholder farms from King Talal Dam and 380 smallholder farms from Kafraïn. The farmers will be selected through a series of consultations between project TPs, WMCs and community representatives, with at least 30% of selected farmers being women. Selection criteria will include: i) representativeness of farmers from across each of the three intervention areas to ensure climate-resilient agricultural practices are demonstrated to as many people as possible, promoting autonomous upscaling; ii) willingness of farmers to adopt innovative agricultural practices; and iii) households whose main livelihood is farming.

139. Sub-activity 2.1.3.3 dependencies:

- Establishment of WMCs under Sub-activity 1.1.4.1 — completed in year 1.
- Establishment of mechanisms required for the implementation of climate-resilient additional livelihoods under Sub-activity 2.1.1.2 — completed in year 2.

Sub-activity 2.1.3.4. Train selected small-scale farmers within the three project sites on the use and maintenance of climate-resilient agriculture techniques, including in-field water conservation and harvesting measures.

140. With support from the MWI, MoA, NARC and project Technical Partners, the smallholder farmers selected under Sub-activity 2.1.3.3 will be trained to implement and maintain climate-resilient agricultural techniques in year 2. The gender-inclusive training will take place via demonstrations on selected farms at each of the three intervention sites. This will include 24 2-day climate-resilient agriculture — including in-field water conservation and harvesting measures — demonstration workshops for 1,200 smallholder farmers during year 2. The demonstrations will be held on selected farms across each of the three demonstration sites. Each workshop will host 50 farmers. Once trained, the selected beneficiary farmers will be supported by the project to implement and manage measures on their farms.

141. An important aspect of the climate-resilient agricultural practices will be the introduction of in-field water conservation and harvesting measures. These measures will include (as informed by consultations with MWI): i) contour ridges for crops; ii) contour bunds for trees; iii) semi-circular bunds; iv) triangular/v-shaped bunds; v) planting pits; vi) Meskat basins; vii) grass, shrub and tree runoff strips; viii) use of AI extension and pest and disease management technologies; and ix) cultivated runoff strips — see Section 8.6.2 of Annex 2 for more detail. The farmers will also receive agricultural inputs to facilitate the implementation of climate-resilient agriculture approaches, including: i) climate-resilient crop and seed varieties (including maize, barley, pearl millet and quinoa amongst others); ii) farming implements such as watering cans, hoes and wheelbarrows; and iii) equipment to implement in-field water harvesting and conservation techniques.

142. Sub-activity 2.1.3.4 dependencies:

- Establishment of mechanisms required for the implementation of climate-resilient additional livelihoods under Sub-activity 2.1.1.2 — completed in year 2.
- Selection of smallholder farmers under Sub-activity 2.1.3.3 — completed in year 2.

Sub-activity 2.1.3.5. Identify private sector champions in the agricultural sector to mentor the beneficiaries of climate-resilient additional livelihoods and agricultural practices to strengthen their capacity for post-harvest processing and marketing.

143. To ensure that beneficiary households (Sub-activities 2.1.3.1 and 2.1.3.2) and smallholder farmers (Sub-activities 2.1.3.3 and 2.1.3.4) are capacitated to market their produce, private sector champions in the agricultural sector (including women) will be identified in year 2 by TPs, with support from WMCs, to mentor the beneficiaries of climate-resilient additional livelihoods and agricultural practices to strengthen their capacity for post-harvest processing to reduce food loss and waste, and entrepreneurship. This will also include standardising product quality, facilitating access to relevant markets and supporting the development of microenterprises where feasible, promoting the sustainability of income-generating livelihoods.

144. Sub-activity 2.1.3.5 dependencies:

- Establishment of WMCs under Sub-activity 1.1.4.1 — completed in year 1.

Output 2.2: Water Fund established in the Jordan Valley to maintain landscape investment for enhanced water resources

Activity 2.1.4. Establish a Water Fund in the Jordan Valley to maintain landscape investment for enhanced water resources.

145. Under the proposed project, a Water Fund will be established as a model for the payment for ecosystem services (PES), in this case, water resources, serving as a mechanism for leveraging financial resources from private (industry and commercial agriculture) and public sector stakeholders (municipalities) for restoration (EbA) interventions in watersheds. This will ensure the ongoing availability of funding for the maintenance and restoration of ecosystems in the three target watersheds and as such promote a continued supply of ecosystem services — particularly those related to water resources. Specifically, the rationale behind this is to provide financial incentives to upstream communities to reduce practices that result in compromised water quality and quantity downstream, and thereby serve as custodians of the ecosystems in these areas. The general model of a Water Fund is presented in the figure below¹⁵³.

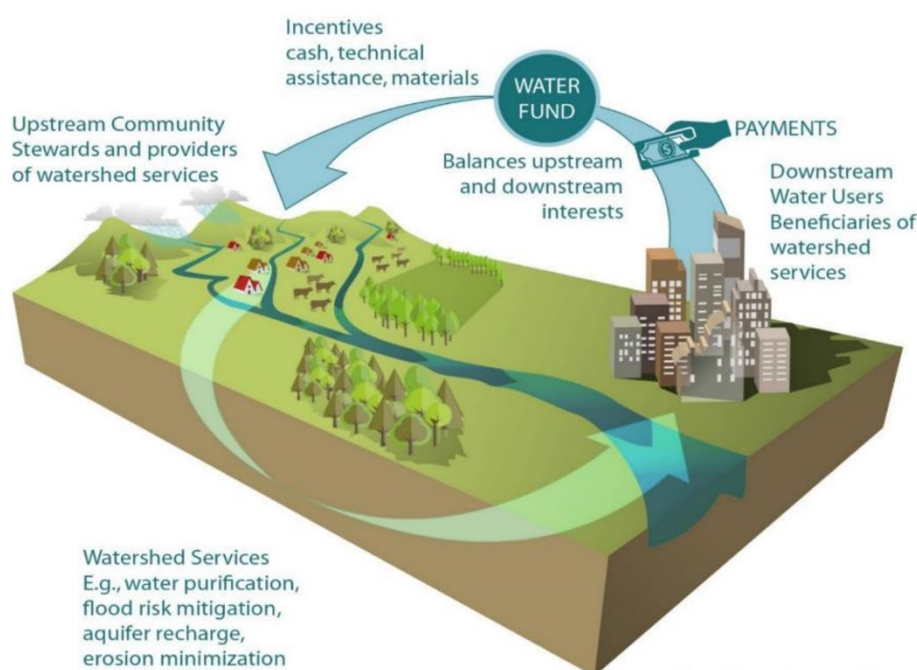


Figure 9. Water Fund model.

146. Several studies undertaken in Jordan have demonstrated a substantial willingness to pay (WTP) for the preservation and provisioning of environmental resources. For example, local communities in the Dead Sea region showed a considerable willingness to pay for the conservation of the Dead Sea Basin — particularly for the provisioning of water for domestic and agricultural consumption¹⁵⁴. This study highlighted that the long-term benefits of improving the environment outweigh the costs, because of the increase in benefits from restoration interventions over time compared with the short-term gains from business-as-usual activities in the region¹⁵⁵. In 2014, an economic valuation of the benefits of ecosystem restoration in Jordan determined the willingness of rangeland field workers and herders to pay for water for their livestock, delivered by truck to their communities¹⁵⁶. Depending on the aridity of the area and the distance to natural sources of water, livestock owners were willing to pay 2 JOD/CM (US\$2.80) on average. The economic valuation shows that this value adequately reflects the societal marginal WTP for water because of a direct correlation with the costs associated with the construction of additional dams in the country¹⁵⁷. In another study, the WTP of pastoralist communities for the restoration of rangeland ecosystems was determined under the implementation

¹⁵³ Source: <https://wlearn.net/resolveuid/9249364e-8d03-4e05-98c7-aafd223cfe0d>

¹⁵⁴ Becker N & Katz DL. 2009. An economic assessment of Dead Sea preservation and restoration. In *The Jordan River and Dead Sea Basin*. Springer, Dordrecht.

¹⁵⁵ Ibid.

¹⁵⁶ Myint MM & Westerberg V. 2015. An economic valuation of a large-scale rangeland restoration project through the Hima system in Jordan. Report for the ELD Initiative by International Union for Conservation of Nature, Nairobi, Kenya.

¹⁵⁷ Ibid.

of the Hima System. WTP was estimated at 120 JOD/month (US\$170) with all fodder sourced from improved landscapes. Because restoration of the landscape results in the replenishment of streams that have dried during the previous 20 years, average WTP of herders increased by 56 JOD (US\$79) per month¹⁵⁸.

147. Anchored in the JVA, the Water Fund will be governed by a multi-stakeholder public-private steering committee that includes representatives of WMCs (established under Sub-activity 1.1.4.1) from the target watersheds, project Technical Partners, governmental partners (such as MWI, MoEnv and MoA), water utilities (such as the Yarmouk Water Company¹⁵⁹), as well as prominent private sector partners in the Jordan Valley. This will provide both political influence, accountability, transparency and societal trust. Direct governance and administrative support will be provided by project Technical Partners. The three primary organisational components of the fund will include: i) a funding mechanism housed within the JVA for the collection and provision of resources for watershed conservation with a focus on climate-resilient ILWRM; ii) a governance mechanism linked to the WMCs and abovementioned implementing partners for planning and decision-making; and iii) a mechanism for watershed management also linked to the WMCs and abovementioned implementing partners to undertake funded activities for conservation and management — this will include an on-granting facility and the Water Fund's grant mechanism. The design and establishment of the Water Fund will take place during project implementation. A climate finance expert will be procured as part of the PMU to oversee the financial aspects of the design and operationalisation of the Water Fund during its first two years, while an experienced fund manager will be engaged by the project to manage the fund.
148. During the establishment, operation and maturity phases of the Water Fund, capital provided by stakeholders, including through PES, can be invested in financial markets through a trust fund, with returns on investment used to provide a continual flow of funds for ongoing restoration and maintenance of watershed ecosystems, and as a result, the continued conservation and attenuation of water resources in the project's intervention areas¹⁶⁰. The use of a trust fund as the foundation for the Water Fund has several benefits, including mitigating cash-flow risks, ensuring that water producers can be certain that the multi-annual commitments of the fund will be fulfilled, and ensuring the credibility of the Water Fund. Another benefit of a trust fund is that it facilitates investment by a variety of investors. Financing strategies to capitalise the Water Fund may include: i) voluntary private donations — including funding from individuals, philanthropic organisations or private corporations with a strategic interest in watershed protection; ii) direct domestic government resources — including financial resources obtained from the country's national budget for natural resources management; iii) multilateral organisations — particularly those with an interest in natural resources management and protection; iv) government fiscal mechanism — including taxes or levies (in the case of Jordan, a portion of existing water levies collected by the MWI/JVA could potentially be diverted to the Water Fund); and v) other sources such as water facilities, hydroelectric companies, or interested productive sector organisations.
149. Medium to long-term investment in the Water Fund will be stimulated via the Fund's long-term financing strategy, which will include securing agreements with major water users who are willing to pay for ongoing ecosystem management in return for the water resources generated — a PES model (including payments for pollinations services to beekeepers such as those supported under Activity 2.1.3). Payers for ongoing ecosystem management would include major water users in downstream areas of the Jordan Valley who would benefit from and rely on increased water resources generated via the Fund's interventions in upstream ecosystems. These include large-scale agricultural producers such as commercial fruit and vegetable producers (the agriculture sector accounts for more than half of all water used in Jordan)¹⁶¹, and the industrial sector (e.g., fertiliser producers, potash and phosphate mining, thermal power plants and cement factories)¹⁶², which although only currently contributes ~3% to the country's overall water demand, continues to grow¹⁶³. Potential payers may include agricultural producers such as the Alfaqih Group¹⁶⁴ and Haddadin Farms¹⁶⁵, large industry including the Manaseer Group¹⁶⁶, Arab Potash Company¹⁶⁷ and Jordan Phosphate Mines Company¹⁶⁸, as well as water utilities such as the Yarmouk Water Company¹⁶⁹.
150. While the Water Fund will only be designed during the first two years of implementation of the proposed project, its broad scope is presented in the diagram below. The presented structure remains flexible and will be refined during

¹⁵⁸ Ibid.

¹⁵⁹ http://www.yw.com.jo/EN/yw_info.aspx

¹⁶⁰ TNC. 2012. Water Funds Conserving Green Infrastructure. Available at: <https://www.nature.org/media/freshwater/latin-america-water-funds.pdf>

¹⁶¹ Ministry of Water and Irrigation. 2017. Jordan water sector — Facts and figures. MWI, Amman.

¹⁶² Government of Jordan. 2014. Jordan's Third National Communication on Climate Change.

¹⁶³ Ministry of Water and Irrigation. 2017. Jordan water sector — Facts and figures. MWI, Amman.

¹⁶⁴ <https://alfaqihgroup.com>

¹⁶⁵ <http://haddadinfarms.com/#home>

¹⁶⁶ <https://www.manaseergroup.com/Home/>

¹⁶⁷ <https://www.arabpotash.com/Default/En>

¹⁶⁸ <http://www.jpmmc.com.jo/Default/En>

¹⁶⁹ http://www.yw.com.jo/EN/yw_info.aspx

the design phase. Finance from the proposed GCF project will be used to design, operationalise and establish the Fund, while long-term funding will be secured through the implementation of the long-term financing strategy that will be developed (see list of potential financing strategies above for capitalising and maintaining the Fund). The Water Fund will not directly finance ILWRM measures implemented by the proposed project (such as Component 2's ecosystem restoration — see below). However, it will contribute to the maintenance and scaling up of ecosystem management measures in the medium to long term.

151. In terms of Water Fund viability, cost-benefit analysis of a reference Water Fund based on a 30-year time horizon, with an investment of US\$ 10 million being disbursed at a rate of US\$ 1 million per year for 10 years, shows that the Fund reaches viability around year 20 after implementation (**Error! Reference source not found.**). While this is a relatively long period, it reflects the spread of investment over a 10-year period¹⁷⁰. In terms of size and replenishment potential of the Water Fund, three potential scenarios are presented below.

- **Low replenishment scenario** — Following an initial GCF investment of US\$ 7,500,000 via the Fund's grant mechanism during project implementation (disbursed at a rate of up to US\$ 1,500,000/year), this is matched by voluntary private donations, donor capitalisation (multilateral fund) and/or public budgets. Essentially, US\$ 1,500,000 per year will continue to be available through the Fund following project closure. Such investment will be supported by the enabling environment created under Component 1, including strengthened governance and capacity for climate-resilient ILWRM in Jordan (Output 1.1) and evidence of ILWRM as an adaptation approach and specific policy recommendations (Output 1.2).
- **Medium replenishment scenario** — This includes the low replenishment scenario described above plus the contribution of one major downstream water user (water utility or private sector). An example of such a water user is the Jordan Valley Authority (JVA). Assuming annual revenue of at least US\$ 15,000,000,¹⁷¹ JVA could conservatively invest US\$ 225,000 (1.5%) into the Water Fund annually, resulting in a total annual investment of US\$ 7,725,000 annually.
- **High replenishment scenario** — This includes the medium replenishment scenario described above, plus a stronger enabling environment supporting policy and subsidy reform in the water sector that enhances investment in landscape management for water resources. Assuming that evidence of ILWRM as an adaptation approach generated under Output 2.1 fosters interest in the Water Fund from multiple major downstream water users, including the private sector, there is potential for an annual investment of at least US\$ 10,000,000 in the Water Fund annually.

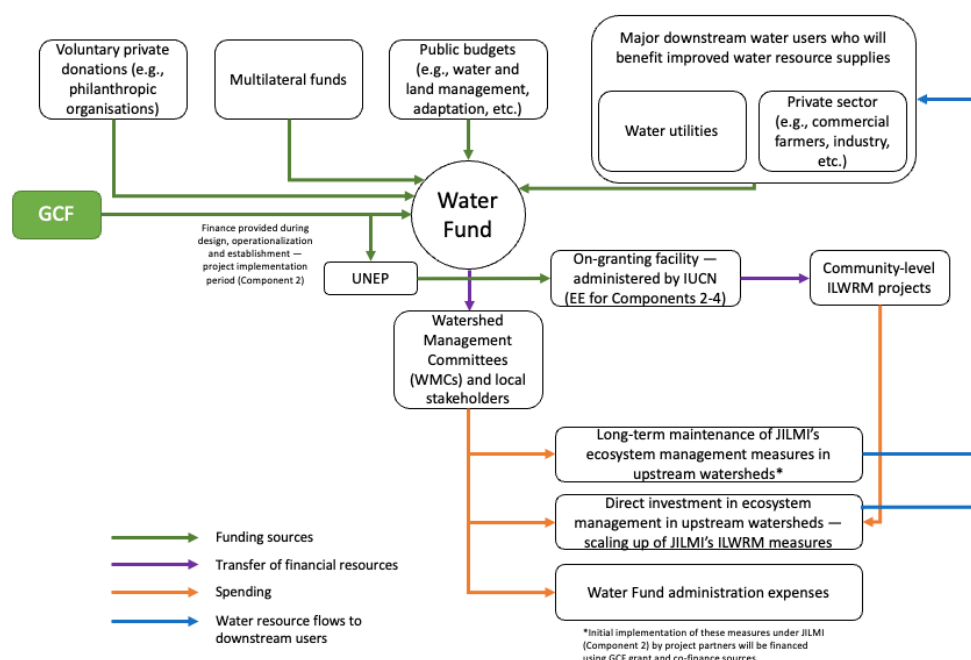


Figure 10. Broad scope of the Water Fund. To be elaborated on and refined during the Fund's design.

¹⁷⁰ TNC. 2015. Upper Tana-Nairobi Water Fund Business Case. Version 2. The Nature Conservancy: Nairobi, Kenya.

¹⁷¹ <https://documents1.worldbank.org/curated/en/275541467993509610/pdf/104504-WPS-Box394877B-PUBLIC-Add-series-WSP.pdf>

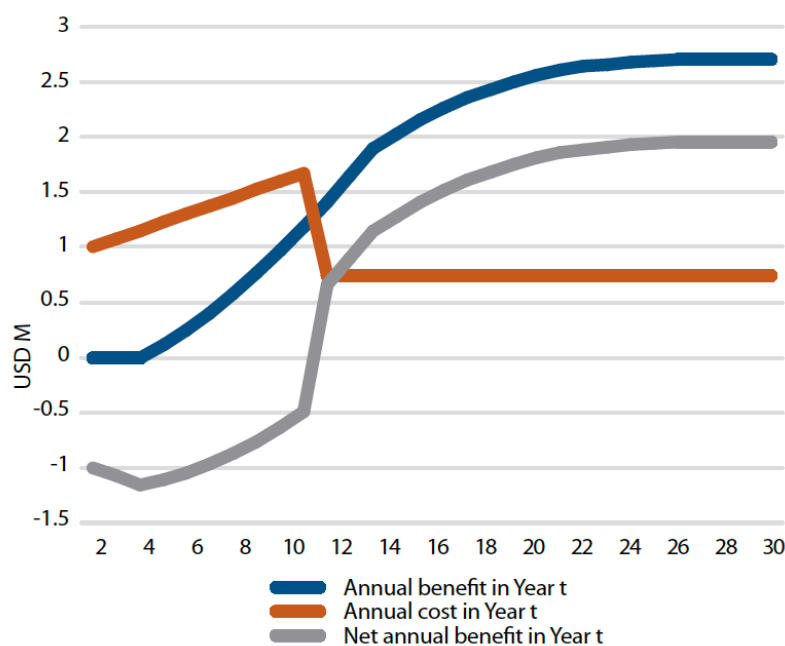


Figure 11. Total annual benefits and costs over time including continued maintenance after 10 years for a reference Water Fund¹⁷².

Sub-activity 2.1.4.1. Design the Water Fund in a participatory manner, including workshops with local communities, institutions, private sector and NGOs — Technical Assistance (TA)

152. During the design phase (first 2 years of project implementation), development of an actionable strategic plan detailing how the Water Fund will: i) build a reputation as a credible, collective action platform; ii) increase its ability to influence public policy by establishing itself as a credible facilitator of multi-stakeholder dialogue and by evolving from simple agreements towards sophisticated legal structures; and iii) create considerable positive impacts on water security through its reputation. A water resources expert procured by the project will undertake a feasibility and situational analysis to inform the design of the Fund. This will, amongst other things, include design studies and baseline surveys. The baseline surveys will be designed to support a future impact assessment, using the 'before-after, control impact (BACI)' approach. This will provide useful recommendations for Water Fund implementation and monitoring. For example, baseline surveys under the Upper Tana-Nairobi Water Fund identified areas with steep and very steep farmlands that lack terraces and areas that lack buffer zones of permanent vegetation along local rivers, but where locals were eager to implement soil erosion control measures. Such detail can assist with the micro-targeting of Water Fund activities¹⁷³. The design studies will also include a willingness to pay (WTP) study to in the Jordan Valley to assess WTP of downstream users for water resources as an ecosystem service generated via Water Fund investments. The study will primarily target major private sector water users (including agricultural producers such as the Alfaqih Group and Haddadin Farms, large industry including the Manaseer Group, Arab Potash Company and Jordan Phosphate Mines Company, as well as water utilities such as the Yarmouk Water Company). A core focus of Water Fund design will be how it can address gender- and youth-related issues, including enhancing their access to finance for climate change adaptation. The design phase will also include the development of eligibility criteria for accessing the Water Fund. Preliminary eligibility criteria are presented in Table 5. ToR for the design of the Water Fund are presented in Annex 32.

153. The design phase will be initiated by project Technical Partners, with technical support from the Nature Conservancy (TNC) and WMCs. During this phase lessons and best practices from other Water Funds around the world will be taken into consideration, such as the Upper Tana-Nairobi Water Fund¹⁷⁴ (see Section 8.5 of Annex 2). Governing partners of the Fund will then seek to leverage financial support from stakeholders and investors (including the private sector) for long-term sustainability. This will include designing the PES model for the Water Fund (including payments for pollinations services to beekeepers such as those supported under Activity 2.1.3), as well as identifying and engaging with downstream private sector stakeholders (such as commercial agriculture and industry) who rely on and will benefit from the water resources as an ecosystem service supplied by the target watersheds. Private sector engagement will also take place as a means to building the financial sustainability and scalability of the fund during its implementation. In addition, consultations

¹⁷² TNC. 2015. Upper Tana-Nairobi Water Fund Business Case. Version 2. The Nature Conservancy: Nairobi, Kenya.

¹⁷³ TNC. 2015. Upper Tana-Nairobi Water Fund Business Case. Version 2. The Nature Conservancy: Nairobi, Kenya.

¹⁷⁴ <https://nairobiwaterfund.org>

with key partners will be used to explore other instruments and strategic partnerships that can be included in the Water Fund. The design phase will also include identifying further opportunities for strengthening linkages and increasing complementarities with GCF's recently approved project for Jordan (FP155, BRCCJ) through consultations with its AE (FAO), EEs and project management unit (PMU).

154. Sub-activity 2.1.4.1 dependencies:
- Establishment of WMCs under Sub-activity 1.1.4.1 — completed in year 1.

Table 5. Preliminary eligibility criteria for accessing the Water Fund (to be further developed during Water Fund design phase.

Activity Scope	Final beneficiaries	Activity target geography
1. New capital investment in green infrastructure that increases the security, quality and/or availability of water for communities in the Jordan Valley, and/or productive use, that also: • Demonstrate public benefit with a national interest element and demonstrated engagement with affected stakeholders; and • Are brought forward with strong support from local stakeholders, such as local authorities and communities (including securing funding contributions). 2. Ecosystem management-related activities deemed appropriate for the target watersheds that contribute positively to water resource quantity and/or quality, such as: • Ecosystem rehabilitation, restoration and conservation; • Gender-responsive climate-resilient additional livelihoods; • Water efficiency and technologies in the agricultural sector; • Fuel sources alternative to biomass that promote a circular economy approach; • Sedimentation management to reduce the pollution of water resources through the revegetation of erosion-prone areas and management of downhill runoff conveyance; • Soil conservation adoption for rural infrastructure development; • Climate-resilient agriculture to support ecosystem management objectives; • Reduction of dry season water demand from rivers, streams, springs and wells by irrigators through encouraging water harvesting practices; • Measures identified in climate-resilient integrated watershed management plans (Activity 1.4); and • Measures identified in community-based water use monitoring and management plans (Activity 3.4). 3. Water infrastructure for the sole benefit of residents in urban areas is ineligible for funding, except to the extent they include a productive economic element and contribute to ILWRM. 4. Activities that supply water for the exclusive use of a single entity or that primarily benefit one major customer are also ineligible for funding. 5. Consistency with national priorities as well as those for the Jordan Valley.	• WMCs (established through Activity 1.4) and their community-level members, including Development Advisory Committees, Stakeholders Advisory Groups, agricultural cooperatives, women's cooperatives, youth groups and Water User Associations — based within the upper Jordan Valley. • Local municipalities — based within the upper Jordan Valley. • NGOs and CSOs that are able to demonstrate credibility and a track record in the water and/or environmental sector and meet the following criteria: ○ Be an NGO committed to Jordan's development priorities, with proposed activities that will be implemented in the Jordan Valley. ○ Be Jordan-based or provide evidence of partnership with Jordanian institutions. ○ Provide evidence of currently valid legal registration under the laws of Jordan, as well as evidence of a certificate to carry out business in Jordan, if necessary. ○ Have appropriate organisational and management capacity, including a governing board. ○ Show the existence of sound financial systems including clear accounting and budgeting standards, financial statements, a transparent budgeting process, audited accounts and other indicators that confirm their capacity to assume fiduciary responsibility for Water Fund resources. ○ Provide evidence of competence, based on past performance, to carry out proposed activities. ○ Demonstrate credibility, knowledge of local values, networks and structures required to carry out the indicated activities. ○ Have their proposals supported by the local government as evidenced by a letter signed by the appropriate official of the government, indicating that government supports the request as being consistent with national priorities. This letter must	• Within the upper Jordan Valley. • Not located on private land. • Area that would not result in physical or economic displacement of local communities. • Area that would not result in decreased food producing potential of the land. • Ecosystems or land that show potential will contribute to improved water resource supplies if placed under improved management. • Area that demonstrates a low safeguards risk based on environmental and social impact assessments. • Area that is anticipated to result in positive benefits for communities in the immediate surroundings.

Activity Scope	Final beneficiaries	Activity target geography
6. Credibility ownership and commitment of the beneficiary. 7. Effectiveness and sustainability of the institutions and investments. 8. Opportunity for effective implementation. 9. Appropriate expected outputs with clear indicators and well-defined targets.	be sent together with the application for support.	

Sub-activity 2.1.4.2. Design an on-granting facility (under UNEP's on-granting accreditation) for the implementation of ILWRM sub-projects. This will include finalizing selection criteria for the evaluation of proposals submitted in response to RFPs — TA

155. An on-granting facility will be designed and established during year 2 of the project. The facility will allow WMCs and their members (including local communities and farmers) to implement sub-projects that are aligned with the proposed project's climate-resilient ILWRM approach and directly or indirectly benefit water resources — additional detail is presented in Table 6 below. The on-granting facility will target sub-projects at the community level which fall into the ILWRM thematic area and contribute to increasing and protecting the Jordan Valley's water resources under climate change conditions. Sub-projects will focus on ecosystem management and stewardship activities that augment water resource supplies for downstream users, who will pay for the ecosystem service provided under the PES model of the Water Fund. A core focus of on-granting facility design will be how it can address gender- and youth-related issues, including enhancing their access to finance for climate change adaptation. The final design of the on-granting facility will be prepared during the Water Fund's design phase in Sub-activity 2.1.4.1. The on-granting facility will function alongside the Water Fund's grant mechanism to support the implementation of ILWRM sub-projects by community-based organisations and non-government organisations. This will help to build up the profile of Water Fund and establish and embed the on-granting practice that will be upscaled by the Water Fund. The Water Fund will be open to other contributions and is expected to take over the on-granting model as the exit mechanism for the GCF-funded on-granting mechanism. IUCN as Executing Entity will administer the on-granting facility with oversight provided by UNEP in its capacity as AE. Funds will be transferred from GCF to UNEP to IUCN for on-granting (see Figure 9). All grants disbursed under the on-granting facility will be non-repayable / non-reimbursable in nature.

Table 6. On-granting facility design features

Type of Support	Qualifying Beneficiaries	Focus of Sub-projects	Maximum amount
Financial resources to implement climate-resilient ILWRM activities in target watersheds	WMCs (established through Activity 1.4) and their community-level members, including Development Advisory Committees, Stakeholders Advisory Groups, agricultural cooperatives, women's cooperatives, youth groups and Water User Associations	Ecosystem management-related activities deemed appropriate for the target watersheds that contribute positively to water resource quantity and/or quality, such as: • Ecosystem rehabilitation, restoration and conservation; • Gender-responsive climate-resilient additional livelihoods; • Water efficiency and technologies in the agricultural sector; • Fuel sources alternative to biomass that promote a circular economy approach; • Sedimentation management to reduce the pollution of water resources through the revegetation of erosion-prone areas and management of downhill runoff conveyance; • Soil conservation adoption for rural infrastructure development; • Climate-resilient agriculture to support ecosystem management objectives; • Reduction of dry season water demand from rivers, streams, springs and wells by irrigators through encouraging water harvesting practices; • Measures identified in climate-resilient integrated watershed	US\$500,000 per year (up to 5 sub-projects) Any remaining funds will be shifted to the following year.

Type of Support	Qualifying Beneficiaries	Focus of Sub-projects	Maximum amount
		management plans (Activity 1.4); and • Measures identified in community-based water use monitoring and management plans (Activity 3.4).	

Sub-activity 2.1.4.3. Officially launch the Water Fund and formalise long-term legal mechanisms — TA

156. Sub-activity 2.1.4.3 will include an official public launch event during year 3, which will create opportunities for attracting membership and investors — including promoting the benefits of a PES model for the private sector — as well as a platform for communicating the Water Fund's vision. During this establishment phase, the Water Fund will become a 'legal entity' and long-term legal mechanisms such as contracts and bylaws will be formalised via the WMCs. In addition, an Annual Operating Plan and Operational Management Readiness will be established¹⁷⁵. In terms of the Water Fund's formal legal establishment:

- Legal form — fund
- Status — public
- Structure — decentralised governance involving contracts between various public and private stakeholders
- Ownership — public
- Establishment formalities — via approval of the Jordanian Legislation and Opinion Bureau followed by that of the Ministerial Cabinet

157. Sub-activity 2.1.4.3 dependencies:

- Establishment of WMCs under Sub-activity 1.1.4.1 — completed in year 1.
- Finalised design of the Water Fund under Sub-activity 2.1.4.1 — completed in year 2.

Sub-activity 2.1.4.4. Operationalise the Water Fund through the development and implementation of a workplan — TA

158. During its operationalisation phase (starting in year 4 of the proposed project), stability within the Water Fund will be established by developing and implementing a workplan that guides: i) the systematic execution of activities; ii) M&E of activities; iii) communication of progress; and iv) improvement of the Water Fund through adaptive management. An experienced Water Fund manager will be hired through a competitive process during the operationalisation phase. Indicative selection criteria for the Water Fund manager are presented below and will be further developed into ToRs during project implementation.

- Relevant degree and 7 years' experience in conservation practice or related field or equivalent combination of education and experience.
- Fundraising experience, including identifying donor prospects and donor cultivation.
- Experience managing complex or multiple projects with strategic goals, including managing finances and coordinating the work of other professionals and partners.
- Experience directing a major program or project of strategic importance, including management/supervision of multi-disciplinary teams and meeting deadlines.
- Experience in partnership development with non-profit partners, community groups, corporate sector and/or government agencies.
- Experience with financial oversight and management of project and program budgets.
- Communicating clearly via written, spoken, and graphical means in Arabic and English and other relevant languages.
- Experience in working with NGO partners, and government agencies.
- Outstanding project management capabilities.
- Experience with communication and presentation skills; ability to persuasively convey the work of the Water Fund to diverse groups government officials, corporate executives, major donors, board members and others who are critical to the Water Fund's success.
- Experience with writing grant and funding proposals.

159. Sub-activity 2.1.4.4 dependencies:

- Establishment of WMCs under Sub-activity 1.1.4.1 — completed in year 1.
- Finalised design of the Water Fund under Sub-activity 2.1.4.1 — completed in year 2.
- Launch of the Water Fund and formalisation of its long-term legal mechanisms under Sub-activity 2.1.4.3 — completed in year 3.

¹⁷⁵ TNC. 2019. Creation. Available at: <https://waterfundstoolbox.org/project-cycle/creation>

Sub-activity 2.1.4.5. Develop and disseminate demand-based RFPs for ILWRM sub-projects under the on-granting facility to WMCs and their members

160. The on-granting facility will be accessed through requests for proposals (RFP) for sub-projects for the implementation of ILWRM measures across the project's three target watersheds. IUCN will launch demand-based RFPs for WMCs and their community-level members from year 3 of the project — including community-based organisations such as women's cooperatives and non-government organisations (Table 6). The RFPs will focus on sub-projects at the community level which fall into the ILWRM thematic area and contribute to increasing and protecting the Jordan Valley's water resources under climate change conditions. IUCN and project TP's will consult with communities via WMCs to identify the types of ILWRM sub-projects that are in demand at the community-level and prepare the RFPs accordingly. This will include a series of design/technical studies to help optimise which set of ILWRM interventions should be implemented via sub-projects and where implementation should take place. Implementing sub-projects will help build community and local contractor capacity in the design and implementation of ILWRM interventions. This approach will foster the buy-in of WMCs and their community-level members. The selected sub-projects will demonstrate tangible results of climate-resilient ILWRM measures at a small-scale, building a case for investment at a larger scale. The RFPs will include clear details on the scope of projects that will be considered for grants and the selection criteria. Additionally, the structure of the proposals will be defined, and a template provided. The project's TP's will provide support to the WMCs and their members (specifically women and youth) for the development of proposals.

161. Sub-activity 2.1.4.5 dependencies:

- Establishment of WMCs under Sub-activity 1.1.4.1 — completed in year 1.
- Finalised design of the on-granting facility and the Water Fund's grant mechanism under Sub-activity 2.1.4.2 — completed in year 2.

Sub-activity 2.1.4.6. Evaluate proposals submitted in responses to the RFPs and select sub-projects to receive grant funding — TA

162. Preliminary selection criteria for the sub-projects are presented in Table 7 and will be finalised by IUCN with guidance provided by UNEP (AE), MoEnv (co-EE), JVA, the national Technical Working Group (TWG), Chief Technical Advisor (CTA) and Water Fund manager during preparation of the RFPs — in addition to the eligibility criteria for accessing the Water Fund in its entirety, which will be developed under Sub-activity 2.1.4.1. Evaluations of proposals submitted in response to the RFPs will be performed undertaken by TP's, private sector stakeholders (downstream water users who would be paying for the increased water resource supplies), the national TWG, the CTA and the project's gender expert, with oversight provided by UNEP in its role as AE.

163. Sub-activity 2.1.4.6 dependencies:

- Dissemination of demand-based RFPs for sub-projects under Sub-activity 2.1.4.5 — ongoing from year 3.
- Establishment of WMCs under Sub-activity 1.1.4.1 — completed in year 1.

Table 7. Preliminary selection criteria for sub-projects under the on-granting facility.

Beneficiaries	Proposal Scope	Proposal Design
• WMCs or community-level members thereof based within the project's three target watershed (Yarmouk, King Talal Dam and Kafraïn). At least 50% representation of women.	• ILWRM-related, such as: ○ Ecosystem rehabilitation, restoration and conservation; ○ Climate-resilient additional livelihoods; ○ Climate resilient agriculture; ○ Measures identified in climate-resilient integrated watershed management plans (Activity 1.1.4); and ○ Measures identified in community-based water use monitoring and management plans (Activity 3.4). • Contributes to water resources attenuation. • Has climate change adaptation benefits. • Located within at least one of the three project's target watersheds	• Aligns with the scope of the RFP. • Has a ILWRM-related objective, clearly described and costed activities, and measurable indicators and targets. • Align with and are informed by guidelines, standards and protocols for climate-resilient ILWRM in the Jordan Valley developed through Activity 1.1.1. • An activity-level budget is presented and the resources requested are sufficient to implement the sub-project. • Intervention sites and beneficiaries are defined. • Stakeholder participation in the design and implementation of the project needs to be clearly described, showing clear ownership of the project. • Clearly described implementation arrangements, including capacity of proposal owners to implement and if any support will be provided by any partners (such as project Technical Partners, government departments, NGOs, etc). • Interventions have a clear scientific, technical, social, legal and/or economic basis.

Beneficiaries	Proposal Scope	Proposal Design
	- Focuses on vulnerable people in the target area (such as women and farmers or people reliant on natural resource-based livelihoods). - Demonstrates gender responsiveness.	- Potential social and environmental safeguards risks related to the inventions have been screened and risk level and mitigation measures align with that presented in the proposed project's ESIA and ESMP (Annex 6). - Complementarity with any past or ongoing initiatives is clearly presented, and any co-financing sources described. - Identifies synergies with other sub-projects under the on-granting facility. - Exit strategy and sustainability approach is defined. - Proposals are gender responsive, promoting the inclusion of women, considering their roles in society and have activities specifically tailored for women and align with the project's Gender Action Plan (GAP) — proposals will be reviewed by JILMI's gender expert to confirm this.

Sub-activity 2.1.4.7. Sign sub-project agreements with beneficiaries of grant support — TA

164. Once proposals have been evaluated, sub-project agreements will be signed between IUCN and beneficiaries (legal entities: NGOs and CBOs) who have been selected to receive grant support from year 3.

165. Sub-activity 2.1.4.7 dependencies:

- Evaluation of proposals under Sub-activity 2.1.4.6 — ongoing from year 3.

Sub-activity 2.1.4.8. Disburse sub-project grants and provide sub-project implementation oversight — On-granting

166. Following the signing of sub-project agreements, sub-project grants will be disbursed to legal entities: NGOs and CBOs (beneficiaries who have signed sub-project agreements) from years 3 to 7 via the on-granting facility by IUCN. Oversight for sub-project implementation will be provided by the project's TPs and CTA.

167. Sub-activity 2.1.4.8 dependencies:

- Signing of sub-project agreements with beneficiaries under Sub-activity 2.1.4.7 — ongoing from years 3.

Sub-activity 2.1.4.9. Develop and implement long-term financing and scaling up strategies for the Water Fund to leverage long-term financing and facilitate upscaling and replication in other areas of the Jordan Valley — TA

168. The final phase (maturation phase) will focus on the long-term viability of the Water Fund through the development and implementation of long-term financing and scaling-up strategies. These strategies will focus on: i) securing long-term financing through fostering private sector buy-in for PES (improved water resources) as identified and engaged during the design phase (aspects of the long-term financing strategy are described in more detail below); ii) integrating the Water Fund into the Jordan Valley's legal framework (institutionalisation) and ensuring long-term ownership; iii) monitoring and evaluating progress systematically and applying adaptive management where required; and iv) securing support from the public to maximise the positive impacts of the Water Fund.

169. The long-term financing strategy for the Water Fund will be distinct from a business case analysis, which only focusses on whether there will be financial benefit for investors. At the heart of the long-term financing strategy will be the Water Fund's PES model, which will be used to foster medium to long-term investment in the Fund from major downstream water users. This will include securing agreements with major water users who are willing to pay for ongoing ecosystem management in return for the water resources generated. Payers for ongoing ecosystem management in upstream watersheds would include major water users in downstream areas of the Jordan Valley who would benefit from and rely on increased water resources generated via the Fund's interventions in upstream ecosystems (see above for additional details). These include large-scale agricultural producers such as commercial fruit and vegetable producers (the agriculture sector accounts for more than half of all water used in Jordan)¹⁷⁶, and the industrial sector (e.g., fertiliser producers, potash and phosphate mining, thermal power plants and cement factories)¹⁷⁷, which although only currently contributes ~3% to the country's overall water demand, continues to grow¹⁷⁸. Potential payers may include agricultural producers such as the Alfaqih Group¹⁷⁹ and Haddadin Farms¹⁸⁰,

¹⁷⁶ Ministry of Water and Irrigation. 2017. Jordan water sector — Facts and figures. MWI, Amman.

¹⁷⁷ Government of Jordan. 2014. Jordan's Third National Communication on Climate Change.

¹⁷⁸ Ministry of Water and Irrigation. 2017. Jordan water sector — Facts and figures. MWI, Amman.

¹⁷⁹ <https://alfaquiigroup.com>

¹⁸⁰ <http://haddadinfarms.com/#home>

large industry including the Manaseer Group¹⁸¹ and Jordan Phosphate Mines Company¹⁸², and water utilities such as the Yarmouk Water Company¹⁸³. The inclusion of such payers in the Water Fund will be tested during project implementation.

170. Also included in the long-term financing strategy will be a detailed model describing how the Water Fund's PES scheme will continue to channel finance into the Fund's grant Mechanism (see below) once GCF finance is no longer available post project. This will comprise details on lessons learned and experiences generated during the project implementation period can be carried over and continue to support the operation of the Water Fund post project. In addition, the development of the long-term financing strategy will consider how the Water Fund, once established and matured, could link to other financial instruments, such as EbA funds¹⁸⁴, and development banks (e.g., the Islamic Development Bank), as well as leverage and unlock additional domestic finance and development banks which would contribute to ensuring long-term financial support for the fund.
171. The long-term financing strategy will also consider how investments in water resources in the country can be linked to and integrated into the Water Fund, as well as potential future water funds in Jordan, supporting its scaling up and replication, and increasing its impact. To strengthen the inclusivity of the long-term financing strategy from a community perspective, WMCs will, through Component 2, identify investment opportunities and develop plans for submission to the Water Fund for accessing finance and stimulating private sector investment. Once submitted to the Water Fund, the plans will be considered for inclusion in the long-term financing strategy. Component 2 activities through which the WMCs can identify investment opportunities and plans are presented below.
- *Activity 2.1.1. Support the implementation of participatory climate-resilient integrated watershed management plans.* Adaptation measures identified in and implemented through the plans (Sub-activity 2.1.1.1) can be prioritised for investment for scaling up and replication. Data and from community-based participatory mechanisms for monitoring and evaluating change in climate resilience and behavioural change (Sub-activity 2.1.3) can be used to identify successful measures and best practices for investment, as well as priority communities.
 - *Activity 2.1.2. Restore ecosystems in the watersheds of the Yarmouk, King Talal Dam and Kafraïn areas of the Jordan Valley.* Protocols for restoration processes developed (Sub-activity 2.1.2.1) can be used to prioritise additional sites and species for planting, supporting targeted investment for effective scaling up and replication. Nurseries established to supply restoration sites (Sub-activities 2.1.2.3 and 2.1.2.4) can be prioritised for additional investment based on seedling demand and production capacity, as well as lessons learned during project implementation — further nursery sites can also be identified for establishment should the need be identified.
 - *Activity 2.1.3. Introduce climate-resilient additional livelihoods and agricultural practices in the project's three intervention areas.* Based on the success of this activity in supporting the success of restoration activities, investment plans can be developed to scale up and replicate this activity in sites identified for restoration in the future.
172. Development of a long-term financing strategy will be undertaken by a financial expert (under the oversight of the Water Fund manager — recruited under Sub-activity 2.1.4.4), who should have prior experience in the design of such a strategy, and should preferably also be involved in long-term financial studies relating to the Water Fund. The long-term financing strategy of the Water Fund will be designed in line with the key principles presented below.
- Securing payers to the fund under its PES model (major water users including commercial agriculture, industry and water utilities).
 - The sources and intended use of the funds will be clearly defined.
 - A mechanism for accountability will be established.
 - Transparency and scrutiny by fund members will be encouraged.
 - The Water Fund will be constituted a legal and authorized financial entity and will obtain all necessary permits in line with regulations of the country.
 - Mechanisms for minimizing administrative costs will be developed.
 - A clear mandate for using funds only for the purpose of the Water Fund will be established.
 - A clear and effective management structure will be developed and approved by all Water Fund members.

¹⁸¹ <https://www.manaseergroup.com/Home/>

¹⁸² <http://www.jpmmc.com.jo/Default/En>

¹⁸³ http://www.yw.com.jo/EN/yw_info.aspx

¹⁸⁴ Such a fund may include the Global EbA Fund, which is administered by UNEP and executed by IUCN: <https://globalebafund.org>

173. Options for scaling up and replicating the Water Fund to other locations of Jordan will also be assessed through the development of a scaling up strategy, aligning closely with the long-term financing strategy. Amongst other things, the scaling up strategy will consider how the Water Fund can contribute to the sustainability of other GCF investments in Jordan's water sector, such as BRCCJ (FP155), including the identification of linkages and complementarities through consultations with its AE (FAO), EEs and project management unit PMU. During the development of the scaling-up strategy, the proposed project will scope the feasibility of replicating the Water Fund approach in other locations of Jordan where downstream water users would benefit from and pay for increased water resource supplies resulting from improved and maintained management of upstream ecosystems. This will include modelling to prioritise where to invest for ecosystem benefits.

Component 3: Improved protection, supply and use of water resources under conditions of climate change

Outcome 3: Enhanced water security for households in target areas

Output 3.1: Improved protection, supply and use of water under climate change conditions.

174. With a baseline of 125 CM/yr of water available per capita — 375 CM/yr below the international water poverty line — water scarcity is at a critical level in Jordan. The current demand for water of 1,412 MCM/yr exceeds the safe yield of renewable water supply by 416 MCM/yr, resulting in abstraction of non-renewable groundwater sources. Moreover, water stress in Jordan has increased over the last decade, with water availability decreasing by 145 CM/yr per person between 2007 and 2017. Water loss through evaporation is a considerable challenge in Jordan, accounting for ~7,636 MCM of rainfall per year¹⁸⁵. This is 94% of the annual surface water budget, higher than the average land evaporation loss, which accounts for 60% of global surface water¹⁸⁶. Jordan's high evaporation rate has ramifications for water levels of the country's rivers — of which 11,376 km are non-perennial and 11,155 km are perennial — and its irrigation canals. Most notably these include the Jordan, Yarmouk and Zarqa Rivers which are 105 km, 50 km and 65 km long, respectively, and the 110 km long King Abdullah Canal. Given the high temperatures and evaporation rates in the country, the use of rainwater harvesting is a particularly effective approach to overcoming challenges related to water scarcity and has historically been extensively used in the country. Indeed, until the development of well drilling and piping in the 1940s, the entire population of Jordan was dependent on rainwater harvesting for domestic uses¹⁸⁷. As central treatment facilities began to supply treated drinking water, the popularity of rainwater harvesting declined¹⁸⁸. However, rainwater harvesting remains the only source of potable water for many rural communities that do not have access to public water supply networks¹⁸⁹. In 2013, there were a total of 250 rainwater harvesting structures in Jordan, including: i) 56 dams; ii) 129 ponds; and iii) 65 pools. During the same year, a total of 75 future water-harvesting projects were being planned, including 29 dams and 46 ponds¹⁹⁰. Further detail on the most common rainwater harvesting techniques in Jordan are provided in Section 8.6 of Annex 2: Feasibility Study.

175. While rainwater harvesting for agricultural purposes in Jordan also has a long history, the practice has only recently received formal attention from the Jordanian government and ordinary farmers — particularly regarding its potential to promote the production of dryland crops in low rainfall regions^{191,192}. Rainwater harvesting techniques have also been used to support the regeneration of degraded rangelands^{193,194}. Indeed, the GoJ has recognised the need for upscaling of rainwater harvesting in the country in its Water Strategy for 2008 to 2022 (described in detail in Section 1.3 of Annex 2: Feasibility Study), with one of the specific goals of the strategy is to promote water-harvesting methods for irrigation¹⁹⁵.

¹⁸⁵ Ministry of Water and Irrigation. 2017. Jordan water sector — Facts and figures. MWI, Amman.

¹⁸⁶ Oki T & Kanae S. 2006. Global hydrological cycles and world water resources. *Science*, 313: 1068–1072.

¹⁸⁷ Majed A-Z, Hazaymeh A. 2012. Urban rainfall harvesting to alleviate water shortages and combat desertification in the arid land of Jordan. *Journal of Arid Land Studies* 22: 87-90.

¹⁸⁸ Abdulla F. 2019. Rainwater harvesting in Jordan: potential water saving, optimal tank sizing and economic analysis. *Urban Water Journal*, pp.1-11.

¹⁸⁹ Radaideh J, Al-Zboon K, Al-Harashsheh A, Al-Adamat R. 2009. Quality assessment of harvested rainwater for domestic uses. *Jordan Journal of Earth and Environmental Sciences* 2: 26-31.

¹⁹⁰ FAO. 2016. Assessment of rainwater harvesting sector in Jordan. Available at: <http://www.fao.org/3/a-i5390e.pdf>

¹⁹¹ Alshawahneh N, Saoub H, Oweis T, Haddad N. Impact of micro-catchment water harvesting on plant diversity in Jordan Badia rangelands 2010: International Dryland Development Commission (IDDC). pp. 273-283.

¹⁹² Karrou M, Oweis T, Ziadat F, Awawdeh F. 2011. Rehabilitation and integrated management of dry rangelands environments with water harvesting. *Community-based optimization of the management of scarce water resources in agriculture in Central and West Asia and North Africa. Aleppo, Syria: ICARDA*.

¹⁹³ Al-Ali M. 2012. Soil water conservation and water balance model for micro-catchment water harvesting system. DPhil, Loughborough University, Loughborough.

¹⁹⁴ Karrou M, Oweis T, Ziadat F, Awawdeh F. 2011. Rehabilitation and integrated management of dry rangelands environments with water harvesting. *Community-based optimization of the management of scarce water resources in agriculture in Central and West Asia and North Africa. Aleppo, Syria: ICARDA*.

¹⁹⁵ FAO. 2016. Assessment of rainwater harvesting sector in Jordan. Available at: <http://www.fao.org/3/a-i5390e.pdf>

176. Under Output 3.1, the quality and quantity of surface and groundwater will be enhanced through the rehabilitation of active freshwater springs, restoration of vegetation and building of natural check dams, as well as the establishment of rockfill dams to better capture rainwater runoff and contribute to aquifer recharge. This will be complemented by introducing rooftop harvesting interventions to communities (female-headed households in particular) designed to reduce their vulnerability to water scarcity. Rooftop rainwater harvesting has been identified as a preferred and viable option for meeting the country's increasing water demand, particularly in urban areas^{196,197}. The potential harvested water from a 100 m² roof ranges from 4–29 CM in areas receiving an annual rainfall of 50–800 mm, which could equate to an average saving of 8% of the total domestic demand if rooftop rainwater harvesting is implemented across Jordan¹⁹⁸. The ability of rooftop rainwater harvesting to save water is attributed to water storage in closed containers that are not subject to the high levels of evaporation that surface water sources are exposed to, thereby limiting water loss and reducing the demand for water within the wider landscape. Potential for rainwater harvesting in the Balqa and Irbid governates within the project area is estimated to be 1.7 and 3.3 MCM/year, respectively^{199,200}. Under the proposed project, 3,000 rainwater harvesting tanks with a capacity of 32 CM each will be installed (at least 50% female-headed households), resulting in a minimum annual yield of 0.28 MCM/year. Rainwater harvesting interventions will therefore reduce the reliance of community members on groundwater (including women) and limit financial costs resulting from digging deeper wells to access lowering water tables.
177. Added to the challenge of water scarcity, Jordan currently imports 96% of its energy from neighbouring countries and is therefore one of the most energy-insecure countries in the world²⁰¹. Renewable energy in Jordan — which primarily consists of solar and wind energy — accounts for ~10% of energy production in the country²⁰². With its high number of sunny days per year, Jordan has considerable potential for the expansion of solar energy projects. By the end of 2020, the country aims to generate ~20% of its energy through renewable sources and increase its solar energy target by more than 65%²⁰³. Multiple solar power projects are planned, however there are numerous challenges to further expansion of solar energy²⁰⁴. The first challenge is the current instability of the transmissions grid, which resulted in the Ministry of Energy and Mineral Resources placing a halt on further renewable energy auctions for projects over 1 MW in 2019. The second challenge is identifying land suitable for solar projects, which depends on the availability of sunlight and the proximity of the site to the transmissions grid.
178. The installation of floating solar panels over the King Abdullah Canal will reduce water lost to evaporation by minimising the exposure of the water body to direct sunlight and wind. This reduction in evaporation increases the efficiency of the surface water body and increases its ability to store water, thereby improving water availability for communities and reducing water demand pressure on the landscape overall. Floating solar, for example, has been found to reduce evaporation from surface water bodies by almost 100% and has the potential to save 8.6 MCM/year from an area of 2.7 million m²²⁰⁵.
179. Outcome 3 will contribute to GCF Outcome A7.0 (strengthened adaptive capacity and reduced exposure to climate risks) and Outcome A8.0 (strengthened awareness of climate threats and risk reduction processes) by improving local communities' access to and use of water for domestic and agricultural purposes, with a focus on women. At the domestic level, small-scale rainwater harvesting interventions will be installed on community buildings — including mosques, churches and community centres — as well as homes (at least 50% female-headed) and schools in project intervention areas²⁰⁶. This will supplement the increased supply of water from the restoration of upstream watershed areas, increasing the resilience of communities to the increased duration, frequency and intensity of droughts under climate change conditions.

¹⁹⁶ Al-Qawasmi O. 2021. Feasibility of rainwater harvesting from residential rooftops in Jordan. *Applied Water Science*, 11: 30.

¹⁹⁷ Saidan MN, Al-Weshah RA & Obada I. 2015. Potential rainwater harvesting: An Adaptation measure for urban areas in Jordan. *American Water Works Association*, DOI: <http://dx.doi.org/10.5942/jawwa.2015.107.0154>.

¹⁹⁸ Abdulla F. 2019. Rainwater harvesting in Jordan: potential water saving, optimal tank sizing and economic analysis. *Urban Water Journal*, DOI:10.1080/1573062X.2019.1648530.

¹⁹⁹ Abu-Zreig M, Hazaymeh A & Shatanawi M. 2013. Evaluation of residential rainfall harvesting systems in Jordan. *Urban Water Journal*, 10: 105–111.

²⁰⁰ Based on rainfall figures of 562 and 370 mm/yr, and potential rooftop area for rainfall harvesting of 3.8 million and 11 MCM, for Balqa and Irbid, respectively.

²⁰¹ The Hashemite Kingdom of Jordan. 2013. Needs Assessment Review of the Impact of the Syrian Crisis on Jordan. Ministry of Planning and International Cooperation. Amman.

²⁰² Franceschini B. 2019. Scaling-up Renewable Energy Development in Jordan: Position Paper. Res4Africa Foundation, Rome.

²⁰³ Rahim NA & Eid N. 2017. Energy projects in Jordan.

²⁰⁴ Franceschini B. 2019. Scaling-up Renewable Energy Development in Jordan: Position Paper. Res4Africa Foundation, Rome.

²⁰⁵ Baradei SE & Sadeq MA. 2020. Effects on solar canals on evaporation, water quality, and power production: An optimization study. Water: MDPI.

²⁰⁶ Relevant interventions are detailed under Section 8.6 of the Feasibility Study, Annex 2.

180. To improve the community-level use and management of water under climate change conditions, gender-responsive community-based water monitoring and management plans will be developed for the target areas. These plans will ensure social accountability for water usage within beneficiary communities. The development and implementation of the plans will be overseen by the WMCs (established under Output 1.1), ensuring complementarity with the climate-resilient integrated watershed management plans developed for the three project target areas (Activity 2.1). WMCs will ensure that target communities' needs and concerns are considered in the development and implementation of water monitoring and management plans (including vulnerable groups such as women and youth). Representatives of target communities (such as women's groups) will be members of the WMCs, providing them with direct inputs into decision-making processes and ensuring that relevant information is made available to them²⁰⁷. To promote the effective implementation of these plans, community members will be informed of their purpose and benefits and will be trained to implement them (including at least 50% women). Gender-responsive additional livelihoods related to the monitoring and management of water in target communities will also be promoted — including plumbing, rainwater harvesting infrastructure installation and maintenance, as well as the installation of solar pumps.

Under the project's delegated execution structure, IUCN will act as Executing Entity for Component 3.

Activity 3.1.1 Implement measures to improve the quantity and quality of ground and surface water supplies within the project sites.

181. At each of the project's three sites, interventions will be implemented to enhance the quality and quantity of ground and surface water through Activity 3.1.1. This will be done through three sub-activities and will include the construction of rockfill dams, vegetation restoration and the rehabilitation of active freshwater springs.

Sub-activity 3.1.1.1 Establish rockfill dams at 12 sites in target area wadis (four per target area) to capture runoff and water from flash floods for managed aquifer recharge (MAR).

182. Managed aquifer recharge (MAR) will be implemented via the construction of rockfill dams to collect and store water from runoff and flash floods. The rockfill dams will be constructed in wadi areas up to 20 m wide with potential catchments of up to 0.9 MCM/yr. With potential infiltration rates of approximately 0.02 mm/s, up to ~0.63 MCM of this amount will contribute to aquifer recharge annually^{208,209}. Four rockfill dams will be constructed in each of the project's three target areas, resulting in a combined annual potential aquifer recharge of ~7.6 MCM/year (additional information on the potential benefits of managed aquifer recharge is presented in Section 11 of Annex 2). This will also include the purchase and installation of piezometers to measure the impacts of the rockfill dams on groundwater levels. Monitoring of groundwater levels associated with each rockfill dam will be undertaken via community-based monitoring as detailed in Sub-activity 2.1.1.3.
183. Site feasibility assessments and design studies for the 12 rockfill dams will be undertaken during year 2 of project implementation by a hydrological engineer, with construction taking place between years 3 and 4. The feasibility process will include the selection of sites using multiple criteria, such as hydrogeology/soil types, climate, land use, land cover, flow availability and rate in streams, water source, water quality, need for permits, licenses or rights pertaining to land use and site suitability for technical design (these criteria are elaborated on in Section 8.12 of Annex 2).
184. Environmental and social risk assessments will also be undertaken as part of feasibility assessments to ensure that negative impacts are avoided. Potential environmental risks associated with the establishment of rockfill dams are *inter alia*: i) noise and air pollution during construction; ii) landscape degradation; iii) reduced downstream water flows affecting local communities and ecosystems; iv) siltation; v) groundwater contamination; and vi) aquifer damage. Potential social risks are *inter alia*: i) disturbance of archaeological sites; ii) land tenure and exclusion of women of water sources; iii) public acceptance; iv) vandalism; v) conflict of interest with local communities regarding water use; vi) safety of infrastructure; and vii) health risks related to water quality. While potential risks have been mentioned above, the risks associated with the rockfill dams to be established by the project are expected to be few, limited in their impact, site-specific, largely reversible and readily addressed by mitigation measures, thereby aligning with the ESS categorization of the project (Category B). Additional details on potential risks associated with the construction of rockfill dams are presented in Section 8.12 of Annex 2 and reflected in Annex 6 along with mitigation measures. Furthermore, an operations and maintenance budget has been allocated to the rockfill dams during the project's implementation period (see Annexes 4 and 21). While not meant to directly provide drinking water but rather facilitate

²⁰⁷ These community representatives will be members of community-based committees already present on the ground in the three project areas, such as Development Advisory Committees, Stakeholders Advisory Groups, agricultural cooperatives, women's cooperatives and Water User Associations.

²⁰⁸ Wolf L, *et al.* 2008. Artificial groundwater recharge in Jordan using various water qualities – Large scale potential map and individual test sites in Jordan. Karlsruhe Institute of Technology.

²⁰⁹ Steinell A. 2012. Guideline for assessment and implementation of managed aquifer recharge (MAR) in (semi-) arid regions: pre-feasibility study for infiltration of floodwater in the Amman-Zarqa and Azraq basins, Jordan. BGR.

MAR, to mitigate any potential water-related health risks, environmental and social risk assessments for the rockfill dams and ancillary/supporting infrastructure (such as small-scale measures planned under Sub-activity 3.1.1.2) will include health assessments, while engineering and O&M plans will include vector control and bio-ecological monitoring, health and safety measures and/or plans.

Sub-activity 3.1.1.2. Implement small-scale measures, including restoration of vegetation and establishment of natural check dams, to capture silt upstream of rockfill dams.

185. To protect the rockfill dams implemented in Sub-activity 3.1.1.1 from sedimentation, small-scale measures will be implemented via project Technical Partners (TPs) in strategic areas upstream from the dams to reduce erosion and capture silt. These measures will include the planting of indigenous vegetation — as per the restoration protocols developed under Sub-activity 2.1.2.1 — and constructing natural check dams from locally available materials. The erosion and silt control measures will be implemented in year 5 of the project and maintained as necessary thereafter by TPs. Potential risks related to these small-scale measures will be assessed via the environmental and social risk measures for rockfill dams under Sub-activity 3.1.1.1.

186. Sub-activity 3.1.1.2 dependencies:

- Development of restoration protocols under Sub-activity 2.1.2.1 — completed in year 1.

Sub-activity 3.1.1.3. Rehabilitate active freshwater springs across the project sites (Yarmouk and Kafraïn).

187. Improved water supply through Sub-activities 3.1.1.1 and 3.1.1.2 will be complemented by the rehabilitation of active freshwater springs in the Yarmouk and Kafraïn sites under the project (with support from project Technical Partners), improving the access of local farmers, livestock owners and communities to water resources. This will include the rehabilitation of approximately five springs at each of the sites during years 6 and 7 of the project. This sub-activity also includes the identification of the springs, construction of small-scale canals and maintenance.

Activity 3.1.2. Install 1 MW of floating solar panels over the King Abdullah Canal to reduce water loss from evaporation.

188. Two sub-activities under Activity 3.1.2 will be aimed at reducing evaporation of surface water resources and increasing electricity supplies through floating solar panel arrays. Using suspended or floating covers has the potential to reduce evaporation from water bodies by ~77–91%²¹⁰. In the lower-lying areas of the project intervention area (Jordan Valley floor), evaporation reduction measures will be implemented on the King Abdullah Canal — located on public land — to reduce surface water evaporation. The floating solar installation will be connected to the local grid (year 3 of the project) to supply additional, renewable power to the MWI's pumping stations located on the King Abdullah Canal. Floating or suspended solar panels have not been used in Jordan before the proposed project but has been used successfully in other countries, making this an innovative aspect of the project. In India, for example, 33,816 solar panels were installed over 7.5 ha of the Narmada irrigation canal, supplying 16.2 million units of power and saving 0.1 MCM/yr of water by reducing evaporation²¹¹. A detailed analysis of alternative evaporation reduction methods for surface water sources is presented in Section 8.7 of Annex 2. This analysis includes a comparison of the costs and benefits, environmental impacts, etc., of floating solar with alternatives such as tree windbreaks, suspended covers, floating covers, destratification and chemical monolayers. The results of the analysis indicate that floating solar has the lowest potential environmental impact and provides the best cost-benefit ratio.

Sub-activity 3.1.2.1. Install 1 MW of floating solar panels over the King Abdullah Canal.

189. Sub-activity 3.1.2.1 will include the establishment of at least 1 MW of floating solar panels over the King Abdullah Canal, which is used to supply water to Amman, as well as local communities and farmers in the northern Jordan Valley²¹². A local electrical engineering firm, with specialisation in large-scale solar installations, will oversee the design and installation during year's 2 and 3 of the project. This intervention will prevent the evaporation of ~0.1 MCM of water annually from the canal and provide 1 MW of power per year to the MWI's pumping stations on the canal²¹³. A co-benefit of the approach is reducing MWI's reliance on fossil-fuel generated energy to supply it with pumping stations, contributing to a transformative change in energy supply across the water sector in Jordan and contributing to a reduction in the sector's GHG emissions.

190. The MWI, through the JVA, will be responsible for operating and maintaining the floating solar installation once in place, and will ensure that the relevant permits for the infrastructure are acquired. According to MoEnv's environmental

²¹⁰ Helfer F, Zhang H & Lemckert C. 2009. Enhancing reservoir management through the use of mechanical evaporation reduction techniques. Society for Sustainability & Environmental Engineering, Melbourne, Australia.

²¹¹ Ibid.

²¹² The KAC is managed by MWI and its subsidiary, the JVA.

²¹³ Youssef YW & Khodzinskaya A. 2019. A review of evaporation reduction methods from water sources. E3S Web of Conferences, 97: 05044.

impact (EIA) regulations, an EIA is only required for solar installations of 5 MW or above²¹⁴. Consequently, an EIA will not be required for this installation. As the floating solar installation will be placed on an artificial water body (King Abdullah Canal), environmental risk is limited, while water quality is expected to benefit through algal growth suppression²¹⁵. The shade provided by floating solar installations assists in reducing the presence of algae blooms in of freshwater bodies. Algae blooms can be dangerous for human health if they develop in a source of drinking water. Algae blooms may also lead to the death of plants and animals living in the body of water²¹⁶. Section 8.7.4 of Annex 2 provides a detailed review of floating solar.

Sub-activity 3.1.2.2. Connect the floating solar installation to the local grid to supply power the MWI's pumping stations on the King Abdullah Canal.

191. Under Sub-activity 3.1.2.2, floating solar panels installed under Sub-activity 3.1.2.1 will be connected to the local grid (year 3 of the project) to supply additional, renewable power to the MWI's pumping stations located on the King Abdullah Canal. Usage authorisations and permits for energy generated from the floating solar panels, as well as for the operation thereof by MWI, will be acquired through the Ministry of Energy and Mineral Resources, who will ensure that the installation and operations meet all local requirements and standards.

192. Sub-activity 3.1.2.2 dependencies:

- Installation of the floating solar panels under Sub-activity 3.1.2.1 — completed between years 2 and 3.

Activity 3.1.3. Introduce small-scale rainwater harvesting interventions in the project's three target areas to improve access to clean water under climate change conditions in a gender-responsive manner.

193. Under Activity 3.1.3 small-scale rooftop rainwater harvesting interventions will be installed at public and private buildings (including homes, at least 50% of which will be female-headed) across the project's three intervention sites. Rooftop rainwater harvesting is a simple measure, standardized in the country under the Uniform Plumbing Code of Jordan²¹⁷. This rainwater harvesting measure ensures that rainwater falling on rooftops is collected in a dedicated cistern/tank, instead of being lost (Figure 12). The activity aligns with Jordan's NDC (2021) and NAP (2020), in which GoJ has included rooftop rainwater harvesting as a priority intervention for climate change adaptation, as well as with Jordan's Water Substitution and Water Reuse Policy (2016).

194. An assessment of rooftop rainwater harvesting in northern Jordan indicated that while acknowledging rooftop rainwater harvesting as a key measure to adapt to water shortages, 46% of the population have not invested in the measure because of the high costs involved (particularly of the storage tank/cistern)²¹⁸. Another study, which surveyed 360 households in the Ajloun governate (situated between the project's two target governates of Irbid and Balqa), revealed that only 14% of households own a rooftop rainwater harvesting system. However, the majority, 81%, of those who do not own a rooftop rainwater harvesting system showed interest in installing one. Approximately 97% of the surveyed households believe that government should provide incentives to subsidize the construction of rooftop rainwater harvesting systems, primarily because of their high initial cost²¹⁹. The need to improve short-term water supplies via rainwater harvesting was also identified during community consultations across the project's target sites (see Annex 7). As a result, Activity 3.1.3 has been included as a grant-funded intervention. However, the project will, through Activity 1.1.2, also recommend policy reform for the use of fiscal incentives for poor households to equip themselves with rooftop rainwater harvesting installations.

195. A 32 CM cistern/tank will be installed at each of the selected 3,000 buildings to supplement and reduce the demand on domestic supplies during dry periods (i.e., serving as a complementary source) at the household level, as well as at community and institutional buildings across the project's three target areas, thereby increasing climate resilience through improved water access²²⁰. The water will be accessible from the tank as needed, via an outlet tap — i.e., it will not be piped into the building. At the household level, assuming that an average household (5 people) would require 500 litres of supplementary water per day for domestic purposes during the dry season, a full 32 CM cistern/tank, would provide water, without any recharge, for a minimum of 65 days. Overall, the activity will result in a

²¹⁴ 5-20 MW requires a preliminary EIA; and 20+ MW requires a full EIA.

²¹⁵ Yao X, Zhang H, Lemckert C, Brook A & Schouten P. 2010. Evaporation reduction by suspended and floating covers: Overview, modelling and efficiency. Urban Water Security Research Alliance Technical Report No. 28.

²¹⁶ Gaidkwad OD & Deshpande UL. 2017. Evaporation control using floating solar PV system and canal roof top solar system. International Research Journal of Engineering and Technology.

²¹⁷ Approved by the Jordan National Building Council's technical committee in 2015.

²¹⁸ Abu-Zreig, M, Ababneh, F, & Abdullah, F. 2019. Assessment of rooftop rainwater harvesting in northern Jordan. Physics and Chemistry of Earth, Parts A/B/C. Vol. 114, 102794. <https://doi.org/10.1016/j.pce.2019.08.002>

²¹⁹ Al-Houri, Z, & Al-Omari, A. 2022. Assessment of rooftop rainwater harvesting in Ajloun, Jordan. Journal of Water Reuse and Desalination (2022) 12 (1): 22-32. <https://doi.org/10.2166/wrd.2021.064>

²²⁰ The use of water collected via rooftop rainwater harvesting systems will not be used for drinking purposes as per national law. Drinking water quality is governed by Jordanian Standard 286 (JS286), which is an adaption of the WHO's guidelines for drinking water quality.

minimum increase in annual water supplies of 0.28 MCM/yr in the target area (more information on rainwater harvesting is available in Section 8.6 of Annex 2). As much of the project's water resource benefits related to ecosystem restoration and MAR will only be realised in the medium to long term, the immediate tangible benefits provided by the introduction of small-scale rainwater harvesting interventions to vulnerable local communities (including groups such as women and youth), is critical to improving their resilience in the short term, and to fostering their support for the project's other interventions, such as ecosystem restoration. Support for the implementation of this activity will be provided by relevant project Technical Partners.

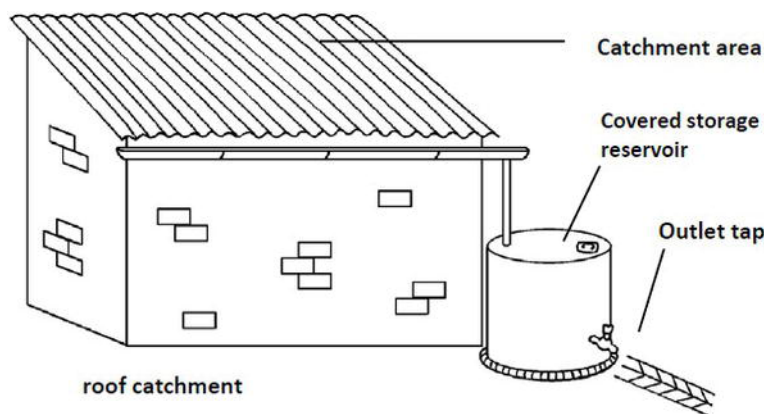


Figure 12. Basic rooftop rainwater harvesting design.

Sub-activity 3.1.3.1. Install small-scale water harvesting interventions at community centres, local institutions and schools at target sites in the Jordan Valley.

196. Under Sub-activity 3.1.3.1, 50 community/religious centres, 50 institutional buildings and 100 schools in the target intervention sites will receive small-scale rooftop rainwater harvesting tanks. The criteria for the selection of public buildings will include their water use, willingness to pay for maintenance costs of the rooftop rainwater harvesting systems, willingness to monitor their water usage and limited access to supplementary domestic water supplies. The tanks will be installed between years 2 and 4 of project implementation.

Sub-activity 3.1.3.2. Install small-scale water harvesting interventions at homes at target sites in the Jordan Valley (100% of women trained in their operation and maintenance).

197. Sub-activity 3.1.3.1 will include the installation of 32 CM rooftop rainwater harvesting tanks on 2,800 private houses (~14,000 people — 8% of the target population; at least 50% female-headed households) across the project's three target sites. Beneficiary household selection criteria will include: i) the appropriateness of buildings for implementation; ii) willingness of owners to pay for maintenance costs; iii) female-headed households; iv) poor households, i.e., those unable to afford the installation of rooftop rainwater harvesting infrastructure; v) households with a person with a disability or more than 6 dependents; and households with limited or no access to supplementary domestic water supplies — i.e., those most vulnerable to water scarcity; and vi) at least 30% female-headed. Installation will take place between years 2 and 4 of project implementation. Local individuals — of which all will be women — will be trained to support the operation and maintenance of these community-shared 32 CM tanks.

Activity 3.1.4. Empower local communities, including women, to monitor and manage local water supplies more effectively.

198. Through three sub-activities under Activity 3.1.4.1, community-level management of water supply will be strengthened across the project's three target sites. This will include female-headed community-based water use monitoring plans, outreach campaigns and the promotion of additional livelihoods that are related to the monitoring and management of clean water. Community engagements under this activity will be facilitated by project TPs and the relevant WMCs, which will include local community members as representatives (at least 50% women). During the project's inception period, the target communities (consulted during the preparatory phase) will be engaged to sensitise them to the project's scope, objectives, expected results, as well as their roles — such as under Activity 3.1.4. These early community engagements (Activity 4.1.1) will also be used to manage the expectations of communities by providing clear project timelines and ensuring a common understanding that the project will not be addressing baseline development challenges, but impacts related to climate change.

Sub-activity 3.1.4.1. Develop gender-responsive community-based water use monitoring and managing plans for project beneficiary communities.

199. Under Sub-activity 3.1.4.1, gender-responsive community-based water use monitoring and management plans will be developed for each target site with the support of project Technical Partners, MWI, WMCs and LDCs — one

each for the three target watersheds during year 5 of project implementation. Sub-plans will also be developed for each district represented in the watersheds — At-Taibah, Al-Wastiyah, Qasabah Irbid and Al-Kurah from the Irbid Governate, and Dir Alla and Ash-Shunah al-Janubiyah districts from the Balqa Governate. The plans will detail the roles and responsibilities of local communities (including women) in the management of water supply and demand, to reduce existing water wastage through leakages, poor quality water infrastructure, malfunctioning meters and illegal connections, as well as strengthen local supplies such as groundwater wells and surface water sources. The development of the plans will be informed by the results of long-term climate analyses and risk assessments.

200. Sub-activity 3.1.4.1 dependencies:

- Establishment of WMCs under Sub-activity 1.1.4.1 — completed in year 1.

Sub-activity 3.1.4.2. Implement a gender-inclusive outreach campaign to raise awareness on the plans developed under sub-activity 3.1.4.1 and train target community members to monitor and manage the use of clean water (100% women).

201. To capacitate community members to implement the plans, annual gender-inclusive training workshops — hosted by MWI — will be held at each intervention site from year two to eight of project implementation. Training workshops will include courses on: i) plumbing; ii) monitoring of water use; iii) monitoring of water quality; and iv) engaging with other community members to reduce wastage. The workshops will target 150 people from each site — including at least 50% female community members, who will also be empowered to become water-wise ambassadors (225 women). The empowerment of female water-wise ambassadors will build on the success of GIZ's Water Wise Women initiative that was implemented in Jordan²²¹. Technical Partners, with support from female water-wise ambassadors, will raise awareness on community-based water use monitoring and management plans to reduce water wastage. This will include collecting information on water resource management interventions that have been demonstrated to be effective in areas with similar water scarcity problems and climatic conditions, which will be presented to local communities during the awareness raising activities. This activity will be implemented between years 5 and 8.

202. The implementation of Sub-activity 3.1.4.2 will be supported by using the knowledge management tools developed under Sub-activity 4.1.2.1 (such as a content repository, a knowledge search tool, a communication tool, social software to facilitate the socialisation of knowledge and a knowledge visualisation tool) to support the training of and awareness raising amongst community members to implement the water use monitoring and managing plans. Training material (such as manuals and videos) will be stored in the content repository, shared via the communication and socialisation tools. The knowledge visualisation tool will be used to visualise the training — i.e., create videos, diagrams, animations, etc. This will ensure that the training material continues to be available once the training workshops have been completed and is also available for use by other stakeholders.

203. Sub-activity 3.1.4.2 dependencies:

- Development of knowledge management tools under Sub-activity 4.1.2.1 — completed during year 1.

Sub-activity 3.1.4.3. Promote additional livelihoods related to the monitoring and management of clean water in target communities (including at least 50% women) through a training course.

204. Community members that excelled in the first year of the training programme implemented under Sub-activity 3.1.4.2 (~25% of participants) will be given the option of receiving further advanced training on plumbing, rainwater harvesting infrastructure installation and maintenance, as well as installation and maintenance of solar pumps. This additional training (year 6) will enable the selected community members to take on additional livelihoods related to water management. Trainees will also receive tools and equipment necessary to practice the livelihoods. The livelihoods introduced under this activity will assist in reducing water wastage, which will limit the over abstraction of water resources that are being augmented by restoration and water harvesting interventions implemented under Output 2.1 and Output 3.1.

205. Sub-activity 3.1.4.3 dependencies:

- Training of target communities to monitor and manage the use of clean water under Sub-activity 3.1.4.2 — ongoing from years 5 to 8.

Component 4: Knowledge management and awareness on climate change adaptation

Outcome 4: Gender-responsive household behavioural change achieved through improved information and outreach

²²¹ https://www.iucn.org/sites/dev/files/import/downloads/wwwi_case_study_may_12_2015.pdf

Output 4.1 Improved knowledge and awareness of climate change adaptation for landscape and water resources management.

206. Strengthened mandates and capacity for climate-resilient ILWRM in Jordan (Outputs 1.1 and 1.2) will be complemented by the improvement of the country's climate change knowledge-management system. This will include the development of knowledge-management tools to facilitate sectoral and cross-sectoral learning and inform the related mainstreaming activities mentioned above. Lessons learned and best practices (including those which are gender-responsive) on adaptation will be made possible through the establishment of a national climate change adaptation knowledge repository. Consequently, Output 4 will contribute to GCF Outcome A8.0 — strengthened awareness of climate threats and risk-reduction processes.
207. The activities conducted under Output 4.1 will build upon baselines established by both government institutions and previous projects concerned with awareness raising and knowledge management. The MoEnv, in seeking to achieve its mission to “maintain and improve the quality of Jordan’s environment, conserve natural resources, and contribute to sustainable development”²²², conducts activities concerned with, inter alia, public awareness and outreach and the support of decision-making through knowledge management. It leads the governance system — including other relevant governmental organisations — in initiating and driving policies, legislation, strategies and monitoring, as well as by mainstreaming environmental policies into all national development plans²²³. To achieve this the MoEnv has set five objectives for the environment sector, notably the development of capacity and institutionalisation of a culture of excellence and raising public awareness and promoting behavioural change for environmental protection²²⁴.
208. To conduct gender-responsive awareness-raising activities, project Technical Partners will be supported by MoEnv’s specialist Outreach and Environmental Awareness unit. In addition to building on and scaling up governmental activities, the proposed project will incorporate undertakings from the HERD project. HERD’s components directly align with the proposed GCF project and in particular Output 4 as they include: i) providing evidence-based technical assistance; ii) strengthening institutions for rangeland governance; iii) upscaling good practices in Sustainable Rangeland Management (SRM); and iv) promoting SRM knowledge-management at global and regional levels. Gender-responsive outreach programmes and improved knowledge-management systems implemented by the proposed project to raise awareness within local communities and support decision-making will complement HERD’s inclusive dialogue with local organisations for climate-resilient ILWRM. Several other initiatives also provide awareness-raising and education activities best practices that will be incorporated by the proposed project. For example, the Sustainable use of Ecosystem Services in Jordan project disseminated appropriate knowledge to the public using a national information system, highlighted the economic importance of biodiversity of ecosystem services and established Jordan as a regional hub for ecosystem services research and awareness. Through the National Programme for Rangeland Rehabilitation, a Pastoral Resource Information and Monitoring and Evaluation Unit was created that developed a number of knowledge tools, including databases, study site maps, land capability plans and vegetation maps for use by rangeland planners and policymakers. Increased awareness at the local, national and regional levels on the importance of biodiversity as implemented by the Integrated Ecosystem Management in the Jordan Rift Valley, while the project Mainstreaming Sustainable Land and Water Management Practices enhanced institutional structures and capacities to promote awareness of sustainable landscape management, including by training over 5,000 farmers and study tours.
209. To prioritise an ILWRM approach, decision-makers need to understand and appreciate the relevant interlinked ecological processes and how they directly produce economic benefits. The relevant information and evidence will be generated by the monitoring and analysis of ILWRM interventions implemented under the project.

Under the project’s delegated execution structure, IUCN will act as Executing Entity for Component 4.

Activity 4.1.1 Implement a gender-responsive climate change adaptation outreach programme across the Jordan Valley.

210. Through six sub-activities under Activity 4.1.1, awareness of climate change and adaptation responses — including climate-resilient ILWRM — among local communities, including local women, in the Jordan Valley will be strengthened. This will improve the understanding of and foster support of those who will be directly affected by the mainstreaming of climate-resilient ILWRM into land-use and development-planning processes. Awareness will be

²²² The Hashemite Kingdom of Jordan. 2017. Jordan’s First Biennial Update Report to the United Nations Framework Convention on Climate Change.

²²³ Ibid

²²⁴ The Hashemite Kingdom of Jordan. Nd. *Ministry of Environment Profile*. Available at: <https://jordan.gov.jo/wps/portal/Home/GovernmentEntities/Ministries/Ministry/Ministry%20of%20Environment?nameEntity=Ministry%20of%20Environment&entityType=ministry>

raised through an outreach programme implemented by local NGOs and project TPs, working closely with community representatives in WMCs. The knowledge management tools developed under Sub-activity 4.1.4.1 (such as a content repository, a knowledge search tool, a communication tool, social software to facilitate the socialisation of knowledge and a knowledge visualisation tool) will be used to support the implementation of the outreach programme and ensure that the information is available post project for continued awareness raising and learning.

Sub-activity 4.1.1.1 Community engagements carried out during the project inception period in the project's three target areas to target communities to the project's scope, objectives and expected results.

211. Community engagements during the project's inception phase (year 1) will focus on introducing and sensitising target communities to the project's scope, objectives and expected results. Specific engagements will be tailored to the needs and availability of different community groups, including women and youth. The focus of these early engagements will be to: i) foster buy-in; ii) ensure that there is a common understanding that the focus of the project is on climate change adaptation, avoiding any misconceptions that it is a poverty-alleviation or environmental protection initiative; iii) manage community expectations; and iv) ensure that the gender-responsiveness of the project is promoted. This will include making all relevant project documentation (such as the Funding Proposal, ESIA and ESMP, Gender Analysis and GAP) available — including translation into Arabic. Stakeholders targeted will include inter alia women, youth, farmers, as well as local decision-makers associated with the agricultural and water sectors in both the upstream and downstream areas of the three project target areas. This sub-activity will be implemented by project TPs, with support provided by MoEnv's specialist Outreach and Environmental Awareness Unit.

Sub-activity 4.1.1.2. Upgrade community centres and local institutions (including NGO centres) in the Jordan Valley, and train local staff so that the centres can serve as climate vulnerability and adaptation information hubs and learning centres.

212. Sub-activity 4.1.1.2 will include equipping community centres and institutions — including mosques, churches, centres, as well as schools — across the Jordan Valley to function as climate change adaptation learning centres and information hubs as permanent sites accessible to local communities (during years 2 to 8 of project implementation). Selected community members from across the Jordan Valley will be trained (at least 50% women), and stationed at adaptation learning centres and hubs, to collect data on the extent of awareness — via a scorecard — on climate change and appropriate related adaptation responses during the implementation of the project's outreach programme (Sub-activity 4.1.1.4). The information will then be consolidated by the relevant project Technical Partners and integrated into the project's M&E system. To equip religious centres, the project will build on the lessons learned and best practices of GIZ's "Religious Leaders Support Water Conservation Project", involving religious leaders in the outreach programme to make use of the local value systems and beliefs.

Sub-activity 4.1.1.3. Public figures identified and trained to champion the case of climate change adaptation in Jordan, including promoting awareness and stimulating local ownership of the project's climate-resilient ILWRM approach.

213. To stimulate interest from the local population in the outreach programme, public figures will be identified and trained to champion the case of climate change adaptation in the country. These public figures, including women, will promote and contribute to the outreach programme through the locally available media, implemented under Sub-activity 4.1.1.5. Training will take place through a 3-day workshop in Amman in year 1 of project implementation. The training will focus on increasing the champions': i) baseline climate change impact and adaptation knowledge; ii) understanding of practical methods and approaches to awareness-raising (including gender-responsive approaches); iii) knowledge on ongoing development challenges which are exacerbated by climate change; and iv) understanding of the proposed project, including objectives and activities. The training will be facilitated by the project's Technical Partners.

Sub-activity 4.1.1.4. Site- and ecosystem-specific gender-responsive climate change adaptation outreach programmes developed for the Jordan Valley, including the usage of locally available media, and the inclusion of schools via the EcoSchools programme.

214. To ensure that the information shared via the outreach programme is contextually relevant to the different areas across the Jordan Valley, it will be tailored to be site- and ecosystem-specific and gender-responsive, through engagements with representatives of local communities and institutions who are members of WMCs. This will include developing an in-depth understanding of target groups (such as women and youth) to ensure that preferred engagement approaches and environments, timings, durations, tools and materials are considered. The programme will include a focus on the benefits and implementation of locally appropriate adaptation approaches (especially EbA as part of an ILWRM approach). At a school-level, the design will build-on ongoing work by Technical Partners. The design of the programme will identify how best locally available media, such as radio, television and social media can be best used to disseminate information. Also included in the design will be a mobile eco-theatre to present climate change issues and adaptation approaches via theatrical performances to relevant stakeholders, including schools. At

a school-level, the design will build-on ongoing work by Technical Partners, such as the EcoSchools Programme. This sub-activity will be implemented by the project's Technical Partners with support provided by MoEnv's specialist Outreach and Environmental Awareness Unit during year 1 of project implementation.

Sub-activity 4.1.1.5. Gender-responsive and youth-inclusive outreach programme implemented to promote awareness of the benefits and implementation of climate change adaptation approaches (especially EbA) among targeted rural communities and relevant stakeholders in the Jordan Valley through locally available media.

215. From years 2 to 8 of the project, based on the development of the outreach programmes in Sub-activity 4.1.1.4, engagement approaches under Sub-activity 4.1.1.5 will include: i) household visits (with a focus on women) to introduce community members to real time climate change concerns related to their daily lives; and ii) capacitating local CBOs to deliver hands-on awareness raising through direct engagements with community members in their households, schools, worship houses and workplace. In addition, locally available media, such as radio, television and social media, will be used to disseminate information. The knowledge management tools developed through the project (Sub-activity 4.1.2.1) will also be used to disseminate the information.

216. Information shared through the outreach programme will include details on the benefits and implementation of locally appropriate adaptation approaches (particularly EbA as part of an ILWRM approach), including those suitable for implementation by women. Moreover, visits to the sites where on-the-ground adaptation interventions are being implemented for local community members and relevant stakeholders will be organised as part of the outreach programme in the project's three target areas. This will promote the use of and support for adaptation interventions under a climate-resilient ILWRM approach and further explain the subsequent benefits. Engagements with farmers will include raising their awareness on the true cost of water to discourage expanding farming of fruits and to consider planting drought-resilient crops with a higher economic value. Moreover, awareness-raising activities will include the presentation of information of best practices in water resource management from other water-scarce regions globally with similar climatic conditions. This will stimulate support from local communities for the interventions implemented through the proposed project.

217. Sub-activity 4.1.1.5 dependencies:

- Development of the outreach programme under Sub-activity 4.1.1.4 — completed during year 1.
- Development of knowledge management tools under Sub-activity 4.1.2.1 — completed during year 1.

Sub-activity 4.1.1.6. Train relevant educational practitioners from the Ministry of Education and tertiary institutions on the integration of climate change adaptation into syllabi and curricula of Jordanian universities.

218. The outreach programme will also include facilitating the integration of climate change adaptation into school syllabi. With oversight from project Technical Partners, educational practitioners from the Ministry of Education will receive training on approaches to include climate change adaptation in the curricula of Jordan's universities. Training will take place via a 5-day training workshop in Amman (in years 2) facilitated by an environmental education expert (with support from Technical Partners) to train relevant educational practitioners (30 people; 50% women) from the Ministry of Education and tertiary institutions on the integration of climate change adaptation into syllabi. Technical Partners will then work with educational practitioners from the Ministry of Education and tertiary institutions to integrate climate change adaptation into the syllabi and curricula of at least 5 universities (University of Jordan, and the University of Science and Technology, Jarash University, Mutah University and Al Balqa' Applied University). This will include the development of new courses and related materials. The integration will take place in year two, with reviews, revisions and updates being made in year 5. This sub-activity will contribute to the long-term sustainability of the outreach programme and promote the autonomous spread of climate change adaptation awareness across the country.

Activity 4.1.2. Improve knowledge management for climate change adaptation in Jordan to support national and sub-national decision-making and build communities of practice.

219. Activity 4.1.2 will target the strengthening of climate change adaptation knowledge management in Jordan. This will be done through five sub-activities aimed at developing knowledge management tools, collecting lessons learned and best practices, long-term monitoring, the establishment of partnerships with academic institutions and the training of users of the knowledge management tools.

Sub-activity 4.1.2.1. Knowledge-management tools developed to facilitate sectoral, cross-sectoral and community learning and mainstreaming of climate change adaptation approaches (including EbA) by policymakers and planners.

220. To facilitate sectoral, cross-sectoral and community learning (including amongst vulnerable groups such as women and youth), capacity building and the mainstreaming of climate-resilient ILWRM (and other adaptation approaches) into Jordan policy-making and planning processes, knowledge-management tools (software, including

AI-based tools) will be developed under Sub-activity 4.1.2.1. in year 1 of project implementation. Investment in knowledge management will enable sharing of information on global best practices, facilitating the gathering and sharing of lessons learned and evidence gathered during the project, and underpinning the integration of climate change adaptation into water resources and land-management policy- and decision-making. Knowledge management tools to be developed include a content repository to manage and share knowledge content, a knowledge search tool, a communication tool to capture and share knowledge, social software to facilitate the socialisation of knowledge (such as a smartphone application) and a knowledge visualisation tool to visually communicate knowledge. As possible, AI will be integrated into the tools. The information and data collected through these tools will be stored in a national climate change adaptation knowledge repository, a publicly accessible online database, which will be housed and maintained by one of the project's Technical Partners — the respective partner will be selected during project implementation. Lessons learned from the restoration of watersheds, improved gender-responsive livelihoods and strengthened local governance practices will also be shared through this national climate change adaptation knowledge repository, ensuring coverage of women and other vulnerable groups' experiences. Access to and use of the knowledge repository by user groups — such as planners, policy- and decision-makers, practitioners and the public (including women) — will be promoted by project Technical Partner websites and social media pages. Within the project, the knowledge management tools will be used to support: i) awareness raising on gender-responsive community-based water use and monitoring plans and training of community members (100% women) to monitor and manage the use of water — Sub-activity 3.1.4.2; and ii) the implementation of the gender-responsive outreach campaign to promote awareness of the benefits of climate change adaptation among rural communities and stakeholder in the Jordan Valley — Sub-activity 4.1.1.5.

Sub-activity 4.1.2.2. Collect lessons and best practices across Jordan (including on the project's ILWRM investments) and between arid states in the Mediterranean region and Middle East, as well as those generated via the project on an annual basis using the knowledge management tools.

221. Knowledge tools developed under Sub-activity 4.1.2.1, including the national climate change adaptation knowledge repository, will be used to inter alia capture and disseminate best practices (including those that promote gender equity), lessons learned, literature and data on climate change adaptation relevant to the Jordanian context, including those related to the project's climate-resilient ILWRM investments. This will include information generated by the proposed project, across Jordan as well as from other arid states in the Mediterranean and Middle East. The collection of lessons learned and best practices will take place from year 2 of project implementation.

222. Sub-activity 4.1.2.2 dependencies:

- Development of knowledge management tools under Sub-activity 4.1.2.1 — completed in year 1.
- Generation of lessons learned via the long-term monitoring plan developed under Sub-activity 4.1.2.3 — completed in year 2.

Sub-activity 4.1.2.3. Design and implement a long-term monitoring plan for the generation of lessons learned on the project's climate-resilient ILWRM investments.

223. Under Sub-activity 4.1.2.3, lessons learned from the project's climate-resilient ILWRM investments and the operationalisation of the Water Fund will be generated through a rigorous, frequent and long-term monitoring plan developed under the project in year 2. These lessons learned will inform Sub-activity 4.1.2.2 and will be captured and disseminated by using the knowledge management tools developed under Sub-activity 4.1.2.1.

Sub-activity 4.1.2.4. Establish partnerships (including the signing of MoUs) with Jordanian academic institutions and research agencies to participate in the implementation of the long-term monitoring plan (Sub-activity 4.1.2.3) and generate relevant research findings and publications to inform innovative landscape-centric approaches to climate change adaptation.

224. Under Sub-activity 4.1.2.4, partnerships with academic institutions will support the generation of new research findings and publications, and new curricula on climate-resilient ILWRM will be developed for relevant learning institutions via the long-term monitoring plan developed under Sub-activity 4.1.2.3. Currently, NCARE is the only government agency involved in land-management research and while some Jordanian universities (University of Jordan, University of Science and Technology, Jarash University, Mutah University and Al Balqa' Applied University) conduct research on land management, they play a minor role in technology transfer or extension services.

225. Sub-activity 4.1.2.4 dependencies:

- Development of the long-term monitoring plan under Sub-activity 4.1.2.3 — completed in year 2.

Sub-activity 4.1.2.5. User groups trained to use knowledge management tools to increase their access to knowledge/information of climate-resilient ILWRM approaches.

226. Through Sub-activity 4.1.2.5, an enabling environment will be created to foster exchange of knowledge and ideas for innovative and sustainable solutions to climate change impacts. This will include training user groups — including planners, policy- and decision-makers and practitioners (at least 30% women) — on the use of the knowledge management tools developed under Sub-activity 4.1.2.1.
227. Sub-activity 4.1.2.5 dependencies:
- Development of knowledge management tools under Sub-activity 4.1.2.1 — completed in year 1.

B.4. Implementation arrangements (max. 1500 words, approximately 3 pages plus diagram)

228. The proposed project will be implemented over an eight-year period. UNEP will be the Accredited Entity (AE) for this project and the Executing Entities (EEs) will be Jordan's Ministry of Environment (MoEnv; Component 1) and the International Union for Conservation of Nature (IUCN; Components 2 to 4). Further details on the suitability of the AE and EEs are presented below.

Accredited Entity (AE)

229. UNEP will be the Accredited Entity (AE) for this project, as designated by the Government of Jordan. The AE will be responsible for overseeing the implementation, financial management, evaluation, reporting and closure of the project, in coordination with the national-level Project Steering Committee (PSC) and a national-level Project Management Unit (PMU). As the AE, UNEP will: i) sign two Project Cooperation Agreements (PCAs; subsidiary agreements), one each with MoEnv and IUCN, to establish clear roles and responsibilities for project execution; ii) ensure that the project is executed in accordance with GCF and UNEP policies and standards; iii) through its on-granting accreditation, oversee the implementation of the on-granting facility and the Water Fund's grant mechanism that will be established (Activity 2.1.5), including ensuring that IUCN signs sub-project agreements with legal entities (as agreed on in the PCA between UNEP and IUCN): NGOs and CBOs; iv) supervise and oversee the implementation of project interventions; v) report on project progress; vi) co-chair the PSC; and vii) ensure that project activities are well coordinated and aligned with national priorities. Through the two PCAs, UNEP will ensure that all the relevant requirements under its Accreditation Master Agreement (AMA) and the Funded Activity Agreement (FAA) are passed down to MoEnv and IUCN and the project is implemented in accordance with such requirements. The PCAs will enable UNEP to directly monitor and supervise project implementation, request MoEnv or IUCN to take remedial actions and ensure that those actions are enforced, as well as to allow GCF to step into the PCAs in accordance with the relevant AMA/FAA (i.e., assume the contractual position of the AE in order to continue operation of the funded activity or to exercise any rights under the PCAs, if deemed necessary by GCF). A Task Manager (TM) will be responsible for project supervision to ensure consistency with GCF and UNEP policies and procedures. This TM will participate in: i) biannual PSC meetings; ii) facilitating interim and final evaluations; iii) preparing Annual Performance Reports and relevant documentation; and iv) technical reviews of project outputs.

Experience and track record of the AE

230. As the AE, UNEP will provide the necessary scientific expertise and technical support for managing multiple socioeconomic and environmental factors affecting ecosystems. UNEP has a broad global portfolio on climate change projects funded through, *inter alia*, the GCF, Least Developed Countries Fund (LDCF), Special Climate Change Fund (SCCF), Adaptation Fund (AF) and bilateral funds such as the European Commission. Through these projects, UNEP has supported national governments and local communities to adapt to climate change. UNEP's knowledge-base is derived from completed and ongoing projects supporting: i) methods and tools for decision-making; ii) prioritising, designing and implementing adaptation interventions; iii) enhancing climate resilience by restoring vulnerable ecosystems that underpin community livelihoods; and iv) monitoring the socio-economic and environmental benefits of adaptation interventions. It will draw upon previous experiences and lessons learned during the implementation of the proposed project.
231. As the GCF AE for the proposed project, UNEP will therefore oversee the efficient and effective delivery of the project's objectives, in alignment with its [Programme Manual](#). UNEP's in-house policies and expertise on gender mainstreaming as well as social and environmental safeguards will contribute towards gender- and environmentally-sensitive project implementation. A UNEP Programme Officer (PO) located in UNEP's West Asia Office (Lebanon) will be responsible for project supervision to ensure consistency with GCF and UNEP policies and procedures. The functions of the PO will include but will not be limited to the following: a) participating in the Annual PSC meetings; b) the clearance of periodic Progress Reports and Project Implementation Reviews; and c) the technical review of project deliverables, d) providing input to periodic readiness portfolio reporting to GCF; and e) preparing requests for disbursements. The project will be operated in accordance with UN regulations, rules and policies and fiduciary management of the project will be managed by UNEP in its role as AE. UNEP will also closely monitor the financial management of the Project using the established Monitoring and Evaluation procedure and financial reporting

mechanism, including an annual audit; and establish internal controls for the Project and project fund management. Within this context, funding allocation mechanisms are managed as per UN rules and procedures, including eligibility criteria, proposal evaluation processes, quality assurance and control, project monitoring and supervision. UNEP is audited annually by the UN Board of Auditors and has established dedicated trust funds for GCF resources.

Executing Entities (EEs)

232. Project execution will be with MoEnv (National EE), and IUCN (Lead EE), for Components 1²²⁵ and 2–4²²⁶, respectively (Table 8). UNEP will sign subsidiary PCAs/grant agreements with each EE (one with MoEnv and one with IUCN) to document their roles and responsibilities with respect to execution of project activities — including the role of IUCN in establishing the Water Fund (Activity 2.1.4). The PCAs between UNEP and EEs will include warranties and caveats by the EEs to *inter alia* ensure compliance with the Anti-Fraud and Anti-Corruption Framework of the United Nations Secretariat, as well as the GCF Policy on Prohibited Practices. An MoU will be signed between the EEs, detailing roles and responsibilities in implementation.

233. Both EEs have the legal capacity and authority to act and enter into contracts independently as required for the proposed project. The responsibility of the EEs will include being accountable to the AE for the effective use of resources during project implementation. All operating policies and procedures will follow the UNEP Programme Manual, which includes provisions for financial management and procurement²²⁷. The EEs, both have the financial management capacity to, and will be responsible for the disbursement of financial resources for the implementation of all project activities related to their relevant components.

Table 8. EE implementing responsibility per JILMI component, output and activity.

Components, Outputs and Activities	Responsible EE
Component 1: Climate change adaptation governance in Jordan	MoEnv
Output 1.1 Strengthened governance and capacity for climate-resilient integrated landscape and water resources management (ILWRM) and Output 1.2: Economic assessments developed quantifying net benefits of ILWRM as an adaptation approach	MoEnv
Activity 1.1.1. Strengthen the mandate and capacity of main government institutions — including MoEnv, MWI and its subsidiaries — for climate-resilient ILWRM in Jordan.	MoEnv
Activity 1.1.2. Draft amendments to Jordan's policy, planning and legal frameworks for agriculture, land, water and natural resources, to mainstream climate-resilient ILWRM as a climate change adaptation approach.	MoEnv
Activity 1.1.3. Generate evidence of the economic benefits of ILWRM to project beneficiaries for use by policymakers and planners.	MoEnv
Activity 1.1.4. Strengthen the mandate and capacity of non-state actors for climate-resilient ILWRM in the Jordan Valley.	MoEnv
Component 2: Integrated landscape and water resources management (ILWRM) under conditions of climate change	IUCN
Output 2.1 Climate-resilient ILWRM practices and ecosystem restoration implemented in three watersheds in the Jordan Valley and Output 2.2: Water Fund established in the Jordan Valley to maintain landscape investment for enhanced water resources	IUCN
Activity 2.1.1 Support the implementation of participatory climate-resilient integrated watershed management plans (developed under Activity 1.4) in the Yarmouk, King Talal Dam and Kafraïn areas of the Jordan Valley.	IUCN
Activity 2.1.2. Restore ecosystems in the watersheds of the Yarmouk, King Talal Dam and Kafraïn areas of the Jordan Valley	IUCN
Activity 2.1.3. Introduce gender-responsive climate-resilient additional livelihoods and agricultural practices in the project's three intervention areas.	IUCN
Activity 2.1.4. Establish a Water Fund in the Jordan Valley to maintain landscape investment for enhanced water resources.	IUCN
Component 3: Improved protection, supply and use of water resources under conditions of climate change	IUCN
Output 3.1 Improved protection, supply and use of water for domestic purposes under climate change conditions	IUCN
Activity 3.1.1 Implement measures to improve the quantity and quality of ground and surface water supplies within the project sites	IUCN
Activity 3.1.2. Install 1 MW of floating solar panels over the King Abdullah Canal to reduce water loss from evaporation	IUCN

²²⁵ Component 1: Climate change adaptation governance in Jordan.

²²⁶ Component 2: Integrated landscape and water resources management (ILWRM) under conditions of climate change; Component 3: Improved protection, supply and use of water resources under conditions of climate change; and Component 4: Knowledge management and awareness on climate change adaptation.

²²⁷ The EEs will be required to comply with UNEP rules, policies and procedures on procurement.

Components, Outputs and Activities	Responsible EE
Activity 3.1.3. Introduce small-scale rainwater harvesting interventions in the project's three target areas to improve access to clean water under climate change conditions in a gender-responsive manner	IUCN
Activity 3.1.4. Empower local communities, including women, to monitor and manage local water supplies more effectively	IUCN
Component 4: Knowledge management and awareness on climate change adaptation	IUCN
Output 4.1 Improved knowledge and awareness of climate change adaptation for landscape and water resources management	IUCN
Activity 4.1.1 Implement a gender-responsive climate change adaptation outreach programme across the Jordan Valley	IUCN
Activity 4.1.2. Improve knowledge management for climate change adaptation in Jordan to support national and sub-national decision-making, and build communities of practice	IUCN

Experience and track record of the EEs

234. As the head ministry for the National Designated Authority (NDA), MoEnv has been identified as the appropriate government entity to fulfil the role of EE for Component 1 of the proposed project, possessing the necessary technical expertise for the efficient and effective management of financial resources for the implementation of project activities. The MoEnv lead the development of the country's Nationally Determined Contribution (NDC), National Communications to the UNFCCC and NAP process, and has been involved in a number of related projects in recent years. This involvement includes managing and delivering national and regional projects focussing on, *inter alia*: i) climate change adaptation; ii) landscape and water sector management; and iii) sustainable development. Moreover, MoEnv's Climate Change Directorate (CCD) oversees all climate change-related activities in Jordan. Some of the initiatives supported by MoEnv over recent years are listed below (more details on these initiatives are provided in Section 7 of Annex 2, Feasibility Study).

- GIZ — Sustainable use of Ecosystem Services in Jordan; Energy and Climate Fund, 2014–2019
- GEF — dRHS Irrigation Technology Pilot Project to face Climate Change impact in Jordan, 2009–2016 (US\$7.6 million)
- IFAD, AFESD, OPEC Fund for International Development and IDRC — Community-based Optimisation of the Management of Scarce Water Resources in Agriculture in Central-West Asia and North Africa (CWANA) Project, 2003–2013
- GEF — Mainstreaming Sustainable Land and Water Management Practices, 2009–2015 (~US\$29.7 million)
- GIZ — Managing Water Resources in Jordan, 2006–2018
- GEF — Integrated Ecosystem Management in the Jordan Rift Valley, 2007–2013 (~US\$12.7 million)
- GEF — Badia Ecosystem and Livelihood Project, 2013–2017 (US\$33.3 million)

235. The MoEnv's mission is to "maintain and improve the quality of Jordan's environment, conserve natural resources and contribute to sustainable development"²²⁸. This is achieved by leading the governance system — including other relevant governmental organisations — in initiating and driving policies, legislation, strategies and monitoring, as well as by mainstreaming environmental policies into all national development plans²²⁹. The MoEnv comprises several specialist units, including the directorates of: i) Climate Change; ii) Nature Protection; iii) Environmental Inspection; iv) Waste Management and Hazardous Substances; v) Licensing and Pollution Prevention; and vi) Outreach and Environmental Awareness. These are supported by the functional directorates of Human Resources and Administrative Affairs, Financial Affairs, and Policies and Institutional Development²³⁰. In 2015, following the recommendations of the National Climate Change Policy, the Climate Change Directorate (CCD) was established in 2014 to serve as the institutional focal point for climate change activities in Jordan, including the country's commitments as a party to the UNFCCC. Additionally, the CCD is the coordinating focal point to global and regional environmental governance systems²³¹. The CCD coordinates high-level policy formulation and implementation, facilitating the participation of relevant stakeholders at all levels. It also oversees development of the internal capacity of MoEnv — as well as the government in general — in addressing the challenges and barriers of climate change²³². CCD is made up of two departments, namely the Emissions Reduction Section and the Adaptation to Climate Change Section. Further detail of the organisational structure of MoEnv can be found in Section 1.3.1 Institutional Framework

²²⁸ Ibid

²²⁹ Ibid

²³⁰ Ministry of Environment. 2017. Organisational Structure. Available at: <http://moenv.gov.jo/En/AboutUS/Pages/OrganizationalStructure.aspx>

²³¹ Ministry of Environment, GIZ. 2015. *Climate Change Governance in Jordan: Towards Policy and Institutional Coordination*. Available at: http://moenv.gov.jo/En/Environmental_Sectors/Atmosphere/Documents/Climate%20Change%20Governance%20in%20Jordan.pdf

²³² Ibid

in Annex 2: Feasibility Study. UNEP, as the Accredited Entity, has undertaken a capacity assessment of the Executing Entity, particularly by reviewing the elements of governance and public accountability. UNEP provides its assurance that the MoEnv has the relevant fiduciary and financial management capacity to act as the Executing Entity. UNEP will undertake a full assessment of the financial management capacity of MoEnv to identify risk elements and to prepare appropriate mitigation measures. UNEP will also closely monitor the financial management of the project using the established M&E procedures and financial reporting mechanisms, including an annual audit; and establish internal controls for the project and project fund management.

236. IUCN's expertise and extensive network provide a solid foundation for a large and diverse portfolio of conservation projects around the world. Combining the latest science with the traditional knowledge of local communities, these projects work to reverse habitat loss, restore ecosystems and improve people's well-being. They also produce a wealth of data and information which feeds into IUCN's analytical capacity. Through its member organisations, IUCN supports and participates in scientific research, promotes and helps to implement national conservation legislation, policies and practices; and funds, operates or manages thousands of field projects worldwide. An example of an initiative supported by IUCN over recent years is (more details on these initiatives are provided in Section 7 of Annex 2, Feasibility Study) the EU-funded Securing Rights and Restoring Lands for Improved Livelihoods project, 2011–2013. The value addition of selecting IUCN (a GCF AE) as EE includes the organisation's presence and footprint in Jordan via its regional office (IUCN-ROWA) which is based in Amman. This means that the project will benefit from the support of the organisation's locally-based Jordanian staff who are familiar with the local context and have experience working with the stakeholders and in the ecosystems included in the project intervention areas. This includes a strong working relationship with the consortium of NGOs that supported the development of this proposal and will function as Technical Partners during project implementation (see below). As a result, the selection of IUCN as EE, as opposed to UNEP for example, is strategic in that it provides a strong operational foundation for the implementation of the project. Additional details on the IUCN's comparative advantage and capacity to function as EE of Components 2 to 4 are presented in Annex 23.

Technical Partners

237. The required expertise to implement project technical activities (Components 2 to 4) will be sourced primarily from members of the Consortium of NGOs, with due care and consideration for the best value for money and best interests of the project. The process of selection and managing technical partners is detailed below.

238. IUCN will develop the Terms of Reference for the specific technical assistance to be provided under relevant project components, which will include specific information on the technical assistance to be delivered, including activities, outputs, expected outcomes and impacts, timeline, deliverables and requirements such as track record including experience with carrying out a project of similar size and scope. The contracting of qualified national and international technical partners will be determined using a competitive selection process, based on track record in delivering the services and best value of money considering entity co-financing to the project. The NGOs in Annex 23 are expected to be the main technical partners. IUCN will be responsible for the competitive selection of technical partners in alignment with UNEP partnership policy and procedures. The selected Technical Partners (TP) will be contractually bound through legal agreements signed with IUCN (IUCN-TP agreements). TPs will produce regular progress and financial reports and will submit deliverables to IUCN. Funds for the implementation of activities will be released by IUCN to the TPs in the form of cash advances (bank transfers) dependent on cleared expenditure reports and progress reports.

Project Steering Committee (PSC)

239. The PSC will comprise representatives from, *inter alia*: i) IUCN and alternating co-chairs, along with UNEP as permanent co-chair; ii) MoEnv's subsidiaries; CCD, WAJ and JVA; iii) MWI; iv) TPs; v) the National Technical Working Group (TWG); and vi) WMCs. The committee will primarily be responsible for providing technical oversight and advisory support. This includes: i) overseeing project implementation; ii) reviewing annual workplans and project reports; and iii) approving any changes to the project's targets, activities or timelines. Biannual PSC meetings will be held²³³ to take management-related and technical decisions, discuss the project's main performance indicators and provide strategic guidance. Any changes made by the PSC to the project's Results Framework or timeline will be communicated to the Project Management Unit (PMU). In addition, the PSC will be responsible for: i) overseeing the PMU; ii) providing technical and administrative functions; and iii) serving as the focal point for engagements and interactions between project stakeholders and partner organisations. ToRs for the PSC will be adopted during the first PSC meeting.

National Technical Working Group

²³³ with ad hoc meetings held as required

240. A National TWG will be established to provide technical advice to National EE, MoEnv, to support the implementation of Component 1's cross-cutting 'soft' interventions that are not site-specific — i.e., the capacity building and mainstreaming activities which are being implemented at a national level. The membership of the National TWG will include entities and agencies involved in the planning of landscape management and water resources, as well as climate change adaptation, such as JVA, WAJ, MoA, MOPIC, MWI and the Jordanian National Commission for Women (JNCW)²³⁴. MoEnv, will work closely with the TWG members (MWI in particular) to ensure efficient implementation. In addition, the National TWG will include local specialists and academics with expertise relevant to the project, such as in climate change adaptation (including EbA) and integrated landscape and water resources management (ILWRM). The National TWG will work closely with the Technical Partners and WMCs to ensure that 'soft' interventions across the Jordan Valley complement the 'hard' interventions that will be implemented at the project's three target sites. Additionally, the TWG, together with TPs and the CTA, will evaluate proposals submitted in response to the RFPs advertised via on-granting facility and the Water Fund's grant mechanism.

Watershed Management Committees (WMCs)

241. To promote the representation and participation of all relevant stakeholders — including government, NGOs, CSOs, local communities, farmers and the private sector — in watershed management in the Jordan Valley, WMCs will be established within each of the project's three target sites (Yarmouk, King Talal Dam and Kafraïn), with at least 50% female representation. ToR for the role and function of WMCs will be developed by project Technical Partners (framework ToR are presented in Annex 28). The presence of WMCs will provide a platform for dialogue between stakeholders in upstream and downstream areas of watersheds, ensuring the coordination of management, planning and implementation. WMCs will ensure that target communities' needs and concerns are considered in the development and implementation of water monitoring and management plans. Representatives of target communities will be members of the WMCs, providing them with direct inputs into decision-making processes and ensuring that relevant information is always made available to them. These community representatives will be members of community-based committees already present on the ground in the three project areas, such as Development Advisory Committees, Stakeholders Advisory Groups, agricultural cooperatives, women's cooperatives and Water User Associations. Once established, the project will work closely with the MoEnv (project Executing Entity) to formally recognize and include WMCs in relevant legislation — including defining their mandate related to working with communities. This will not only reinforce their contribution to the project and strengthen the participation of local stakeholders (communities in particular) but contribute to their sustainability and a paradigm shift in community participation in development projects.

242. The responsibilities of the WMCs will include: i) acting as focal points to facilitate interactions between the PSC, PMU, National TWG and stakeholders (specifically local communities) at a local level; ii) providing guidance to the PSC and PMU; iii) having representation on the PSC; and iv) coordinating the implementation of the proposed project's interventions in each of the three target sites. Such local-level structures, like WMCs, are important to build capacity and structures in support of ILWRM (including EbA), as well as to directly manage the watershed-level interventions under the project. The WMCs will serve as important cross-sectoral coordination bodies at the site-level, which currently do not exist. In addition, WMCs will serve as channels through which climate information, including forecasts and early warnings will be disseminated to local communities. JVA and local governors will co-chair the WMCs (with direct support being provided by project Technical Partners), convening quarterly meetings to discuss ongoing activities, as well as any plans for future activities related to watershed management.

243. Several activities and sub-activities are dependent on the formal establishment of the WMCs. These include 1.1.4.2, 1.1.4.4, 1.1.4.5, 2.1.1.3, 2.1.2.2, 2.1.3.2, 2.1.4.1, 2.1.3.3, 2.1.3.5, 2.1.4.1, 2.1.4.3, 2.1.4.4, 2.1.4.5, 2.1.4.6, and 3.1.4.1 — dependencies have been mapped across the implementation timeline in further detail in Annex 5. Focus will therefore be placed on ensuring that their inclusion in legislature, constitution and operationalisation is prioritised and finalised in year 1 of project implementation.

Project Management Unit (PMU)

244. A gender-balanced PMU will be established by IUCN and MoEnv in year one of project implementation to coordinate the implementation of project interventions with the AE, EEs and other implementing partners. The PMU will be hosted within MoEnv and will consist of: i) a Project Coordinator (PC); ii) a Financial Officer; iii) a Procurement Officer; iv) an Environmental and Social Safeguards Officer (ESO); v) a Communications Officer (CO); vi) a Monitoring Officer; vii) a Gender Officer; viii) a Policy and Institutional expert; ix) an Administrative Officer (AO); and x) a Chief Technical Advisor (CTA). The recruitment process will ensure that the PMU is gender balanced. The PC will be a full-time staff member, recruited competitively, who will be responsible for the daily on-the-ground implementation and management of the project. The PC will be primarily responsible for ensuring that the project achieves the targets

²³⁴ To ensure the inclusion of gender expertise in decision-making bodies.

set out in the Logical Framework (Section E) to the required quality standards, and within the specified time and budget allocations. To achieve these targets, the PC will: i) report directly to the PSC on project management-related matters; ii) manage the project in accordance with the specified workplans and allocated budget; iii) ensure that all project interventions are implemented according to GCF and UNEP guidelines; iv) manage and disburse financial resources allocated for the project during project implementation; v) work closely with national and local authorities (including municipalities) to ensure that the project is managed effectively and that the needs of all beneficiary groups are considered; and vi) oversee the efficient and effective information- and knowledge-transfer to relevant project partners. The PC will be required to fulfil these roles for the duration of the project and upon submission of the final evaluation report — as well as any other required documentation — to the GCF and UNEP. The Financial Officer, Procurement Officer and AO will report directly to the PC. Their responsibilities will include: i) providing administrative, logistical, technical and financial support to the PMU; ii) compiling reports on the disbursement procedures for the project and anticipated requirements for project funding during the implementation phase; and iii) ensuring that all project-related tasks are undertaken according to UNEP and GCF standards. The Financial and Procurement Officers in particular, will manage the financial transactions for the project outputs and activities, which will be implemented in line with the established project workplan and national priorities.

245. A Chief Technical Advisor (CTA) will be employed on a full-time basis and be hosted within the PMU — this position will be recruited via the IUCN. The CTA will primarily be responsible for providing technical support on climate-proofing interventions that will be implemented under the project to ensure that these interventions meet their targets, as specified in the Logical Framework (Section E). In addition, the CTA will be responsible for: i) supporting the annual planning process and budgets; ii) providing monitoring and operational support to the project; iii) coordinating and supervising the work of specialist technical consultants who will contribute to specific deliverables associated with each project output; iv) providing biannual reports to the PMU and PSC on project progress, performance towards objectives and recommendations; v) evaluating proposals submitted in response to the RFPs advertised via the on-granting facility and Water Fund's grant mechanism (Activity 2.1.5) — together with TPs and the TWG; and vi) supervising, in collaboration with TPs and WMCs, sub-projects under the on-granting facility and the Water Fund's grant mechanism.

Figure 13 below shows the management structure for the proposed project.

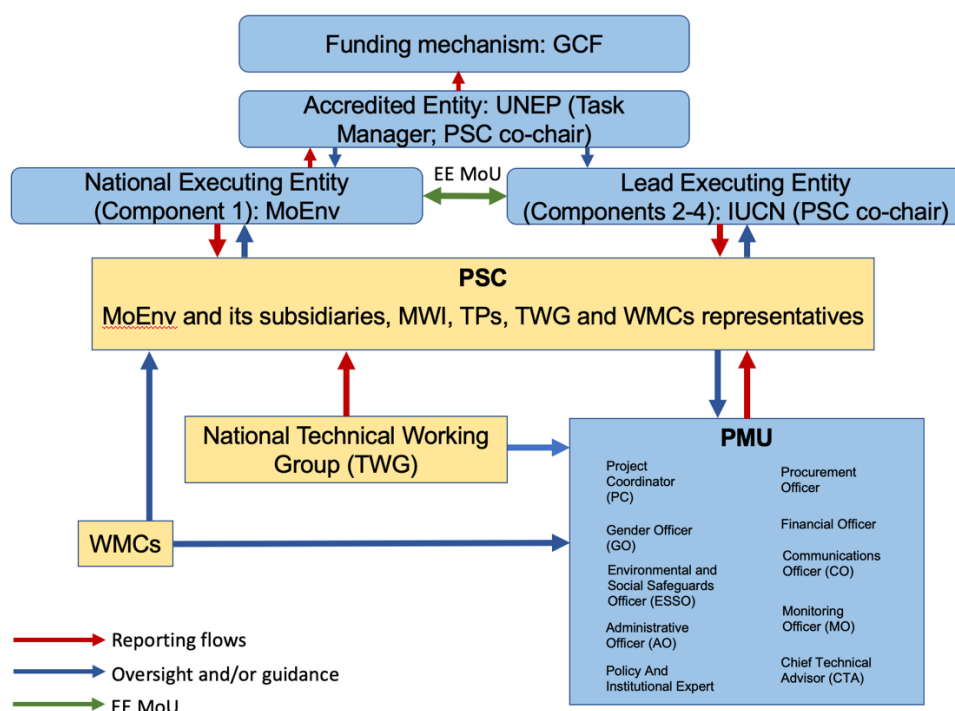


Figure 13. Project management structure for the proposed project. The blue lines represent oversight and guidance, whilst red lines representing reporting flows.

Flow of funds and contracting arrangements

246. Figure 14 shows the project flow of funds and contractual arrangements. The green arrows indicate the flow of funds, orange arrows reporting, while the blue arrows show the contractual or legal arrangements. Included are the types of agreements between the relevant parties. PCAs will be signed by UNEP and each EE (one per EE) and one

MoU will be signed between the two EEs (MoEnv and IUCN). GCF funds will flow from UNEP to both EEs, and both will be required, as per the PCAs, to submit expenditure reports to UNEP — IUCN will consolidate reporting from MoEnv into one expenditure report to be submitted to UNEP. Through its PCA with UNEP, IUCN will, under UNEP's oversight as AE, establish the on-granting facility. The MoU between the two EEs will outline roles and responsibilities, their contribution to the project as a whole, and reporting lines. GCF proceeds will flow directly to each EE via UNEP. IUCN will sign legal agreements (IUCN-TP agreements) with TPs (including members of the Consortium of NGOs, refer to Annex 23). IUCN, through the on-granting facility, will launch RFPs for sub-projects at the community level which fall into the ILWRM thematic area and contribute to increasing and protecting the Jordan Valley's water resources under climate change conditions. Preliminary selection criteria for the sub-projects are presented in Table 7 and will be finalised by IUCN with guidance provided by UNEP, MoEnv, JVA, the national TWG and CTA during the preparation of the RFPs. Sub-project agreements will be signed between IUCN and beneficiaries. Additional detail on the sub-projects is provided in the description of Activity 2.1.5 in Section B.3.

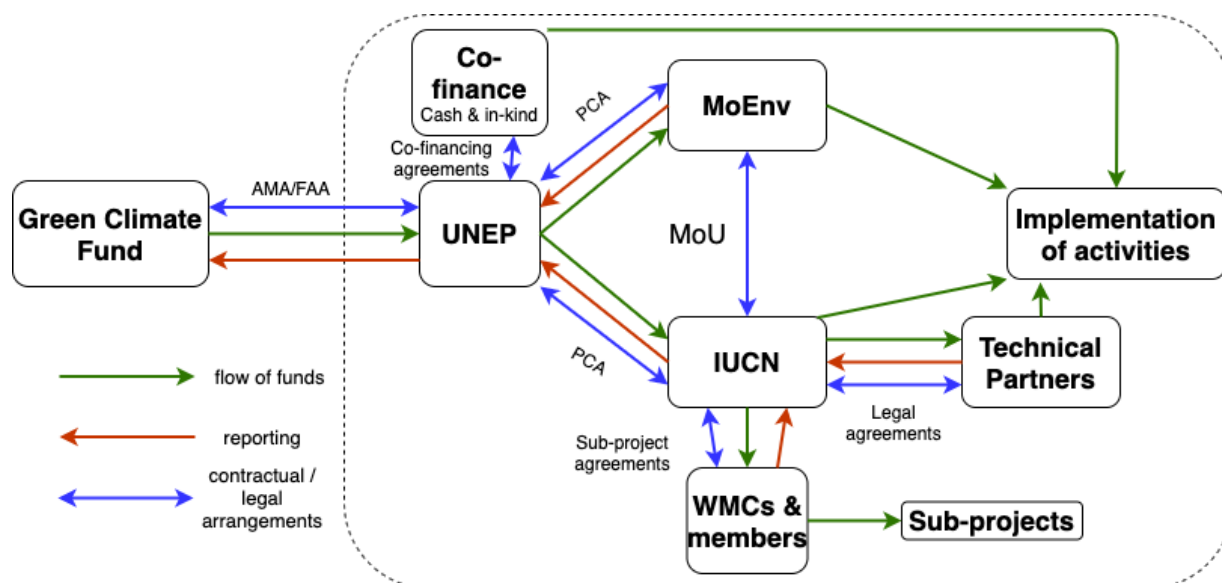


Figure 14. Flow of funds and contractual and legal arrangements for project implementation.

B.5. Justification for GCF funding request (max. 1000 words, approximately 2 pages)

247. Jordan is ranked as the fifth most water-stressed country in the world²³⁵, with only 125 CM/yr of water available per person²³⁶. Climate change will further exacerbate water stress in Jordan through decreased annual rainfall, higher annual temperatures and more frequent, extreme and prolonged droughts. As a result of the country's existing vulnerability to critical water stress, even a minor decrease in water supply resulting from climate change impacts is expected to have severe implications for the country's water security.

248. As water is already scarce in Jordan, the additional impact of a change in climatic conditions at any scale could have severe consequences on climate-resilient development and the health and wellbeing of the country's most vulnerable people. Furthermore, many low cost, accessible measures to increase water availability have already been deployed. Consequently, it is not expected that there are many easily implementable, low-cost options to increase water availability remaining. However, the benefits of enhancing water availability are likely to be large, especially in the context of climate change. As a result, there is a strong economic argument for exploring interventions that increase water availability which cannot easily be provided by markets and the current enabling environment. This is especially the case where expected climatic conditions make the probability of upscaling existing measures seem particularly high and where success is very likely.

249. In consideration of the above, addressing the impact of water stress on Jordan's inhabitants, ecosystems and economy requires a climate change adaptation response based on an integrated landscape and water resources management (ILWRM) approach for land and water resources. For Jordanians whose livelihoods are threatened by climate change, ILWRM offers a pathway towards more resilient livelihoods, based on sustainable resource use that takes into account the interconnections between different ecosystems, hydrological systems and various uses within

²³⁵ WRI. 2019. Aqueduct baseline water stress rankings. Available at: <https://www.wri.org/aqueduct/>.

²³⁶ Hammouri N, Adbulla F, Qdais HA & Freiwani M. 2015. Assessing the impacts of climate change on water resources of Jordan. World Water Congress XV. International Water Resources Association (IWRA), Edinburgh.

a landscape. However, while the climate change adaptation and mitigation potential of ILWRM is clear, its implementation in Jordan through the JILMI Project represents a major investment.

250. To this end, the Government of Jordan (GoJ) is seeking US\$44,904,203 of GCF grant financing to achieve this objective by reducing the negative impacts of climate change on water resources through a landscape-level approach. The funding will be used to implement climate-resilient interventions with a focus on, *inter alia*: i) ecosystem restoration; ii) managed aquifer recharge; iii) rainwater harvesting; iv) evaporation reduction; v) diversified natural resource livelihoods; and vi) strengthened governance. There are multiple factors which compound to make a very strong rationale for investing in ILWRM, particularly with a strong ecosystem-based adaptation (EbA) component. These are summarised below.

- Market failures: markets tend to be skewed toward hard infrastructure as they often do not account for any negative impacts of hard infrastructure and the wider benefits of ecosystem restoration and associated ecosystem services are undervalued.
- Barriers to entry for alternative technologies: buyers may not have the technical knowledge to understand and be able to compare ecosystem-based approaches with traditional man-made alternatives.
- Transaction costs: economically viable options are prevented from coming forward at a landscape scale without coordination as a result of the costs of bringing stakeholders together and strengthening governance processes.

251. These factors provide a clear rationale for investing in an enabling environment that advances ILWRM in Jordan as a climate change adaptation response and demonstrates the potential to further reduce water scarcity through approaches which remain novel or unexploited. This is particularly the case for approaches with high upfront costs and where there is difficulty with regards to accessing credit. Such activities have the potential to be scaled up once their potential or the ability of investment to generate returns is proven within the country, given expectations of increased renewable water deficits (resulting from climate change) and consequent increases in the demand for solutions.

252. Although Jordan is classified as an upper middle-income country²³⁷, the country continues to face major economic burdens including the continued influx of refugees from unstable states in the region, conflicts in major trading partners of Iraq and Syria, and the ongoing COVID-19 pandemic. Such burdens will continue to compound the country's growing public debt, which increased to ~94% of the GDP in 2019 from 65.4% in 2011²³⁸, totalling US\$40.2 billion in January 2019²³⁹. This debt has been attributed largely to: i) successive governments adopting an expansionist fiscal policy²⁴⁰; ii) the provision of considerable subsidies for staple goods; iii) a reduction in foreign aid, which has forced the country to borrow externally; iv) increased welfare spending for citizens and refugees; v) excessive spending by the National Electric Power Company (NEPCO) and the Water Authority of Jordan (WAJ); and vi) low tax collections — nearly 95% of the population is not subject to income tax. Compounding Jordan's public debt is a high trade deficit (~26% in 2015) — mainly owing to a reliance on energy and food imports — and domestic fiscal deficit (6.6% in 2015, although down from 12.7% in 2012)²⁴¹. In terms of the refugee influx, it is estimated that up to 3 million refugees (predominantly from Syria) are currently hosted by Jordan (total population of 10.2 million), placing increased strain on financial and natural resources in the country. The direct cost of Syrian refugees for the Jordanian government from 2011 to 2016 was ~US\$4.2 billion²⁴². The international community has recognised the strain placed upon Jordan's economy and infrastructure, directing resources to alleviate some of that pressure, although matching these resources with needs on-the-ground is increasingly difficult²⁴³. The conflicts in Syria and Iraq, that are fuelling the influx of refugees, have also resulted in a considerable increase in Jordan's trade deficit²⁴⁴, by disrupting exports that had relied on Syria as a major intermediary route to other important destinations such as Lebanon, Europe, Turkey and Central Asia²⁴⁵. From 2011, border crossings with Syria and Iraq were severely limited and between 2015 and late 2018 were completely closed, causing a profoundly negative impact on

²³⁷ Source: <https://datahelpdesk.worldbank.org/knowledgebase/articles/906519-world-bank-country-and-lending-groups>

²³⁸ Source: <https://tradingeconomics.com/jordan/government-debt-to-gdp>

²³⁹ Asharq Al-Awsat. 2019. Jordan's National Debt Exceeds \$40 Bn. Available:

<https://aawsat.com/english/home/article/1642336/jordan%E2%80%99s-national-debt-exceeds-40-bn>.

²⁴⁰ Wherein the state reduces tax rates and/or increases government spending to increase domestic demand for goods and services.

²⁴¹ Idris I (ed). 2016. *Economic Situation in Jordan*. University of Birmingham. Birmingham.

²⁴² Alshoubaki We & Harris M. 2018. The impact of Syrian refugees on Jordan: A framework for analysis. *Journal of International Studies*. 11: 154-179.

²⁴³ Amjad R, Aslan J, Borgnäs E, Chandran D, Clark E, Ferreira dos Passos A, Joo J, Mohajer O (eds). 2017. Examining barriers to workforce inclusion of Syrian refugees in Jordan. International Labour Office. Geneva.

²⁴⁴ Appendino M, Cerisola M, Martin E, Salman F, Tiffin A, Morra P, Sobrinho N, Wagner H (eds). 2017. *Jordan. Selected issues*. International Monetary Fund. Washington, DC.

²⁴⁵ Ibid.

Jordanian exports reaching their final markets²⁴⁶. Exports to Iraq — once Jordan's major export market, accounting for almost a fifth of domestic exports and up to ~US\$1.2 billion a year — have declined by more than 50% since the closing of the Iraqi-Jordanian border²⁴⁷. Such economic stresses are likely to continue in the short to medium term, reducing the amount of public finance available for climate change adaptation. Consequently, the demonstration of the benefits of investments in adaptation approaches such as ILWRM to decision-makers in Jordan will remain limited, preventing a shift to climate-resilient sustainable development.

253. Jordan has recently seen the approval of GCF FP155 — Building resilience to cope with climate change in Jordan through improving water use efficiency in the agriculture sector (BRCCJ) — which includes US\$ 25 million in grant finance for the GCF. However, there is little overlap between the proposed project and BRCCJ and opportunity for great complementarity. While both projects have a strong focus on water, their approaches differ with BRCCJ focusing on water use efficiency in the agriculture sector, and the proposed project focusing on ILWRM. Furthermore, the projects are being implemented in different areas of the country, BRCCJ in the Dead Sea Basin and the proposed project in the northern Jordan Valley — a detailed analysis of similarities, differences and complementarities is presented in Annex 29. This increases the scale of impact that GCF resources will have in Jordan, with the two projects collectively contributing to water security and climate resilient sustainable development in Jordan. As both projects focus on water, this further highlights the need for a paradigm shift in how water resources in Jordan are managed under conditions of climate change, the multitude of different, yet complementary, approaches which are required to do so, and the major funding gap that stands in the way of achieving adaptation priorities in the country.
254. Given the funding gap between Jordan's climate priorities and its public finances, adaptation will require a shift in national planning and budgeting (alongside international funding)²⁴⁸. Without GCF support, Jordan will continue to operate under the existing paradigm of an uncoordinated and compartmentalised model of land and water resources management that is not climate resilient. Under the current scenario, Jordan's dwindling water resources will become increasingly vulnerable to the projected impacts of climate change. This will result in: i) reduced quantity and quality of water resources and water insecurity; ii) food insecurity; iii) increased desertification and degradation of rangelands and impacts on pastoral and rural communities' livelihoods; iv) decreased supply of ecosystem goods and services; and v) increased conflict over water and other natural resources. Through GCF funding, however, these negative impacts of climate change will be avoided and a paradigm shift in the country towards climate-resilient sustainable development will be achieved²⁴⁹. This includes transforming from a compartmentalised, sector-specific approach to landscape management to one that is integrated, participatory and adaptation focussed, contributing to climate-resilient sustainable development in the country. As a result, there will be: i) increased budgets allocated to adaptation; ii) multiple landscapes under climate-resilient ILWRM; iii) improved protection supply and use of water for domestic purposes under climate change conditions; and iv) behavioural change.

B.6. Exit strategy (max. 500 words, approximately 1 page)

255. A framework of the proposed project's Exit Strategy is presented in the paragraphs that follow, focusing on the role of the Water Fund, sustainability through technical and institutional capacity building, social and environmental sustainability, and financial sustainability. This framework will be further developed by the PMU during the last two years of project implementation through: i) taking stock of all project interventions for which O&M will be required in the long-term; ii) identifying which stakeholders will take over O&M of relevant interventions and signing agreements with them (if not already done so during implementation); iii) developing long-term O&M plans for the identified ILWRM interventions (if not already part of implementation); developing a scaling up and replication strategy for the project's interventions; and iv) identifying sources of finance to cover the costs of long-term O&M, as well as to scale up and replicate its ILWRM interventions.
256. A key financial source for the long-term O&M and scaling up and replication of the project's ILWRM interventions in the Water Fund (Activity 2.1.4). The Water Fund will be designed specifically to generate financial flow to contribute to the sustained management of project interventions, as well as to fund their scaling up and replication. This will ensure that the benefit flows (particularly those related to water resources) generated through the implementation of the project are sustained. Included in the Water Fund's design will be a grant mechanism to support the implementation of subprojects within the project's ILWRM approach by WMCs and their members. During its

²⁴⁶ Reuters. 2018. Jordan and Syria reopen Nassib border crossing. Available here: <https://www.reuters.com/article/us-mideast-crisis-syria-jordan/jordan-and-syria-reopen-nassib-border-crossing-idUSKCN1MP0L4> Accessed: 26 Aug 2019

²⁴⁷ Appendino M, Cerisola M, Martin E, Salman F, Tiffin A, Morra P, Sobrinho N, Wagner H (eds). 2017. *Jordan. Selected issues*. International Monetary Fund. Washington, DC.

²⁴⁸ Source: https://assets.publishing.service.gov.uk/media/5d30a131ed915d2ff003b781/619__Jordan_Environment_Policies_and_Engagement.pdf

²⁴⁹ A detailed description of project activities is presented in Sections B.3 and E.6 of the Funding Proposal.

establishment phase, the Water Fund will become a 'legal entity' and long-term legal mechanisms such as contracts and bylaws will be formalised via the WMCs, ensuring its sustainability. During the Water Fund's maturation phase, options for scaling up to other areas of Jordan will also be assessed through the development of a scaling up strategy. The scaling up strategy will link closely with a long-term financing strategy that will detail how financial resources for the Fund's ongoing operation post GCF involvement will be leveraged from public and private sources. The long-term financing strategy will include a detailed model describing how the Water Fund's PES scheme will channel finance into the grant mechanism (which will include grant and results-based payments mechanisms). In addition, the development of the long-term financing strategy will consider how the Water Fund, once established and matured, could link to other financial instruments, such as EbA funds²⁵⁰, and development banks (e.g., the Islamic Development Bank), as well as leverage and unlock additional domestic finance and development banks which would contribute to ensuring long-term financial support for the fund — this will draw from experiences in other parts of the world. The leveraging of commitments from other financial instruments and partners will be fostered through proving the concept of the Water Fund through the project and developing the necessary enabling conditions for its long-term success through the activities of Component 1. As the Water Fund is designed, established and matures, the project's Exit Strategy for continued investment in and management of its ILWRM interventions will be built around it, resulting in the development of a rigid plan by project closure. It should be noted that the realisation of the scaling-up scenarios presented in Section D.2. are not reliant on the Water Fund. The contribution of the project to the creation of an enabling environment (Component 1) that supports adaptation at scale is critical as it provides the necessary conditions. This includes strengthening the mandate and capacity of main government institutions and non-state actors for ILWRM in Jordan (Activities 1.1.1 and 1.1.4); mainstreaming ILWRM as an adaption approach into regulatory frameworks for land, water, agriculture and natural resources (Activity 1.1.2); and generating evidence of economic benefits of ILWRM (as an adaptation approach) for use by policy makers and planners (Activity 1.1.3). Through its policy-related activities in Component 1, the project will support budget reforms, increasing the allocation of public budgets to adaptation (ILWRM in particular), which would support upscaling of the interventions in the medium to long term. Additional details on the Water Fund's contribution to financial sustainability are provided below.

257. The financial, technical and social sustainability of interventions has been accounted for during the design process to deliver optimal climate change adaptation benefits to vulnerable communities in Jordan beyond the lifetime of the proposed project. Details of the specific steps taken to ensure long-term sustainability are presented below.

Sustainability through technical and institutional capacity building

258. The project will support the development of an enabling environment for the implementation of a climate-resilient ILWRM approach in Jordan, by strengthening governance and building institutional capacity at the national and local levels in a gender-responsive and -inclusive manner, focusing government attention, catalysing opportunities for mainstreaming, and addressing policies, plans and investment patterns that are currently barriers to adaptation in the sector. Institutional strengthening includes extending the mandate of the Jordan Valley Authority (JVA) to facilitate the uptake of ILWRM principles in the decision-making process for sustainable socio-economic development. Specifically, interventions under Activity 1.1.2 will include training of: i) policy- and decision-makers (at least 50% women) on the integration of climate resilient ILWRM into policies, plans and strategies; ii) national and sub-national planners on climate-resilient ILWRM planning; and iii) staff from relevant national and sub-national institutions (at least 50% women) on the use of climate data, the results of climate analyses, and risk assessment in policy- and decision-making as well as disaster risk management. These interventions will strengthen institutional capacity for adaptation actions and enhance the technical capacity of stakeholders for the planning and implementation of climate-resilient ILWRM. Furthermore, through building the capacity of local partners such as MoEnv, MWI, MoA, and NGOs (Activities 1.1.1 and 1.1.4) for climate-resilient ILWRM, in-country capacity for accessing climate finance, including developing adaptation proposals for GCF support, will be enhanced. Technical capacity of stakeholders will also be strengthened as part of Activity 2.1.3, which involves training small-scale farmers in the three project sites on the use and maintenance of water conservation and harvesting measures (including at least 50% female-headed households).
259. To ensure that Jordan's regulatory system is aligned with the overall objective of enhancing the resilience of its population to climate change, the country's policy framework will be strengthened under Activity 1.1.2 — specifically through the incorporation of climate-resilient ILWRM. This will be achieved through the drafting of amendments to the country's relevant legislation (such as Jordan's Decentralisation Programme and National Development Plan), which will ensure that the ILWRM approach is sustained as part of the government's development strategy under future climate conditions. The above activities will contribute to the sustainability of the project, particularly in supporting the development of an enabling environment for climate-resilient ILWRM in Jordan, by addressing

²⁵⁰ Such a fund may include the Global EbA Fund, which is administered by UNEP and executed by IUCN: <https://globalebafund.org>

priorities identified in national policies and strategies (see Section 10 of Annex 2: Feasibility Study for specific details). This will focus government attention, catalysing opportunities for mainstreaming, and addressing policies, plans and investment patterns that are currently barriers to adaptation in the water sector. Targeting national and local processes in important national planning documents and policies, will also enhance project scalability and replicability. In addition, the GoJ will include the monitoring and analysis of the project's on-the-ground interventions in their ongoing operations to contribute to the capacity building at the policy- and decision-making level and policy reform process under Activity 1.1.2 to integrate climate-resilient ILWRM into Jordan's planning, policy-making and budgeting process. This will facilitate the embedding of coordination and operation of these mechanisms and structures in budgets and mandates beyond the life of the project. These plans will also include long-term funding commitments for enhancing the ecological infrastructure and maintaining the physical infrastructure associated with livelihood interventions. Improved knowledge-management — such as the storage and dissemination of lessons learned and gender-responsive best practices on adaptation — under Activity 4.1.2 will complement policy reform through benefitting national and sub-national decision-making for climate change adaptation. The abovementioned interventions will also ensure that best practices and lessons learned are incorporated into future adaptation actions.

Social and environmental sustainability

260. The proposed project has been developed and will be executed through a gender-inclusive participatory and consultative process, which will enable all relevant stakeholders to contribute to the design and implementation of Jordan's climate change adaptation interventions. This approach will allow the project to be country-driven by enabling stakeholders at all levels to take ownership of project interventions (including vulnerable groups such as women and youth), negating the need for a top-down approach to management and ensuring the long-term sustainability of interventions. The gender-responsive outreach programme undertaken as part of Activity 4.1.1 will secure the buy-in of stakeholders at the community level, which will ensure the continued maintenance of on-the-ground interventions. The outreach programme will also include facilitating the integration of climate change adaptation into school syllabi. This will promote the autonomous spread of adaptation awareness across the country, encourage the long-term maintenance and ownership of the project's interventions, and provide a platform for the youth to enrol for climate change-related studies at tertiary institutions in the future, and thereby contribute to climate-resilient sustainable development in Jordan.
261. Under Activity 1.1.4, gender-inclusive WMCs will be established at each target watershed to facilitate dialogue between stakeholders in upstream and downstream areas, and further strengthen project ownership at the community level. This will ensure that local communities take ownership of interventions and that all stakeholders' (including women) considerations are accounted for in decision-making that affects water resources they rely on in the long term, which is a paradigm shift in community participation in development projects. Once established, the project will work closely with the MoEnv (project Executing Entity) to formally recognize and include WMCs in relevant legislation — including defining their mandate related to working with communities. This will not only reinforce their contribution to the project and strengthen the participation of local stakeholders (communities in particular) but contribute to their sustainability and a paradigm shift in community participation in development projects.
262. The passive restoration of ecosystems (Activity 2.1.2) will also contribute to both social and environmental sustainability as it is a traditional form of community-based natural resource management that places rangeland management into the purview of local pastoralist communities. This will similarly be the case for women-led community-managed nurseries established under Activity 2.1.2 that will afford local participants a greater sense of ownership in restoration interventions, increasing long-term care and maintenance of restored areas and improving survival rates of seedlings. The condition of ecological infrastructure to be established under active restoration interventions will be maintained by protecting the ecosystems, minimising human pressures and promoting natural ecological processes that facilitate recovery of ecosystems from a degraded state. Physical infrastructure, including rockfill dams and floating solar installations, will be maintained by ensuring that regular inspections take place, and that required repair work is undertaken timeously.
263. The full socioeconomic and environmental benefits investments into ecosystem restoration as an adaptation approach are often only realised a decade or more after implementation. For this reason, the long-term data sets from the project will be archived within Jordanian research institutions and the long-term monitoring of the GCF project's investments will also be institutionalised. Similarly, detailed long-term O&M plans will be developed for all ecological and physical infrastructure constructed by the project and embedded in the budgets of relevant government institutions. Government funding for O&M will include: i) financing of ongoing restoration, conservation and management of those ecosystems restored by the project; and ii) maintaining the mechanisms and tools established by the project.

264. An extensive gender-responsive stakeholder engagement plan, detailed in Annex 7, has also been developed to ensure that the proposed participatory approach is continued during implementation, fostering stakeholder buy-in and the long-term ownership of project interventions. Continued stakeholder engagements — including recommended participants from, *inter alia*, government ministries, the private sector, NGOs and representatives of local communities (including women's groups); periods for engagements and a description of engagements required such as training, consultation and validation workshops — have been integrated into each relevant activity. This will ensure that all stakeholders are included in the implementation of relevant project activities which will contribute to the long-term sustainability of interventions. Gender-responsive community engagements during the project's inception phase will focus on introducing and sensitising target communities to the project's scope, objectives and expected results. The focus of these early engagements will be to: i) encourage buy-in; ii) ensure that there is a common understanding that the focus of the project is on climate change adaptation, avoiding any misconceptions that it is a poverty-alleviation or environmental protection initiative; and iii) manage community expectations. This will include making all relevant project documentation available — including translation into Arabic.
265. Engagement of the private sector will centre on the Water Fund, focusing on fostering private sector buy-in for PES (improved water resources) allowing for the continued funding of climate-resilient ILWRM. This will include raising awareness on investing in climate-resilient ILWRM, via the Water Fund, in the long-term to climate-proof private sector enterprises and how PES will facilitate this. In addition, the private sector will be engaged via the proposed project climate change adaptation outreach programme under Output 3.1, to highlight the impacts of climate change on water resources and their operations, as well as the adaptation approaches being promoted to overcome current and future challenges.
266. The impact and sustainability of adaptation interventions in Jordan will be further enhanced through the proposed project's focus on ensuring that gender-equal benefits are experienced by beneficiaries. Gender-responsive targets and objectives for all project activities have therefore been set in the project's Gender Action Plan (GAP; Annex 8) to ensure that project interventions are not gender-biased. This will contribute to the ownership of and involvement in project activities by both women and men which, in turn, will result in gender-responsive adaptation benefits being conferred beyond the project lifetime.

Financial sustainability

267. The project's recommendations on amendments to Jordan's policy framework under Activity 1.1.2 will promote the allocation of sufficient funds to climate change adaptation via government budgets and ensure domestic sources of finance are used for future adaptation initiatives. Where project activities have ongoing costs, either related to operation or maintenance these have been examined through the economic analysis (Annex 3) to ensure that the beneficiaries of the interventions are sufficiently concentrated and benefit enough on an ongoing basis to have the incentive to ensure these costs are covered. In most cases this is both obvious and clear. For example, the expected value of electricity generated by solar installations far outweighs the anticipated ongoing costs, and the funds to pay for this should be generated directly; likewise, the alternative natural resource-based livelihoods examined are anticipated to be profitable at a household, over and above ongoing cost and hence at a household level maintenance of equipment etc will be incentive compatible. Similarly given anticipated levels of water scarcity, the value of rainwater harvested from public building and household is likely to exceed the costs of maintenance by a factor of five, implying a direct incentive for households / public authorities to dedicate the time / money to ensure the potential to harvest rainwater continues.
268. Where there are more stakeholders involved both in delivering and receiving benefits, the key for financial stability will be having a mechanism in place to transfer resources from the beneficiaries to those bearing the costs of operation and maintenance. This is the case for the continued restoration and maintenance of ecosystems and rockfill dams for managed aquifer recharge in targeted watersheds. The value of additional ground water is expected to significantly exceed the ongoing costs of ecosystem restoration and dam maintenance, and the Water Fund is designed to channel resources from those who are willing and able to pay for increased ground water supplies to those whose actions within the landscape can deliver them. The operation of the Water Fund will be sustained through its operation and maturation phases that will be implemented under the project (Activity 2.1.4). The operation phase includes developing and implementing a work plan that guides: i) the systematic execution of activities; ii) M&E of activities; iii) communication of progress; and iv) improvement of the Water Fund through adaptive management. Results expected under this phase include an annual planning cycle that is aligned with the strategic plan, the ongoing implementation of interventions, development of a long-term strategy, routine review and communication of results as well as the application of corrective actions. Maturity of the fund will ensure its long-term viability. Safeguarding a sustainable and considerable impact on water security, the project will: i) secure long-term financing through fostering the support of the private sector for PES under the Water Fund; ii) integrate the Water Fund into the country's legal framework; iii) monitor and evaluate progress systematically and apply adaptive

management where required; and iv) secure support from the public to maximise the positive impacts of the Water Fund. At the heart of the long-term financing strategy will be the Water Fund's PES model, which will be used to foster medium to long-term investment in the Fund from major downstream water users. This will include securing agreements with major water users who are willing to pay for ongoing ecosystem management in return for the water resources generated. Payers for ongoing ecosystem management in upstream watersheds would include major water users in downstream areas of the Jordan Valley who would benefit from and rely on increased water resources generated via the Fund's interventions in upstream ecosystems. Under the maturation phase, long-term financing will be committed, routine reports released to document the Water Fund's ongoing impacts, the influence of the Water Fund demonstrated, and positive public opinion generated. As a result, financial sustainability beyond project finance is designed into the activities covered.

C. FINANCING INFORMATION

C.1. Total financing

(a) Requested GCF funding (i + ii + iii + iv + v + vi + vii)		Total amount		Currency		
		44,904,203		million USD (\$)		
GCF financial instrument		Amount	Tenor	Grace period	Pricing	
(i)	Senior loans	<u>Enter amount</u>	<u>Enter years</u>	<u>Enter years</u>	<u>Enter %</u>	
(ii)	Subordinated loans	<u>Enter amount</u>	<u>Enter years</u>	<u>Enter years</u>	<u>Enter %</u>	
(iii)	Equity	<u>Enter amount</u>	<u>Enter years</u>		<u>Enter % equity return</u>	
(iv)	Guarantees	<u>Enter amount</u>				
(v)	Reimbursable grants	<u>Enter amount</u>				
(vi)	Grants	44,904,203				
(vii)	Result-based payments	<u>Enter amount</u>				
(b) Co-financing information		Total amount		Currency		
		15,605,748		million USD (\$)		
Name of institution		Financial instrument	Amount	Currency	Tenor & grace	Pricing
Government of Jordan		<u>In kind & Grant</u>	1,000,000	<u>million USD (\$)</u>	<u>Enter years</u> <u>Enter years</u>	<u>Enter%</u>
Consortium of NGOs		<u>In kind & Grant</u>	14,605,748	<u>million USD (\$)</u>	<u>Enter years</u> <u>Enter years</u>	<u>Enter%</u>
<u>Click here to enter text.</u>		<u>Options</u>	<u>Enter amount</u>	<u>Options</u>	<u>Enter years</u> <u>Enter years</u>	<u>Enter%</u>
<u>Click here to enter text.</u>		<u>Options</u>	<u>Enter amount</u>	<u>Options</u>	<u>Enter years</u> <u>Enter years</u>	<u>Enter%</u>
(c) Total financing (c) = (a)+(b)		Amount		Currency		
		60,509,951		million USD (\$)		
(d) Other financing arrangements and contributions (max. 250 words, approximately 0.5 page)		Annex 30 includes a detailed list of Parallel Finance sources for the JILMI, the proposed GCF-financed project. These sources include ongoing projects and initiatives implemented or supported by the Consortium of NGOs who have contributed to the development of the project and who will function as Technical Partners during project implementation. The total amount of Parallel Finance for the proposed project is USD 13,133,000. The list of Parallel Finance sources presented in Annex 30 is made up of resources that will flow alongside the GCF resources requested for JILMI, but that are earmarked for other outcomes, which may be consistent with general mitigation and adaptation solutions, and ultimately contribute to the broader objectives and goals of the proposed project. These resources are not part of the GCF funding proposal for this project (i.e.,				

not included in the budget), and as such will not be tracked nor reported on as GCF impacts.

C.2. Financing by component

Component	Indicative cost Options	GCF financing		Co-financing		
		Amount Options	Financial Instrument	Amount Options	Financial Instrument	Name of Institutions
Component 1: Climate change adaptation governance in Jordan	<u>USD 2,338,394</u>	<u>USD 1,616,864</u>	<u>Grants</u>	<u>USD 621,530</u> <u>USD 100,000</u>	In kind & Grant	<u>Government of Jordan</u> <u>Consortium of NGOs</u>
Component 2: Integrated landscape and water resources management (ILWRM) under conditions of climate change	<u>USD 32,925,452</u>	<u>USD 22,395,865</u>	<u>Grants</u>	<u>USD 273,970</u> <u>USD 10,255,617</u>	In kind & Grant	<u>Government of Jordan</u> <u>Consortium of NGOs</u>
Component 3: Improved protection, supply and use of water resources under conditions of climate change	<u>USD 15,891,345</u>	<u>USD 12,758,414</u>	<u>Grants</u>	<u>USD 3,132,931</u>	In kind & Grant	<u>Consortium of NGOs</u>
Component 4: Knowledge management and awareness on climate change adaptation	<u>USD 3,879,500</u>	<u>USD 3,229,500</u>	<u>Grants</u>	<u>USD 650,000</u>	In kind & Grant	<u>Consortium of NGOs</u>
M&E	<u>USD 3,272,840</u>	<u>USD 3,272,840</u>	<u>Grants</u>	<u>NA</u>	<u>NA</u>	<u>NA</u>
PMC	<u>USD 2,202,420</u>	<u>USD 1,630,720</u>	<u>Grants</u>	<u>USD 571,700</u>	<u>In-kind & Grant</u>	<u>GoJ, Consortium of NGOs</u>
Indicative total cost (USD)	<u>USD 60,509,951</u>	<u>USD 44,904,203</u>		USD 15,605,748		

GCF and Co-financing split by activity is presented in the table below. Addition details on co-financing can be found in Annexes 4, 13 and 25.

Components, Outputs and Activities	GCF grant	Co-finance	Co-financier
Component 1: Climate change adaptation governance in Jordan	USD 1,616,864	USD 721,530	GoJ and Consortium of NGOs
Output 1.1 Strengthened governance and capacity for climate-resilient integrated landscape and water resources management (ILWRM) and Output 1.2: Economic assessments developed quantifying net benefits of ILWRM as an adaptation approach			
Activity 1.1.1. Strengthen the mandate and capacity of main government institutions — including MoEnv, MWI and its subsidiaries — for climate-resilient ILWRM in Jordan.	USD 51,078	USD 15,500	GoJ
Activity 1.1.2. Draft amendments to Jordan's policy, planning and legal frameworks for agriculture, land, water and natural resources, to mainstream climate-resilient ILWRM as a climate change adaptation approach.	USD 142,331	USD 62,000	GoJ
Activity 1.1.3. Generate evidence of the economic benefits of ILWRM to project beneficiaries for use by policymakers and planners.	USD 323,705	USD 107,630	GoJ and Consortium of NGOs
Activity 1.1.4. Strengthen the mandate and capacity of non-state actors for climate-resilient ILWRM in the Jordan Valley.	USD 222,952	USD 528,000	GoJ
Component 2: Integrated landscape and water resources management (ILWRM) under conditions of climate change	USD 22,416,865	USD 10,529,587	

Components, Outputs and Activities	GCF grant	Co-finance	Co-financier
<u>Output 2.1 Climate-resilient ILWRM practices and ecosystem restoration implemented in three watersheds in the Jordan Valley and Output 2.2: Water Fund established in the Jordan Valley to maintain landscape investment for enhanced water resources</u>			GoJ and Consortium of NGOs
Activity 2.1.1 Support the implementation of participatory climate-resilient integrated watershed management plans (developed under Activity 1.4) in the Yarmouk, King Talal Dam and Kafraïn areas of the Jordan Valley.	USD 837,000	USD 250,000	Consortium of NGOs
Activity 2.1.2. Restore ecosystems in the watersheds of the Yarmouk, King Talal Dam and Kafraïn areas of the Jordan Valley	USD 5,671,278	USD 9,115,617	Consortium of NGOs
Activity 2.1.3. Introduce gender-responsive climate-resilient additional livelihoods and agricultural practices in the project's three intervention areas.	USD 4,058,095	USD 890,000	Consortium of NGOs
Activity 2.1.4. Establish a Water Fund in the Jordan Valley to maintain landscape investment for enhanced water resources.	USD 8,541,000	USD 273,970	GoJ
Component 3: Improved protection, supply and use of water resources under conditions of climate change	USD 12,737,414	USD 3,132,931	Consortium of NGOs
<u>Output 3.1 Improved protection, supply and use of water for domestic purposes under climate change conditions</u>			
Activity 3.1.1 Implement measures to improve the quantity and quality of ground and surface water supplies within the project sites	USD 2,548,000	USD 2,300,000	Consortium of NGOs
Activity 3.1.2. Install 1 MW of floating solar panels over the King Abdullah Canal to reduce water loss from evaporation	USD 1,645,000	NA	NA
Activity 3.1.3. Introduce small-scale rainwater harvesting interventions in the project's three target areas to improve access to clean water under climate change conditions in a gender-responsive manner	USD 5,700,000	NA	NA
Activity 3.1.4. Empower local communities, including women, to monitor and manage local water supplies more effectively	USD 871,694	USD 832,931	Consortium of NGOs
Component 4: Knowledge management and awareness on climate change adaptation	USD 3,229,500	USD 650,000	Consortium of NGOs
<u>Output 4.1 Improved knowledge and awareness of climate change adaptation for landscape and water resources management</u>			
Activity 4.1.1 Implement a gender-responsive climate change adaptation outreach programme across the Jordan Valley	USD 1,968,970	USD 75,000	Consortium of NGOs
Activity 4.1.2. Improve knowledge management for climate change adaptation in Jordan to support national and sub-national decision-making, and build communities of practice	USD 434,000	USD 575,000	Consortium of NGOs

C.3. Capacity building and technology development/transfer (max. 250 words, approximately 0.5 pages)

C.3.1 Does GCF funding finance capacity building activities?	Yes ☒	No ☐
C.3.2. Does GCF funding finance technology development/transfer?	Yes ☒	No ☐

Technology transfer

269. Under Outputs 2.1 and 3.1 of the proposed project, technology transfer will be facilitated through the: i) climate-resilient restoration and management of ecosystems in target watersheds in the Jordan Valley as part of an EbA approach (Activity 2.1.2); ii) introduction of gender-responsive climate-resilient natural resource-based livelihoods (Activity 2.1.3); iv) introduction of small-scale water-harvesting interventions (Activity 3.1.3); and v) implementation of floating solar panels (Activity 3.1.2).

Capacity-building interventions

270. Technical and institutional capacity of governmental and non-governmental actors involved in landscape management, as well as local communities (at least 50% female participation) for climate-resilient Integrated Landscape and Water Resources Management (ILWRM) will be strengthened under all three of the project's outputs. In addition, Output 1.1 will include interventions with the objective of strengthening the policy and legal framework for adaptation — with a focus on ILWRM — in Jordan. These interventions will involve enhancing the mandate and capacity of key stakeholders for gender-responsive participatory climate-resilient ILWRM in the Jordan Valley (Activity 1.1.1 and 1.1.3). Under this activity, training of stakeholders (including at least 50% women) on climate-resilient watershed management (IWM) will be undertaken, and guidelines, standards and protocols for IWM developed. The output will also include drafting of amendments to Jordan's land and water-management policy, planning and legal frameworks for the promotion of climate-resilient ILWRM as an EbA solution (Activity 1.1.2). Under this activity training will be provided to policy- and decision-makers (at least 50% women) on integration of climate-resilient ILWRM practices into policies, plans and strategies. Training of relevant national and sub-national planners (at least 50% women) on climate-resilient ILWRM planning will also be undertaken. In addition, staff from relevant national and

sub-national institutions (at least 50% women) will be trained to use climate data, risk assessments, and the results of climate analyses, in policy-and decision-making processes as well as in planning and disaster risk management.

271. Capacity-building under Output 2.1 will include training of local communities and small-scale farmers (up to 75% women) on the climate-resilient additional livelihoods and improved farming methods. Under Output 3.1, local communities will be trained to use and maintain water harvesting technologies for domestic use (up to 50% female-headed households). In addition, community members (at least 50% women) will be trained in additional livelihoods related to the monitoring and management of water.

D. EXPECTED PREFORMANCE AGAINST INVESTMENT CRITERIA

This section refers to the performance of the project/programme against the investment criteria as set out in the GCF's Initial Investment Framework.

D.1. Impact potential (max. 500 words, approximately 1page)

272. The proposed project will contribute to the GCF's fund-level adaptation objective of increased climate-resilient sustainable development by facilitating a shift towards conserving and augmenting water resources in Jordan. This shift will be realised via climate-resilient ILWRM. Along with improving the sustainability of water resources, ILWRM will climate proof agricultural and natural resource-based livelihoods, improving the climate resilience of local communities. This will be accomplished through the GCF's fund-level impacts and outcomes presented below²⁵¹.

GCF Fund-level Impacts²⁵²

- ARA 1 — Most vulnerable peoples and communities;
- ARA 2 — Health, well-being, food and water-security; and
- ARA 4 — Ecosystems and ecosystem services.

273. The project will strengthen the mandates and capacities of relevant Jordanian institutions to implement climate-resilient ILWRM. This will be achieved through several elements of project design. First, proposed amendments to land-management policy, planning and legal frameworks will be drafted to promote climate-resilient ILWRM. The drafting of these amendments will be preceded by the training of planners, as well as policy- and decision-makers on using ILWRM for planning and decision-making processes in the water and land sectors. The training will lay the foundation for the preparation of policy briefs with recommendations for the integration of ILWRM into relevant policies and plans. Second, Watershed Management Committees (WMCs) will be established in target watersheds to foster participatory ILWRM and dialogue between stakeholders²⁵³. Third, training on ILWRM, will be delivered to relevant stakeholders to strengthen capacity for the participatory implementation of the approach as a climate change adaptation measure. Following this training, participatory ILWRM plans will be developed for the project's three target watersheds. Finally, ILWRM will be facilitated through the development of land-use guidelines, standards and protocols.

274. Moreover, under the proposed project, a Water Fund will be established to facilitate investment in ongoing climate-resilient management and restoration of upper watershed areas. This will not only promote the continued maintenance of restoration activities and improvement of water resources in these areas but also foster a participatory approach to watershed management. Water use and supply will be improved at the domestic and agricultural levels by: i) introducing small-scale water harvesting infrastructure at the community- and household-level; ii) empowering communities through education and training to monitor and manage water supply and demand; and iii) introducing in-field water conservation and harvesting measures on smallholder farms.

275. The project will strengthen the adaptive capacity of vulnerable communities (including groups such as women and youth) in the Jordan Valley through the: i) climate-resilient restoration of degraded watershed areas, as well as diversification and strengthening of livelihoods in a gender-responsive manner; and ii) improved protection and provision of water resources under climate change conditions. The ecosystem restoration interventions will result in an improved supply of ecosystem services — including the regulation of water resources and reduction of soil erosion. In addition, ecosystem restoration combined with managed aquifer recharge (MAR) will result in improved rates of aquifer recharge. These interventions will be complemented by the installation of 1 MW of floating solar panels over the King Abdullah Canal, reducing evaporative water loss (see Section 8.7.3 of Annex 2). Additionally, rainwater-

²⁵¹ Detailed information on the assumptions and estimates of the impact potential of the proposed project are provided in the project's results framework and economic analysis (Section E of this document and Annex 3, respectively).

²⁵² A detailed description of the project activities that will be implemented under each output to achieve the GCF project outcomes is presented in Section E.6 of the Funding Proposal.

²⁵³ Framework ToR describing the role and functions of WMCs are provided in Annex 28 and will be further developed by project Technical Partners during project inception.

harvesting interventions will enhance the provision of clean water resources to communities with a focus on female-headed households (see Section 8.6 of Annex 2). As a result, reduced water stress will buffer local communities both upstream and downstream of project sites to the exposure of climate risks such as droughts. The project will also increase adaptive capacity by climate-proofing agricultural livelihoods and identifying additional gender-responsive climate-resilient natural resource-based livelihoods. Moreover, additional gender-responsive livelihoods related to the monitoring and management of water in target communities will be adopted — including plumbing and installation of solar pumps.

276. Climate change adaptation awareness will be increased through the implementation of a gender-responsive outreach programme targeting local institutions and communities. In addition, access for decision-makers to relevant information related to effective adaptation will also be improved. The outreach programme will include: i) equipping community centres and local institutions to serve as climate vulnerability and information hubs; ii) identifying a public figure to champion the programme nationally; and iii) enhancing awareness of appropriate adaptation responses. At the institutional level, knowledge-management tools will be developed to facilitate the mainstreaming of gender-responsive adaptation best practices and lessons learned into decision-making processes. This will be complemented by the establishment of an adaptation-knowledge repository to increase access to best practices and lessons learned.

Adaptation impact

277. Project interventions will directly benefit a total of 246,942 people (~2.2% of the Jordanian population) — 135,736 men (55%) and 111,206 women (45%) — in the Jordan Valley by reducing their climate vulnerability. Of this total: i) 62,446 (30,607 women and 31,839 men) are from the Yarmouk site — Khaled Bin Al Waleed and Al Seru municipalities; ii) 105,841 (45,179 women and 60,662 men) are from the King Talal Dam site — New Ardah, Deir Alla and M'addi municipalities; and iii) 78,655 (35,420 women and 43,235 men) are from the Kafraïn site — Shouna Alwasta and Mahes municipalities. Direct beneficiaries include all people who will benefit directly from a project activity or the result of an activity. For example, under Output 3.2, 1,200 smallholder farming households (~6,000 people — an average of 5 people per smallholder household) across the three sites will benefit from on-farm interventions, while 2,800 households across the sites (~14,000 people) will benefit directly from rainwater harvesting infrastructure implemented on houses. Direct beneficiaries also include all vulnerable peoples in the project's three target areas, as they are all expected to benefit from the augmentation of water resources under Outputs 2.1, 2.2 and 3.1, either via improvements in water supplies for domestic and/or agricultural usage. Under Outcome 3, all 172,179 people living in the municipalities of the three target sites (and related municipalities) listed above will benefit from the improved protection, supply and use of water for domestic purposes under climate change conditions. Specifically, they will directly benefit from enhancements in the quality and quantity of ground and surface water through Activity 3.1.1. This will be achieved through the construction of rockfill dams to support managed aquifer recharge and the rehabilitation of active freshwater springs. As integrated land and water resources management interventions (ILWRM) are scaled up via the Water Fund (Activity 2.1.4), it is expected that an additional 4,000 ha of land (bringing the total to 12,000 ha) will be placed under improved management for water resources benefits, with at least one additional (adjacent) municipality per watershed site benefitting directly. Consequently, an additional 74,763 people (41,604 men and 33,159 women)²⁵⁴ are expected to benefit directly from the project. See Annex 33 for additional beneficiary calculation details.

278. 756,000 (48% women) people across the broader Jordan Valley will benefit indirectly from: i) the attenuation of water resources in the northern Jordan Valley; ii) strengthened local governance to implement climate-resilient ILWRM; iii) reduced water loss through evaporation from the King Abdullah Canal; and iv) improved awareness of the impacts of climate change and the adaptation options to address them. Indirect beneficiaries include people in the northern Jordan Valley who will not directly benefit from a project activity, or result of an activity. This includes all people located outside of the project's three target sites in the northern Jordan Valley.

279. Interventions at the regulatory and institutional level will indirectly benefit the Jordanian population as a whole, amounting to ~10 million people. Project interventions, such as the establishment of a Water Fund in the Jordan Valley, will also facilitate public and private sector investment in climate-resilient ILWRM. The Water Fund will serve as a mechanism to sustain the climate-resilient management and restoration of the Jordan Valley's watershed areas in the long term by maintaining and attenuating water resources under conditions of climate change. Public and private sector beneficiaries of improved water resources will contribute to their maintenance through financial contribution to the Water Fund. This will contribute to creating an enabling environment for future upscaling and replication of climate change adaptation interventions locally, nationally and regionally.

Table 9. Number of direct and indirect beneficiaries per output and activity (45% women).

Outputs and Activities	Direct	Indirect
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²⁵⁴ https://dosweb.dos.gov.jo/DataBank/Population/Population_Estimares/Municipalities.pdf

Output 1.1 Strengthened governance and capacity for climate-resilient integrated landscape and water resources management (ILWRM)	240 (50% women)	756,000 (48% women; 36% youth)
Activity 1.1.1. Strengthen the mandate and capacity of main government institutions — including MoEnv, MWI and its subsidiaries — for climate-resilient ILWRM in Jordan.	60	756,000
Activity 1.1.2. Draft amendments to Jordan's policy, planning and legal frameworks for agriculture, land, water and natural resources, to mainstream climate-resilient ILWRM as a climate change adaptation approach.	60	756,000
Activity 1.1.4. Strengthen the mandate and capacity of non-state actors for climate-resilient ILWRM in the Jordan Valley.	120	756,000
Output 2.1 Climate-resilient ILWRM practices and ecosystem restoration implemented in three watersheds in the Jordan Valley and Output 2.2: Water Fund established in the Jordan Valley to maintain landscape investment for enhanced water resources	246,942 (45% women, 36% youth)	756,000 (48% women, 36% youth)
Activity 2.1.1 Support the implementation of participatory climate-resilient integrated watershed management plans (developed under Activity 1.4) in the Yarmouk, King Talal Dam and Kafraïn areas of the Jordan Valley.	-	756,000
Activity 2.1.2. Restore ecosystems in the watersheds of the Yarmouk, King Talal Dam and Kafraïn areas of the Jordan Valley	172,179	756,000
Activity 2.1.3. Introduce gender-responsive climate-resilient additional livelihoods and agricultural practices in the project's three intervention areas.	9,750	-
Activity 2.1.4. Establish a Water Fund in the Jordan Valley to maintain landscape investment for enhanced water resources.	74,763	756,000
Output 3.1 Improved protection, supply and use of water for domestic purposes under climate change conditions	172,179 (45% women, 36% youth)	756,000 (48% women, 36% youth)
Activity 3.1.1 Implement measures to improve the quantity and quality of ground and surface water supplies within the project sites	172,179	-
Activity 3.1.2. Install 1 MW of floating solar panels over the King Abdullah Canal to reduce water loss from evaporation	6,000	756,000
Activity 3.1.3. Introduce small-scale rainwater harvesting interventions in the project's three target areas to improve access to clean water under climate change conditions in a gender-responsive manner	14,000	-
Activity 3.1.4. Empower local communities, including women, to monitor and manage local water supplies more effectively	450	-
Output 4.1 Improved knowledge and awareness of climate change adaptation for landscape and water resources management	-	756,000 (48% women, 36% youth)
Activity 4.1.1 Implement a gender-responsive climate change adaptation outreach programme across the Jordan Valley	-	756,000
Total	246,942 (45% women, 36% youth)²⁵⁵	756,000 (48% women, 36% youth)

Potential water resource benefits

280. Under the proposed project, water supplies in the Jordan Valley will be improved through the implementation of several interventions across the project area. These interventions will include: i) the restoration of watershed ecosystems; ii) managed aquifer recharge; iii) rooftop and in-field rainwater harvesting; and iv) the installation of floating solar panels. The benefits of each of these interventions to Jordan's water supply are summarised in Table 10 and the narrative below. Further details are provided in Section 8 of Annex 2: Feasibility Study.

Table 10. Potential impact of project interventions on water supply.

Intervention	Impact on water resources per year by project end (CM)	Potential impact on water resources per year after 25 years (CM)	Beneficiary
Ecosystem restoration over 12,000 ha	400,000	8,300,000	Domestic users and farmers
Managed aquifer recharge via rockfill dams	7,600,000	136,100,000	Domestic users and farmers
Rooftop rainwater harvesting	300,000	4,000,000	Domestic users

²⁵⁵ Direct and indirect beneficiaries under all outputs are from the same groups (246,942 and 756,000, respectively), hence the numbers across the outputs are not additive.

Evaporation reduction via floating solar panels	800,000	14,400,000	Farmers
Total	9,100,000	162,800,000	

Climate-resilient restoration in watersheds

281. Increasing vegetative cover across the project's intervention sites will reduce runoff and the subsequent loss of topsoil from watersheds caused by soil erosion. This will decrease sedimentation levels in downstream rivers or dams and lower the rate at which their storage capacity is reduced. For example, the Hima system has the potential to increase the storage capacity of the King Talal Dam by ~7,687,300 CM over a period of 25 years through a reduction in sediment loading of 5,490,931 tonnes²⁵⁶. Using similar interventions across the project's 12,000 ha (Activity 2.1.2) could similarly increase dam storage capacity by 2.2 MCM over 25 years. Upscaling this intervention over 500,000 ha is predicted to increase the storage capacity of Jordan's dams by 157.5 MCM over the same period. Improved vegetative cover will also result in greater surface roughness, reducing runoff and the speed of overland water flow, improving water infiltration, and consequently, soil moisture content and aquifer recharge rates. Total aquifer recharge is expected to increase by up to 35.3 CM/ha/yr²⁵⁷. This extrapolates to an additional average of 0.18 MCM of water infiltrating into aquifers annually²⁵⁸. Additional benefits of the increased infiltration will include the reduced rates of evaporation, reduced potential of flooding in downstream areas, and particularly, sustained basal flow of rivers, which will reduce the need of farmers downstream to extract water for irrigation from underground sources during dry periods.

Managed aquifer recharge (MAR)

282. The project's ILWRM approach will also include the construction of small-scale water harvesting infrastructure for MAR in wadi areas to collect and store water from runoff and flash floods (Activity 3.1.1.1). Interventions will include rockfill dams of up to 20 m wide with potential catchments of up to 0.9 MCM/yr. With potential infiltration rates of ~0.02 mm/s, up to ~0.63 MCM of this amount will contribute to annual aquifer recharge²⁵⁹. Four rockfill dams will be constructed across each of the project's three target areas (12 dams in total), resulting in a combined annual potential aquifer recharge of ~7.6 MCM/yr. Additional information on MAR is available in Section 11.2 of Annex 2.

Rainwater harvesting

283. Under Activity 2.1.3, in-field water conservation and harvesting measures will be implemented on smallholder farms. These interventions are not intended to improve aquifer recharge, but rather to increase soil moisture levels to support improved agricultural production and reduce pressure on underground water sources. Rooftop harvesting will be implemented across the project target area in the Jordan Valley (Activity 3.1.3). Potential for rainwater harvesting in the Balqa and Irbid governates is estimated to be 1.7 and 3.3 MCM/year, respectively^{260,261}. Under the proposed project, 3,000 rainwater harvesting tanks (with a capacity of 32 CM per tank) will be installed, resulting in a minimum annual yield of 0.28 MCM/year. More information relating to in-field and rooftop rainwater-harvesting interventions is presented in Section 8.6 of Annex 2.

Construction of floating solar panels (evaporation reduction)

284. Evaporation reduction interventions will include the establishment of at least a 1 MW floating solar panel installation over the King Abdullah Canal within the project intervention area (Activity 3.1.2). This will prevent the evaporation of 800,000 CM of water annually from the canal and provide ~1 MW of power to the MWI's pump stations on the canal²⁶².

285. Overall, the project's interventions are expected to improve water supply in the project implementation area in the Jordan Valley by up to 9.3 MCM/year by the end of project implementation (eight years) and potentially up to 83.8 MCM/year after 25 years. While this increase is relatively small in comparison to the current and projected deficits, there is considerable scope for upscaling these interventions across the country through the enabling environment for climate-resilient ILWRM that will be created by the project.

D.2. Paradigm shift potential (max. 500 words, approximately 1 page)

²⁵⁶ Myint MM & Westerberg V. 2014. An economic valuation of a large-scale rangeland restoration project through the Hima system in Jordan. Report from the ELD Initiative by International Union for Conservation of Nature, Nairobi.

²⁵⁷ ELD. 2015. Economics of Land Degradation Initiative: An economic valuation of a large-scale rangeland restoration project through the Hima system in Jordan.

²⁵⁸ The majority of water percolates to the shallow aquifer (~86%), with the rest ending up in the deep aquifer (~14%).

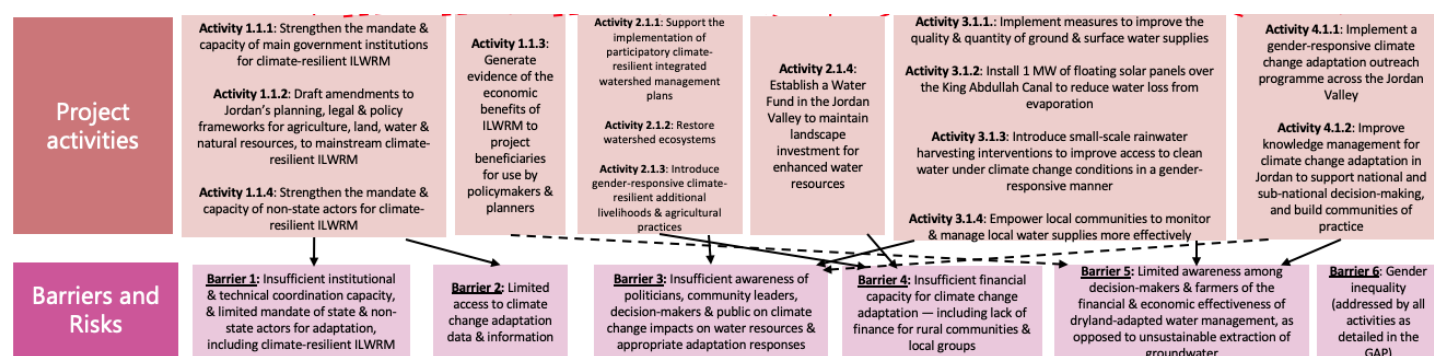
²⁵⁹ Steinel A. 2012. Guideline for assessment and implementation of managed aquifer recharge (MAR) in (semi-) arid regions: pre-feasibility study for infiltration of floodwater in the Amman-Zarqa and Azraq basins, Jordan. BGR.

²⁶⁰ Abu-Zreig M, Hazaymeh A & Shatanawi M. 2013. Evaluation of residential rainfall harvesting systems in Jordan. *Urban Water Journal*, 10: 105–111.

²⁶¹ Based on rainfall figures of 562 and 370 mm/year, and potential rooftop area for rainfall harvesting of 3.8 million and 11 million m², for Balqa and Irbid, respectively.

²⁶² Youssef YW & Khodzinskaya A. 2019. A review of evaporation reduction methods from water sources. E3S Web of Conferences, 97: 05044.

286. Transformative change to climate-resilient integrated land and water resources management (ILWRM) in Jordan is currently inhibited by several barriers (as detailed in Section B.2.: Theory of Change), including: i) Insufficient institutional and technical coordination capacity, as well as limited mandates of state and non-state actors for climate change adaptation, including climate-resilient ILWRM; ii) Limited access to climate change adaptation information and data; iii) Insufficient awareness by politicians, community leaders, decision-makers and the public of the impacts of climate change and appropriate adaptation responses; iv) Insufficient financial capacity for climate change adaptation; and v) Limited awareness amongst decision-makers and farmers of the financial and economic effectiveness of dryland-adapted water management, including the management of green water, in Jordan as opposed to unsustainable abstraction of groundwater. The project intended approach to removing these barriers is summarised in Figure 15 below — additional detail is provided in the project's Theory of Change in Section B.2.
287. Through the removal of the barriers presented above, the project will result in: i) strengthened governance and capacity for climate-resilient ILWRM; ii) climate-resilient ILWRM practices and ecosystem restoration implemented in three watersheds in the Jordan Valley; iii) improved protection, supply and use of water for domestic purposes under climate change conditions; iv) improved knowledge and awareness of climate change adaptation for land and water resources management; and v) enhanced availability of financial resources for long-term operations and maintenance of interventions, and for upscaling EbA across other parts of Jordan. The outcomes of these results will be increased public budgets for adaptation, 8,000 ha of land under climate-resilient ILWRM, improved protection, supply and use of water for domestic purposes under climate change conditions and household behavioural change through improved information outreach. This will catalyse an institutional paradigm shift within Jordan's water sector from a compartmentalised, sector-specific approach to land and water resources management to one that is integrated, participatory and adaptation focussed, contributing to climate-resilient sustainable development in the country. For example, networks of non-state actors, including NGOs, will be leveraged as constituencies of change through their participation in, inter alia, WMCs — to facilitate dialogue and participatory watershed management between upstream and downstream stakeholders (Activity 1.1.4) — and outreach programmes that will serve as awareness-raising campaigns (Activity 4.1.1). A key transformative aspect of the project will be the establishment of a Water Fund (Activity 2.1.4), which will leverage GCF resources to crowd-in private and public sector investment into EbA, thereby generating financial flows for the maintenance and upscaling of climate-resilient ILWRM interventions implemented by the project. The Water Fund will also serve as model that can be replicated in other areas of the country, transforming the water sector's approach to ILWRM. Importantly, the project will work closely with the recently approved GCF-financed "Building resilience to cope with climate change through improving water use efficiency in the agriculture sector (BRCCJ)" or FP155²⁶³ (see Section B.1., Table 1 and Annex 29). The project's goal is to increase the climate resilience of water management systems and the farming community in four governorates located in the Dead Sea Basin. The different, yet complementary, scope of BRCCJ allows the two projects to collectively contribute to the long-term climate-resilient transformation of Jordan's water sector to one that promotes ILWRM and improves agricultural water use efficiency, supported by innovative financing models such as the Water Fund.
288. In terms of the addressing of gender issues, the project will support transformational change both in the lives of female beneficiaries and in prompting gender mainstreaming in national climate change adaptation policy-making and implementation. The project seeks to promote gender equality by including actions and targets which will increase women's participation, representation and awareness, as well as ensuring that training programmes are gender-responsive and -inclusive. A primary objective of the project with regards to gender is increased resilience to climate change of at least 77,080 women in the project area. This will be achieved through direct improvement of women's capacities for improved water supply and demand management through climate-resilient ILWRM practices.



²⁶³ GCF. 2021. Building resilience to cope with climate change in Jordan through improving water use efficiency in the agriculture sector (BRCCJ). Available at: <https://www.greenclimate.fund/project/fp155>

Figure 15. Project approach to addressing barriers to transformational change in land and water resources management in Jordan as presented in the Theory of Change (Section B.2.).

Upscaling potential of proposed water resource augmentation interventions

289. Several scenarios, ranging from conservative to optimistic, have been presented in the table below for the upscaling potential of each of the identified interventions. Potential impact (in MCM) under each of the scenarios has been estimated after 8-, 20- and 25-year periods. Ecosystem restoration (12,000 ha under the with-project scenario) is the only intervention with an increase in cumulative impact over time, while the other interventions have fixed impact potentials, which do not change over time. Should the proposed project's ecosystem restoration interventions be upscaled across the entire northern Jordan Valley (1,147,600 ha), it is predicted that annual streamflow/water yield will increase by ~16.8% (~186.5 MCM/year) compared with values under the BAU scenario in the area. Similarly, annual percolation changes are projected to increase by ~10.6% (~49.4 MCM/year) under an upscaling scenario. This indicates the considerable contribution that ecosystem restoration could have of the water resources of the northern Jordan Valley in the long term — more information on the upscaling potential of the project's ecosystem restoration interventions is presented in Section 10.7 of Annex 2. Jordan's current water deficit is 224 MCM/yr. By 2040, it is expected to reach 687 MCM/yr. With regards to the extrapolations: the potential cumulative increase in water resources over 25 years by upscaling the project interventions is 34 MCM under a conservative scenario, and 344 MCM under an optimistic scenario (Table 11). The scaling up scenarios presented in Table 11, are indicative of what is possible should an enabling environment for climate-resilient ILWRM in Jordan be established and the required conditions created for the proposed interventions to be implemented at scale (see the description of Component 1 in Section B.3. for more detail on the project's approach to strengthening the enabling environment for climate-resilient ILWRM).

290. In terms of the managed aquifer recharge via rockfill dams, there may be potential limitations related to their scalability. These are related to the suitability of sites and potential social and environmental risks related to this type of intervention. Details on the selection of sites for rockfill dams, including criteria and potential risks to consider are presented in Annex 2: Section 8.12. Site feasibility assessments would therefore be required to confirm the scaling up potential as presented below.

Table 11. Potential impact of project interventions on water supply under project and upscaling scenarios.

Intervention	Scenario (project and scaling up: 1–4) ²⁶⁴	Coverage	Impact on water supply per year after 8 years (MCM)	Potential cumulative impact on water supply after 20 years (MCM)	Potential cumulative impact on water supply after 25 years (MCM)
Ecosystem restoration	Project	12,000 ha	0.4	6.1	8.3
	1 ²⁶⁵	50,000 ha	1.8	26	34
	2 ²⁶⁶	100,000 ha	3.5	51	69
	3 ²⁶⁷	250,000 ha	8.8	128	172
	4 ²⁶⁸	500,000 ha	18	256	344
Managed aquifer recharge (rockfill dams)	Project	12 dams	7.6	98	136
	1	20 dams	13	164	227
	2	50 dams	32	410	567
	3	100 dams	63	819	1,134
	4	200 dams	126	1,638	2,268
Rooftop rainwater harvesting	Project	3,000 tanks	0.3	2.6	4
	1	15,000 tanks	0.5	5	7
	2	90,000 tanks	3	32	44
	3	180,000 tanks	6	69	94
	4	400,000 tanks	13	163	216

²⁶⁴ Scenarios: 1 – conservative scaling up scenario; 2 – mildly conservative scaling up scenario; 3 – optimistic scenario; 4 – best case scenario. Calculations of groundwater recharge under different scaling scenarios were undertaken using figures in Table 7 in Annex 3 and assuming that upscaling would achieve the same recharge per unit area as the project area of 12,000 hectares.

²⁶⁵ 1 – Scaling up is done conservatively at a limited number of potential sites within 20-25 years (10% – ecosystem restoration and rockfill dams; 4% – rainwater harvesting; 25% – floating solar panels).

²⁶⁶ 2 – Scaling up is done with a mildly conservative approach at an increased number of potential sites within 20-25 years (20% – ecosystem restoration and rockfill dams; 22.5% – rainwater harvesting; 37.5% – floating solar panels).

²⁶⁷ 3 – Scaling up is done optimistically at up to half of potential sites within 20-25 years (50% – ecosystem restoration and rockfill dams; 45% – rainwater harvesting; 50% – floating solar panels).

²⁶⁸ Scaling up takes place in the majority of available sites (50-100% – ecosystem restoration, rockfill dams, rainwater harvesting and floating solar panels).

Intervention	Scenario (project and scaling up: 1–4) ²⁶⁴	Coverage	Impact on water supply per year after 8 years (MCM)	Potential cumulative impact on water supply after 20 years (MCM)	Potential cumulative impact on water supply after 25 years (MCM)
Evaporation reduction (floating solar panels)	Project	1 MW	0.8	10	14
	1	5 MW	1.6	21	29
	2	7.5 MW	3.2	42	58
	3	10 MW	6.4	83	115
	4	20 MW	13	166	230
Combined benefit of interventions under each scenario	Project		9	117	163
	1		16	215	297
	2		41	534	737
	3		84	1099	1515
	4		169	2223	3059

Potential for upscaling, replication and strengthened regulatory frameworks

291. The proposed project has been designed to maximise the potential for upscaling and replication of interventions to enhance the impact of the project across Jordan and the surrounding region. This potential will be reached through the establishment of standards and protocols for the adoption of climate-resilient integrated landscape and water resources management (ILWRM) interventions to be implemented under this project that can be readily scaled up and replicated at the national and sub-national level. Potential for upscaling, replication and strengthened regulatory frameworks through the proposed project will be realised primarily through its innovative approach to ILWRM. This approach focuses on the integration of landscape and water resources management into a single approach for climate change adaption in the Jordan Valley, ensuring complementarity between interventions that were historically implemented in isolation. Upscaling, replication and the strengthening of regulatory frameworks will be supported by *inter alia*: i) introducing a Water Fund as a long-term financing mechanism (an approach not yet introduced in Jordan); ii) developing land use guidelines, standards and protocols for climate-resilient integrated watershed management in the Jordan Valley addressing of current and future climate risks through improved use of the results of climate risk analyses; iii) training staff from relevant national and sub-national institutions to use climate data, and the results of climate analyses and risk assessments in policy- and decision-making processes, as well as planning and disaster risk management; iv) training policy- and decision-makers, and planners on the integration of climate-resilient ILWRM practices into policies, plans and strategies; v) reviewing Jordan's national land-management policies, plans and legislation to identify gaps and opportunities for the consideration of climate-resilient ILWRM as an adaptation approach; vi) implementing a climate change adaptation outreach programme across the Jordan Valley; vii) improving knowledge management for climate change adaptation in Jordan to support national and sub-national decision-making, and build communities of practice; and viii) generating evidence of the economic benefits of ILWRM to project beneficiaries for use by policymakers and planners.

292. The widescale adoption of an innovative ILWRM approach to be delivered under the proposed project will be supported by: i) amendments to Jordan's water and land management policy, planning and legal frameworks; ii) strengthening the mandate and capacity of major government agencies and non-state actors for ILWRM; iii) developing land-use guidelines, standards, protocols and participatory climate-resilient watershed management plans for the Jordan Valley that will maximise the social, environmental and economic benefits of ILWRM and can be replicated in other areas; iv) establishing an outreach programme championed by several public figures, to promote climate change adaptation in Jordan amongst all demographics; and v) improving the creation or strengthening of knowledge generation and collective learning in the areas of ILWRM and EbA across local communities in the Jordan Valley. Specifically, knowledge can be spread autonomously to surrounding areas and communities through the mobility of migrant workers and WMC members, thereby creating opportunities for scaling up and replication. Moreover, the project will develop a training programme for all relevant state, non-state, municipal, NGO and community stakeholders on climate-resilient ILWRM, thereby building technical and institutional capacity for the application of ILWRM principles. This will facilitate the wide-scale replication of climate-resilient ILWRM at the national and sub-national level. Long-term sustainability of the upscaling strategy will be ensured through the establishment of a Water Fund under the project, which will serve as a mechanism for leveraging financial resources for the restoration of watersheds and for sustaining the supply of ecosystem services from these areas. The Water Fund will allow for downstream stakeholders who are benefitting from the increased water supply to channel finances to upstream stakeholders, facilitating the ongoing maintenance and upscaling of EbA interventions such as ecosystem restoration and other interventions as detailed in Section E.6. Please see the description of Activity 2.1.4 under Section B.3 for a discussion of implications for downstream and

upstream beneficiaries of the Water Fund. During the Water Fund's maturation phase, options for scaling up and replication in other areas of Jordan will also be assessed through the development of a scaling up strategy. The scaling up strategy will link closely with a long-term financing strategy that will detail how financial resources for the Fund's ongoing operation post GCF involvement will be leveraged from public and private sources. The long-term financing strategy will include a detailed model describing how the Water Fund's PES scheme will continue to channel finance into the grant mechanism (which will include grant and results-based payments mechanism) once GCF finance is no longer available post project. The establishment and demonstration of a Water Fund through the proposed project can catalyze the establishment of similar funds, not only in Jordan, but also in the Middle East and North Africa (MENA) region as a whole, where very few Water Funds have been establishment. This will potentially change the approach to water resources management under climate change conditions across the MENA region.

Potential for knowledge sharing and learning

293. Jordan's National Capacity Self-Assessment (NCSA) report published in 2007, identified low capacity for knowledge management as a significant constraint to climate change adaptation. Moreover, several other capacity gaps exist in areas related to knowledge sharing and awareness, including *inter alia*: i) the absence of a national database and monitoring system of desertification; ii) the lack of a national system for knowledge management and data processing for monitoring and reporting on biodiversity; and iii) insufficient long-term awareness and education programmes for biodiversity management²⁶⁹. The proposed project will improve current knowledge-management systems for climate change adaptation in Jordan by developing several knowledge-management tools that will increase policymakers, planners, and local communities' (including women and youth) access to information on climate-resilient ILWRM approaches and their benefits. This will include the establishment of a national knowledge repository of lessons learned and gender-responsive best practices for climate change adaptation, focussing on the ILWRM approach proposed under this project. These knowledge-management tools will enhance the technical capacity of decision-makers to integrate ILWRM into national and sub-national plans and strategies. This technical capacity which will be further strengthened through the training of user groups in the use of these knowledge management tools. Moreover, the centralised nature of these tools will build communities of practice, thereby improving the generation and dissemination of knowledge under the project and other adaptation initiatives through the social nature of learning between groups of people with common concerns (such as women's groups), who can learn from each other as they interact regularly. The accessibility and reach of the knowledge systems implemented through this project will be enhanced through the implementation of a gender-responsive outreach programme within the Jordan Valley on climate-resilient ILWRM. The outreach programme will include the establishment of local knowledge hubs to improve awareness of the impacts of climate change, both generally and more specifically on water resources, and appropriate adaptation responses among vulnerable communities (including women and youth). To support the outreach programme, local staff and public figures will be trained to provide information through the community centres and to champion the case for climate change adaptation, respectively. Additionally, relevant educational practitioners (including 50% women) will be trained on integrating climate change adaptation into the syllabi and curricula of Jordanian universities. The training of local staff, public figures and educational practitioners to support the outreach programme assist in the scaling up and sustainability of knowledge sharing beyond the proposed project's scope.

Potential for the creation of an enabling environment

294. The creation of an enabling environment for IWRM in Jordan is essential for the sustainability of the abovementioned upscaling and replication strategies. The enabling environment will encompass strengthened governance and institutional capacity for climate change adaptation within which interventions implemented under the proposed project are sustained in the long term, with support from all levels of government. To achieve this, the project will establish WMCs within each of the three target watersheds which will facilitate dialogue and cooperation between upstream and downstream stakeholders and ensure the continuation of an integrated approach to land management at the project sites in the long-term. The effectiveness of the WMCs will be further enhanced by strengthening the mandate and capacity of major government agencies — including the Secretary Generals of the MoEnv, MoA and Ministry of Water and Irrigation (MWI) — and non-state actors to coordinate an integrated approach to watershed management. The continued funding of climate-resilient IWRM in Jordan will be facilitated by the mainstreaming of this adaptation approach into the national budget through the recommendations made during project implementation for amendments to relevant land-management legislation. In addition, the establishment of a Water Fund will generate sustainable financial resources to support the continued funding of project interventions at the local level.

D.3. Sustainable development (max. 500 words, approximately 1 page)

295. The proposed project will contribute towards achieving nine of the 17 Sustainable Development Goals (SDGs), namely: SDG 2 — Zero hunger; SDG 3 — Good health and well-being; SDG 5 — Gender equality; SDG 6 — Clean water and

²⁶⁹ Ministry of Environment. 2007. National Capacity Self Assessment for Global Environmental Management (NCSA) – Jordan. Available at: <https://www.thegef.org/sites/default/files/ncta-documents/544.pdf>

sanitation; SDG 8 — Decent work and economic growth; SDG 10 — Reducing inequality; SDG 12 — Responsible consumption and production; SDG 15 — Life on land; and SDG 16 — Peace, justice, and strong institutions. In addition, project interventions will produce numerous environmental, social and economic co-benefits through a gender-responsive and -responsive development approach, as described below.

Environmental, social and economic co-benefits

296. The climate-resilient restoration of watershed ecosystems along with the implementation of integrated landscape and water resources management (ILWRM) in project intervention areas will yield numerous environmental, social and economic benefits, which are presented below and summarised in Table 12.

- **Strengthened flood attenuation.** Jordan's current sporadic rainfall and intense storm events result in flash flooding and widespread soil erosion²⁷⁰. This causes multiple problems for local communities, such as deaths, damage to infrastructure, and destruction of croplands, both as a direct and indirect result of flooding. These disasters lead to the disruption of water purification systems and the contamination of sewage disposal systems, which in turn increases the likelihood of epidemics caused by vector-, water- and food-borne diseases²⁷¹. Frequent flooding of downstream areas is also caused by increased rates of runoff and overland flow during heavy rainfall events, which is partly attributable to degradation of ecosystems. This is expected to be intensified by the impacts of climate change. By improving vegetative cover in restoration areas across the project's intervention sites, there will be greater rainfall interception, reduced volume of runoff and speed of overland water flow compared with a BAU scenario. This will lead to a significant reduction in the flooding of downstream areas and the associated impacts.
- **Enhanced food security, livelihood resilience and biodiversity through the restoration of degraded habitats.** Despite its high land- and water-usage, the agricultural sector in Jordan supplies around only 19% of the country's food requirements, with the average Jordanian family spending ~40% of its income on food, much of which is imported. Additionally, with a baseline demand for fodder of ~790,000 tonnes/yr across the country, production of fodder within Jordan is unable to match demand. National biodiversity is being negatively impacted by the expansion of agricultural and pastoral land because of low yields of major crops and decreasing livestock and agricultural productivity, as well as, *inter alia*, unsustainable hunting, tourism and deforestation. The ILWRM approach will enhance the conservation of biodiversity and ecosystems, which include pastoral rangelands and areas surrounding agricultural land, thereby increasing the availability of fodder for livestock²⁷². For example, after two years of passive Hima restoration in the Zarqa River basin, the dry yield of cereals for fodder in a rainfed systems increased from an average of 40 kg/ha/yr to 113 kg/ha/yr. The overall avoided costs for supplementary fodder purchases using this system across 109,000 ha of the Zarqa River Basin is likely to amount to US\$23 million after 25 years.
- **Decreased pollution of water resources which will filter runoff, prevent erosion and slow sedimentation.** Not only will decreased sedimentation of dams through reduced runoff and erosion increase the storage capacity of dams, but it will also reduce the associated maintenance costs and increase the lifespan of these structures. Additionally, the promotion of climate-resilient agricultural practices, such as permaculture, will reduce the need for fertilisers that negatively contribute to pollution levels. By improving infiltration through active and passive restoration, freshwater and riparian habitats will be less affected by pollutants such as heavy metals or fertilisers which cause eutrophication. Moreover, soil quality will improve, and health risks posed by nitrates, which may lead to numerous gastrointestinal illnesses, diarrhoea, cancer and methaemoglobinemia^{273,274}, will be reduced. Increasing the water quality of existing water sources will be complemented by improving access to better health and sanitation through the enhanced supply of clean water as a result of the introduction of small-scale rainwater-harvesting interventions. Moreover, by providing alternative clean water sources in the form of these rainwater harvesting interventions, conflict over resources between host and refugee communities is expected to decrease.
- **Energy generation and reduced evaporation of water from floating solar panels.** Without protective coverings, water storage facilities are affected by incoming solar radiation and wind action, resulting in evaporation losses. Floating covers will be used to substantially negate these processes and trap water vapour below the cover, increasing humidity and further preventing evaporation. This method of solar energy generation has not previously been used in Jordan, but has considerable potential for contributing to the country's aim of increasing its production of renewable energy. This will contribute to reducing MWI's reliance on fossil-fuel generated energy to supply it with pumping stations, contributing to a transformative change in energy supply across the water sector in Jordan to one that is low emission. The floating solar system to be installed over the King Abdullah Canal can produce 1 MW of power and

²⁷⁰ Weather Online. 1999. Jordan. Available at: <https://www.weatheronline.co.uk/reports/climate/Jordan.htm>.

²⁷¹ Ibid.

²⁷² NWRM. No date. Afforestation of reservoir catchments. Available at: http://nwrn.eu/sites/default/files/nwrn_ressources/f3_-_afforestation_of_reservoir_catchments_0.pdf

²⁷³ Methaemoglobinemia or 'blue baby syndrome' mainly affects infants and affects the blood's ability to carry oxygen to various part of the body.

²⁷⁴ Jamaludin N, Sham SM & Ismail SMS. 2013. Health risk assessment of nitrate exposure in well water of residents in intensive agricultural area. *American Journal of Applied Sciences* 10: 442–448.

save up to 0.8 MCM of water annually through reduced evaporation²⁷⁵. Moreover, this will create employment and income-generating opportunities through the introduction of alternative livelihoods such as solar panel installation and maintenance.

Table 12. Cross-cutting benefit streams.

Benefit stream	Co-benefits		
	Environmental	Social	Economic
Strengthened flood attenuation	Reduced sedimentation and exposure of soils.	- Reduced direct deaths. - Reduced disease epidemics. - Improved drinking water quality.	- Reduced damage of property and crops. - Increased lifespan and capacity of dams.
Enhanced biodiversity	Enhanced habitats for local wildlife species.	Continuation of cultural and aesthetic value associated with biodiversity of ecosystems.	Benefits associated with prevention of ecosystem degradation such as carbon sequestration and water regulation.
Decreased pollution of water resources	- Slowing of sedimentation. - Improved soil quality. - Reduced eutrophication.	- Reduced health risks posed by nitrates. - Improved drinking water quality.	- Increased lifespan and capacity of dams. - Positive effects on household economy through reduced healthcare costs.
Improved access to clean water	NA	- Improved health and sanitation. - Reduced conflict over water resources.	Positive effects on household economy through reduced healthcare costs.
Strengthened food security	NA	Improved health.	Reduced reliance on food imports.
Increased resilience of pastoral livelihoods to climate change	Rehabilitation of rangeland ecosystems, resulting in increased extent, and enhanced biodiversity and soil quality.	Reduced conflict over water resources for livestock. Currently, there is increased potential for conflict resulting from Jordan's domestic agricultural policy to shift water-use away from crops, to support traditional livestock owners²⁷⁶.	- Improved quality of livestock products as a result of improved condition of animals related to enhanced fodder and water supplies and quality. - Positive effects on household economy through reduced purchasing of supplementary fodder supplies.
Energy generation from floating solar panels	Reduced environmental impact from the generation of energy from non-renewable sources.	Generation of additional livelihoods including solar panel installation and maintenance.	Production of ~1 MW of power annually.

Gender-responsive development impacts

297. The proposed project will support a transformation in the water sector to one in which women are included in the decision-making for and management of water resources at the community level, including being involved in monitoring of water resource use and maintenance of community-level water infrastructure. This will allow for their specific water-related climate-change adaptation needs to be addressed. The project's community-level gender-responsive approach also includes promoting gender-inclusive climate-resilient livelihoods that will benefit female-headed households (can be implemented at home and allow for the strengthening of adaptive capacity). Tailored awareness-raising and outreach programmes will also allow women to become leaders in the promotion of climate change adaptation in their communities.

- Given the prevailing gender and social norms in Jordan, as well as challenges related to gendered vulnerability to climate impacts (see Section 1.2 in Annex 2), the proposed project's objective of climate-resilient ILWRM will result in gender-responsive sustainable development. The gender-responsive planning, resource-allocation and

²⁷⁵ Youssef YW & Khodzinskaya A. 2019. A review of evaporation reduction methods from water sources. *E3S Web of Conferences*, 97: 05044.

²⁷⁶ Vasquez O, Khraishy M (eds). 2016. *Market overview and guide to Jordanian market requirements*. USDA Foreign Agricultural Service. Amman.

implementation of the proposed project is a high priority for the Jordanian government. The provision of gender-equal benefits to men and women, as outlined in the gender action plan (GAP), will contribute to equal ownership of project activities and will result in gender-responsive adaptation benefits being conferred beyond the project lifetime. The GAP additionally outlines gender-disaggregated impacts, outcomes and output statements, gender-responsive activities, and gender-disaggregated performance indicators and targets. As a result, the proposed project will contribute to the achievement of several of the sustainable development goals — most notably SDG 5 (Gender equality).

- The project's GAP outlines actions to reduce the potential to perpetuate or worsen gender inequality in Jordan, by *inter alia*, ensuring : i) women's participation in training equal at least 50% of the total number of participants; ii) revisions to Jordan's water and land-management policy, planning and legal frameworks address women's needs and interests; iii) at least 50% of the total population of women and youth in the targeted areas will participate in relevant project interventions and planning processes; iv) women and vulnerable groups participate in the consultation process for climate-resilient restoration of watersheds; v) 100% of female household beneficiaries are trained on the operation and maintenance of small-scale water-harvesting interventions; and vi) 20% of women farmers are represented by the women's groups as water-wise ambassadors.
- Disseminating climate change knowledge products to all user groups through a centralised platform will help to overcome gendered accessibility to information and resources, thereby ensuring gender-responsive climate change adaptation in Jordan.
- Inclusion of gender considerations in all the policies reviewed as well as acknowledgement of gender differentials within government institutions in response to climate risks will ensure the continued and long-term impact of gender-affirming actions and increased climate resilience in Jordan.

D.4. Needs of the recipient (max. 500 words, approximately 1 page)

Vulnerability of the country

298. Climate change will exacerbate the existing water stress in Jordan through decreases in annual rainfall, higher annual temperatures and more frequent, extreme and prolonged droughts. Because of the country's existing vulnerability to critical water stress, even a minor decrease in water supply resulting from these climate change impacts is expected to have severe implications for the country's water security. This water stress will particularly impact agricultural productivity, resulting in the need for increased food imports²⁷⁷ to alleviate food insecurity. This impact is placing an increasing financial burden on the Jordanian government and the country's inhabitants. The financial impacts are compounded by the increasing cost of extracting usable water from declining water resources. For example, the over-extraction of groundwater and reduced recharge rates of groundwater basins have resulted in lowering water tables in many areas, causing farmers to invest more money and resources into drilling deeper wells. Because of the high cost of drilling deeper wells to reach water that is decreasing in quality, many farmers abandon wells, and consequently also the livelihoods that depend on water from them.

Vulnerability of beneficiary groups

299. The project target area is located within the Jordan Valley — the primary agricultural region of the country. The majority of local communities in the region depend on agriculture for their livelihoods. As agricultural activities are vulnerable to temperature fluctuations, increased exposure to droughts and greater precipitation variability, crop productivity in the Jordan Valley is directly affected by climate change. Under future climate conditions, decreased agricultural yields will reduce the income potential for local communities who already face high poverty rates²⁷⁸ (see Section B.1). Moreover, women in Jordan are particularly vulnerable to the economic impacts of climate change because of high levels of gender inequality in the country, which disadvantages and underrepresents women in the labour force and decision-making roles²⁷⁹.

300. Future climate change will additionally present multiple health risks to Jordan's most vulnerable inhabitants, including children, women and refugees. Climate change impacts on health include an increase in the incidence of heat stress-related illnesses and exposure to diseases from reduced water quality²⁸⁰. Also, the health of the population is adversely

²⁷⁷ Currently, Jordan imports three times (~US\$1.4 billion) more agricultural products than it exports (~US\$563 million), a deficit that is compounded during drought years when production decreases. This deficit and its associated impact on the agricultural sector's contribution to the economy is expected to be exacerbated under future climate scenarios (as described in Section 3 of the Feasibility Study), as water supplies will be further constrained.

²⁷⁸ Poverty rate is the percentage of people whose income falls below the poverty line, which is taken as half the median household income of the total population. A higher poverty rate indicates a population with a higher proportion of individuals below the poverty line.

²⁷⁹ OECD Development Centre. 2019. Social Institutions & Gender Index Country Report – Jordan. Available at: <https://www.genderindex.org/wp-content/uploads/files/datasheets/2019/JO.pdf>.

²⁸⁰ USAID. 2016. Climate change risk profile: Jordan.

impacted by reduced food security related to lower agricultural yields²⁸¹. As yields decline, the total quantity of food available in the country decreases, causing food prices to fluctuate. Consequently, higher prices lead to a reduction in the diversity of food that households can afford, causing malnutrition²⁸². The capacity of vulnerable inhabitants to address the impacts of climate change described above is constrained by limited awareness of the public, politicians and decision-makers of the impact of climate change and potential adaptation options (Barrier 3, Section B.2). Additionally, there is limited awareness amongst decision-makers and farmers of the economic and financial benefits of dryland-adapted water management (Barrier 5, Section B.2).

301. Project interventions will directly benefit a total of 246,942 people (~2.2% of the Jordanian population) — 135,736 men (55%) and 111,206 women (45%) — in the Jordan Valley by reducing their climate vulnerability. Of this total: i) 62,446 (30,607 women and 31,839 men) are from the Yarmouk site — Khaled Bin Al Waleed and Al Seru municipalities; ii) 105,841 (45,179 women and 60,662 men) are from the King Talal Dam site — New Ardah, Deir Alla and M'addi municipalities; and iii) 78,655 (35,420 women and 43,235 men) are from the Kafraïn site — Shouna Alwasta and Mahes municipalities. Direct beneficiaries include all people who will benefit directly from a project activity or the result of an activity. For example, under Output 3.1, 1,200 smallholder farming households (~6,000 people — an average of 5 people per smallholder household) across the three sites will benefit from on-farm interventions, while 2,800 households across the sites (~14,000 people) will benefit directly from rainwater harvesting infrastructure implemented on houses. Direct beneficiaries also include all vulnerable peoples in the project's three target areas, as they are all expected to benefit from the augmentation of water resources under Outputs 2.1, 2.2 and 3.1, either via improvements in water supplies for domestic and/or agricultural usage. Under Output 3.1, all 172,179 people living in the municipalities of the three target sites (and related municipalities) listed above will benefit from the improved protection, supply and use of water for domestic purposes under climate change conditions. As integrated land and water resources management interventions (ILWRM) are scaled up via the Water Fund (Activity 2.1.4), it is expected that an additional 4,000 ha of land will be placed under improved management for water resources benefits, with at least one additional (adjacent) municipality per watershed site benefitting directly. The 4,000 ha is related to total water attenuation potential for the three watersheds, in addition to the 8,000 ha to be restored directly by the project (see Table 27 and figures 116-118 in Annex 2). The rainfall attenuation potential represents the slowing of overland water flows which reduces the amount of water running off the land and increases the amount of time available for water infiltration, and hence groundwater recharge. The Water Fund will scale the set of restoration and land management interventions in Component 2 to target the entire 12,000 ha with potential for restoration to support water attenuation. Consequently, an additional 74,763 people (41,604 men and 33,159 women)²⁸³ are expected to benefit directly from the project.
302. 756,000 (48% women) people across the broader Jordan Valley will benefit indirectly from: i) the attenuation of water resources in the northern Jordan Valley; ii) strengthened local governance to implement climate-resilient ILWRM; iii) reduced water loss through evaporation from the King Abdullah Canal; and iv) improved awareness of the impacts of climate change and the adaptation options to address them. Indirect beneficiaries include people in the northern Jordan Valley who will not directly benefit from a project activity, or result of an activity. This includes all people located outside of the project's three target sites in the northern Jordan Valley.

Socioeconomic status

303. Jordan has a substantial public debt totalling US\$40.2 billion in 2019 (96.7% of GDP²⁸⁴), which is attributed to: i) successive governments adopting an expansionist fiscal policy²⁸⁵; ii) the provision of considerable subsidies for staple goods such as bread in an attempt to improve the food security of vulnerable groups in the country²⁸⁶; iii) a reduction in foreign aid, which has resulted in the country borrowing externally; iv) increased welfare spending for citizens and refugees; v) subsidisation of the National Electric Power Company (NEPCO) and the Water Authority of Jordan (WAJ);

²⁸¹ Approximately 25% of those living below the poverty line in Jordan rely on agriculture for subsistence and as a source of income. Although food is currently still readily available in Jordan, the slowing economy and declining agricultural productivity are increasingly threatening food security for many households. The average Jordanian family already spends ~40% of its income on food.

²⁸² Bawadi HA, Tayyem RF, Dwairy AN & Al-Akour N. 2012. Prevalence of food insecurity among women in northern Jordan. *Journal of Health, Population, and Nutrition*, 30: 49.

²⁸³ https://dosweb.dos.gov.jo/DataBank/Population/Population_Estimares/Municipalities.pdf

²⁸⁴ CEIC. 2019. Jordan Government Debt: % of GDP. Available at: <https://www.ceicdata.com/en/indicator/jordan/government-debt--of-nominal-gdp>

²⁸⁵ Encouraging economic development by decreasing taxes and/or increasing government spending.

²⁸⁶ In 2018, bread subsidies were lifted, resulting in considerable price rises.

Reuters, 2008. Jordan ends bread subsidy, doubling some prices, to help state finances Available at: <https://www.reuters.com/article/us-jordan-economy-subsidies-bread/jordan-ends-bread-subsidy-doubling-some-prices-to-help-state-finances-idUSKBN1FF2CP>

and vi) low tax collections²⁸⁷. Moreover, limited opportunities exist for the public and private sector to access financial resources for addressing climate change impacts, resulting in a constrained capacity of Jordan's national budget to prioritise adaptation. This is compounded by a high trade deficit resulting from a reliance on energy and food imports²⁸⁸ as well as conflicts in neighbouring Syria and Iraq that disrupt exports from the country²⁸⁹. External investment is, therefore, critical to increase the resilience of Jordan's population to climate change given the insufficient financial capacity of the country for climate change adaptation (Barrier 4, Section B.2).

Institutional capacity needs

304. Jordan's Third National Communication to the United Nations Framework Convention on Climate Change (UNFCCC)²⁹⁰, as well as its National Capacity Self-Assessment (NCSA)²⁹¹, identified limited technical, institutional and financial capacity as major constraints inhibiting the implementation of adaptation interventions in the country (Barrier 1, Section B.2). In particular, the Jordanian government has limited capacity to mainstream climate change adaptation into policy- and decision-making, as well as into relevant financial mechanisms accessible by the public and private sector. This is largely as a result of, *inter alia*: i) insufficient access to knowledge on climate change adaptation — including on how to effectively implement, monitor and maintain climate change adaptation measures — and the resulting benefits (Barrier 2, Section B.2); ii) the absence of formalised processes to enforce such mainstreaming; and iii) limited expertise in relevant and effective adaptation technologies.

D.5. Country ownership (max. 500 words, approximately 1 page)

305. The proposed project is well-aligned with Jordan's national priorities, having been designed in collaboration with, and validated by, project stakeholders at all levels across *inter alia* the water, agriculture and environmental sectors. These priorities, and their complementarity with the proposed project, are outlined below.

- The focus of the **Aligned National Action Plan to Combat Desertification in Jordan 2015–2020** is to decrease the degradation of landscapes, thereby enhancing the resilience of the population by strengthening and diversifying their livelihoods. The plan is aimed at the facilitation of sustainable economic development in Jordan whilst mainstreaming climate change adaptation into landscape management. The proposed project will be aligned with the Plan's overall focus on combating desertification, and in particular with its objective of promoting the mainstreaming of climate-resilient landscape management.
- The proposed GCF project is well-aligned with the 2020 Strategic Goals of Jordan's **National Biodiversity Strategy and Action Plan 2015–2020 (NBSAP)**. Specifically, this alignment pertains to: i) enhancing the national governance of biodiversity by facilitating the participation of stakeholders at all levels as well as through mainstreaming and integration of biodiversity conservation into planning and policy frameworks; ii) reducing the impacts of pressures on biodiversity — including habitat destruction, political conflicts, and tourism; iii) maintaining protected areas, priority species and genetic resources; iv) enhancing the national understanding of dryland ecosystem benefits to national resilience, economic sustainability and local livelihoods; and v) developing biodiversity knowledge as the main tool for enabling behavioural change at the cultural level, to inform policy and decision-making support.
- The **National Climate Change Policy 2013–2020** advances strategic objectives, measures and instruments to facilitate Jordan's adaptation to climate change in priority areas — including water, coastal zones, agriculture and food security, health, tourism, biodiversity and socioeconomic status and poverty. The proposed project aims to address relevant goals of the policy, including: i) achieving a climate-resilient Jordan that is well prepared for the future impacts of climate change; ii) maintaining healthy, sustainable and resilient communities; iii) maintaining sustainable water and agricultural resources; and iv) conserving thriving and productive ecosystems. The objective of the policy is to build the adaptive capacity of communities and institutions in Jordan — with a specific focus on gender equality and addressing the needs of vulnerable groups — through increasing the resilience of ecosystems, as well as water and agricultural resources, to climate change and optimising climate change mitigation opportunities.
- The use of integrated approaches to natural resource management is a fundamental principle of the **Climate Change Policy for a Resilient Water Sector (2016)**. The Policy describes the Integrated Water Resources Management (IWRM) approach upon which **Jordan's National Water Strategy (2023)** is based (alignments follow). IWRM and Environmental protection: i) Goal 1 - Sustainably manage groundwater resources to restore safe yield levels and protect Jordan's aquifers; ii) Goal 2 - Sustainably manage and protect surface water resources and infrastructure; iii) Goal 4 - Increase water use efficiency in households, tourism, industry, and other key business sectors. Energy efficiency and renewable energy in the water sector: i) Goal 2 - Expand the use of renewable energy across water sector operations

²⁸⁷ Asharq Al-Awsat. 2019. Jordan's National Debt Exceeds \$40 Bn. Available at: <https://aawsat.com/english/home/article/1642336/jordan%E2%80%99s-national-debt-exceeds-40-bn>.

²⁸⁸ Idris I (ed). 2016. Economic Situation in Jordan. University of Birmingham. Birmingham.

²⁸⁹ Appendino M, Cerisola M, Martin E, Salman F, Tiffin A, Morra P, Sobrinho N, Wagner H (eds). 2017. Jordan. Selected issues. International Monetary Fund. Washington, DC.

²⁹⁰ Jordan's Third National Communication to the UNFCCC. 2014.

²⁹¹ UNDP. 2007. Jordan's National Capacity Self-Assessment (NCSA).

to lower costs and improve financial performance of the sector. Sector governance and institutional development: i) Goal 1 - Restructure water institutions to ensure that they have the legal mandates, staff capacity, resources, and incentives to modernize the sector in order to enhance accountability and increase public trust; ii) Goal 2 - Strengthen organizational capacity and management practices within water institutions to be responsive and resilient to dynamic sector management needs; iii) Goal 3 - Strengthen sector-level strategic planning, management, monitoring, evaluation, and reporting; iv) Goal 4 - Promote gender, inclusion, and youth empowerment throughout the water sector. Innovation, technology and private sector engagement: i) Goal 3 - Private sector participation increased to improve operational efficiency and sustainability, introduce innovation and technology, access higher flexibility in execution, improve risk management, and access alternative funding. Climate change resilience: i) Goal 1 - Jordan's water sector is resilient to the changing quantity and quality of water resources and to the level of threat to sector infrastructure and operations from climate change; ii) Goal 2 - Jordan's water strategy fully reflects the need to adapt to and mitigate against the impacts of climate change.

- Gender-responsive management of watersheds, agriculture and rangelands is acknowledged as central to Jordan's climate change adaptation strategy in the country's **updated Nationally Determined Contribution (NDC) (2021)**. The project aligns with and directly responds to the updated NDC's adaptation priority areas of: i) water resource management, by enhancing the enhancing water quantity, quality and use efficiency of water across the targeted regions; ii) agriculture and food security, by facilitating the introduction of climate-resilient agricultural practices; and iii) gender mainstreaming, by supporting gender-responsive livelihood diversification and climate change adaptation.
- The project is aligned with the priority sectors for climate change adaptation identified in Jordan's **National Adaptation Plan (NAP)** which builds upon the adaptation of the priorities of the NDC, but with gender added as a core component.
- The project is aligned with the **National Strategy for Women in Jordan (2020–2025)**, including supporting its goals: i) women and girls are able to exercise their economic, political and human rights and freely lead and participate in society free of gender-based discrimination; ii) women and girls enjoy a life free of all forms of gender-based violence; iii) positive gender norms, roles and attitudes support gender equality and women empowerment; and iv) institutions are executing and sustaining policies, structures and services that support girls and women empowerment in alignment with Jordan's national and international commitments.

306. The MoEnv (Component 1) which will be a co-EE with IUCN (Components 2 to 4), is well-suited to the NDA role for Jordan, as it leads the country's environmental governance system — including other relevant governmental organisations — in initiating and driving projects, policies, legislation, strategies, and monitoring. Under MoEnv, the Climate Change Directorate (CCD) coordinates high-level policy formulation and implementation, facilitating the participation of relevant stakeholders at all levels to ensure uptake and buy-in amongst policy- and decision-makers.

Use of domestic accountability mechanisms

307. In 2015, following the recommendations of the National Climate Change Policy, the Climate Change Directorate (CCD) was established to serve as the institutional focal point for climate change activities in Jordan, including the country's commitments as a party to the UNFCCC. Additionally, the CCD is the coordinating focal point to global and regional environmental governance systems²⁹². The CCD coordinates high-level policy formulation and implementation, facilitating the participation of relevant stakeholders at all levels. It also oversees development of the internal capacity of MoEnv — as well as the government in general — in addressing the challenges and barriers of climate change²⁹³. Via its Adaptation to Climate Change Section, CCD, which will be a member of the PSC, will enforce accountability related to the project's policy and institutional capacity-building activities (see Component 1's description in Section B.3.).

308. Outside of the project's governance structure accountability during implementation will be supported by the National Committee on Climate Change (NCCC), which is convened and chaired by MoEnv, and includes the Ministries of Agriculture, Health, Water and Irrigation, Energy and Mineral Resources, as well as other ministries and participants from public institutions, civil society, and academia. The NCCC was established in 2001 and is the highest coordination and supervising body for climate policy in Jordan²⁹⁴. The NCCC's responsibilities, which will contribute to maintaining accountability for the project, include: i) supervising and supporting the realisation of UNFCCC commitments in Jordan in accordance with national interests; ii) supervising and assuring the development and execution of legal, regulatory

²⁹² Ministry of Environment, GIZ. 2015. *Climate Change Governance in Jordan: Towards Policy and Institutional Coordination*. Available:

http://moenv.gov.jo/En/Environmental_Sectors/Atmosphere/Documents/Climate%20Change%20Governance%20in%20jordan.pdf

²⁹³ Ibid.

²⁹⁴ Combaz, E. 2019. Jordan's environmental policies and engagement on climate change. K4D Helpdesk Report. Available: Hashemite Kingdom of Jordan. 2018. Jordan Integrated Landscape Management Initiative (JILMI). Project Preparation Funding Application.

https://assets.publishing.service.gov.uk/media/5d30a131ed915d2ff003b781/619_Jordan_Environment_Policies_and_Engagement.pdf

and institutional arrangements and frameworks; iii) acting as a national advisory body, providing overall institutional and technical guidance; iv) leading climate change adaptation and mitigation efforts, including the integration of adaptation and gender into national development strategies and plans; and v) providing feedback on, and overseeing, climate change programmes and projects, in terms of institutional and technical aspects²⁹⁵.

309. An additional domestic mechanism that will support accountability during project implementation is the NAP's Monitoring, Review and Learning (MRL) system²⁹⁶. The MRL system will be used to assess and report on progress made on addressing the impacts of climate change in alignment with the NAP's goals and objectives (such as through the proposed project), to support learning, transparency and accountability. This includes assessments of progress in the implementation of: i) adaptation policies, plans and/or investments; ii) activities enabling the implementation of adaptation measures, or amplifying their impacts; and iii) adaptation measures, including their actual contribution to climate resilience. As the CCD will oversee the implementation of the NAP MRL, its linking with and use for accountability within the proposed project's M&E process will be ensured.

Stakeholder engagement

310. The proposed project was designed following an extensive consultative process with MoEnv — which is responsible for the overall management of Jordan's GCF portfolio — and the consortium of NGOs²⁹⁷, as well as all other relevant project stakeholders²⁹⁸. All stakeholder consultations were held in Arabic (where necessary, translation was provided) and gender sensitivities were considered. Stakeholders were mapped and consulted at the national level (24 stakeholders) and local level across the three target areas (South Shounah – 38 stakeholders, Upper King Talal Dam – 21 stakeholders, and Yarmouk – 33 stakeholders). Details of the stakeholder engagement process are summarised below, with further detail presented in Annex 7: Summary of Consultations and Stakeholder Engagement Plan.

311. Stakeholder consultations took place during four in-country missions. In October 2018, an initial scoping mission was undertaken to discuss and agree on an approach to refining the proposed project's rationale and design. This mission included a workshop and meetings with the consortium of NGOs and MoEnv, as well as an expedition across the Jordan Rift Valley to view potential project intervention sites and to gain an understanding of baseline conditions and challenges faced by local communities. Following the scoping mission, a second in-country mission was undertaken in April 2019 and included a two-day technical meeting. Attendees included representatives from the United Nations Environment Program (UNEP), NGOs that constitute the consortium and additional project consultants. The primary objectives of the meeting were to: i) discuss and agree on the refined scope and rationale of the project; ii) discuss and develop a common understanding on what is required to finalise the design of the project; iii) discuss the approach to the identification and selection of project intervention sites; and iv) discuss and agree on a plan for the remainder of the project preparation facility (PPF) phase. In addition, individual meetings were held with the members of the consortium of NGOs to identify how their expertise will best contribute to project design and implementation.

312. A third in-country mission was undertaken in July 2019 to: i) hold an official project inception workshop with all relevant stakeholders; ii) discuss project design; iii) confirm the selection of project intervention areas; iv) discuss environmental and social safeguards (ESS) and gender requirements of the project design, including the development of required assessments and plans; v) discuss project financing — including the approach to the Financial and Economic Appraisal (FEA) and potential sources of co-financing; and vi) visit the Jordan Valley to view potential project intervention sites, collect additional baseline information (including on barriers to adaptation), consult with local stakeholders (state and non-state actors and local communities) and identify potential adaptation interventions. The inception mission was followed by a series of consultations and site visits by a local consultant team²⁹⁹ to gather additional information required to inform project design. An environmental and social impact assessment (ESIA), environmental and social management plan (ESMP), gender analysis and action plan, and an adaptation needs assessment were developed to assist design of the proposed project. In addition, the location and details of project intervention sites in the Jordan Valley were confirmed. Stakeholders consulted during this period included local communities, NGOs, sectoral experts³⁰⁰ and state and non-state institutions, as well as national-level stakeholders. Additionally, stakeholder needs have been accounted for by ensuring ministerial reports and reviews have been considered. For example, the MoA's 2011 review of agriculture

²⁹⁵ Hashemite Kingdom of Jordan. 2018. Jordan Integrated Landscape Management Initiative (JILMI). Project Preparation Funding Application.

²⁹⁶ Source: http://www.moenv.gov.jo/ebv4.0/root_storage/ar/eb_list_page/final_draft_nap-2021.pdf

²⁹⁷ Including the following NGOs: i) Royal Botanical Garden (RBG); ii) Future Pioneers Empowering Communities (FPEC); iii) Horizons for Green Development; iv) International Union for Conservation of Nature and Natural Resources (IUCN), Regional Office for West Asia (ROWA); v) Jordan Hashemite Fund for Human Development (JOHUD); vi) Princess Alia Foundation (PAF); vii) Jordanian Royal Society for the Conservation of Nature; and viii) Royal Marine Conservation Society (JREDS).

²⁹⁸ including representatives from *inter alia* the Ministry of Agriculture (MoA), Ministry of Water and Irrigation (MWI), the Jordan Valley Authority (JVA), Water Authority of Jordan (WAJ), Ministry of Planning and International Cooperation (MOPIC), National Agricultural Research Centre (NARC), as well as participating NGOs and local communities.

²⁹⁹ Including ESS, gender and adaptation specialists.

³⁰⁰ *Inter alia* ecologists, botanists, hydrologists, social scientists.

in Jordan identified priority areas, which the project will contribute to by: i) addressing the risk and impact of droughts — by reducing drought impact through improved water resource management and supply under climate change conditions (Outputs 1.1, 1.2, 2.1, 2.2, 3.1 and 4.1); ii) increasing domestic production of grain crops, fodder and seed production — via introducing climate-resilient agricultural practices (Output 2.1); iii) revising legislation and regulations — via supporting the mainstreaming of climate-resilient ILWRM into policy and legislation (Output 1.1); and iv) developing and protecting ecosystem wealth — through restoration interventions (climate-resilient) and improved management (Outputs 1.1, 2.1 and 2.2).

313. An additional round of community consultations was held between October and November 2020 to explore and document local community perceptions, opinions, concerns and priorities on the scope of the proposed project, as part of the project's validation process. The group of respondents interviewed by the research team was a mixed social group with stronger representation and focus on farmers from both upstream and downstream areas of the Yarmouk and King Talal Dam target areas, and only downstream areas from the Kafraïn target area. The research sample was also almost equally distributed between women and men and a slight tilt towards men as they represent the majority of local community members involved in farming. A significant number of young people (i.e., youth groups with ages 18–35) represented almost half of the people interviewed with particular focus on unemployed men and women, and early retirees from both genders. Table 13 below summarizes the names of the settlements visited by the research team, their associated municipalities, the number of interviews conducted in each settlement, the number of respondents approaches in total and classified by gender. Recommendations and concerns that arose from these consultations have been used to update relevant aspects of project design across the Funding Proposal package. A report detailing the community consultations is presented in Annex 7: Stakeholder Consultations and Engagement Plan.

Table 13. Number of interviews and respondents within the three target areas.

No.	Target Area	No of Interviews and %	Males and %	Females and %	Total Respondents
1	Kafraïn Area	15 (27%)	26 (52%)	24 (48%)	50
2	King Talal Dam Area	23 (41%)	34 (45%)	41 (55%)	75
3	Yarmouk Area	18 (32%)	27 (50%)	27 (50%)	54
Totals		56 (31%)	87 (49%)	92 (51%)	179

314. An activity-level stakeholder engagement plan for the implementation period of the proposed project is presented in Annex 7. A project-level Grievance Redress Mechanism (GRM), which will be made accessible to all stakeholders, has been prepared and is presented in Annex 6. Below, the proposed roles of stakeholders in the implementation of the project's activities are summarised.

Table 14. Proposed roles of stakeholders across project activities.

Stakeholder group	Development role	Proposed role during project
Component 1		
<i>Activity 1.1.1. Strengthen the mandate and capacity of main government institutions for climate-resilient ILWRM in Jordan</i>		
Main government institutions involved in water resources and landscape management (MoEnv, MWI, MLI, MLA, DLS, AGM, ASEZA, MoA, WAJ and JVA)	Environmental, land and water resources management	- Contribute to the development of land use guidelines, standards and protocols for climate-resilient ILWRM - Receive training on use of climate data, and the results of climate analyses and risk assessment in policy- and decision-making processes, and planning and disaster risk management
Governate (Irbid, Balqa) and district staff (At-Taibah, Al-Wastiyah, Qasabah Irbid, Al-Kurah, Dir Alla, Ash-Shunah al-Janubiyah)		Receive training on use of climate data, and the results of climate analyses and risk assessment in policy- and decision-making processes, and planning and disaster risk management
Gender mainstreaming actors JOHUD, FPEC, Gender Unit (Ministry of Planning and International Cooperation), Gender Unit (Ministry of Social Development)	Governmental and NGO gender mainstreaming	- Gender mainstreaming into policy amendment processes - Gender mainstreaming into and gender representation within WMCs
<i>Activity 1.1.2. Draft amendments to Jordan's policy, planning and legal frameworks for agriculture, land, water and natural resources, to mainstream climate-resilient ILWRM as a climate change adaptation approach</i>		
Policy- and decision-makers from institutions such as MoEnv, JVA, MoA, WAJ and MWI	Environmental, land and water resources management	- Receive training on integration of climate-resilient ILWRM into policies, plans and strategies - Support reviews of Jordan's national land-management policies, plans and legislation to identify gaps and opportunities for the consideration of climate-resilient ILWRM - Validation of policy briefs with recommendations for the integration of climate-resilient ILWRM into national policies, plans and strategies

National and sub-national development and land use planners		- Receive training on climate-resilient ILWRM planning - Support reviews of Jordan's national land-management policies, plans and legislation to identify gaps and opportunities for the consideration of climate-resilient ILWRM
Technical partners (IUCN, RSCN, JREDS, PAF, HGD, JOHUD, RBG and FPEC)	NGO	Support reviews of Jordan's national land-management policies, plans and legislation to identify gaps and opportunities for the consideration of climate-resilient ILWRM
Gender mainstreaming actors JOHUD, FPEC, Gender Unit (Ministry of Planning and International Cooperation), Gender Unit (Ministry of Social Development)	Governmental and NGO gender mainstreaming	- Gender mainstreaming into policy amendment processes - Gender mainstreaming into and gender representation within WMCs
Activity 1.1.3. Generate evidence of the economic benefits of ILWRM to project beneficiaries for use by policymakers and planners		
WMCs	Local-level land and water resource management	Support collection of data
Jordanian research institutions	Academia	Support data collection, collation and storage
Gender mainstreaming actors JOHUD, FPEC, Gender Unit (Ministry of Planning and International Cooperation), Gender Unit (Ministry of Social Development)	Governmental and NGO gender mainstreaming	Ensure identification of gender-related benefits
Activity 1.1.4. Strengthen the mandate and capacity of non-state actors for climate-resilient ILWRM in the Jordan Valley		
Local government, local communities, Technical Partners (NGOs), Private Sector, Farmers, CSOs	Local-level land and water resource management	Members of WMCs
WMCs	Local-level land and water resource management	- Receive training on climate-resilient ILWRM - Support the development of climate-resilient integrated watershed management plans
Sub-national governmental institutions	Environmental, land and water resources management	- Receive training on use of climate data, and the results of climate analyses and risk assessment in policy- and decision-making processes, and planning and disaster risk management - Support the development of climate-resilient integrated watershed management plans
Technical Partners	NGOs	Receive NGO-specific training on ILWRM
Gender mainstreaming actors JOHUD, FPEC, Gender Unit (Ministry of Planning and International Cooperation), Gender Unit (Ministry of Social Development)	Governmental and NGO gender mainstreaming	Ensure capacity building is gender responsive and inclusive.
Component 2		
Activity 2.1.1 Support the implementation of participatory climate-resilient integrated watershed management plans in the Yarmouk, King Talal Dam and Kafraïn areas of the Jordan Valley		
Technical partners	NGO	- Implementation of integrated watershed management plans - Support the design and establishment of mechanisms for the implementation of climate-resilient additional livelihoods and agricultural practices
WMCs	Local-level land and water resource management	- Support implementation of integrated watershed management plans - Support the establishment and implementation of participatory mechanisms for monitoring and evaluating change in community climate resilience and behavioural change
Local communities		Support the implementation of climate-resilient watershed management measures
NARC	Agricultural research	Support the design and establishment of mechanisms for the implementation of climate-resilient additional livelihoods and agricultural practices
Local Development Committees	Community-level development	Support the establishment and implementation of participatory mechanisms for monitoring and evaluating change in community climate resilience and behavioural change
Gender mainstreaming actors JOHUD, FPEC, Gender Unit (Ministry of Planning and International Cooperation), Gender Unit (Ministry of Social Development)	Governmental and NGO gender mainstreaming	Ensure that gender considerations are captured in the planning processes and that engagements are gender inclusive
Activity 2.1.2. Restore ecosystems in the watersheds of the Yarmouk, King Talal Dam and Kafraïn areas of the Jordan Valley		
Technical partners	NGO	Oversee the development of protocols for climate-resilient restoration, including baseline assessments

		Oversee implementation of restoration activities
MoEnv	Environmental management	Support the development of protocols for climate-resilient restoration, including baseline assessments
Jordanian research institutions	Academia	Oversee the development of protocols for climate-resilient restoration, including baseline assessments
Local communities	Local resource users	Oversee the development of protocols for climate-resilient restoration, including the contribution of traditional knowledge
RBG	NGO	Oversee central and community-managed nurseries that will supply seedlings for restoration interventions
WMCs	Local-level land and water resource management	Support implementation of restoration activities
CSOs and CBOs	Community-level development	
Local communities	Local resource users	Hired to assist with restoration activities
Gender mainstreaming actors JOHUD, FPEC, Gender Unit (Ministry of Planning and International Cooperation), Gender Unit (Ministry of Social Development)	Governmental and NGO gender mainstreaming	Ensure restoration activities are gender inclusive
Activity 2.1.3. Introduce gender-responsive climate-resilient additional livelihoods and agricultural practices in the project's three intervention areas		
Technical partners	NGO	- Provide guidance on the uptake of climate-resilient alternative livelihoods and agricultural practices - Support the selection of beneficiary households - Promote and train beneficiary households on beekeeping, vegetable gardening and climate-resilient agricultural practices - Identify private sector champions to mentor beneficiaries on climate-resilient additional livelihoods and agricultural practices
NARC	Agricultural research	- Provide technical guidance on the uptake of climate-resilient alternative livelihoods and agricultural practices - Support the training of small-scale farmers on climate-resilient agricultural practices
WMCs	Local-level land and water resource management	- Support the selection of beneficiary households - Promote and train beneficiary households on beekeeping and vegetable gardening and climate-resilient agricultural practices - Support the identification of private sector champions to mentor beneficiaries on climate-resilient additional livelihoods and agricultural practices
MWI	Water and irrigation management	Support the training of small-scale farmers on climate-resilient agricultural practices
MoA	Agricultural management	
Gender mainstreaming actors JOHUD, FPEC, Gender Unit (Ministry of Planning and International Cooperation), Gender Unit (Ministry of Social Development)	Governmental and NGO gender mainstreaming	Ensure that livelihoods are gender inclusive and responsive
Activity 2.1.4. Establish a Water Fund in the Jordan Valley to maintain landscape investment for enhanced water resources		
JVA	Local development in the Jordan Valley	- Housing of Water Fund - Support establishment and implementation of Water Fund
Technical Partners	NGO	- Support establishment and implementation of Water Fund - Stakeholder consultations - Leverage financial support - Support design and establishment of on-granting facility and the Water Fund's grant mechanism; - Assist WMCs with the identification of sub-projects under the on-granting facility and the Water Fund's grant mechanism; - and preparation of RFPs - Support WMCs and their members to develop proposals in response to RFPs under the on-granting facility and the Water Fund's grant mechanism; - Support evaluation of proposals submitted in response to RFPs under the on-granting facility and the Water Fund's grant mechanism; - Provide oversight for implementation of sub-projects under the on-granting facility
WMCs	Local-level land and water resource management	- Water Fund Governance - Support establishment and implementation of Water Fund - Stakeholder consultations - Leverage financial support - Beneficiaries of Water Fund's on-granting facility
TNC	NPO	Support design, establishment and implementation of Water Fund
UNEP	GCF AE	Support design and establishment of on-granting facility and the Water Fund's grant mechanism;

		- Assist WMCs with the identification of sub-projects under the on-granting facility and preparation of RFPs and the Water Fund's grant mechanism; - Oversee evaluation of proposals submitted in response to RFPs under the on-granting facility and the Water Fund's grant mechanism; - Support the development of long-term financing and scaling up strategies for the Water Fund
Gender mainstreaming actors JOHUD, FPEC, Gender Unit (Ministry of Planning and International Cooperation), Gender Unit (Ministry of Social Development)	Governmental and NGO gender mainstreaming	Ensure that women have access to the on-granting facility and the Water Fund's grant mechanism and are supported to access finance
Component 3		
<i>Activity 3.1.1. Implement measures to improve the quantity and quality of ground and surface water supplies within the project sites</i>		
Local communities	Local resource users	Consulted during the ESIA process to be undertaken for the selection of rockfill dam sites
WMCs	Local-level land and water resource management	
Technical partners	NGO	- Implement and maintain small-scale measures to prevent sedimentation of rockfill dams - Support the rehabilitation of active freshwater springs
<i>Activity 3.1.2. Install 1 MW of floating solar panels over the King Abdullah Canal to reduce water loss from evaporation</i>		
MWI and JVA	Water and irrigation management	Operate and maintain floating solar installation
Ministry of Energy and Mineral Resources	Energy and mining management	Provide usage authorisations for the energy generated from floating solar panels
<i>Activity 3.1.3. Introduce small-scale rainwater harvesting interventions in the project's three target areas to improve access to clean water under climate change conditions in a gender-responsive manner</i>		
Technical partners	NGO	Support the selection of households and buildings for the installation of small-scale water harvesting interventions
Local women	Local resource users	Trained to support the operation and maintenance of small-scale water harvesting interventions
Gender mainstreaming actors JOHUD, FPEC, Gender Unit (Ministry of Planning and International Cooperation), Gender Unit (Ministry of Social Development)	Governmental and NGO gender mainstreaming	Ensure that women are the primary beneficiaries.
<i>Activity 3.1.4. Empower local communities, including women, to monitor and manage local water supplies more effectively</i>		
Local communities	Local resource users	- Support the development of community-based water use monitoring and management plans - Trained on the implementation of community-based water use monitoring and management plans - Empowered to become water-wise ambassadors (female community members) - Receive further advanced training on plumbing, rainwater harvesting installation, and the installation and maintenance of solar pumps (selected members)
Technical partners	NGO	
WMCs	Local-level land and water resource management	Support the development of community-based water use monitoring and management plans
LDCs	Local development	Support the development of community-based water use monitoring and management plans
Gender mainstreaming actors JOHUD, FPEC, Gender Unit (Ministry of Planning and International Cooperation), Gender Unit (Ministry of Social Development)	Governmental and NGO gender mainstreaming	Ensure that activities are gender inclusive and responsive
Component 4		
<i>Activity 4.1.1 Implement a gender-responsive climate change adaptation outreach programme across the Jordan Valley</i>		
Technical partners	NGO	- Support the implementation of the outreach programme - Identify and train public figures to champion the case of climate change adaptation in Jordan
WMCs	Local-level land and water resource management	
MoEnv's specialist Outreach and Environmental Awareness Unit	Outreach and environmental awareness	

Community centres and institutions	Religious centres and education	Equipped to be climate change adaptation learning centres and information hubs in local communities
Ministry of Education and tertiary institutions	Education and academia	Trained on the integration of climate change adaptation into syllabi and curricula of Jordanian universities
Gender mainstreaming actors JOHUD, FPEC, Gender Unit (Ministry of Planning and International Cooperation), Gender Unit (Ministry of Social Development)	Governmental and NGO gender mainstreaming	Ensure that the outreach programme is gender inclusive and responsive
<i>Activity 4.1.2. Improve knowledge management for climate change adaptation in Jordan to support national and sub-national decision-making and build communities of practice</i>		
Technical partners	NGO	- House and maintain knowledge management tools developed by the project - Promote access to the use of the climate change adaptation knowledge repository by user groups
Planners, policy-makers and practitioners	Policy-making, planning and implementation	Use of the climate change adaptation knowledge repository
Jordanian academic institutions and research agencies	Academia and research	Establishment of partnerships to implement the project's long-term monitoring programme and generate research findings and publications

D.6. Efficiency and effectiveness (max. 500 words, approximately 1 page)

315. The Government of Jordan is requesting US\$44,904,203 in grant finance from the GCF to implement climate-resilient integrated landscape and water resources management (ILWRM) as part of a national response to climate change. This will enable Jordan to balance its water demand with water supply in the long term despite predicted climate change impacts. The GCF investment will be supported by the Government of Jordan (GoJ) and Consortium of NGOs, which have committed approximately US\$15,605,748 in co-financing and will be used to overcome the barriers to increasing the resilience of Jordan's land and water resources management to climate change threats. While the co-finance will be used to cover all baseline costs — such as business-as-usual repairs or maintenance to water supply infrastructure necessary to ensure the sustainability of climate-proofing interventions, GCF funding will solely be used to address the additional impacts of climate change. In addition to co-finance, parallel finance of US\$13,133,000 has been identified for the project. The parallel finance consists of resources included under aligned initiatives and projects supported by the Consortium of NGOs (see Annex 30: Parallel Finance). With the support of GCF grant funding, the proposed project will deliver several adaptation benefits in Jordan that will contribute to achieving a paradigm shift in the country's approach to land and water resources management towards the achievement of climate-resilient sustainable development³⁰¹. The GCF and co-financing partners combined amount of US\$60,515,256 in financing, supported by parallel finance of US\$13,133,000, will be used efficiently and effectively to, *inter alia*: i) create an enabling environment for the use of ILWRM to increase the resilience of communities in Jordan to the impacts of climate change by improving the management and augmentation of water resources; ii) strengthen the capacity of state and non-state actors — including local communities — to plan and implement climate-resilient ILWRM; and iii) implement scalable and replicable on-the-ground adaption interventions including ecosystem restoration (the benefits of ecosystem restoration to Jordanian society far outweigh the costs^{302,303}), managed aquifer recharge, evaporation reduction through the installation of floating solar panels, as well as rainwater harvesting and water-conservation measures, to ultimately improve Jordan's water supply. The primary benefits of the ecosystem restoration interventions will be water related. These include: i) increased aquifer recharge, thereby increasing water supplies under climate change conditions; and ii) reduced sedimentation of surface water sources, resulting in an improvement in water volumes and quality. This is because decreased sedimentation of dams will positively impact both their lifespan and storage capacity. Similarly, water quality and quantity and the functioning of waterways will be enhanced because of increased filtration of runoff into soils, which prevents pollution of waterways. For example, restoration interventions implemented over a period of 25 years to reduce sedimentation of the King Talal Dam will reduce average annual sediment loading in the dam from a baseline of 3,605,775 tonnes/ha to 3,386,137 tonnes/ha, which will result in an increase in the dam's storage capacity from ~282,000 MCM to ~307,500 MCM. The impacts of sedimentation on storage capacity of dams and reservoirs is discussed in more detail in Section 1.5.2 of the Feasibility Study.

316. Project interventions will also target local communities (including smallholder farmers) who will benefit from the installation of rooftop rainwater harvesting systems and the introduction of climate-resilient livelihoods, as well as pastoralists who will see an increase in fodder productivity. These interventions have been designed to build upon and

³⁰¹ A detailed description of project activities is presented in Sections B.3 and E.6 of the Funding Proposal.

³⁰² Becker N & Katz DL. 2009. An economic assessment of Dead Sea preservation and restoration. In: The Jordan River and Dead Sea Basin. Springer, Dordrecht.

³⁰³ Myint MM & Westenberg V. 2015. An economic valuation of a large-scale rangeland restoration project through the Hima system in Jordan. *Report for the ELD Initiative by International Union for Conservation of Nature, Nairobi, Kenya.*

scale up existing practices and technologies that have been shown to be effective in Jordan. For example, in-field rainwater harvesting structures have been used to improve soil structure and decrease soil erosion rates by increasing soil organic matter and intercepting runoff water at relatively short-distance intervals^{304,305}. These structures have also successfully supported the re-establishment of drought-tolerant forage shrubs in degraded rangelands³⁰⁶ and an economic analysis, undertaken for the Badia Benchmark Project, demonstrated that planting of shrubs with harvested rainwater produced an economic internal rate of return (EIRR) estimated at ~17%, with the addition of the contribution of environmental benefits³⁰⁷. Rainwater harvesting and water conservation interventions will be introduced on local smallholder farms in the project target areas to increase the resilience of local farmers to increasing drought conditions³⁰⁸. Beneficiaries will subsequently be trained to use and maintain the rainwater harvesting and conservation measures to maximise their benefits and sustainability. The cost-effectiveness of the proposed project interventions as well as a summary of international adaptation best practices and lessons learned used to inform the project's development are presented below.

Cost-effectiveness of project interventions

317. Jordan experiences interconnected climate phenomena that increase the scarcity of water resources in an already water-stressed country. The expected occurrence of more frequent and prolonged droughts — a function of reductions in Jordan's average annual rainfall and increases in the average annual temperatures projected under climate change scenarios — will worsen water stress. Decision-making on the type of climate change adaptation options that the Jordanian government should employ to respond to increased water scarcity, should be informed by an understanding of the cost of adaptation measures, as well as the associated potential avoided costs of project interventions. In order to inform decision-making on adaptation interventions, a full economic analysis across all four project outputs has been undertaken using a cost benefits analysis (CBA) approach. This economic analysis, assuming a 25-year lifetime for the project and a discount rate of 5%, is presented in Annex 3: Economic Feasibility Assessment (EFA). The results for the main economic summary statistics for the proposed project are summarised below.

Net Present Value (NPV)	105,279,874 USD
Benefit / Cost Ratio (BCR)	3.00
Economic Rate of Return (ERR)	25%
Payback period (years)	9
Total project cost (USD, undiscounted costs incurred in years 1-8)	60,509,951

318. The project is comparable to the cost of similar interventions within Jordan. For example, the GCF-funded project entitled 'Building Resilience to Cope with Climate Change in Jordan Through Improving Water Use Efficiency in the Agriculture Sector (BRCCJ) (2021–2029) will cost US\$33.3 million³⁰⁹ and the GEF-funded project entitled 'Mainstreaming Sustainable Land and Water Management Practices' (2009–2015) cost ~US\$30 million^{310,311}. The overall BCR is 3, meaning the economic benefits of implementing the project are expected to exceed the GCF investment costs by a factor 3. On this basis, there is a clear economic case for implementing the JILMI project and for GCF grant funding to finance it. The cumulative economic benefits of implementing the JILMI project are expected to exceed the project costs shortly after project closure (i.e., the payback period of 9 years is equal to the project

³⁰⁴ Critchley W, Siegert K, Chapman C, Fink M. 1991. *Water harvesting: A manual for the design and construction of water harvesting schemes for plant production*. Rome: Food and Agriculture Organization of the United Nations (FAO).

³⁰⁵ Sprong C. 2019. Soil water dynamics affected by micro rainwater harvesting structures in Jordanian Badia restoration context. MSc thesis, Universiteit Utrecht, Utrecht.

³⁰⁶ Karrou M, Oweis T, Ziadat F, Awawdeh F. 2011. Rehabilitation and integrated management of dry rangelands environments with water harvesting. *Community-based optimization of the management of scarce water resources in agriculture in Central and West Asia and North Africa. Aleppo, Syria: ICARDA*.

³⁰⁷ Ibid.

³⁰⁸ Relevant interventions are detailed under Section 8.6 of the Feasibility Study, Annex 2.

³⁰⁹ GCF. 2021. Building resilience to cope with climate change in Jordan through improving water use efficiency in the agriculture sector. Available at: <https://www.greenclimate.fund/project/fp155#details>.

³¹⁰ The GEF Trust Fund. 2008. Mainstreaming Sustainable Land and Water Management Practices: Project Identification Form. Available at: <https://www.thegef.org/project/menarid-mainstreaming-sustainable-land-and-water-management-practices>.

³¹¹ GEF Independent Evaluation Office. 2016. Mainstreaming Sustainable Land and Water Management Practices: Terminal Evaluation Review form. Available at: <https://www.thegef.org/project/menarid-mainstreaming-sustainable-land-and-water-management-practices>.

timeline of 8 years). This adds further support to the economic rationale for project funding, as the JILMI project is expected to deliver net economic benefits almost immediately after all project activities have been completed (and before the impact of any scaling up has started to deliver significant benefits).

319. The ERR for the JILMI project is relatively high at 25%. This is considerably higher than the ERR hurdle rate of most international donor agencies. For example, the Millennium Challenge Corporation (MCC) of the USA employs a hurdle rate of 10% ERR for projects³¹². The ERR for the MCC Jordan Compact project is estimated at 19%, indicating the high potential economic value of the proposed JILMI project in the context of climate change adaptation and water scarcity issues in Jordan³¹³. According to the sensitivity analysis presented in the EFA, the economic summary statistics given above are robust to changes in input values. For example, the analysis is sensitive to reducing the value of groundwater to agriculture. However, when this is reduced by 30%, the ERR for the project is still favourable at 22%. Furthermore, the value of 0.5 USD/CM of water is well supported via other studies and literature. A second key parameter is the additional groundwater recharge from the rockfill dams. Given that there is more uncertainty around this quantity, it has been reduced by 50% for the sensitivity analysis. However, this still results in a favourable ERR for the project — with a rate of 20% overall. It is also shown that if no scale up of activities is achieved via Outputs 1 and 4, the ERR remains favourable at 17%. Overall, this project therefore represents a highly cost-effective spend for the GCF.
320. The proposed project will additionally reduce ecosystem pressures more widely than considered via the economic analysis. For example, the avoided degradation of land that would accompany the project's EbA interventions and be exacerbated via climate change, will have considerable co-benefits that are not captured in the economic statistics presented. Such co-benefits include: i) strengthened flood attenuation, reducing the related impacts — such as loss of life and damage to infrastructure — on downstream communities; ii) enhanced food security, livelihood resilience and biodiversity through the restoration of degraded ecosystems; and iii) decreased pollution of water resources which will filter runoff, prevent erosion and slow sedimentation. Additional details on the anticipated co-benefits of the proposed project are presented in Section D.3 of this Funding Proposal.
321. Moreover, the proposed project will also ensure an enhanced capacity for cross-sectoral coordination and planning in the GoJ, through monitoring and analysis of the sustainable development benefits of the project's ILWRM interventions during project implementation (Activity 1.1.3). The monitoring and analysis will consider: i) avoided costs — impacts or losses that would otherwise occur in the absence of ecosystem services, such as reduced water resource quantity and quality; and ii) replacement costs — the implication of implementing 'grey' artificial systems (as opposed to 'green' nature-based solutions) that would be required to deliver the same services as ecosystems. GoJ will include the monitoring and analysis in their ongoing operations. This will facilitate the embedding of coordination and operation of these mechanisms and structures in budgets and mandates beyond the life of the project. These plans will also include long-term funding commitments for enhancing the ecological infrastructure and maintaining the physical infrastructure associated with livelihoods interventions. By recognising the positive environmental, economic and social co-benefits to communities of ILWRM, cross-sectoral cooperation and integrated planning to conserve and augment water resources will be fostered.
322. The opportunity costs associated with the JILMI are minimal and short term. They mainly relate to exclusion of illegal livestock grazing practices at the ecosystem restoration sites. Given that the fodder-production production potential at these sites will become increasingly reduced as a result of expected land degradation processes (including illegal grazing), there is an opportunity for the improved management of these areas as the project's restoration interventions strengthen fodder production, allowing for legal sustainable grazing of livestock.

Potential to leverage public and private finance

323. Under Activity 2.1.4, a Water Fund will be designed to channel resources from those who are willing and able to pay for increased ground water supplies to those whose actions within the landscape can deliver them. The Water Fund will be designed to stimulate public and private investments into climate-resilient ILWRM as an adaptation approach, enhancing the local ownership of interventions targeted at water resource attenuation under climate change conditions, such as ecosystem restoration, MAR and floating solar installations. During the Water Fund's maturation phase, a long-term financing strategy that will detail how financial resources for the Fund's ongoing operation post GCF involvement will be leveraged from public and private sources. The long-term financing strategy will include a detailed model describing how the Water Fund's PES scheme will continue to channel finance into grant mechanism (which will include grant and results-based payments mechanisms) once GCF finance is no longer available post

³¹² Millennium Change Corporation. 2021. Guidelines for economic and beneficiary analysis. Available at:

<https://www.mcc.gov/resources/story/story-cdg-guidelines-for-economic-and-beneficiary-analysis>

³¹³ Millennium Change Corporation. 2021. Jordan compact economic rates of return. Available at: <https://www.mcc.gov/where-we-work/jordan-compact>

project. Public and private finance will also be leveraged through the policy reforms to be supported by the project, including the integration of climate-resilient ILWRM into Jordan's planning, policy-making and budgeting process. This will facilitate the embedding of coordination and operation of these mechanisms and structures in budgets and mandates beyond the life of the project. These plans will also include long-term funding commitments for enhancing the ecological infrastructure and maintaining the physical infrastructure associated with livelihood interventions.

324. Implemented in parallel with the proposed project will be the recently approved GCF project; FP155 "Building resilience to cope with climate change through improving water use efficiency in the agriculture sector (BRCCJ)"³¹⁴. BRCCJ's approach of promoting water use efficiency in the agriculture sector in the Dead Sea Basin complements the proposed project's climate-resilient ILWRM approach in the northern Jordan Valley, with both falling under the country's GCF Programme. These parallel investments will collectively contribute to climate-resilient sustainable development in Jordan.

Application of best practices in land and water resources management

325. Best practices from similar projects as outlined in Section B.1 Climate context were used to inform the design of climate-proofing interventions to be implemented under the proposed project. Lessons learned from numerous baseline investments into climate change adaptation in Jordan were also used to guide the design of all project interventions. Such lessons include appropriate mechanisms to ensure that project activities are implemented in a participatory, gender-inclusive and sustainable way. The particular international best practices and lessons learned that were applied to the design of the proposed project and that will contribute to the efficiency and effectiveness of its interventions are summarised below. Specific details on all relevant best practices and lessons learned are presented in Section 8 of Annex 2: Feasibility Study.

326. International best practices and lessons learned incorporated into the design of the project include:

- ensuring that landscape planning, implementation of policies, strategies and monitoring processes are integrated and iterative to account for the dynamic nature of landscape processes;
- acknowledging the need to engage stakeholders in an equitable manner to fairly distribute benefits and incentives of landscape management;
- facilitating participatory monitoring and adaptive management by ensuring that information is widely accessible and not managed by any single stakeholder;
- incorporating traditional knowledge into training for smallholders, pastoralists and local communities on climate-resilient agricultural practices, water monitoring and the adoption of alternative livelihoods;
- drawing on regional experiences and resources to increase institutional capacity to respond to the impacts of extreme events;
- applying an integrated, systemic and coordinated approach to disaster response through the establishment of formalised communication channels;
- developing appropriate strategies for securing financial resources for project development and implementation;
- effectively managing the available financial and human resources to ensure the efficient and cost-effective implementation of project interventions;
- undertaking regular monitoring and evaluation of climate-adaptive interventions to ensure that the most effective and appropriate solutions are being implemented under future conditions of climate change;
- engaging and collaborating extensively with all relevant project stakeholders to encourage buy-in from national- and local-level decision-makers, therefore contributing to the sustainability of proposed adaptation interventions over the long term; and implementing effective financial and project management strategies to ensure the efficient use of financial resources and avoid delays during the implementation phase.

³¹⁴ A brief complementarity analysis of the two projects is presented in Section B.1., with more detail provided in Annex 29.

E. LOGICAL FRAMEWORK

E.1. Project/Programme Focus

- ☐ Reduced emissions (mitigation)
- ☒ Increased resilience (adaptation)

E.2. GCF Impact level: Paradigm shift potential (max 600 words, approximately 1-2 pages)

Assessment Dimension	Current state (baseline)		Potential target scenario (Description)	How the project/programme will contribute (Description)
	Description	Rating		
Scale	Currently (2023), landscape and water resources management in Jordan is compartmentalised and sector specific, lacking an integrated approach. This has resulted in an unsustainable model of sectoralised water resource management where landscape-level ecosystem resilience is not prioritized, increasing the vulnerability of communities to climate change.	<u>Low</u>	The paradigm shift would involve a move away from a business-as-usual model of land and water resources that is uncoordinated and compartmentalised to one that is integrated across Jordan's landscapes, economic sectors, government ministries and departments, municipalities, non-government organisations and civil society, as well as into water resources and land-management policy. This will result in increased water resource supplies at the landscape level under drying conditions.	The project will result in strengthened governance and capacity for climate-resilient ILWRM (locally and nationally) and cumulatively improve water resource supplies by up to 154.5 million CM over 25 years through ecosystem restoration, and improving the protection, supply and use of water at the community level, before any replication takes place, contributing significantly to water security. The scaling up strategy has various strands, as follows: i) a Water Fund as a long-term financing mechanism (an approach not yet introduced in Jordan); ii) developing land use guidelines, standards and protocols for climate-resilient integrated watershed management in the Jordan Valley; iii) trained staff in national and sub-national government institutions on climate-resilient ILWRM practices into policies, plans and strategies; iv) amendments to Jordan's policy, planning and legal frameworks relevant to ILWRM; v) an outreach programme across the Jordan Valley; vi) improving knowledge management for climate change adaptation in Jordan to support national and sub-national decision-making, and build communities of practice.

Replicability	An ILWRM approach to climate change adaptation in the water has not yet been demonstrated in Jordan, so replication is not yet possible. This includes the use of a Water Fund serving as a mechanism for leveraging financial resources for the attenuation of water resources at a landscape level.	<u>Low</u>	If a case for the benefits (water resources and economic) of an ILWRM approach to climate change adaptation in the water sector can be demonstrated, including the contribution of a Water Fund to generating financial flows into the maintenance and scaling up of the approach, then it can be replicated in other areas of Jordan and the region.	Climate-resilient ecosystem restoration along with investments in the protection supply and use of water at the community level as part of a climate-resilient ILWRM to improve water resource supplies, along with a working model to increase private and public investments into adaptation (Water Fund), can inform efforts to replicate the intervention in other areas of Jordan and the region.
Sustainability	While the Kingdom of Jordan is committed to climate-resilient sustainable development, including water security, the mandate of state and non-state actors for adaptation is impeded by gaps in relevant policy, planning and legislature frameworks. In addition, the allocation of public budgets to adaptation is still insufficient due to ongoing economic strains on the country, landscape management does not yet consider an integrated approach and there is limited knowledge and awareness of landscape-level approaches to adaptation in the water sector such as ILWRM.	<u>Low</u>	The potential target scenario would see improved governmental support for a landscape-level approaches to adaptation in the water sector, supported by policy, increased capacity and sufficient funding (from public and private sources), resulting in the attenuation of water resources under future climate scenarios. Knowledge availability and behaviour change at government and community levels, will support this, with the country as a whole supporting an ILWRM approach to adaptation.	The mandates and capacity of state and non-state institutions to implement landscape-level adaptation approaches for enhancing water resource supplies, such as ILWRM, will be strengthened. This will be supplemented by stimulating increases in public budget allocations through providing evidence of the economic benefits of an ILWRM approach to adaptation, along with the establishment of a Water Fund to leverage finance for sustained financial flows into landscape-level approaches to water resources management. Additionally, gender-inclusive behavioural change will be fostered through improved information outreach and knowledge management.

E.3. Fund-level impacts

GCF Result Area	IRMF Indicator	Means of Verification (MoV)	Baseline	Target		Assumptions / Note
				Mid-term	Final ³¹⁵	
<u>Total Beneficiaries</u>	<u>Core 2: Direct and indirect beneficiaries reached</u>	Baseline report and project monitoring (interim and final evaluations) ³¹⁶ informed by: i) Household surveys and	Direct beneficiaries: 0 Indirect beneficiaries: 0	Direct beneficiaries: 61,735 people (27,781 female; 33,954 male)	Direct beneficiaries: 246,942 people (111,206 female; 135,736 male)	<u>Assumptions:</u> Relevant activities are not delayed and implemented on time.

³¹⁵ The final target means the target at the end of project/programme implementation period. However, for core indicator 1 (GHG emission reduction), please also provide the target value at the end of the total lifespan period which is defined as the maximum number of years over which the impacts of the investment are expected to be effective.

³¹⁶ Means of Verification for Fund-level Impact and Outcome indicators will be triangulated with the baseline surveys that will be undertaken by project consultants, and at project mid-term and project termination.

		questionnaires; and ii) Field surveys to confirm uptake and income generation of climate-resilient livelihood options, as well as improved quality and quantity of water being used by beneficiaries.		Indirect beneficiaries: 378,000 (177,660 female; 200,340 male)	Indirect beneficiaries: 756,000 (355,320 female; 400,680 male)	By project mid-term at least half of the targeted indirect beneficiaries have been reached through the awareness-raising campaign.
<u>ARA1 Most vulnerable people and communities</u>	<u>Core 2: Direct and indirect beneficiaries reached</u>	Baseline report and project monitoring (interim and final evaluations) ³¹⁷ informed by: i) Household surveys and questionnaires; and ii) Field surveys to confirm uptake and income generation of climate-resilient livelihood options, as well as improved quality and quantity of water being used by beneficiaries.	Direct beneficiaries: 0 Indirect beneficiaries: 0	Direct beneficiaries: 61,735 people (27,781 female; 33,954 male) Indirect beneficiaries: 378,000 (177,660 female; 200,340 male)	Direct beneficiaries: 246,942 people (111,206 female; 135,736 male) Indirect beneficiaries: 756,000 (355,320 female; 400,680 male)	<u>Data sources and methodologies applied for estimating baselines and targets</u> Direct beneficiaries include all people who will benefit directly from a project activity or the result of an activity. This includes: i) 32,000 (15,496 women and 15,864 men are from the Yarmouk site — Khaled Bin Al Waleed municipality; ii) 80,000 (38,400 women and 41,600 men are from the King Talal Dam site — New Ardah and Deir Alla municipalities; and iii) 52,000 (22,960 women and 27,040 men) are from the Kafraïn site — Shouna Alwasta municipality. Direct beneficiaries include all people who will benefit directly from a project activity or the result of an activity.
<u>ARA2 Health, well-being, food and water security</u>	<u>Core 2: Direct and indirect beneficiaries reached</u>	Baseline report and project monitoring (interim and final evaluations) ³¹⁸ informed by: i) Household surveys and questionnaires; and ii) Field surveys to confirm uptake and income generation of climate-resilient livelihood options, as well as improved quality and quantity of water being used by beneficiaries.	Direct beneficiaries: 0 Indirect beneficiaries: 0	Direct beneficiaries: 61,735 people (27,781 female; 33,954 male) Indirect beneficiaries: 378,000 (177,660 female; 200,340 male)	Direct beneficiaries: 246,942 people (111,206 female; 135,736 male) Indirect beneficiaries: 756,000 (355,320 female; 400,680 male)	Indirect beneficiaries include people benefitting from: i) the attenuation of water resources in the northern Jordan Valley; ii) strengthened local governance to implement climate-resilient ILWRM; iii) reduced water loss through evaporation from the King Abdullah Canal; and iv) improved awareness of the impacts of climate change and the adaptation options to address them. Indirect beneficiaries include people in the northern Jordan Valley who will not directly benefit from a project activity, or result of an activity. This includes all people located outside of the project's three target sites in the northern Jordan Valley.

³¹⁷ Means of Verification for Fund-level Impact and Outcome indicators will be triangulated with the baseline surveys that will be undertaken by project consultants, and at project mid-term and project termination.

³¹⁸ Means of Verification for Fund-level Impact and Outcome indicators will be triangulated with the baseline surveys that will be undertaken by project consultants, and at project mid-term and project termination.

<u>ARA1 Most vulnerable people and communities</u>	<u>Supplementary 2.1: Beneficiaries (female/male) adopting improved and/or new climate-resilient livelihood options</u>	Baseline report and project monitoring (interim and final evaluations)³¹⁹ informed by: i) Household surveys and questionnaires; and ii) Field surveys to confirm uptake and income generation of climate-resilient livelihood options, as well as improved quality and quantity of water being used by beneficiaries.	0 people benefitting from the adoption project interventions to support diversified, climate resilient livelihoods in agriculture, beekeeping and vegetable gardening.	4,875 people (47% female; 53% male) benefitting from the adoption of additional climate resilient livelihoods in beekeeping and vegetable gardening.	9,750 people (47% female; 53% male) benefitting from the adoption of additional climate resilient livelihoods in beekeeping and vegetable gardening³²⁰.	<u>Assumptions</u> Local communities actively take up livelihood options introduced through the project. Livelihood options are successful and generate income. Climate resilient livelihood strategies will improve the resilience of ecosystems, ecosystem services and people depending on those systems. Water resource attenuation interventions are successful. <u>Data sources and methodologies applied for estimating baselines and targets</u> Baselines were estimated through consultations with local partners and target communities, and profiling during intervention site selection. The baselines will be validated during project inception. Livelihood targets were estimated based on the number of community members that will directly benefit from additional livelihoods (beekeeping – 150 households; vegetable gardening – 600 households; climate-resilient agriculture – 1,200 households, using an estimate of 5 people per household). Water supply targets were benefitted from estimating the number of beneficiaries located in the project's three target sites that are expected to benefit, based on their reliance of local water resources.
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³¹⁹ Means of Verification for Fund-level Impact and Outcome indicators will be triangulated with the baseline surveys that will be undertaken by project consultants, and at project mid-term and project termination.

³²⁰ Although not included here as targets (because of the difficulty of attribution), most of the people in the project areas engaging in agricultural livelihoods stand to benefit from the enhanced water resources and ecosystem services (erosion control, flood attenuation etc.) provided by the “non-livelihood” ILWRM interventions (Outputs 2 and 3).

						Data source: Jordanian Department of Statistics
<u>ARA2 Health, well-being, food and water security</u>	<u>Supplementary 2.3: Beneficiaries (female/male) with more climate-resilient water security</u>	Baseline report and project monitoring (interim and terminal evaluations) ³²¹ informed by: i) Household surveys and questionnaires; and ii) Field surveys to confirm improved access by beneficiaries to a reliable and safe water supply.	0 households benefitting from improved quantity and quality of ground and surface water supplies within the project sites.	0 households benefitting from improved quantity and quality of ground and surface water supplies within the project sites — no change from baseline as the project's water resource benefits would not have materialised yet	32,800 households (~5 people per household; 47% women) benefitting from improved quantity and quality of ground and surface water supplies within the project sites.	<u>Assumptions</u> Water resource attenuation interventions are successful. Water safety will be established following national standard no. JS 286:20185³²². Measurements will be made according to the measurement approach in the methodology and applying random sampling. The water source will be defined as “reliable” in cases when water is available for at least 3 hours every day throughout a calendar year. <u>Data sources and methodologies applied for estimating baselines and targets</u> Baseline and targets calculated using estimated number of beneficiaries present in the project's three target sites based on data from the Jordanian Department of Statistics, as well as statistics presented by UNICEF's WASH programme, which indicate that ~50.3% of Jordanians only receive water 1 day per week³²³.
<u>ARA4 Ecosystems and ecosystem services</u>	<u>Core 4: Hectares of natural resources brought under improved low-emission and/or climate-resilient management practice</u>	Field surveys including GIS mapping of changes in land cover	0 hectares rehabilitated and/or restored	3,200 ha of watershed ecosystems in the northern Jordan Valley protected and strengthened in response to climate change and variability	12,000 ha of watershed ecosystems in the northern Jordan Valley protected and strengthened ³²⁴ in response to	<u>Assumptions</u> Restoration interventions are successful and supported by local stakeholders through the reduction of non-climatic drivers of ecosystem degradation (supported by the introduction of alternative fuel sources and renewable and energy-efficient technologies across project sites).

³²¹ Means of Verification for Fund-level Impact and Outcome indicators will be triangulated with the baseline surveys that will be undertaken by project consultants, and at project interim and project termination.

³²² Available at: <https://www.unhcr.org/jo/wp-content/uploads/sites/60/2019/08/Annex-A-1-Technical-Regulation-of-Jordan-Standards-and-Metrology-Organization.pdf>

³²³ Source: <https://www.unicef.org/jordan/water-sanitation-and-hygiene>

³²⁴ Through restoration and improved land and water resources management.

				(1,600 ha per year from year 3 of implementation)	climate ³²⁵ change and variability	As integrated land and water resources management interventions (ILWRM) are scaled up via the Water Fund (Activity 2.4), it is expected that an additional 4,000 ha of land will be placed under improved management for water resources benefits, with at least one additional (adjacent) municipality per watershed site benefitting directly. The 4,000 ha is related to total water attenuation potential for the three watersheds, in addition to the 8,000 ha to be restored directly by the project (see Table 27 and figures 116-118 in Annex 2). The rainfall attenuation potential represents the slowing of overland water flows which reduces the amount of water running off the land and increases the amount of time available for water infiltration, and hence groundwater recharge. The Water Fund will scale the set of restoration and land management interventions in Component 2 to target the entire 12,000 ha with potential for restoration to support water attenuation.
<u>ARA4 Ecosystems and ecosystem services</u>	<u>Supplementary 4.1: Hectares of terrestrial forest, terrestrial non-forest, freshwater and coastal marine areas brought under resoration and/or improved ecosystems</u>	Field surveys including GIS mapping of changes in land cover	0 hectares rehabilitated and/or restored	3,200 ha of watershed ecosystems in the northern Jordan Valley protected and strengthened in response to climate change and variability (1,600 ha per year from year 3 of implementation)	12,000 ha of watershed ecosystems in the northern Jordan Valley protected and strengthened ³²⁶ in response to climate ³²⁷ change and variability	<u>Data sources and methodologies applied for estimating baselines and targets</u> GIS mapping and stakeholder consultations to identify areas with potential for ecosystem restoration and/or rehabilitation.

E.4. GCF Outcome level: Enabling environment (IRMF core indicators 5-8 as applicable)

³²⁵ Although not enumerated here (because of the difficulty of attribution) the provision of alternative fuel sources, and efficient and renewable energy technologies, as well as the improved protection, supply and use of water for domestic purposes will also have indirect positive impacts on the scope of resilient ecosystems. Additionally, improved agricultural outputs may reduce pressure on ecosystems, by reducing land use change or the need for bridging income in the event of failing yields.

³²⁶ Through restoration and improved land and water resources management.

³²⁷ Although not enumerated here (because of the difficulty of attribution) the provision of alternative fuel sources, and efficient and renewable energy technologies, as well as the improved protection, supply and use of water for domestic purposes will also have indirect positive impacts on the scope of resilient ecosystems. Additionally, improved agricultural outputs may reduce pressure on ecosystems, by reducing land use change or the need for bridging income in the event of failing yields.

Core Indicator	Baseline context (description)	Rating for current state (baseline)	Target scenario (description)	How the project will contribute	Coverage
<u>Core Indicator 5: Degree to which GCF investments contribute to strengthening institutional and regulatory frameworks for low emission climate-resilient development pathways in a country-driven manner</u>	Currently, an unsustainable model of sectoralised water resource management exists in Jordan that does not prioritise landscape-level ecosystem resilience. However, adaptation of the country's population through conservation and adaptive measures in land and water resources management can be achieved through an integrated approach to conserving Jordan's water resources under climate change conditions. Baseline gaps identified in the National Water Strategy (2016–2025) include that it does not reflect the initiated decline of available freshwater, presuming that alternative water supplies would increase in availability from 2025 (which is at odds with preceding and proceeding National Communications to the UNFCCC). Addressing this is part of the Jordan Water Management Initiative's (2016–2021) scope, by increasing the accountability and sustainability of water sector management and governance. In addition, the National Water Strategy highlights gaps in how the GoJ manages water demand in the agricultural sector, as well as specific interventions proposed to address these gaps. The development of a legislative framework on climate change was identified as a gap through Vision 2025 and the EDP — which emphasised the need for a legal framework to regulate national climate adaptation activities, in addition to enhancing the capacities of national institutions and experts to address climate change impacts. In addition, the Climate Change Policy for a Resilient Water Sector aims to bridge the gap between priority areas to support policy coherence by building upon other policies and plans in the water sector.	<u>low</u>	Broadened mandate of state and non-state actors for climate change adaptation — including climate-resilient ILWRM. Integration of adaptation into relevant legislation and policy, and improved funding for adaptation. Enhanced cross-sectoral coordination and dialogue related to adaptation and watershed management, particularly between the water and agriculture sectors. Enhanced coordination in the implementation of climate change adaptation interventions across the country.	Under Component 1, The mandates of the institutions involved in climate-resilient ILWRM will be strengthened through its integration into relevant policies, plans and legislature. Guidelines, standards and operational protocols will be developed to facilitate a standardized and coordinated approach to climate resilient ILWRM. 24 days of training to national level government staff and 24 days of training for sub-national staff (2 Governorates and 6 Districts) will be administered over the 8 years. Draft amendments to Jordan's policy and legal frameworks for agriculture, land, water will be developed.	<u>National level (one country)</u>
<u>Core indicator 8: Degree to which GCF investments</u>	Constrained access of decision-makers to relevant tools, knowledge and data on climate change	<u>medium</u>	Enhanced access of decision-makers to relevant tools,	Under Component 1, evidence of the economic benefits of ILWRM	<u>National level (one country)</u>

<u>contribute to effective knowledge generation and learning processes, and use of good practices, methodologies and standards</u>	impacts and appropriate adaptation responses. Currently, there is a gap in Jordan between research and policy-making on climate change adaptation. The MoEnv, in seeking to achieve its mission to “maintain and improve the quality of Jordan’s environment, conserve natural resources, and contribute to sustainable development”, conducts activities concerned with, inter alia, public awareness and outreach and the support of decision-making through knowledge management. It leads the governance system — including other relevant governmental organisations — in initiating and driving policies, legislation, strategies and monitoring, as well as by mainstreaming environmental policies into all national development plans. To achieve this the MoEnv has set five objectives for the environment sector, notably the development of capacity and institutionalisation of a culture of excellence and raising public awareness and promoting behavioural change for environmental protection.		knowledge and data on climate change impacts and appropriate adaptation responses. Removal of the current gap between research and policy-making on climate change adaptation to enable effective and informed responses to the impacts of climate change across the country.	to project beneficiaries for use by policymakers and planners will be generated by the monitoring and analysis of ILWRM interventions implemented under the project. A full impact study of the project will be conducted in the project’s final two years. The long-term data sets from the project will be archived within Jordanian research institutions. Under Component 4, community centres mosques and schools across the Jordan valley will be equipped to function as adaptation learning centres and information hubs accessible to local communities. Community-based organizations will build awareness. A range of media will be used to deliver an outreach strategy. Educational practitioners will receive training on approach to include climate change adaptation into the curricular of Jordan’s tertiary education. A national climate change adaptation knowledge repository will be used to capture and disseminate best practices, lessons learned, literature and data on climate change adaptation relevant to the Jordanian context.	
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E.5. Project/programme specific indicators (project outcomes and outputs)

Project/programme results (outcomes/ outputs)	Project/programme specific Indicator	Means of Verification (MoV)	Baseline	Target		Assumptions / Notes
				Mid-term	Final	
Outcome 1. Increased public budgets for adaptation	Budgeted plans of Ministries involved in the project submitted to Ministry of Finance	Change analysis of budgeted plans of Ministries involved in the project	0	10% additional budget for adaptation in 2 Ministry/ Ministerial Department that scale up the implementation of integrated water and land adaptive measures	30% additional budget for adaptation in 2 Ministry/ Ministerial Departments that scale up the implementation of integrated water and land adaptive measures	Government culture promotes evidence-based policies and plans
Output 1.1. Strengthened governance and capacity for climate-resilient integrated landscape and water resources management (ILWRM)	Integrated watershed management plans for the three target watersheds developed over iterative planning cycles	Plans developed and validated	0 plans	3 plans	3 plans	Planners are willing to engage in cross sectoral planning and budgeting.
	Number of staff from national government institutions with increased capacity for planning and implementing climate-resilient ILM	Training reports Key informant interviews	0 staff	60 staff (18 female; 42 male)	60 staff (18 female; 42 male)	Relevant institutions and staff actively engage in trainings
	Number of sub-national staff from national government institutions and WMC members with increased capacity for planning and implementing climate-resilient ILM	Training reports Key informant interviews	0 staff 0 WMC members	30 staff (15 female; 15 male) 90 WMC members (45 female; 45 male)	30 staff (15 female; 15 male) 90 WMC members (45 female; 45 male)	Relevant institutions and staff actively engage in trainings WMCs and members actively engage in trainings
Output 1.2. Economic assessments developed quantifying net benefits of ILWRM as an adaptation approach	Report on the evidence of economic benefits of climate-resilient ILWRM	Reports	TBD: Evidence base limited	Raw data collection ongoing	1 report on the evidence for the economic benefits of climate-resilient ILWRM	Monitoring and analysis implemented as planned Report will be used to inform plans which will be used to

						leverage funding for the scaling up and replication of project climate-resilient ILWRM interventions
Outcome 2. Climate-resilient ILWRM practiced in three Jordan Valley watersheds	Hectares of land under climate-resilient ILWRM	Field surveys including GIS mapping of changes in land cover	0 hectares rehabilitated	3,200 ha of watershed ecosystems in the northern Jordan Valley protected and strengthened in response to climate change and variability (1,600 ha per year from year 3 of implementation)	8,000 ha of watershed ecosystems in the northern Jordan Valley protected and strengthened ³²⁸ in response to climate change and variability (1,600 ha per year from year 3 of implementation)	Restoration interventions are successful and supported by local stakeholders
Output 2.1. Climate-resilient ILWRM practices and ecosystem restoration implemented in three watersheds in the Jordan Valley.	# people engaged in adapted and sustainable ways of managing land resources.	Household surveys	0 people	4,875 people (at least 40% female)	9,750 people (at least 40% female)	People are open to new ways of managing land use
Output 2.2. Water Fund established in the Jordan Valley to maintain landscape investment for enhanced water resources	Water Fund operationalised	Key informant interviews Reports Water Fund documents	0 operational Water Fund	1 finalised Water Fund design	1 operationalised Water Fund	Water Fund is supported by local stakeholders and the private sector
Outcome 3. Enhanced water security for households in target areas	# households benefitting from cleaner and accessible water resources.	Household surveys based on random sampling plan in the three project sites Key informant interviews	0 households	0 households	32,800 households	Water resource attenuation interventions are successful
Output 3.1. Improved protection, supply and use of water for	Community-based water resources management plans developed.	Counts of water resources management plans	0 plans	0 plans	3 plans (one each for the three target watersheds, with sub-plans for each district)	Technical capacity is available in Jordan to install and maintain the technologies.

³²⁸ Through restoration and improved land and water resources management.

³²⁹ Although not enumerated here (because of the difficulty of attribution) the provision of alternative fuel sources, and efficient and renewable energy technologies, as well as the improved protection, supply and use of water for domestic purposes will also have indirect positive impacts on the scope of resilient ecosystems. Additionally, improved agricultural outputs may reduce pressure on ecosystems, by reducing land use change or the need for bridging income in the event of failing yields.

domestic purposes under climate change conditions					represented in the watersheds - At-Taibah, Al-Wastiyah, Qasabah Irbid and Al-Kurah from the Irbid Governate, and Dir Alla and Ash-Shunah al-Janubiyah districts from the Balqa Governate)	
	# of installations that promote aquifer recharge and harvest rainwater	Site visits Household surveys	0 installations	1,504 installations	3,012 installations	Technical capacity is available in Jordan to install and maintain the technologies.
	# of MW of floating solar installed to reduce evaporation along the King Abdullah II canal	Site visits	0 MW	500 KW	1 MW	Technical capacity is available in Jordan to install and maintain the technologies.
Outcome 4. Gender-responsive household behavioural change achieved through improved information and outreach	Number of people exhibiting behaviour change towards climate-resilient practices through awareness raising	Household surveys based on random sampling methods Focus group discussions	0	Approximately 380,000 people (including 179,420 females and 200,580 males) at behaviour change level 2 (behaviour change scorecard) ^{330,331}	Approximately 760,000 people (including 358,840 females and 401,160 males) at behaviour change level 2–3	People understand the messages and are motivated to act on them
Output 4.1. Improved knowledge and awareness of climate change adaptation for landscape and water resources management.	# of people reached through climate change information hubs and learning centres, and outreach campaigns	Evaluation of outreach campaign based on focus group discussions Key informant interviews	0	Approximately 380,000 people (including 179,420 females and 200,580 males)	Approximately 760,000 people (including 358,840 females and 401,160 males)	People actively listen to the outreach campaign

³³⁰ This behaviour change scorecard will have three levels and will measure the extent of awareness of climate change and appropriate related responses amongst target communities in the northern Jordan Valley. Level 0: Climate change and appropriate related responses not understood, or not considered; Level 1: Climate change and appropriate related responses partially understood and considered when implementing livelihood activities and using water resources; Level 2: Climate change and appropriate related responses fully understood and considered when implementing livelihood activities and using water resources.

³³¹ Selected community members from across the Jordan Valley will be trained, and stationed at climate change adaptation learning centres and hubs, to collect data on the extent of awareness — via a scorecard — on climate change and appropriate related adaptation responses during the implementation of the project's outreach programme (Activity 4.1). The information will then be consolidated by the relevant project Technical Partner (TP) and integrated into the project's M&E system.

Project/programme co-benefit indicators						
Project/programme results (outcomes/ outputs)	Project/programme specific Indicator	Means of Verification (MoV)	Baseline	Mid-term target	Final target	Assumptions / Note
Co-benefit 2: Reduced GHG emissions	# GHG emissions (tCO ₂ eq) reduced as a result of improved land use management practices	Field surveys including GIS mapping and remote sensing of changes in land cover EX-Ante Carbon-balance Tool (EX-ACT)	0 tCO ₂ eq	107,478 tCO ₂ eq	537,390 tCO ₂ eq	Assuming improved land use management activities commence in year three of project implementation, resulting in approximately - 107,478 tCO ₂ eq reduced annually (see Annex 31). Lifetime target (25 years) is 2,686,959 tCO ₂ eq.

E.6. Project/programme activities and deliverables

Activities	Description	Sub-activities	Deliverables
Component 1: Climate change adaptation governance in Jordan			
<u>Output 1.1: Strengthened governance and capacity for climate-resilient integrated landscape and water resources management (ILWRM) and Output 1.2. Economic assessments developed quantifying net benefits of ILWRM as an adaptation approach</u>			
<i>Activity 1.1.1 Strengthen the mandate and capacity of main government institutions — including MoEnv, MWI and their subsidiaries³³² — for climate-resilient ILWRM in Jordan</i>	To improve the mainstreaming of climate-resilient ILWRM into Jordan's relevant policy-making and planning processes, staff from relevant national and sub-national institutions (such as the Climate Change Directorate and National Committee on Climate Change) will be trained to interpret and apply climate data, as well as the results of climate analyses and risk assessments. The training, which will include two three-day workshops (for 20 people, with at least 50% female representation) will also focus on the use of this data for disaster risk management. Trained individuals will then work with experts procured by the project to review relevant policies and prepare policy briefs with recommendations for the integration of climate-related data into policies, plans and disaster-risk management. This training will contribute to the	1.1.1.1: Develop land use guidelines, standards and protocols for climate-resilient ILWRM in the Jordan Valley. 1.1.1.2: Train staff from relevant national and subnational institutions to use climate data, and the results of climate analyses and risk assessments in policy- and decision-making processes, as well as planning and disaster risk management.	1.1.1.1.a. One set of climate-resilient ILWRM guidelines and protocols 1.1.1.2.a. 30 staff trained (30 national, 34 sub-national; at least 50% women)

³³² Such as the Jordan Valley Authority (JVA).

	sustainability of this mainstreaming momentum, after project end. The policies will be presented and validated at the same workshop that will be hosted by the NCCC for the validation of policy briefs on climate-resilient ILWRM. Land use guidelines, standards and protocols will be developed for climate-resilient integrated watershed management in the Jordan Valley, including roles and responsibilities of all relevant actors.		
<i>Activity 1.1.2. Draft amendments to Jordan's policy, planning and legal frameworks for agriculture, land, water and natural resources, to mainstream climate-resilient ILWRM as a climate change adaptation approach</i>	Recommendations will be made via policy briefs for the integration of climate resilient ILWRM into relevant policies, plans and legislature, including Jordan's Decentralisation Programme and National Development Plan. To ensure that this is done effectively, several capacity-building actions targeting national- and local-level decision-makers and planners will be implemented. These include training: i) policy and decision-makers on the mainstreaming of climate-resilient ILWRM; ii) planners on climate-resilient ILWRM planning; and iii) staff from relevant national and sub-national institutions to use climate data, and the results of climate analyses and risk assessments in decision-making processes, as well as planning and disaster risk management. The capacity build through these trainings will facilitate the preparation of policy briefs with recommendations for the integration of climate-resilient ILWRM practices into relevant policies, plans, strategies and legislature. The policy briefs will be submitted to the National Committee of Climate Change (NCCC) for validation.	1.1.2.1: Train policy- and decision-makers on the integration of climate-resilient ILWRM practices into policies, plans and strategies. 1.1.2.2: Train relevant national and sub-national planners on climate-resilient ILWRM planning. 1.1.2.3: Review Jordan's national land-management policies, plans and legislation to identify gaps and opportunities for the consideration of climate-resilient ILWRM (including EbA, climate analyses and risk assessments) as an adaptation approach. 1.1.2.4: Prepare policy briefs with recommendations for the integration of climate-resilient ILWRM practices, and the results of climate analyses and risk assessments into relevant national policies, plans and strategies. 1.1.2.5: Convene a workshop to present the policy briefs prepared under 1.1.2.4 to representatives from relevant ministries and departments (with at least 50% female participation) for validation.	1.1.2.1.a. 175 people trained (at least 50% women) 1.1.2.2.a. 60 people trained (at least 50% women) 1.1.2.3.a. At least four policies/plans/legislation reviewed 1.1.2.4.a. At least 4 draft policy briefs 1.1.2.5.a. At least 20 workshop participants (50% female) 1.1.2.5.b. At least four validated policy briefs
<i>Activity 1.1.3. Generate evidence of the economic benefits of ILWRM to project beneficiaries for use by policymakers and planners</i>	To prioritise and ILWRM approach, decision-makers need to understand and appreciate the relevant interlinked ecological processes and how they directly produce economic benefits. The relevant information and evidence will be generated by the monitoring and analysis of ILWRM interventions implemented under the project. The monitoring and analysis would consider: i) avoided costs — impacts or losses that would otherwise occur in the absence of ecosystem	1.1.3.1: Gather primary and secondary data from project interventions, including a 1-day stakeholder engagement workshop at each site. 1.1.3.2: Analyse monitoring data from ILWRM interventions implemented under the project to identify benefit streams for valuing ILWRM benefits (including ecosystem services) under different climate scenarios.	1.1.3.1.a. One consultant report identifying benefit streams for valuing ILWRM benefits. 1.1.3.1.b. One stakeholder engagement workshop report identifying benefits derived from local ecosystem services. 1.1.3.2.a. One consultant report identifying data streams and modelling methodology for valuing ILWRM benefits.

	services, such as reduced water resource quantity and quality; and ii) replacement costs — the implication of implementing 'grey' artificial systems (as opposed to 'green' nature-based solutions) that would be required to deliver the same services as ecosystems. Where feasible, monitoring of the following sustainable development benefits will be included: flood attenuation, enhanced biodiversity, decreased pollution of water resources, improved access to clean water, strengthened food security, increased climate-resilience of pastoral livelihoods and energy generation from floating solar.	1.1.3.3: Identify data streams and modelling methodology for valuing ILWRM benefits. 1.1.3.4: Carry out valuation modelling exercises of benefit streams identified under 1.3.1. 1.1.3.5: Prepare full and summary reports of the analysis and results, including briefing notes for decision-makers highlighting evidence of ILWRM as an adaptation approach and specific policy recommendations.	1.1.3.3.a. Captured primary and secondary data on project interventions. 1.1.3.4.a. Results of valuation modelling exercise. 1.1.3.5.a. Two consultant reports (summary and full) of analysis and results and one set of briefing notes for decision-makers. 1.1.3.5.b. One summary report for decision-makers on the economic benefits of ILWRM, including policy recommendations.
<i>Activity 1.1.4. Strengthen the mandate and capacity of non-state actors for climate-resilient ILWRM in the Jordan Valley</i>	To ensure a participatory and cross-sectoral approach to landscape management, Watershed Management Committees (WMCs) will be established at each of the project's three target areas. The WMCs will be chaired by the JVA and include representatives from all stakeholders involved in landscape management, including local communities. The technical capacity of WMC members to perform their roles (as defined in the climate-resilient integrated watershed management guidelines, standards and protocols) will be strengthened through a tailored training programme developed under the project by local experts. Once trained, the three WMCs will be assisted by land use planners (trained under Activity 1.2) and adaptation experts to develop participatory climate-resilient integrated watershed management plans for their respective watersheds.	1.1.4.1: Establish Watershed Management Committees (WMCs), under the leadership of the JVA and local governors, at each of the three target watersheds to facilitate dialogue and participatory watershed management between upstream and downstream stakeholders. This will include working with MoEnv for the formal recognition of WMCs in relevant literature - including defining their responsibility towards working with communities. 1.1.4.2: Develop a training programme on climate-resilient integrated watershed management, including EbA, targeting the WMC members, including JVA, LDCs, local leaders, decentralization committees, local councils and other relevant stakeholders. 1.1.4.3: Train relevant watershed stakeholders (WMC members) on climate-resilient ILWRM and its economic benefits based on programme develop under Activity 1.1.4.2. 1.1.4.4: Train staff from relevant sub-national institutions to use climate data, and the results of climate analyses and risk assessments in policy- and decision-making processes, as well as planning and disaster risk management.	1.1.4.1.a. Three WMCs established and formally recognized in relevant legislature. 1.1.4.2.a. One climate-resilient integrated watershed management training programme for 90 stakeholders. 1.1.4.3.a. At least 90 people trained (at least 50% women). 1.1.4.4.a. 30 staff trained (at least 50% women). 1.1.4.5.a. Three integrated watershed management plans. 1.1.4.6.a. One ILWRM capacity-building programme for NGOs. 1.1.4.6.b. Capacity of eight NGOs (IUCN, RSCN, JREDS, PAF, HGD, JOHUD, RBG and FPEC) strengthened through the ILWRM capacity-building programme.

		1.1.4.5: Develop participatory climate-resilient integrated watershed management plans for the three target watersheds. 1.1.4.6: Develop and implement an ILWRM capacity building programme for relevant NGOs.	
Component 2: Integrated landscape and water resources management (ILWRM) under conditions of climate change			
<u>Output 2.1: Climate-resilient ILWRM practices and ecosystem restoration implemented in three watersheds in the Jordan Valley</u>			
<i>Activity 2.1.1. Support the implementation of participatory climate-resilient integrated watershed management plans (developed under Activity 1.4) in the Yarmouk, King Talal Dam and Kafraïn areas of the Jordan Valley.</i>	The climate-resilient integrated watershed management plans developed for each of the project intervention sites (Activity 1.1.4) will be implemented under the support and guidance of project Technical Partners. These plans will be implemented by WMCs once they have completed their watershed management training under Activity 1.1.4. This will afford local communities management rights over their local ecosystems, incentivising the enforcement of land-management laws, resulting in reduced human-driven ecosystem degradation and increased support for restoration activities. The implementation of the plans will include (as identified and detailed in the plans): i) implementing climate-resilient watershed management interventions in collaboration with local communities and NGOs; and ii) establishing the mechanisms via which climate-resilient natural resource-based additional livelihoods and climate-resilient agricultural practices on smallholder farms will be implemented (Activity 2.1.4).	2.1.1.1: Implement climate-resilient watershed management (WSM) measures at relevant project sites. 2.1.1.2: Establish mechanisms for the implementation of gender-responsive climate-resilient additional livelihoods and agricultural practices under Activity 2.4. 2.1.1.3: Establish and implement community-based participatory mechanisms for monitoring and evaluating change in community climate resilience and behavioural change, including community fora.	2.1.1.1.a. 8,000 ha of watershed ecosystems in the northern Jordan Valley protected and strengthened in response to climate change and variability. 2.1.1.2.a. Three (one per target site) gender-responsive community climate-resilient additional livelihood and agricultural practice implementation models including marketing approaches. 2.1.1.3.a. Three (one per target watershed) community groups established to monitor changes in community climate resilience.
<i>Activity 2.1.2. Restore ecosystems in the watersheds of the Yarmouk, King Talal Dam and Kafraïn areas of the Jordan Valley.</i>	Restoration of forest, rangeland, riparian and wadi ecosystems in these watershed areas will be informed by the development of site- and ecosystem-specific restoration protocols. These will include recommended climate-resilient plant species, water harvesting and conservation measures and maintenance guidelines. The target ecosystems will then be restored to strengthen the supply of ecosystem services, such as water, fodder and NTFPs, to buffer the impacts of climate change on the lives and livelihoods of local communities.	2.1.2.1: Develop protocols for the climate-resilient restoration approaches identified for each project site, based on baselines assessments. 2.1.2.2: Implement climate-resilient restoration of degraded watersheds at project sites, 3,178 ha within the Yarmouk site, 2,309 ha within the King Talal Site, and 2,512 ha within the Kafraïn site. 2.1.2.3: Establish central nursery at the RGB and community-managed nurseries (including the provision of seed and seedlings of species	2.1.2.1.a. Three baseline assessments and three sets of climate-resilient restoration protocols (one for each site) 2.1.2.2.a. 8,000 ha of restored ecosystems (~1,600 ha per year from year 3 of implementation) 2.1.2.3.a Central and community-managed nurseries 2.2.4a Small-scale rainwater harvesting interventions at restoration sites

		identified under 2.1.2.1) at each intervention site to grow the seedlings required for climate-resilient restoration. 2.1.2.4: Implement small-scale rainwater harvesting interventions at restoration sites to enhance the success of restoration interventions.	
<i>Activity 2.1.3. Introduce gender-responsive climate-resilient additional livelihoods and agricultural practices in the project's three intervention areas.</i>	Climate-resilient additional livelihoods and agriculture practices will be introduced to reduce the dependency of beneficiary communities on livelihoods that are heavily reliant on water resources and responsible for ecosystem degradation. This approach will complement ecosystem restoration under Activity 2.1.2 through both reducing unsustainable degradative practices and promoting livelihoods that are reliant on healthy ecosystems. As a result, the conservation as well as ongoing restoration of the ecosystems in the target watersheds will be promoted. An important aspect of the climate-resilient agricultural practices will be the introduction of in-field water conservation and harvesting measures.	2.1.3.1: Technical partner site visits to promote and train local communities on beekeeping as a climate-resilient additional livelihood to incentivize ecosystem restoration. 2.1.3.2: Technical partner site visits to promote and train communities (at least 75% women) on vegetable gardening through hydroponics and permaculture (wicking beds) as an additional livelihood. 2.1.3.3: Technical partner site visits to promote the uptake of climate-resilient agricultural practices on 1,200 smallholder farms across the three intervention areas to improve productivity under climate change conditions. 2.1.3.4: Train selected small-scale farmers within the three project sites on the use and maintenance of climate-resilient agriculture techniques, including in-field water conservation and harvesting measures. 2.1.3.5: Identify private sector champions in the agricultural sector to mentor the beneficiaries of climate-resilient additional livelihoods and agricultural practices to strengthen their capacity for post-harvest processing and marketing.	2.1.3.1.a. Beekeeping as an additional livelihood at 150 households (at least 75% female headed). 2.1.3.2.a. Vegetable gardening as an additional livelihood at 600 households (at least 75% female headed). 2.1.3.3.a. Three training and engagement sessions (1 per site) to stimulate the uptake of and train communities on the use of climate-resilient agricultural practices — 1,200 households trained (at least 30% female headed). 2.1.3.4.a. Climate-resilient agricultural practices, including in-field water conservation and harvesting measures at 1,200 households (at least 30% female headed). 2.1.3.5.a. Private sector champions from the agricultural sector identified and mentoring 1,200 smallholder farmers (at least 30% women) on post-harvest processing and marketing.
<u>Output 2.2: Water Fund established in the Jordan Valley to maintain landscape investment for enhanced water resources</u>			
<i>Activity 2.1.4. Establish a Water Fund in the Jordan Valley to maintain landscape investment for enhanced water resources</i>	Under the project, a Water Fund will be established as a mechanism through which financial resources for restoration (EbA) interventions in watersheds will be leveraged. By This will ensure the ongoing availability of funding for the maintenance and restoration of	2.1.4.1: Design the Water Fund in a participatory manner, including workshops with local communities, institutions, private sector and NGOs.	2.1.4.1.a. Situation Analysis; Design Studies (technical analyses to inform design decisions); a Water Fund strategic plan; Monitoring and Evaluation plan; and

	ecosystems in the three target watersheds and therefore a continued supply of ecosystem services, water resources in particular. The objective of the fund is to leverage investment into the restoration of watersheds to reduce the need to build wastewater treatment plants for the attenuation of dwindling water supplies. The rationale behind this is to provide financial incentive to upstream communities to reduce practices that result in compromised water quality and quantity downstream, and to serve as custodians of the ecosystems in these areas. The Water Fund will be a model for the payment for ecosystem services (PES), leveraging financial from private (industry and commercial agriculture) and public sector stakeholders (municipalities) for investment in the restoration of watershed ecosystems in the project areas. The steps in setting up the Water Fund are described below. Design — During this phase, an actionable strategic plan will be developed, and an on-granting facility designed as well as the Water Fund's grant mechanism. Leveraging of financial support from stakeholders and investors. Establishment — This phase includes an official public launch event, which will create opportunities for attracting membership and investors as well as a platform for communicating the Water Fund's vision. Operation — During this phase, stability within the Water Fund will be established by developing and implementing a work plan. Maturity — The final phase will focus on the long-term viability of the Water Fund.	2.1.4.2: Design an on-granting facility for the implementation of ILWRM sub-projects. This will include finalising selection criteria for the evaluation of proposals submitted in response to RFPs. 2.1.4.3: Officially launch the Water Fund and formalise long-term legal mechanisms. 2.1.4.4: Operationalise the Water Fund through the development and implementation of a workplan. 2.1.4.4: Develop and disseminate demand-based RFPs for ILWRM sub-projects under on-granting facility to WMCs and their members. 2.1.4.6: Evaluate proposals submitted in responses to the RFPs and select sub-projects to receive grant funding. 2.1.4.7: Sign sub-project agreements with beneficiaries of grant support. 2.1.4.8: Disburse sub-project grants and provide sub-project implementation oversight. 2.1.4.9: Develop and implement long-term financing and scaling up strategies for the Water Fund to leverage long-term financing and facilitate upscaling and replication in other areas of the Jordan Valley.	Resourcing Agreements with donors and investors. 2.1.4.2.a. On-granting facility design report. 2.1.4.2.b. Finalised selection criteria for the evaluation of proposals submitted in response to RFPs. 2.1.4.3.a. One Annual Operating Plan and Operational Management Readiness. 2.1.4.4.a. One Annual workplan. 2.1.4.5.a.: Demand-based RFPs 2.1.4.6.a. Proposal evaluation reports. 2.1.4.6.b. List of proposals to receive grant support. 2.1.4.8.a. Sub-project implementation workplans and progress reports. 2.1.4.9.a. One long-term financing strategy. 2.1.4.9.b. One scaling up strategy.
Component 3: Improved protection, supply and use of water resources under conditions of climate change			
<u>Output 3.1: Improved protection, supply and use of water for domestic purposes under climate change conditions</u>			
<i>Activity 3.1.1. Implement measures to improve the quantity and quality of ground and surface water supplies within the project sites</i>	Managed aquifer recharge (MAR) will be implemented via the construction of rockfill dams to collect and store water from runoff and flash floods. Four rockfill dams will be constructed across in each of the	3.1.1.1: Establish rockfill dams at 12 sites in target area wadis (four per target area) to capture runoff and water from flash floods for managed aquifer recharge (MAR).	3.1.1.1.a. 12 rockfill dams. 3.1.1.2.a. 12 small-scale silt reduction measures.

	project's three target areas. To protect the rockfill dams from sedimentation, small-scale measures will be implemented in strategic areas upstream from the dams to reduce erosion and capture silt. These measures will include the planting of indigenous vegetation and constructing natural check dams from locally available materials.	3.1.1.2: Implement small-scale measures, including restoration of vegetation and establishment of natural check dams, to capture silt upstream of rockfill dams. 3.1.1.3: Rehabilitate active freshwater springs across the project sites (Yarmouk and Kafra).	3.1.1.3.a. 10 rehabilitated springs.
<i>Activity 3.1.2. Install 1 MW of floating solar panels over the King Abdullah Canal to reduce water loss from evaporation</i>	In the lower-lying areas of the project intervention area (Jordan Valley floor) evaporation reduction measures will be implemented on the King Abdullah Canal to reduce surface water evaporation. These measures will include the establishment of at least 1 MW of floating solar panels over the King Abdullah Canal. The electricity generated will be used to provide power to the MWI's pumping stations for the canal, reducing their reliance on fossil fuel-generated energy.	3.1.2.1: Install 1 MW of floating solar panels over the King Abdullah Canal. 3.1.2.2: Connect the floating solar installation to the local grid to supply power the MWI's pumping stations on the King Abdullah Canal.	3.1.2.1a. 1 MW of floating solar panels installed over the King Abdullah Canal. 3.1.2.2.a. Floating solar installation connected to the local grid, supplying power to the one MWI pumping station.
<i>Activity 3.1.3. Introduce small-scale rainwater harvesting interventions in the project's three target areas to improve access to clean water under climate change conditions in a gender-responsive manner</i>	Under Activity 3.1.3. small-scale rooftop rainwater harvesting interventions will be installed at various buildings across the project's three intervention sites. These buildings will include 2,800 private houses (~14,000 people — 8% of the target population), 50 community/religious centres, 50 institutional buildings and 100 schools — 3,000 buildings in total. Beneficiary households will be selected based on the appropriateness of buildings for implementation, willingness of owners and water availability — i.e. those most vulnerable to water scarcity. A 32 CM cistern will be installed at each of these buildings to supplement and reduce the demand on domestic supplies during dry periods. This will result in a minimum increase in annual water supplies of 0.28 MCM/year in the target area.	3.1.3.1: Install small-scale water harvesting interventions at community centres, local and schools institutions at target sites in the Jordan Valley. 3.1.3.2: Install small-scale water harvesting interventions at homes at target sites in the Jordan Valley (100% of women trained in their operation and maintenance).	3.1.3.1/2.a. Small-scale water harvesting interventions on 3,000 buildings.
<i>Activity 3.1.4. Empower local communities, including women, to monitor and manage local water supplies more effectively</i>	To improve the community-level use and management of water under climate change conditions, community-based water monitoring and management plans will be developed for the target areas. These plans will ensure social accountability for water usage within beneficiary communities. To promote the effective implementation of these plans, community members will be made aware of their	3.1.4.1: Develop gender-responsive community-based water use monitoring and managing plans for project beneficiary communities. 3.1.4.2: Implement a gender-inclusive outreach campaign to raise awareness on the plans developed under sub-activity 3.1.41 and train	3.1.4.1.a. Three gender-responsive community-based water use monitoring and managing plans (one each for the three target watersheds, with sub-plans for each district represented in the watersheds - At-Taibah, Al-Wastiyah, Qasabah Irbid and Al-Kurah from the Irbid Governate, and Dir Alla

	purpose and benefits, and trained to implement them. Additional livelihoods related to the monitoring and management of water in target communities will also be promoted — including plumbing, rainwater harvesting infrastructure installation and maintenance, as well as installation of solar pumps.	target community members to monitor and manage the use of clean water (100% women). 3.1.4.3: Promote additional livelihoods related to the monitoring and management on clean water in target communities (including at least 50% women) through a training course.	and Ash-Shunah al-Janubiyah districts from the Balqa Governorate) 3.1.4.3.a. 38 community members per site (114 in total) practicing additional livelihoods related to water monitoring and management.
Component 4: Knowledge management and awareness on climate change adaptation			
<u>Output 4.1: Improved knowledge and awareness of climate change adaptation for landscape and water resources management.</u>			
<i>Activity 4.1.1 Implement a gender-responsive climate change adaptation outreach programme across the Jordan Valley</i>	An outreach programme will be implemented to raise awareness on climate changes and appropriate adaptation responses — including climate-resilient ILWRM — across the Jordan Valley. This outreach programme will use community centres – including mosques, churches and NGO centres – as information hubs and learning centres, as well as local media (radio and television) to disseminate information to local communities.	4.1.1.1: Community engagements carried out during the project inception period in the project's three target areas to target communities to the project's scope, objectives and expected results. 4.1.1.2: Upgrade community centres and local institutions (including NGO centres) in the Jordan Valley, and train local staff so that the centres can serve as climate vulnerability and adaptation information hubs and learning centres. 4.1.1.3: Public figures identified and trained to champion the case of climate change adaptation in Jordan, including promoting awareness and stimulating local ownership of the project's climate-resilient ILWRM approach. 4.1.1.4: Site- and ecosystem-specific gender-responsive climate change adaptation outreach programmes developed for the Jordan Valley, including the usage of locally available media, and the inclusion of schools via the EcoSchools programme. 4.1.1.5: Gender- and youth inclusive outreach programme implemented to promote awareness of the benefits and implementation of climate change adaptation approaches (especially EbA) among targeted rural communities and relevant	4.1.1.2.a. Community centres equipped as climate vulnerability and adaptation information hubs and learning centres. 4.1.1.3.a. Climate change adaptation champions in Jordan to promote awareness of climate change adaptation in Jordan and stimulate local ownership of the project's climate-resilient ILWRM approach. 4.1.1.4.a. One gender-responsive climate change adaptation outreach programme developed. 4.1.1.5.a. Gender- and youth-inclusive adaptation outreach programmes at implemented three sites. 4.1.1.6.a. At least 10 educational practitioners (50% women) trained. 4.1.1.6.b. Climate change adaptation integrated into the syllabi and curricula of at least 5 Jordanian universities.

		stakeholders in the Jordan Valley through locally available media. 4.1.1.6: Training relevant educational practitioners from the Ministry of Education and tertiary institutions on and the integration of climate change adaptation into syllabi and curricula of at least five Jordanian universities (University of Jordan and the University of Science and Technology, Jarash University, Mutah University and Al Balqa' Applied University).	
<i>Activity 4.2.1 Improve knowledge management for climate change adaptation in Jordan to support national and sub-national decision-making, and build communities of practice</i>	Improving the country's climate change knowledge-management system will include the development of knowledge-management tools to facilitate sectoral and cross-sectoral learning, and inform the related mainstreaming activities mentioned above. Capturing of and access to lessons learned and best practices on adaptation — including those generated through the proposed project — will be made possible through the establishment of a national climate change adaptation knowledge repository.	4.2.1.1: Knowledge-management tools developed to facilitate sectoral, cross-sectoral and community learning and mainstreaming of climate change adaptation approaches (including EbA) by policymakers and planners. 4.2.1.2: Collect lessons and best practices across Jordan (including on the project's ILWRM investments) and between arid states in the Mediterranean region and Middle East, as well as those generated via the project on an annual basis using the knowledge management tools (develop under sub-activity 4.2.1.1). 4.2.1.3: Design and implement a long-term monitoring plan for the generation of lessons learned on the project's climate-resilient ILWRM investments. 4.2.1.4: Establish partnerships (including the signing of MoUs) with Jordanian academic institutions and research agencies to participate in the implementation of the long-term monitoring plan (Sub-activity 4.2.3) and generate relevant research findings and publications to inform innovative landscape-centric approaches to climate change adaptation. 4.2.1.5: User groups trained to use knowledge management tools to increase their access to	4.2.1.1.a. Knowledge-management tools (software) including a climate change knowledge repository to manage and share knowledge content, a knowledge search tool, a communication tool to communicate knowledge and capture knowledge generated by communication, social software to facilitate the socialization of knowledge and a knowledge visualization tool to visually communicate knowledge. 4.2.1.2.a. Data base of lessons learned and best practices on adaptation, including ILWRM. 4.2.1.3.a. Long-term monitoring plan implemented and generating lessons learned. 4.2.1.4.a. Partnerships established with at least 1 academic and 1 research institution (including the signing of MoUs). 4.2.1.4.b. At least 2 research reports and 2 publications. 4.2.1.5.a. 3 user groups (planners, policy- and decision-makers, and practitioners) trained using knowledge management tools.

		knowledge/information of climate-resilient ILWRM approaches.	
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E.7. Monitoring, reporting and evaluation arrangements (max. 500 words, approximately 1 page)

327. The Project Coordinator (PC), under the oversight of the UNEP Task Manager, will be responsible for monitoring progress against the project's results-based indicators during implementation. For on-the-ground monitoring of project interventions, a full time Monitoring Officer will be employed to conduct and coordinate monitoring and evaluation (M&E) activities for the project. At the start of project implementation, the Monitoring Officer will design and operationalise a performance monitoring framework to track the project's progress towards achieving its targets. This will include: i) measuring performance against the project indicators (Sections E.3–5) to evaluate the progress of the project; ii) reporting the project's performance to the PSC and PMU; and iii) providing technical support to the PC. Additionally, the Monitoring Officer will be responsible for overseeing and monitoring the application of gender-disaggregated indicators and ensuring that all gender targets are met. At particular milestones throughout the project — including annual performance reports, interim evaluations and at project conclusion — the project team will undertake evidence-gathering exercises to verify the progress. These internal reports will be further validated by reviews at the mid-term and end of the project implementation, as described below. UNEP will be responsible for managing the interim and the final evaluation of the project.

Interim Evaluation

328. The UNEP Task Manager will oversee the process of hiring an external independent consultant to carry out the interim evaluation, who will provide an assessment of project performance against its targets at the project's mid-point. This will be a formative exercise and will include analysing whether the project is on track, what problems and challenges the project is encountering, and what corrective actions are required so that the project will achieve its intended outcomes by project completion in the most efficient and sustainable way. The PSC will participate in the interim evaluation and develop a management response to the evaluation recommendations along with an implementation plan. It is the responsibility of the UNEP Task Manager to monitor whether the agreed recommendations are being implemented during the remainder of the project's operational life.

Final Evaluation

329. An independent consultant will be contracted by the UNEP Evaluation Office to conduct the final evaluation at the end of project implementation. UNEP's Evaluation Office (EO) will be responsible for overseeing the final evaluation at the end of project implementation — which is a summative evaluation — and will liaise with the UNEP Task Manager throughout the process. An independent assessment of project performance against standard evaluation criteria (e.g., strategic relevance, effectiveness, efficiency, likelihood of impact and sustainability) will be made based on documentary evidence, stakeholder interviews and, in most cases, a field mission. The evaluation criteria will be aligned with the criteria contained in the GCF Evaluation Policy. Each evaluation criterion will be rated using a six-point rating scheme, and a weighted average will be determined to provide an overall performance rating for the project as a whole. Where there are any differences in ratings between the evaluation team and the Evaluation Office a final determination will be made by the Evaluation Office when the evaluation report is finalised.

330. The draft final evaluation report will be sent to project stakeholders during a commenting process managed by the Evaluation Office. Formal comments on the report will be shared by the Evaluation Office in an open and transparent manner. This evaluation report will be publicly disclosed and will be followed by a recommendation on compliance processes. The costs for results monitoring and performance evaluation are included in the project budget (Annex 3) under budget lines 95, 96, 127, 128, 167 and 180.

331. The EEs (MoEnv and IUCN) will submit semi-annual progress reports and quarterly financial statements to the PMU and the PMU will consolidate the reports and submit them to UNEP as the AE. In turn, UNEP will submit annual performance reports and semi-annual financial reports to GCF. The detailed reporting timelines are as follows:

332. Under the PCAs, the EEs will be required to report to UNEP as follows:
- a. Progress reports: by 30 July for January to June;
 - b. Annual Performance reports on or before 1 February;
 - c. Quarterly financial reports by 15 January, 15 April, 15 July, and 15 October;
 - d. Annual audited statements by 30 April;
 - e. Final report: within 3 months of Programme completion.
333. UNEP (AE) reports to the GCF:
- a. Annual Performance Reports by 1 March;
 - b. Semi-annual Financial Information by 1 March and 30 Sept;
 - c. Mid-Term Assessment report: halfway through Programme;
 - d. Final APR: within 6 months of Programme completion;
 - e. Final Evaluation report: within 12 months of Programme completion.

Monitoring will also be undertaken by the AE through supervision visits and field missions to track implementation progress and challenges and strategically plan the way forward.

F. RISK ASSESMENT AND MANAGEMENT

F.1. Risk factors and mitigations measures (max. 3 pages)

Selected Risk Factor 1		
Category	Probability	Impact
Technical and operational	Medium	Medium
Description		
High staff turnover could impede capacity building, the retention of skills and knowledge, and knowledge-sharing across the relevant institutions.		
Mitigation Measure(s)		
4. To mitigate the potential impact of a high staff turnover, communities and district authorities at different levels will be involved in the implementation of the proposed project. As a result, knowledge regarding project planning and implementation processes will be spread across a range of individuals, thereby reducing the impact on the project if individuals leave their respective organisations. Another method of negating high staff turnover is to account for this factor in the selection criteria for the hiring of project staff. For example, individuals with diverse skill sets and a professional life expectancy of more than 10 years should be encouraged to apply — particularly young people with sufficient relevant experience. This will ensure that staff members remain in their positions for longer periods of time and that institutional knowledge is retained over the project's lifespan. 5. Under the proposed project, institutional stability will be provided by the Project Steering Committee (PSC) and technical working groups (TWG) established as part of the project management arrangements (see Section B.4 for more detail). In addition, a knowledge-sharing platform will be developed under Activity 4.1.2 to ensure that access to information on best practices and lessons learned related to climate change adaptation (including ILWRM) is available to all new staff. The training programme developed under Activities 1.1.1 and 1.1.3, will be made available to all relevant stakeholders to facilitate the appropriate training of all new staff, further contributing to the mitigating of this risk.		
Selected Risk Factor 2		
Category	Probability	Impact
Technical and operational	Medium	Medium
Description		
The implementation of proposed project interventions will cease after the conclusion of the project.		
Mitigation Measure(s)		
336. An Exit Strategy has been developed to ensure the proposed project's long-term sustainability and to decrease the likelihood that this risk will occur (Section B.6). The primary aspects of the Exit Strategy include: i) the integration of climate-resilient Integrated Landscape and Water Resources Management (ILWRM) into policies, strategies and plans, to ensure the maintenance, replication and upscaling of project interventions post project (Activity 1.1.2); ii) capacity building at national and local, governmental and non-governmental levels, to ensure that the maintenance of project activities is sustained (Activities 1.1.1 and 1.1.3); iii) the improvement of knowledge management for climate change adaptation in Jordan, facilitating the maintenance and upscaling of project interventions after the conclusion of the project (Activity 4.1.2); and iv) the establishment of a Water Fund at the project interventions sites, which will leverage funding for the maintenance and continued restoration of upstream watershed area in the Jordan Valley (Activity 2.1.4).		
Selected Risk Factor 3		
Category	Probability	Impact
Technical and operational	Low	Medium
Description		
337. Community trust in the project falters leading to conflict, lack of ownership and ineffectiveness of interventions. This may result from insufficient coordination between implementing agencies during project implementation, limited integration of local knowledge into project activities and/or lack of effective benefit sharing (for example, large farmers may gain control over certain project interventions, excluding others).		

Mitigation Measure(s)

338. The proposed project's development has involved extensive engagement with local-level stakeholders to ensure their support for project activities. Extensive engagements with project partners and stakeholders will continue during project implementation, as detailed in the project's Stakeholder Engagement Plan (Annex 6).
339. The development of standard operating procedures (SOPs) by the project's Executing Entities (EEs) — MoEnv and IUCN — and the PMU that guide the project's approach to working with target communities. The SOPs will be included in the contracts of all Technical Partners, Contractors, Consultants, etc., engaged under the project. Adherence to SOPs will be monitored by the project's safeguards officer.
340. The development and signing of agreements between target communities and project EEs related to implementation. Any Technical Partners, Contractors, Consultants, etc., will also be bound by these agreements. The agreements will outline commitments and obligations of the project to communities.
341. Project implementation activities will build-on and/or draw from existing local committees/groups, systems and procedures to ensure local stakeholder participation. This will be complemented by WMCs established under Component 1 of the project, which will facilitate vertical as well as horizontal coordination and collaboration between local-level stakeholders and project partners. The presence of WMCs will ensure that target communities' needs and concerns are considered in the development and implementation of water monitoring and management plans. Representatives of target communities will be members of the WMCs, providing them with direct inputs into decision-making processes and ensuring that relevant information is always made available to them. These community representatives will be members of community-based committees already present on the ground in the three project areas, such as Development Advisory Committees, Stakeholders Advisory Groups, agricultural cooperatives, women's cooperatives, youth groups and Water User Associations).
342. A detailed stakeholder mapping analysis to ensure that all relevant stakeholders (including community groups are represented in WMCs). The representation of and engagement with communities will be built into the mandate/ToR of WMCs (a framework ToR for WMCs is presented in Annex 28). WMCs will ensure that on-the-ground activities (such as ecosystem restoration, Outputs 2.1 and 2.2) are only undertaken after resources users, who have traditional rights of access and use of the land, have been consulted and agreed to the interventions. Additionally, WMCs will monitor benefit sharing under the project's activities, ensuring that no stakeholder group, such as large commercial farmers, gains control over certain project interventions, excluding others such as local communities. Fostering the formal recognition of WMCs in relevant environmental legislature (via MoEnv) will ensure their recognition and authority, contributing to the representation of communities and building their trust in the project. Moreover, the local experience and footprint of the consortium of NGOs involved in project execution will contribute to successful engagement with local-level stakeholders. These NGOs will operate under the guidance of one of the project's Executing Entities.
343. Community support for interventions will also be fostered by undertaking of baseline assessments at each of the restoration sites, including consultations with local resource users to draw on their knowledge and foster their FPIC for the interventions. In addition, local resource users will be consulted during the development of restoration protocols to ensure that their traditional knowledge is considered.
344. Should access by local resource users to restoration sites be restricted, then suitable — and agreed upon — alternatives or compensations arrangements needs to be developed, negotiated with local resource users, and formally adopted before the commencement of any field interventions. During periods when resource users may have limited access to restoration sites, they will be identified as restoration workers so that they are able to continue earning an income. These individuals will also be targeted for the additional climate-resilient livelihoods introduced through the project. In the case of livestock owners, animal feed will be provided via the project, should restriction of access to grazing areas be encountered.
345. The impact of awareness raising (Output 4.1) will be monitored under the outcome-level indicator for this output through the assessing of behavioural change of project beneficiaries. This will be facilitated by the establishment of community-based participatory mechanisms for monitoring and evaluating change in climate resilience, as well as behavioural change (Activity 2.1.1).

Selected Risk Factor 4

Category	Probability	Impact
Governance	Medium	Medium

Description		
Limited coordination between implementing partners reducing the efficiency of interventions.		
Mitigation Measure(s)		
346. Although experience with multi-sectoral approaches is limited in Jordan, there are encouraging signs that their recognition is rising and that multiple players realise their synergetic value, particularly with regard to climate change (for example, through the Climate Change Policy, and NCCC). The project's implementation strategy will build on this momentum and provide a catalytic impetus through hands-on implementation experience on the ground. This, in combination with the project's multi-stakeholder implementation approach, will foster coordination. Relevant aspects of the project's implementation strategy are presented below. • A project management unit and steering committee will be established to coordinate implementation (Further details on the implementation arrangements are presented in Section C.7). • A national-level TWG, as well as local-level TWG's at each of the project intervention sites will be established to facilitate coordination during implementation. 347. The relevant institutions were involved in the design of the project activities and have agreed on all roles and responsibilities. A monitoring and evaluation plan has been designed for the project, which will provide a platform for monitoring the performance of implementing partners and will ensure that these institutions and agencies are fulfilling their responsibilities during the proposed project's implementation period. This will be supported by regular meetings/engagements between implementing partners to facilitate information sharing.		
Selected Risk Factor 5		
Category	Probability	Impact
Other	Low	Medium
Description		
348. Private sector stakeholders — for example profit-driven large-scale agriculture and industry — may not have the same goals or interests as other stakeholders such as small-scale farmers. This will lead to misalignment in priorities between stakeholders, inhibiting the realization of benefits and the effective implementation of project interventions. For these private sector stakeholders, short-term profits gained from continued business-as-usual practices may be a higher priority than the long-term benefits gained from improved quality and quantity of water resources enabled by restoration and water attenuation interventions implemented under Outputs 2.1, 2.2 and 3.1.		
Mitigation Measure(s)		
349. Under the proposed project, due diligence of potential private sector partners will be undertaken; concrete and clear roles and responsibilities for each partner will be established and strict adherence to role and responsibilities will be ensured.		
Selected Risk Factor 6		
Category	Probability	Impact
Other	Medium	High
Description		
Conflicts in neighbouring countries and the potential influx of additional refugees could influence government priorities, resulting in delays in project implementation and reduce the likelihood of project sustainability.		
Mitigation Measure(s)		
350. While the project will not prevent possible conflicts in neighbouring countries, nor a resulting influx of refugees, the inclusion of both state and non-state actors on the PSC, TWGs and as implementing partners, will contribute to the continued implementation of project activities should this risk materialize		
Selected Risk Factor 7		
Category	Probability	Impact
Governance	Low	Medium
Description		

Changes in political leadership at local level, as well as within ministries and other participating governmental institutions could result in delays in implementation.

Mitigation Measure(s)

351. As a constitutional monarchy, Jordan's government is considered stable — particularly in comparison to other countries in the region. Even with more frequent changes in governing parties through elections and via personnel shifts in ministries, which are to be anticipated, the project will be fully anchored through the involvement of multiple ministries and institutions, as well as the leading role played by the consortium of NGOs during implementation, ensuring project continuity without major interruptions. The integrated approach to project implementation also aims for strong involvement of stakeholder groups and collaborative engagement of various administrative entities, facilitating continued implementation of interventions in the event of changes in the political situation in the country. Specific mitigation measures which have been built into the projects design are mentioned below.

- Regular engagement with and capacity building of non-political actors — such as NGOs, government agency officials, staff and local community members — will be prioritised during project implementation.
- Formulation of local ordinances (including TWGs and WMCs), mainstreaming of project interventions into local planning, policy-making and budgeting systems will ensure that political issues will have negligible impact on the sustainability of project interventions.

Selected Risk Factor 8

Category	Probability	Impact
Governance	Low	Medium

Description

352. Risks related to AML/CFT and Prohibited Practices compliance are under review by UNEP. UNEP notes that there are no UN sanctions or embargoes listed against MoEnv and there appears to be no evidence of engagement in prohibited practices by MoEnv. In addition, there is no individual or entity listed on any United Nations Security Council (UNSC) sanctions lists, including the UN Consolidated Sanctions list will be involved in any manner with the project or its activities, either as a counterparty, implementer, or beneficiary.

Mitigation Measure(s)

353. UNEP will undertake a full assessment of the financial management capacity of MoEnv to identify risk elements and to prepare appropriate mitigation measures. UNEP will also closely monitor the financial management of the Project using the established monitoring and evaluation procedures and financial reporting mechanisms, including an annual audit; and establish internal controls for the Project and project fund management.

354. The Project Cooperation Agreements (PCAs) between UNEP and EEs will include warranties and caveats by the EEs to inter alia ensure compliance with the Anti-Fraud and Anti-Corruption Framework of the United Nations Secretariat, as well as the GCF Policy on Prohibited Practices.

Selected Risk Factor 9

Category	Probability	Impact
Other	Medium	Medium

Description

Resistance by water users, land holders and local institutions to adaptation measures, energy-efficient and renewable energy technologies, climate-resilient additional livelihoods and agricultural methods.

Mitigation Measure(s)

- Consultations with stakeholders — including water users, land holders and local institutions — during project development included raising awareness on the proposed ILWRM measures and the need for their support and buy-in (compliance) for the project to be a success. Details on these consultations are presented in Annex 7: Stakeholder consultations and engagement plan.
- Awareness-raising activities under the proposed project's climate change adaptation outreach programme in the Jordan Valley will contribute to compliance with ILWRM measures through the sharing of information on benefits and fostering ownership of the project's interventions by local stakeholders.
- Under Activity 1.1.4, Watershed Management Committees (WMCs) will be established at each of the proposed project's three intervention areas. The WMCs will ensure that local stakeholders (including local communities) are represented at both decision-making and implementation levels with regards to ILWRM. This will ensure that their

concerns are considered and that they are not excluded from accessing benefits, promoting compliance with the ILWRM measure implemented by the project.

- Adequate training on the introduced methods will be provided to ensure that beneficiaries are able to implement them effectively.
- Climate-resilient agricultural methods will be introduced on part the land of beneficiary farmers as a way to diversify their methods, not completely change them. This will allow for time for benefits to be realized after which they can then decide if they want to upscale. Farmers already practicing the same or similar methods at the project sites will be enlisted to demonstrate their use and raise awareness on their benefit.

Selected Risk Factor 10

Category	Probability	Impact
Other	Low	Medium

Description

Non-compliance by resource users and administrators with the project's ESMP.

Mitigation Measure(s)

- The ESMP will be made available to all stakeholders, including translation into Arabic.
- During the implementation of interventions, stakeholders will be made aware of relevant aspects of the ESMP and compliance will be monitored by the ESS officer along with implementation partners.

Selected Risk Factor 11

Category	Probability	Impact
Technical and operational	Medium	Medium

Description

Execution delays as a result of procurement and recruitment challenges or problems, particularly in relation to Components 2 and 3. Execution delays could occur during implementation, leading to potential changes in the project's implementation plan, design and budget under the GCF's Policy on Restructuring and Cancellation.

Mitigation Measure(s)

- The project's procurement plan (Annex 10) will be reviewed and further detailed during project inception to mitigate any procurement delays. The plan will be reviewed and updated at least annually during project implementation to ensure that lessons and challenges are considered.
- Recruitment of PMU staff, consultants, firms, equipment, etc. required for the project will be handled by a full time and experienced procurement officer based in the PMU who will also be supported by a financial officer to ensure that payment are made on time and via the appropriate mechanisms. These officers will also be supported by procurement, recruitment and financial experts from as well as the use of systems of IUCN to avoid any delays.

Selected Risk Factor 12

Category	Probability	Impact
Technical and operational	Medium	Medium

Description

Gender targets are not met and gender-related benefits do not accrue as expected. This may result from: women being excluded from participation in certain activities by men based on cultural norms and land tenure; the timing of activities not suiting the availability of women given their responsibilities at home including child minding; certain work being perceived as not suitable for women; harassment of women at project sites where on-the-ground interventions involving beneficiaries are taking place.

Mitigation Measure(s)

A full-time gender officer will be seated in the PMU to oversee the implementation and monitoring of all gender-related activities, including the tracking of progress against relevant targets in the logical framework and costed gender action plan. In addition, the project's TPs include NGOs who work directly with women in the target areas. These TPs will directly support the implementation of gender-related activities, ensuring that they are implemented appropriately based on their experience and allowing for adaptive management as needed to ensure targets are met and benefits accrue.

Gender-responsive sensitisation to the participation and roles of women in relevant activities will take place both at project inception (Sub-activity 4.1.1.1) as well as prior to the commencement of relevant activities to ensure awareness and understanding across beneficiary groups regarding the project's gender targets and gender-related benefits. The project's gender officer (supported by TPs) will ensure that all relevant information is shared with beneficiary groups and that any concerns are addressed. All beneficiary groups will also be made aware of the project's GRM, including how to access it.

The implementation of relevant activities will be tailored to ensure that timing is suitable to female beneficiaries and that childcare is provided as needed. Where certain work may not be appropriate for women to undertake, this work will be allocated to men, with other roles being allocated to women in consultation with relevant stakeholders. The project's gender officer, project staff, TPs and implementation partners will ensure that harassment is mitigated by providing oversight at all project intervention sites where beneficiaries are working. Access to the project's GRM will also be made as convenient as possible.

Please see additional details presented in the Gender Action Plan (GAP) in Annex 8.

G. GCF POLICIES AND STANDARDS

G.1. Environmental and social assessments (max. 750 words, approximately 1.5 pages)

355. The project is being proposed as a Category B or 'moderate risk' according to the GCF project classification. This classification is because any adverse environmental or social risks of project activities will have few impacts that are site-specific, irreversible or not readily addressed through mitigation measures. Justification for this is provided in the Environmental, Social and Economic Review Note (ESERN), Environmental and Social Management Plan (ESMP), Environmental and Social Impact Assessment (ESIA) and Social Safeguards Assessment in Annex 6. The main risks anticipated are summarised below in Table 15, along with mitigation measures to address them.

Table 15. Summary of environmental and social risks and mitigation measures.

Summary of risks	Mitigation factors or measures and management strategies
Environmental	
The risk of changes to the landscape and natural drainage systems.	Through the correct selection of intervention sites related to project activities and the adoption of appropriate implementation protocols providing details of the interventions before implementation, substantial changes to biophysical features of the landscape will be avoided or kept to an absolute minimum. International best practices for ecosystem restoration will be adopted. Project planning and implementation teams will be trained on guidelines and techniques to ensure that these best practices are adhered to.
The risk of solid and liquid waste generation.	Procedures and products will be used that reduce the production of liquid or solid waste — including the limited use of single-use plastics and the use of water efficient appliances. Proper liquid and solid waste containers will be installed and workers will be trained on correct disposal procedures. All construction machinery will receive periodic maintenance to ensure efficient performance. The maintenance will be undertaken offsite in specialised maintenance stations to avoid the release of liquid waste. All construction or domestic liquid and solid waste will be collected and disposed of following the environmental legislative requirements. This includes the recycling and reuse of as much solid waste as possible, the avoidance of solid waste burning, and waste removal by professional waste management companies. All temporary solid waste dumping sites will be rehabilitated after construction is complete.
The risk of air and noise pollution.	To reduce emissions, the transportation of excavated material over long distances will be avoided. This will be accomplished by the synchronization of excavating and filling processes, which will additionally reduce the accumulation of any excavated material near the intervention site. Excavation machinery will be maintained to ensure efficiency performance, which will reduce both air and noise pollution. Activities will be avoided at night to reduce noise generation during these times.
The risk of natural ecosystem deterioration and fragmentation.	A detailed baseline assessment of target watersheds has been undertaken and will be used to inform decisions related to the design and planning of interventions. In addition to this, government-approved procedures will be used to screen and assess potentially adverse project impacts. Any facilities required for project management and support will be located as far as possible from ecologically sensitive areas. Workers will be trained on ecosystem monitoring and assessments using a set of clear and specific ecological indicators.
The risk of disturbance of ecological corridors and the removal of vegetation.	Major ecological corridors have been mapped to inform project interventions. All workers will be informed on the locations and sensitivities of these ecological corridors. Signs will be placed around ecologically sensitive areas within the project implementation sites. The removal of vegetation will be prohibited or kept to an absolute minimum. Specific instructions and safeguards for natural vegetation will be included in all contracts and work orders.
The risk of disturbance of wildlife, including breeding or migratory birds.	Signs will be placed in project sites prohibiting the killing or disturbance of wildlife. Stakeholders will be trained on the importance of wildlife and the correct method of dealing with their presence. Cooperation with law enforcement agencies will be ensured for temporary protection schemes at project sites.
The risk of the deterioration of genetic resources.	Areas of importance for genetic resources have been identified as part of the baseline assessments. Signs will be posted in these areas highlighting their ecological importance and stakeholders will be trained on the relevance of genetic resources.
The risk of the introduction of invasive species.	The selection criteria for intervention sites will include special provision for exotic or invasive species. Interventions plans will include the eradication of all exotic and invasive species from project sites, after which no planting of exotic species will be allowed.
Social	
The risk of damage to known and unknown archaeological sites.	A cultural assessment of all new sites for which information is not currently available will be undertaken. Any construction or excavation activities will be halted if any antiquities or archaeological items are discovered and will be reported to the Director of the Department of Antiquities or the nearest General Security Center. No removal or change to cultural sites or

	artefacts will be permitted in project intervention sites. A special protection scheme with strong law enforcement arrangements will be applied for all cultural sites.
Risks to the disturbance of local traditions and customs.	Special training will be provided to all contractors and workers on the respect of local tradition and customs. Local communities will be able to submit complaints or report any intrusion on their local customs and norms through the provision of a clear and easy reporting mechanism.
The risk of involuntary resettlement	Output two of the project will include a temporary restriction of the use of degraded forests and rangelands by local communities. This restriction will gradually be removed as restoration and rehabilitation activities progress. The project will not affect official and clearly recognized land-tenure arrangements. The proposed project Grievance Redress Mechanism (GRM; see Annex 6) will ensure local users have the ability to express their concerns with regards to any violation of their access and user rights which have not been explicitly recognized in the project design.
Risks to labour and working conditions.	The project will involve small-scale work activities and temporary employment. To prevent the potential of forced or child labour, the project's procurement plan will strictly adhere to UNEP and the GCF's procurement principles, as well as national laws.
Risks to gender equality	The project incorporates a Gender Analysis and Action Plan with specific gender-targeted activities built into the project design, to ensure that the different interests, roles and responsibilities of both women and men are assessed during the planning and implementation phases.

356. The risks considered in the ESERN risk screening checklist have been further elaborated in the Environmental and Social Management Plan (ESMP) in Annex 6. This demonstrates how these risks have been considered in the fundamental design of the project and any additional mitigation measures that may be required. In addition, given the structure of the project, whereby the exact siting and detailed design of the project interventions are being determined as part of the project itself (through the integrated landscape and water resources management processes), the specific nature of any residual environmental and social risks that remain will be assessed as part of project implementation.

357. Risks will be managed as described in the list below.

- Undertaking planning, design and implementation processes with close involvement of an experienced environmental and social safeguards specialist.
- Ensuring the project officers (forestry, livelihoods, agriculture and water) recruited onto the project are sufficiently experienced in safeguards relevant to their specialist technical areas.
- Establishing methodologies for the environmental and social assessment of each of the activities under the project.
- Ensuring the M&E process includes adequate review of environmental and social risks (this will be done by an M&E specialist with specific experience in reviewing environmental and social considerations).
- Assigning formal roles and responsibilities for environmental and social management across the project team.
- Establishing a reporting procedure for environmental and social concerns related to the interventions.
- Incorporating environmental and social safeguard considerations in the training, capacity building and technical assistance delivered through the project.

358. The environmental and social management processes will draw upon the following documents:

- GCF Environmental and Social Management System;
- UNEP safeguards systems;
- summaries of stakeholder consultations;
- baselines of environmental and social conditions;
- project ESMP, including specific management frameworks and baselines (Annex 6);
- project-level GRM (see Annex 6); and
- project Gender Assessment and Action Plan (Annex 8).

G.2. Gender assessment and action plan (max. 500 words, approximately 1 page)

359. During project preparation, a gender assessment was undertaken to determine the differential vulnerability to, and impact of, climate change on women and men in Jordan, focusing on the Jordan Valley in particular. From the findings of this assessment and an analysis of the proposed project solution, a gender action plan (GAP) was developed. To design this gender analysis as part of preparation for a GAP, a rapid review was undertaken of supporting documents related to gender mainstreaming in climate change, as well as previous gender analyses conducted by other donors in Jordan — in addition to data available from the Department of Statistics (DoS). The gender assessment also identifies gender considerations that are relevant to the project and examines potential gender mainstreaming opportunities. Stakeholder consultations were undertaken and focus group discussions (FGDs) were

carried out in three target areas selected to benefit from the project with a total of 60 participants, of which more than 50% were women. This was done to highlight gender and social norms and to identify key entry points for women's participation in project activities within the study area.

360. The Government of Jordan has established that gender-responsive planning, resource-allocation and implementation in the context of the Jordan Integrated Landscape Management Initiative (JILMI) will be a priority. Therefore, supporting transformational change both in the lives of female beneficiaries as well as in prompting gender mainstreaming in national climate change policy-making and implementation was put forward as a priority in stakeholder consultations. In addition to describing men and women's relative vulnerability and adaptation strategies to climate change, the gender assessment and problem analysis identified risks and opportunities relating to gender. Specific findings of the analysis are described below.

- Only 30% of the FGD women participants knew about climate change and its disproportionate impacts on women and children.
- The majority of FGD male participants (80%) said that the climate change conditions will affect both men and women in the same way.
- Men control access to resources in which they own the key productive assets such as land, livestock and medium to large businesses. For climate change interventions particularly in drylands, this means that activities which include the use of productive resources like land or livestock should include both men and women in decision-making. Secondly, any proposed land use change relating to dryland conservation should include both local men and women as key stakeholders.
- Access to credit is a key constraint in improving women's economic productivity in the selected areas. Therefore, the project will focus on improving women's access to credit through the Water Fund — which will be established under Activity 2.4 — to pay for ongoing watershed restoration in the upper-catchment areas of the project's three target sites.
- In the targeted areas, rates of women's illiteracy, poverty and unemployment are high, which make them more vulnerable to climate change impacts than men. Gender-responsive trainings provide opportunities to support vulnerable groups in the targeted areas, particularly women-headed households.
- More investment is required to improve women's participation in alternative livelihoods, to ensure that their rights and interests are recognised.
- The proposed project will provide support to the Ministry of Environment to activate gender mainstreaming actions stated in the climate change policy and involve women in the decision-making process.
- Increased collaboration and partnership with CSOs, NGOs and the entities working on climate change issues will increase awareness of gender and environmental sustainability challenges in the targeted areas. The objective of this awareness-raising should be to increase women's knowledge and equal rights, in particular. In addition, CSOs are expected to support gender mainstreaming by involving whole communities — including men and local leaders — to overcome sensitivity to the traditional divisions of labour that may help to achieve better social acceptability and inclusion for project activities.

361. To mitigate against introducing gender biases amongst project participants and beneficiaries during implementation and after project completion, the GAP outlines several interventions to ensure that the project does not unintentionally contribute to perpetuating or worsening gender equality and gender-based climate vulnerability in the project areas. The GAP seeks to promote gender equality by including actions and targets which will increase women's participation, representation and awareness, as well as ensuring that training programmes are gender-responsive and -inclusive. It includes clear targets, sex-disaggregated data collection, gender-responsive design features and quantifiable performance indicators to promote women's participation and benefits from the project. The project will also:

- ensure gender inclusion in all the policies reviewed;
- ensure sustainability and long-term impact of gender-affirming actions (please see the actions listed below);
- provide capacity building for government institutions in acknowledging gender differentials in responding to climate risks; and
- increase social accountability as described in the list of actions below.

362. To achieve the objectives, several actions have been identified which will be part of the GAP. These are listed below.

- Ensure women's participation in trainings of at least 30% of total number of participants.
- Ensure that revisions to Jordan's land-management policy, planning and legal framework are gender-responsive by including sections addressing women's needs and interests.
- Ensure women and youth participation of 50% of the total population in the targeted areas.
- Ensure participation of women CSOs in each targeted area.

- Introduce success stories highlighting the impact of women and other vulnerable groups participation in the climate change adaptation.
- Include at least two women's CSOs in each of the targeted areas.
- Ensure women and vulnerable groups participate in the consultation process for climate-resilient restoration of watersheds.
- Ensure 100% of female household beneficiaries are trained on the operation and maintenance of small-scale water-harvesting interventions.
- Create women's groups as water-wise ambassadors via the community-based water use management and monitoring system.
- Ensure 20% of women farmers are represented by the women's groups as water-wise ambassadors.
- Create funding mechanisms for women.

363. A primary objective of the project with regards to gender is increased resilience to climate change of at least 77,080 women in the project area. This will be achieved through direct improvement of women's capacities for improved water supply and demand management through climate-resilient ILWRM practices. The project will employ a full-time Gender Officer, who will be situated within the Project Management Unit (PMU), to oversee the implementation of the GAP, as well as provide inputs and conduct gender mainstreaming during project implementation. The full Gender Assessment and project-level gender action plan are included as Annex 8.

G.3. Financial management and procurement (max. 500 words, approximately 1 page)

364. The financial management and procurement within the project will be guided by UN financial regulations, rules and procedures, as well as UNEP's Programme Manual. The financial rules of UNEP, which follow International Public Sector Accounting Standards (IPSAS), are promulgated pursuant to the Financial Regulations and Rules of the UN. Within this context, funding-allocation mechanisms are managed as per the UN framework, including eligibility criteria, proposal evaluation processes, quality assurance and control, project monitoring and supervision. UNEP is audited annually by the UN Board of Auditors and has established dedicated trust funds for GCF resources.

365. The funding of a project will be established through a distinct grant within the Trust Fund, with the project itself being set up in the UNEP Enterprise Resource Planning (ERP) as a "Work Breakdown Structure" (WBS), which is itself further broken down into "Work Breakdown Structure Elements" (WBSEs), organised by output and outcome. The grant of the project will be linked to the WBSEs to fund the activities contributing to the delivery of specific outputs (as per the disbursement schedule detailed in the term sheet). The fee income will be managed through a dedicated grant independently and separately of the GCF project grant funds. In line with UNEP procedures, the project will appoint a Financial Officer within the PMU who will be responsible for monitoring, reporting on and approving requests for funds on a quarterly basis for the activities executed by the Ministry of Environment (MoEnv). Reports to summarise the disbursement and projected demands for project funding will be prepared and submitted to a UNEP Task Manager who will conduct project supervision, in line with reporting standards and methodologies applied in past projects, such as those implemented using GEF modalities. The UN financial regulations and rules require the segregation of duties, and safeguards to ensure compliance with UN financial rules and regulations. In addition, a Fund Management Officer will be appointed to assist UNEP's Task Manager with all financial monitoring and supervision functions. All procurement will be undertaken in line with UN procurement regulations, rules and policies. UNEP's modalities for project implementation, results in funds being transferred in tranches to the EEs (MoEnv and IUCN) once the EEs have satisfied the conditions that are defined under the legal/cooperation instrument — including the Project Cooperation Agreement (PCA) and Internal Cooperation Agreement (ICA), to be signed between UNEP's Ecosystems Division and MoEnv. The PCA will include specific obligations for the EEs on financial management and reporting and will require periodic financial reporting to follow international financial and auditing standards. The PCA specifically requires the audit be undertaken by a recognised firm of certified public accountants or, for governments, by a government auditor. This auditor should state whether the GCF proceeds were covered by the scope of the audit.

366. As a GoJ authority, MoEnv follows the government's financial and procurement rules and standards. During the proposed project's inception phase, UNEP will undertake a thorough assessment of MoEnv's capacity to undertake procurement in line with UN regulations, rules and processes. This assessment will guide the procurement monitoring plan which will be agreed upon between UNEP and MoEnv. The assessment will be undertaken following project approval but prior to project implementation. It will include assessments of elements of governance and public accountability such as, inter alia, review of the existence and quality of policies, legal and institutional framework, and systems supporting transparency, accountability and controls, especially in the use of donor funds. The outcome of the assessment will determine the threshold for procurement that is undertaken by MoEnv, which will be reflected in the procurement monitoring plan. This plan is an annex to the PCA and will be shared with GCF upon signature, as required by the Funded Activity Agreement (FAA). The procurement monitoring plan will also be reviewed periodically. The project's investments in equipment will be undertaken in accordance with UN procurement procedures. Finally, in

line with the UNEP Programme Manual, MoEnv will be requested to provide an annual compliance audit covering all aspects of the project execution including review of all expenditures incurred during the financial year.

G.4. Disclosure of funding proposal

- ☒ No confidential information: The accredited entity confirms that the funding proposal, including its annexes, may be disclosed in full by the GCF, as no information is being provided in confidence.
- ☐ With confidential information: The accredited entity declares that the funding proposal, including its annexes, may not be disclosed in full by the GCF, as certain information is being provided in confidence. Accordingly, the accredited entity is providing to the Secretariat the following two copies of the funding proposal, including all annexes:
- full copy for internal use of the GCF in which the confidential portions are marked accordingly, together with an explanatory note regarding the said portions and the corresponding reason for confidentiality under the accredited entity's disclosure policy, and
 - redacted copy for disclosure on the GCF website.

The funding proposal can only be processed upon receipt of the two copies above, if containing confidential information.

H. ANNEXES

H.1. Mandatory annexes

☒	Annex 1	NDA no-objection letter(s) (template provided)
☒	Annex 2	Feasibility study - and a market study, if applicable
☒	Annex 3	Economic and/or financial analyses in spreadsheet format
☒	Annex 4	Detailed budget plan (template provided)
☒	Annex 5	Implementation timetable including key project/programme milestones (template provided)
☒	Annex 6	E&S document corresponding to the E&S category (A, B or C; or I1, I2 or I3): (ESS disclosure form provided) ☐ Environmental and Social Impact Assessment (ESIA) or ☐ Environmental and Social Management Plan (ESMP) or ☐ Environmental and Social Management System (ESMS) ☐ Others (please specify – e.g. Resettlement Action Plan, Resettlement Policy Framework, Indigenous People's Plan, Land Acquisition Plan, etc.)
☒	Annex 7	Summary of consultations and stakeholder engagement plan
☒	Annex 8	Gender assessment and project/programme-level action plan (template provided)
☒	Annex 9	Legal due diligence (regulation, taxation and insurance)
☒	Annex 10	Procurement plan (template provided)
☒	Annex 11	Monitoring and evaluation plan (template provided)
☒	Annex 12	AE fee request (template provided)
☒	Annex 13	Co-financing commitment letter, if applicable (template provided)
☒	Annex 14	Term sheet including a detailed disbursement schedule and, if applicable, repayment schedule

H.2. Other annexes as applicable

☒	Annex 15	Evidence of internal approval (template provided)
☐	Annex 16	Map(s) indicating the location of proposed interventions
☐	Annex 17	Multi-country project/programme information (template provided)
☐	Annex 18	Appraisal, due diligence or evaluation report for proposals based on up-scaling or replicating a pilot project
☐	Annex 19	Procedures for controlling procurement by third parties or executing entities undertaking projects financed by the entity
☐	Annex 20	First level AML/CFT (KYC) assessment
☒	Annex 21	Operations manual (Operations and maintenance)
☒	Annex 22	Climate change adaptation assessment report
☒	Annex 23	Consortium Members Comparative Advantage
☒	Annex 24	Site Selection Results
☒	Annex 25	Co-financing analysis
☒	Annex 26	Problem Tree

☒	Annex 27	Sample ToR PSC and PMU
☒	Annex 28	Framework ToR WMCs
☒	Annex 29	JILMI and BRCCJ Comparison
☒	Annex 30	Parallel Finance
☒	Annex 31	EXACT Mitigation Potential Results
☒	Annex 32	Framework Water Fund Design ToR
☒	Annex 33	Beneficiary Calculations

** Please note that a funding proposal will be considered complete only upon receipt of all the applicable supporting documents.*

JORDAN INTEGRATED LANDSCAPE MANAGEMENT INITIATIVE (JILMI)

ANNEX 1

NDA NO-OBJECTION LETTER





Ministry of Environment

Ref.No 2-7-1320

Date 15-2-2021

Mr. Yannick Glemarec
Executive Director
The Green Climate Fund ("GCF") Secretariat
G-Tower, 123 Central Street
Songdo-dong, 206840, Republic of Korea

Re: Funding proposal for the GCF by United Nation Environmental Program (UNEP) regarding Jordan Integrated Landscape Management Initiative (JILMI)

Dear Mr. Yannick Glemarec,

We refer to the project titled *Jordan Integrated Landscape Management Initiative (JILMI)* in Jordan as included in the funding proposal submitted by **United Nation Environmental Program (UNEP)** to us on 18 January 2021.

The undersigned is the duly authorized representative of the Ministry of Environment, the National Designated Authority/focal point of Jordan. Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to the project as included in the funding proposal.

By communicating our no-objection, it is implied that:

- (a) The government of Jordan has no-objection to the project as included in the funding proposal;
- (b) The project as included in the funding proposal is in conformity with the national priorities, strategies and plans of Jordan;
- (c) In accordance with the GCF's environmental and social safeguards, the project as included in the funding proposal is in conformity with relevant national laws and regulations.

We also confirm that our national process for ascertaining no-objection to the project as included in the funding proposal has been duly followed.

We also confirm that our no-objection applies to all projects or activities to be implemented within the scope of the project.

We acknowledge that this letter will be made publicly available on the GCF website.

Kind Regards,

Eng. Belal Shqarin

Directorate of Climate Directorate

Dr. Mohamed Khashashneh

Secretary General

Environmental and social safeguards report form pursuant to para. 17 of the IDP

Basic project or programme information	
Project or programme title	Jordan Integrated Landscape Management Initiative (JILMI)
Existence of subproject(s) to be identified after GCF Board approval	Yes
Sector (public or private)	Public
Accredited entity	United Nations Environment Programme (UNEP)
Environmental and social safeguards (ESS) category	Category B
Location – specific location(s) of project or target country or location(s) of programme	Hashemite Kingdom of Jordan
Environmental and Social Impact Assessment (ESIA) (if applicable)	
Date of disclosure on accredited entity's website	Monday, May 20, 2024
Language(s) of disclosure	English and Arabic
Explanation on language	Arabic is the official language of Jordan, and English is widely spoken in Jordan.
Link to disclosure	English: https://open.unep.org/docs/gcf/UNEPFP_Jordan_Annex_6_A_ESIA_and_ESMP_May%2024_English.pdf Arabic: https://open.unep.org/docs/gcf/UNEPFP_Jordan_Annex_6_A_ESIA_and_ESMP_May_24_Arabic.pdf
Other link(s)	https://open.unep.org/project/GCF-GCF-25950
Remarks	An ESIA consistent with the requirements for a Category B project is contained in <i>Annex 6: Environmental and Social Impact Assessment and Environmental and Social Management Framework</i> of the funding proposal package.
Environmental and Social Management Plan (ESMP) (if applicable)	
Date of disclosure on accredited entity's website	Thursday, June 6, 2024
Language(s) of disclosure	English and Arabic
Explanation on language	Arabic is the official language of Jordan, and English is widely spoken in Jordan.
Link to disclosure	English: https://open.unep.org/docs/gcf/UNEPFP_Jordan_Annex_6_A_ESIA_and_ESMP_May%2024_English.pdf Arabic: https://open.unep.org/docs/gcf/UNEPFP_Jordan_Annex_6_A_ESIA_and_ESMP_May_24_Arabic.pdf
Other link(s)	https://open.unep.org/project/GCF-GCF-25950
Remarks	An ESMP consistent with the requirements for a Category B project is contained in <i>Annex 6: Environmental and Social</i>

	<i>Impact Assessment and Environmental and Social Management Framework</i> of the funding proposal package.
Environmental and Social Management (ESMS) (if applicable)	
Date of disclosure on accredited entity's website	N/A
Language(s) of disclosure	N/A
Explanation on language	N/A
Link to disclosure	N/A
Other link(s)	N/A
Remarks	N/A
Any other relevant ESS reports, e.g. Resettlement Action Plan (RAP), Resettlement Policy Framework (RPF), Indigenous Peoples Plan (IPP), Indigenous Peoples Planning Framework (IPPF) (if applicable)	
Description of report/disclosure on accredited entity's website	Indigenous Peoples Planning Framework (IPPF)/ Thursday, June 6, 2024
Language(s) of disclosure	English and Arabic
Explanation on language	Arabic is the official language of Jordan, and English is widely spoken in Jordan.
Link to disclosure	English: https://open.unep.org/docs/gcf/UNEPFP_Jordan_Annex_6_B_IPPF_May%2024_English.pdf Arabic: https://open.unep.org/docs/gcf/UNEPFP_Jordan_Annex_6_B_IPPF_Arabic.pdf
Other link(s)	N/A
Remarks	An Indigenous Peoples Planning Framework (IPPF) is contained as Annex 6B.
Disclosure in locations convenient to affected peoples (stakeholders)	
Date	Monday, May 20, 2024
Place	Summaries of the Environmental and Social Impact Assessment and Environmental and Social Management Framework were made available to stakeholders during the stakeholder consultations. The full translated copies in Arabic and English were shared on the orientation meetings with local communities in the project's sites from north to south through JOHUD centers; Princess Basma Center, Sheikh Hussein, Mansoura, and Al-Kafrain on 20th of May 2024. In addition, in the Royal Botanic Garden through the Tal Al Rumman Women Society in 19th of May 2024 in a meeting with local community there. In 21st of May 2024, the Jordan Valley Authority announced the Environmental and Social Impact Assessment and Environmental and Social Management Framework mentioned above of the project in their offices across the JVA. Now the Environmental and Social Impact Assessment and Environmental and Social Management Framework

	mentioned above version is available in all centres around the project areas and in the designated ministries there. The following are the contact details of the centers/ organizations that disclosed the project's Environmental and Social Impact Assessment and Environmental and Social Management Framework mentioned above. The Royal Botanic Garden – Tal Al Rumman http://royalbotanicgarden.org/ Email: mshudiefat@royalbotanicgarden.org Email: info@royalbotanicgarden.org Address: P.O. Box 99, Amman 11910, Jordan Jordan Hashemite Fund for Human Development (JOHUD) https://www.johud.org.jo/ Email: duaa@johud.org.jo Email: sameer.a@johud.org.jo Address: info@johud.org.jo Email: Bashar.g@jaah.org.jo Main Offices: Queen Zain AlSharaf Institute for Development Al Hashmi Al Shamali. Admin Offices: 127 Madina Street, Amman. Jordan Valley Authority www.mwi.jva.gov.jo Email: hisham_al-hisa@mwi.gov.jo Email: Zohuire_Albawna@mwi.gov.jo Address: Obada bin Al-Samit Street, Building No. 15 - Shmeisani – Amman Fax: +962 6 5689916
Date of Board meeting in which the FP is intended to be considered	
Date of accredited entity's Board meeting	N/A
Date of GCF's Board meeting	Monday, July 15, 2024

Note: This form was prepared by the accredited entity stated above.

Secretariat's assessment of FP232

Proposal name:	Jordan Integrated Landscape Management Initiative (JILMI)
Accredited entity:	United Nations Environment Programme
Country:	Jordan
Project/programme size:	Medium

I. Overall assessment of the Secretariat

- The funding proposal is presented to the Board for consideration with the following remarks:

Strengths	Points of reflection
Comprehensive approach: Jordan Integrated Landscape Management Initiative employs an integrated land and water resources management approach, aiming to reduce the negative impacts of climate change on water resources through improved ecosystem management, managed aquifer recharge, rainwater harvesting, evaporation reduction, improved agricultural practices and strengthened governance.	The structure of the water fund will be designed during the first two years of the implementation. This offers a unique opportunity to factor in further consultations and ensure a very robust design of the fund.
Wide-reaching impact: the project targets significant environmental and social outcomes, including the enhancement of water security for households, increased resilience of ecosystems and direct benefits to 246,942 people in the Jordan Valley, reducing their climate vulnerability.	
Collaborative funding and execution: the project involves collaboration between the Government of Jordan, various non-governmental organizations and GCF, with substantial co-financing. This collective effort enhances project credibility and resource pooling.	
Alignment with Sustainable Development Goals: the projects aligns with multiple Sustainable Development Goals, addressing urgent adaptation needs, ecosystem restoration and food security, all of which are crucial for Jordan's resilience to climate change impacts.	

Scalability and replicability: the project is designed with the potential for replication and scaling across different regions, which could amplify its impact and serve as a model for similar arid regions facing water scarcity and climate change challenges.	
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2. The Board may wish to consider approving this funding proposal in accordance with the term sheet agreed between the Secretariat and the accredited entity, and, if considered appropriate, subject to the conditions set out in Annex II of document GCF/B.39/02.

II. Summary of the Secretariat's assessment

2.1 Project background

3. Jordan is considered one of the most drought-prone countries in the world, with agricultural consumption accounting for the majority of the water resources available in the country. Jordan's current food systems face significant challenges, with climate change having an irreversible impact on current and future food production as well as a direct negative impact on the ecosystems and their ability to maintain and support the hydrological cycle in the country.

4. Farmers and communities face challenges in rapidly transitioning to the less drought-prone, diversified and more resilient local food systems necessary to secure future food production, while simultaneously managing, supporting and restoring the ecosystems in their crucial role in driving the hydrological cycle.

5. The proposed project seeks to overcome these climate challenges through focusing on ecosystem restoration, improving local food systems and, through improved governance and water management strategies, facilitating a sustainable landscape-led approach with long-term benefits. The project presents strong adaptation impacts with significant mitigation co-benefits.

6. The targeted interventions are estimated to reach 246,942 direct and 756,000 indirect beneficiaries across three target watersheds in the Jordan Valley – Yarmouk, King Talal Dam and Kafraín – achieving ecosystem restoration aimed at water resources benefits across 12,000 ha.

7. The total project financing is USD 60,509,951, of which a GCF grant of USD 44,904,203 is requested. Co-financing of USD 15,605,748 has also been secured to support the project. The co-financing will be provided by the Government of Jordan and a consortium of non-governmental organizations (NGOs) in the form of grants and in-kind contributions. In addition to the co-financing, the project brings parallel finance worth USD 13,133,000. The project is submitted under environmental and social safeguards category B, and the total duration of the project is eight years.

2.2 Component-by-component analysis

Component 1: Climate change adaptation governance in Jordan

Outcome 1: Increased public budgets for adaptation (total costs: USD 2,338,394; GCF costs USD 1,616,864, or 69 per cent)

8. Component 1 will play a strategic role in creating the enabling policy and governance environment to facilitate future investments in the food and ecosystems sectors for the project lifetime and beyond. The component will implement interventions seeking to mainstream integrated landscape water resources management (ILWRM) into key regulatory frameworks in Jordan in the land, water, agriculture and natural resources sectors. This will support the creation of sound and practical evidence for the economic benefits of ILWRM investments.
9. The component will establish key protocols, standards and guidelines on ILWRM for the site-specific conditions of the Jordan Valley across the three targeted watersheds to ensure full coordination and strategic approaches across these sites. Currently there are insufficient practices, capacity and knowledge on the ground to systematically embed ILWRM approaches, and hence component 1 plays an important role in ensuring a sustainable roll-out of the proposed ILWRM interventions. These protocols, standards and guidelines will target both State and non-State actors in the country and is based on a highly participatory approach in securing the most practical and sustainable use and implementation of the tools and guidelines.
10. These key materials will be the basis for a number of capacity-building interventions across the three watersheds and further help to form the basis for draft amendments to the current policy landscape on planning and legal frameworks for agriculture, land, water and natural resources to ensure that ILWRM innovations are well represented and embedded. This will help to ensure that all future investments and innovations in the sectors are built on the solid ground of ILWRM strategic ideas and evidence.
11. The component further seeks to establish water management committees, which will be a core part of the proposed four components and their successful implementation as well as the long-term transition of the food and ecosystems sectors envisaged by the project. The water management committees will be composed of a solid representation of all key stakeholders in the three watersheds and will ensure a sound and practical solution platform where key solutions and practical advice will be conveyed and scaled up. The component will be financed by GCF resources with co-financing from the Government of Jordan.

Component 2: Integrated landscape and water resources management under conditions of climate change

Outcome 2. Climate-resilient integrated landscape and water resources management practised in three Jordan Valley watersheds (total costs: USD 32,925,452; GCF costs USD 22,395,865, or 68 per cent)

12. Component 2 of the project emphasizes the implementation of climate-resilient ILWRM across three watersheds in the Jordan Valley, aiming to enhance adaptation to climate change through ecosystem restoration. A key strength of this component is its holistic approach to managing environmental degradation by integrating various restoration practices that focus on both mitigating current damage and preventing future vulnerabilities. This approach is operationalized through outcome 2, which promotes climate-resilient ILWRM practices, and output 2.1, where specific techniques and ecosystem restorations are implemented.
13. In particular the component will establish and design a water fund to operate as a key mechanism for financial resources to support restoration activities, ecosystem-based adaptation. The architecture of the fund will be developed during the first two years of the project enabling a solid and robust design building on a model for payment for ecosystem services.
14. The component also addresses critical issues such as vegetation loss and soil erosion exacerbated by climate change, employing both passive and active restoration methods. Passive

restoration includes measures such as removing sources of degradation, while active restoration involves re-establishing vegetation and afforestation.

15. Community involvement is central to the project, with local communities participating actively in restoration activities. This not only fosters community ownership of the initiatives but also helps in building local capacity through training and employment opportunities, particularly for women and other marginalized groups. Gender-responsive climate-resilient agricultural techniques will be implemented on smallholder farms to increase productivity under climate change conditions, providing immediate benefits for local communities, while at the same time reducing the pressure on degraded ecosystems undergoing restoration.

Component 3: Improved protection, supply and use of water resources under conditions of climate change

Outcome 3: Enhanced water security for households in target areas (total costs: USD 15,891,345; GCF costs 12,758,414 or 80 per cent)

16. This component will aim to achieve enhanced water security for households in target areas. The main output will be improved protection, supply and use of water under climate change conditions. Key interventions under this component will include (i) rehabilitation of active freshwater springs and building of natural check dams; (ii) establishment of rockfill dams to capture rainwater run-off and contribute to aquifer recharge; and (iii) introduction of rooftop rainwater harvesting systems, specifically in communities that are vulnerable to water scarcity, primarily focusing on female-headed households. These interventions aim to improve both the quality and quantity of surface water and groundwater, subsequently enhancing water security for households. The implementation of rooftop rainwater harvesting is projected to save a significant amount of the total domestic demand if applied across Jordan.

17. Furthermore, the installation of floating solar panels over the King Abdullah Canal should reduce water loss due to evaporation by minimizing exposure to direct sunlight and wind, thus increasing water storage efficiency, improving water availability and reducing overall water demand. Gender-responsive community-based water monitoring and management plans will also be developed for target areas to ensure better management and sustainable use of water resources. This will involve the wider community, ensuring that needs and concerns are considered and that information is made available to them. Training and livelihood support related to water management will also be promoted, including skills such as plumbing, rainwater harvesting infrastructure installation and maintenance, and installation of solar pumps.

18. Overall, this component targets improving communities' resilience to droughts, contributing to increased adaptive capacity, and enhancing awareness of climate threats and risk reduction processes.

Component 4: Knowledge management and awareness on climate change adaptation

Outcome 4: Gender-responsive household behavioural change achieved through improved information and outreach (total costs USD 3,879,500; GCF costs USD 3,229,500, or 83 per cent)

19. The strategic goal of this component is the consolidation and enhancement of Jordan's climate change knowledge as well as the creation of knowledge management tools to facilitate cross-sectoral learning. This would be achieved through the establishment of a national repository of climate change adaptation knowledge with a view to strengthening awareness of climate threats and risk-reduction processes.

20. Ultimately, the successful execution of the component will contribute to the adoption of an integrated land and water resource management approach, incorporating gender-responsive elements and promoting local, national and regional awareness of the importance and benefits of biodiversity. Therefore, the performance of this component crucially depends on effective communication strategies, community engagement and the development of evidence-based knowledge management systems.

III. Assessment of performance against investment criteria

3.1 Impact potential

Scale: Medium to high

21. The adaptation impact potential of the project is significant, focusing on enhancing the resilience of Jordan's water resources and agricultural systems to climate change. The project aims to significantly enhance the capacity of Jordan to adapt to climate change, particularly in terms of water resources management and agriculture. The project is expected to directly benefit 246,942 individuals in the Jordan Valley by reducing their vulnerability to climate-related impacts. Additionally, it will indirectly benefit another 756,000 people across the northern Jordan Valley through improved water quality and quantity, stronger governance for integrated land and water resources management, and enhanced resilience to the impacts of climate change. It is expected that 12,000 ha land will be placed under improved management for water resources benefits, with at least one additional (adjacent) municipality per watershed site benefiting directly. Interventions at the regulatory and institutional level will indirectly benefit the Jordanian population as a whole, amounting to approximately 10 million people.

3.2 Paradigm shift potential

Scale: High

22. The proposed project's vision for paradigm shift in Jordan's management of land and water resources under climate change conditions is centred on transforming a fragmented system into an integrated, participatory and climate-resilient framework. The initiative aims to catalyse transformative change by addressing the institutional, technical and financial barriers that have historically hindered ILWRM in Jordan. The project's theory of change is robust, detailing mechanisms through which the identified barriers (e.g. insufficient institutional coordination, limited access to climate change adaptation information and financial constraints) will be addressed. By strengthening governance, enhancing technical capacity and improving financial mechanisms through the creation of a water fund, the project seeks to enable a systemic shift. This shift is expected to improve resilience and adaptability across various sectors and communities, focusing on inclusive approaches that engage women and other marginalized groups. The project outlines clear pathways for replication and scaling, emphasizing the upscaling potential of interventions such as ecosystem restoration and managed aquifer recharge. The project addresses sustainability through the establishment of the water fund, which is designed to secure ongoing financial resources for the maintenance and expansion of climate-resilient practices. This financial mechanism, coupled with the training of stakeholders and the integration of climate-resilient ILWRM into national policies, underscores a commitment to sustaining outcomes over the long term. The project effectively identifies and proposes solutions to the most binding constraints to systemic change in Jordan's water sector. By targeting institutional and awareness barriers and leveraging a participatory approach to engage a broad range of stakeholders, the project is well positioned to foster a supportive environment for sustained change.

3.3 Sustainable development potential

Scale: High

23. The proposed project holds substantial potential for sustainable development across several domains. It has several economic, social and environmental co-benefits. Regarding economic co-benefits, the project aims to offer community members training in areas such as plumbing, installation and maintenance of rainwater harvesting infrastructure and setting up solar pumps. This capacity-building could significantly contribute to job creation and boost income levels, particularly among the women trained. In terms of social co-benefits, improved access to water is likely to have a ripple effect on health and safety. It could reduce incidences of waterborne diseases and cut the time spent on water collection, especially for women and girls. These saved hours could then be channelled towards educational pursuits, thus potentially enhancing school attendance rates. Furthermore, the upgrades in sanitation facilities that the project envisions could result in additional social co-benefits. As far as environmental co-benefits are concerned, the use of rainwater harvesting systems and solar energy should contribute towards improved air, water and soil quality, not to mention conservation and biodiversity. The project's focus on using renewable water sources and minimizing water loss due to evaporation presents clear environmental benefits.

24. A vital aspect of the project is its emphasis on gender empowerment. It seeks to bridge gender inequalities by building capacities, involving women in water management committees, and targeting a significant percentage of interventions – such as rainwater harvesting technology – to benefit female-headed households. This focus on empowerment can lead to greater decision-making power for women, equitable access to resources and enhancement of livelihood opportunities.

25. In addition to these benefits, the project aligns well with the following Sustainable Development Goals (SDGs): SDG 1 (No poverty), SDG 3 (Good health and well-being), SDG 4 (Quality education), SDG 5 (Gender equality), SDG 6 (Clean water and sanitation), SDG 7 (Affordable and clean energy), SDG 13 (Climate action) and SDG 15 (Life on land).

3.4 Needs of the recipient

Scale: High

26. The project proposal addresses the financial, economic, social and institutional needs of Jordan, particularly in the context of barriers to accessing climate-related finance. It outlines the country's challenges in securing domestic and international funds for climate adaptation and mitigation. By focusing on strategic interventions such as improving water management and enhancing local governance, the project directly tackles these barriers, ensuring a clear pathway to alleviating the financial and knowledge constraints identified.

27. The proposal highlights Jordan's pronounced vulnerability to climate change due to its critical water stress conditions, specifying the impact of reduced rainfall, increased temperatures and frequent droughts on the nation's water supply and agricultural productivity. The project's targeted approach in the Jordan Valley aims to protect and improve water resources, which is crucial for the local agricultural communities who are at the front line of climate impacts. This is aligned with addressing the direct needs of these vulnerable groups, including women and children, enhancing their resilience to climate-induced challenges.

28. Institutional capacity constraints are also a significant focus of the proposal, acknowledging Jordan's limited ability to integrate climate change adaptation into policy-making and sectoral financial planning. The project aims to fill these gaps by fostering better access to knowledge and technical expertise, thus empowering policymakers and practitioners to implement effective climate responses. This comprehensive consideration of Jordan's socioeconomic and institutional landscape ensures that the project not only mitigates immediate climate vulnerabilities but also builds long-term resilience.

3.5 Country ownership

Scale: High

29. The proposed project demonstrates strong alignment with Jordan's nationally determined contributions and various national strategies, demonstrating robust country ownership. It directly corresponds with national plans such as the Aligned National Action Plan to Combat Desertification 2015–2020 and the National Biodiversity Strategy and Action Plan 2015–2020, aiming to enhance biodiversity, mitigate desertification and integrate climate resilience into agricultural and landscape management. Furthermore, it supports the goals of the National Climate Change Policy 2013–2020 by focusing on increasing the adaptive capacity of water and agricultural resources and conserving ecosystems, which are critical areas under Jordan's climate adaptation strategy. This alignment ensures that the project is not only relevant but also enhances the likelihood of successful implementation and sustainability.

30. The institutional framework for the project involves key national authorities such as the Ministry of Environment (MoEnv) and the Climate Change Directorate, which coordinate policy implementation and stakeholder engagement. This structure underpins the project's foundation in Jordan's environmental governance, enhancing its integration into national policies and strategies. The establishment of the Climate Change Directorate as a focal point for climate change activities and its role in coordinating this project ensures high-level policy support and alignment with Jordan's commitments under the United Nations Framework Convention on Climate Change. Such integration is crucial for the project's success and its ability to influence broader policy and institutional changes.

31. Extensive stakeholder engagement has been a core component of the project's development, ensuring that the project reflects the needs and inputs of a broad range of stakeholders, including government bodies, NGOs and local communities. Detailed consultations, including in-country missions and workshops, have been conducted to refine the project's design and objectives, ensuring that they are well informed by local conditions and challenges. This thorough engagement process supports the project's relevance and feasibility, reflecting a well-rounded approach to understanding and addressing the specific climate vulnerabilities and adaptation needs of Jordan.

3.6 Efficiency and effectiveness

Scale: Medium to high

32. The total investment from GCF is USD 44,904,203, supplemented by USD 15,605,748 in co-financing from the Government of Jordan and consortium of NGOs, with an additional USD 13,133,000 in parallel finance. This financing structure indicates a substantial leverage of additional funding relative to the GCF contribution, emphasizing the project's efficiency in mobilizing financial resources for large-scale climate adaptation initiatives.

33. In terms of cost-effectiveness, the project presents a favourable economic profile. The economic analysis predicts a net present value of USD 105,279,874 and an economic rate of return of 25 per cent, which significantly exceeds typical thresholds for such projects, indicating a high level of economic justification for the investment. The benefit–cost ratio of 3.00 further underscores the project's effectiveness, suggesting that the economic benefits substantially outweigh the costs. These figures are grounded in well-established practices and reflect a thoughtful application of adaptation measures tailored to Jordan's specific climate challenges, particularly in water resources management.

34. Furthermore, the proposal details the application of best practices in ILWRM, which are essential for addressing Jordan's critical water scarcity issues exacerbated by climate change. The project leverages lessons learned from similar initiatives and incorporates adaptive management strategies, participatory monitoring and stakeholder engagement, ensuring that interventions are both sustainable and community-focused. This approach is expected to enhance resilience across vulnerable communities and ecosystems, demonstrating a comprehensive application of international best practices tailored to local conditions.

IV. Assessment of consistency with GCF safeguards and policies

4.1 Environmental and social safeguards

35. **Project overview.** The project aims to reduce the negative impacts of climate change on water resources in Jordan through an ILWRM approach. It has four component covering enhanced governance, resource management practices such as restoration, protection and use of water resources coupled with knowledge management and climate change awareness and outreach activities. The environmental and social co-benefits of the project include reduced sedimentation as a result of flood attenuation, enhanced biodiversity due to improved habitats, decreased water pollution due to reduced eutrophication, sedimentation and improved soil quality. Social co-benefits include improved health and sanitation due to better access to clean water and improved conflict management on water resources.

36. **Environmental and social risk category and safeguards instrument.** The project falls under environmental and social safeguards category B classification. The Secretariat confirms the risk category, which is also within the environmental and social risk accreditation level of the accredited entity (AE). Its site-specific interventions are small scale, and while potential negative impacts may vary, they are expected to be minor, dispersed over an area and reversible. Additionally, the development and execution of integrated watershed management plans, which can potentially negatively affect specific groups or sectors, will be conducted through a consultative and participatory approach. This ensures that any potential adverse impacts are carefully considered and addressed during the planning phase. Consequently, the environmental and social risks associated with this project are assessed as low to moderate. Since the final details of proposed activities (e.g. plans, protocols, specific locations of interventions) will still be determined through further engagement with stakeholders during the implementation of the project, an environmental and social management framework (ESMF) has been prepared.

37. **Compliance with GCF environmental and social safeguards (ESS) standards.** The following discussion describes the environmental and social risks and/or impacts of the project and the proposed mitigation measures.

38. **ESS1 (Assessment and Management of Environmental and Social Risks and Impacts).** The ESMF provides an assessment of the potential social and environmental impacts and risks of the project activities. It provides background information about the physical, biological and socioeconomic profile of the project area, including information about water resources and waste management. The ESMF has assessed that the project will have mainly positive impacts on the biophysical and human environment. The negative impacts have been identified and their corresponding mitigation measures provided.

39. The ESMF enumerates the country laws and regulations and international conventions that are of relevance to certain aspects of the project. It also discusses the biophysical and socioeconomic profile of the target areas, as well as the procedures and requirements for assessing and managing the environmental and social impacts and risks of individual subprojects/activities under the project. An environmental and social management system will also be established during the operation of the water fund.

40. **ESS2 (Labour and Working Conditions).** The project will involve small-scale work activities which would create temporary employment. The ESMF has identified the issues of child labour and migrant workers (e.g. trafficking in persons) as potential sources of risks which will be managed by adhering to the procurement principles and national law requirements of the AE. Measures include maintaining regular inspection of occurrence and cases of trafficking in persons, including the recording and reporting procedures for documenting and reporting

such cases. Constant monitoring of the labour status and compliance of contractors with national labour law regulations will also be implemented.

41. **ESS3 (Resource Efficiency and Pollution Prevention).** The project is expected to significantly contribute to resource efficiency in terms of fuel and water use as addressing these issues is a major aim of the project. In terms of pollution, the project is not expected to generate significant amounts of pollutants apart from minor noise, dust, air emissions from vehicles, sedimentation and small amounts of solid and liquid wastes during construction. These can be addressed through standard mitigation measures that can be implemented by the contractors. Pollution from pesticides and other agrochemicals are also expected to be reduced in the long run as a result of the adoption of integrated pest management principles and sustainable practices in the agricultural production to be supported under the project.

42. **ESS4 (Community Health, Safety and Security).** The project poses construction-related safety hazards to nearby residents, visitors and passers-by. However, these are expected to pose only minor risks to community health and safety; the construction of rockfill dams and installation of solar panels are the only major works under this project and are not large-scale operations and will not be concentrated in one area. The risk of accidents at construction sites and routes will be addressed through ensuring that safety measures rules and regulations are applied on site and in transport routes.

43. Changes in the water regime of the area could also alter the ecological balance and result in negative impacts such as resurgence of vector-borne diseases. The increased availability of water due to the construction of rockfill dams could create an environment where vector-borne diseases could proliferate. This aspect will be further explored in the conduct of the environmental and social impact assessment and, if necessary, integrated vector management approaches will be implemented. Potential stakeholder conflicts will be addressed within the project's consultative mechanisms such as those of the watershed management committees and participatory monitoring processes.

44. **ESS5 (Land Acquisition and Involuntary Resettlement).** The project is not expected to involve displacement or relocation of people and the infrastructure development activities are likely to be undertaken in government-owned facilities (e.g. King Abdullah Canal for the 1 MW floating solar panels) or in public lands (e.g. wadi areas for rockfill dams). However, there is a possibility of local communities experiencing temporary restriction of use or access to the forests and rangelands being subjected to rehabilitation. The temporary restrictions are expected to be relaxed as restoration and rehabilitation activities progress. The project will aim for a maximum level of local user engagement and participation the development of land-use restrictions/plans/schemes based on mutually acceptable agreements. Moreover, the project will provide alternative or additional livelihood options. The project's grievance redress mechanism (GRM) will also ensure that local users can express their concerns with regard to any restriction of access and/or failure to respect their user rights.

45. **ESS6 (Biodiversity Conservation and Sustainable Management of Living Natural Resources).** In the long run, the project is deemed to have mostly positive impacts on the biological environment, particularly in terms of conservation of fragile habitats, enhancing biodiversity and conservation of genetic resources. These positive impacts are expected as a result of the adoption of ecosystem-based approaches in restoration efforts; effective management of existing protected areas and hotspots; promotion of nature-based alternative livelihoods; adoption of sustainable agricultural practices; and effective enforcement of appropriate policies and regulations, among others. Increased activities in the project area, particularly around the site of the facilities during construction and later during operation will have some negative impacts on natural habitats, including vegetation loss; alteration of surface morphology and natural water run-off schemes; and wildlife disturbance. The construction activities and subsequent operation and maintenance of facilities may also result in vegetation loss, littering and disturbance to wildlife around the facilities. It is therefore recommended that

the formulation of the integrated landscape and watershed management plans be based on a thorough ecological study and robust ecological monitoring for each watershed and that an adaptive management approach can also be employed, when needed.

46. **GCF Indigenous Peoples Policy and ESS7 (Indigenous Peoples).** The presence of Indigenous Peoples in the area of the project is probable, and it has been confirmed in the vicinity areas, which and an Indigenous Peoples Planning Framework has been provided. Potential impacts on Indigenous Peoples include temporary restrictions on access to land and natural resources and changes in the current use patterns of their territories. Restrictions on land and resource access will only proceed once the free, prior and informed consent is secured, and the modalities of implementation are consulted and agreed upon with Indigenous Peoples. Documentation and diffusion of Indigenous knowledge in a platform is subject to securing the free, prior and informed consent, and the attribution of intellectual property rights to the knowledge holders. It is recommended that the AE engage with the GCF Indigenous Peoples Advisory Group on the relevant Indigenous Peoples' issues that may be encountered.

47. **ESS8 (Cultural Heritage).** Jordan is renowned for its abundant historical and cultural heritage sites. Hence there is a risk that project activities could inadvertently damage artifacts or historical sites. The safeguards screening conducted by the AE has confirmed this risk. To address this, the ESMF required the conduct of comprehensive cultural assessments for each project development site where information about the presence of cultural heritage resource is unavailable. It also required that construction contract documents incorporate provisions allowing for the suspension of construction and excavation activities in the event of any discoveries of antiquities or archaeological items. Chance finds will be promptly reported to the Director of the Department of Antiquities or the nearest general security centre.

48. **Sexual exploitation, abuse and harassment (SEAH).** The ESMF identified the contextual risks related to gender-based violence (GBV) and SEAH in Jordan, underpinned by the Syrian refugee crisis and increasing vulnerability of the displaced populations. The AE also identified that insufficient legal frameworks and enforcement contribute to the persistence of GBV and SEAH in the target country. Against such contextual factors, the project may exacerbate SEAH risks owing to various factors, such as the project's close interaction with highly vulnerable groups, construction activities to be undertaken and the use of security forces. The project will integrate SEAH consideration across all project activities to identify and mitigate the project-driven SEAH risks, including to assessed and addressed specific SEAH risks at the subproject level in line with the SEAH due diligence process outlined in the ESMF and as part of the environmental and social management system for the water fund. The AE will develop a methodology for assessing potential GBV and SEAH risks during the project implementation and during the project operation. The ESMF requires a zero-tolerance policy for all employees or implementing partners. The AE will provide training and capacity-building support at the community level and operate a parallel process in the GRM for SEAH-specific complaints in a survivor-centred manner. The AE has also identified and will leverage SEAH service provisions based on a network of GBV and SEAH survivor support organizations that are active in the Jordan Valley.

49. **Implementation arrangements.** The project will be overseen by a project steering committee and implemented by a national-level project management unit (PMU). Hosted by MoEnv, the PMU will consist of specialists including an Environmental and Social Safeguards Officer and a Gender Officer. The Environmental and Social Safeguards Officer will be engaged as the lead technical advisor on environmental and social management matters. The role includes overall implementation of the ESMF and oversight of safeguards across the project components. This includes the implementation and oversight of specific mitigation actions as described in the ESMF; coordination and implementation of additional safeguards assessments and safeguards instruments; monitoring and reporting of the environmental and social performance of the project; assessment and capacity-building towards executing entities and/or

implementing partners; and coordination with other members of the PMU, such as the Monitoring and Evaluation Officer and the Gender Officer where required.

50. **Stakeholder engagement and information disclosure.** The project was crafted through an extensive consultation process involving various stakeholders encompassing workshops, technical meetings, interviews and field visits and is detailed in annex 7 to the funding proposal (Stakeholder Consultations and Engagement Plan). The interactions involved a wide spectrum of participants, including NGOs, government bodies, local communities and stakeholders at both the national and the local level, including farmers from the three designated project areas.

51. The ESMF and the Stakeholder Consultations and Engagement Plan also provided a structured plan for engaging stakeholders throughout the project's execution. Prior to initiating any on-the-ground activities, a comprehensive Stakeholder Engagement Plan will be prepared by the PMU. A framework outlining key components and requirements for this implementation phase Stakeholder Engagement Plan has been provided, including an outline of activity-level stakeholder engagements which aligns with the participatory planning requirements that have been integrated into the design of project activities such as the development and implementation of the ILWMP.

52. **Grievance redress mechanism.** A project-level GRM, which will be made accessible to all stakeholders, has been prepared and presented as part of the ESMF. The mechanism will utilize various communication channels (email, telephone, contact person, personal address) and would consist of four levels. Concerns can be lodged at the lowest appropriate levels, starting at the PMU, followed by the Representative of the International Union for Conservation of Nature (IUCN), the IUCN regional office level and the IUCN Head of Oversight. The Safeguard Specialist at the PMU will act as the GRM focal person. The Project Manager of the PMU will be responsible for documenting and reporting on any grievances received and how they were addressed. In line with the GCF Indigenous Peoples Policy, the GCF indigenous peoples focal point will be available for assistance at any stage, including before a claim has been made.

4.2 Gender policy

53. The AE provided a gender assessment and action plan that complies with the GCF gender policy requirements. The project acknowledges the critical role that women play as agents of change in climate resilience, while also noting that women are disproportionately affected by climate change with unequal access to finance, resources and assets, barriers to decision-making and limited mobility, to name but a few disadvantages. The gender action plan provides interventions that address the challenges faced by women in ILWRM sectors and covers baseline indicators, targets, timelines, gender and social expertise and budgets. The interventions included in the gender action plan reflect the findings of desk reviews, stakeholder consultations and recommended actions by the Gender Unit of the Ministry of Water and Irrigation.

54. The project will focus on addressing the practical needs of women by including interventions through the water fund for the Jordan Valley, which focuses on improving local women's access and benefits from credit and grant financing to support ILWRM initiatives in their own communities. Further, it will work on enhancing women's agency in dealing with climate change risks by including activities targeted exclusively at women. The Water Wise Women initiative is central to the gender action plan and includes measures to support women's empowerment and decision-making capabilities, including strengthening rural women's voices and leadership capacities to advocate for gender-sensitive strategies and policies related to adaptation to climate change.

55. The AE will hire a full-time social inclusion and gender expert to mainstream gender throughout the project along with gender consultants to support the implementation of specific gender action plan activities. The gender action plan is inclusive of women, female-headed households, youth and the elderly. It includes measures to mitigate GBV and provides women with access to grievance mechanisms.

56. The main risks associated with the success of the gender action plan are the entrenched sociocultural norms that have led to Jordan being consistently ranked as one of the countries with the lowest gender equity; and lack of land and resource tenure for women that threatens the sustainability of the impact of the project's investments on women's empowerment and well-being.

4.3 Risks

4.3.1. Overall programme assessment (medium risk)

57. The Secretariat assesses overall programme risk as medium.

4.3.2. Accredited entity/executing entity capability to execute the current programme (low risk)

58. Project execution will be by the national executing entity and national designated authority, MoEnv, and the lead executing entity IUCN, which is also an AE to GCF. Both executing entities have experience of operating in the country.

4.3.3. Programme-specific execution risks (medium risk)

59. Political risk: political unrest due to ongoing conflict in the region may spill over into Jordan with subsequent safety risks and may lead to political instability and loss of project assets. As mitigants, the AE adequately identified this risk and will inform GCF of any adverse change in the risk assessment during implementation on a regular basis. Furthermore, the AE has privileges and immunities in the Host Country.

60. Operationalization of the water fund: a weak appetite to co-invest in the water fund may result in insufficient liquidity of the revolving resources that are required and intended, on a revolving resources basis, for delivering replicability and scalability to target communities and water conservation efforts in other parts of the country. However, the Secretariat takes comfort from the inclusion in the term sheet of the delivery of a detailed structure and strategy for the water fund subject to a condition prior to the third disbursement to be reviewed and approved by the Secretariat.

4.3.4. Compliance risk (medium risk)

61. The project addresses water programs in Jordan. Generally, the proposed activities themselves do not pose an unusually high level of risks for money laundering. The Government of Jordan has made great strides in its efforts to combat money laundering and terrorist financing, resulting in the removal of Jordan from the FATF "great list" in recent months. Nevertheless, there are as noted above, a number of conflicts and instances of political unrest in the region that would merit continued scrutiny by the AE in monitoring for, and addressing, potential instances and other vulnerabilities. To that extent, the AE has indicated its ability to undertake those precautions.

62. Based on a review of the proposed activities of the project, and taking into account, the potential risks that regional conflicts may have with regard to money laundering, terrorist financing, and related prohibited practices, and taking into account the proposed measures of

the AE to monitor and take action to prevent such instances, the Compliance risk for this project is rated as “MEDIUM”.

4.3.5. GCF portfolio concentration risk (low risk)

63. In the case of approval, the impact of this proposal on the GCF portfolio concentration in terms of results area and single proposal is not material.

4.3.6. Recommendation

64. It is recommended that the Board consider the above factors in its decision.

Summary risk assessment	
Overall project/programme	Medium
Accredited entity/executing entity capability to implement the project/programme	Low
Project-specific execution	Medium
GCF portfolio concentration	Low
Compliance	Medium

4.4 Fiduciary

65. The United Nations Environment Programme (UNEP) will serve as the AE overseeing the project, with MoEnv and IUCN acting as executing entities. MoEnv will manage component 1, while IUCN will handle components 2 to 4. Responsibilities for the AE include overseeing implementation, financial management, evaluation, reporting and project closure in coordination with a national-level project steering committee and PMU. UNEP will ensure compliance with the Accreditation Master Agreement and the funded activity agreement, monitoring financial management through established procedures, including annual audits and internal controls.

66. The project will adhere to United Nations regulations, with UNEP managing fiduciary aspects. Financial management and procurement will follow the UNEP Programme Manual, guided by International Public Sector Accounting Standards and United Nations financial rules. Each executing entity will sign subsidiary agreements, clarifying roles and ensuring compliance with United Nations anti-fraud frameworks and GCF policies. The legal capacities of MoEnv and IUCN allow them to independently contract and manage financial resources for their respective components.

67. A Financial Officer within the PMU will oversee financial operations, ensuring compliance with United Nations financial regulations. This includes quarterly fund requests and reporting, supervised by a UNEP Task Manager. Project funding will be disbursed in tranches to executing entities upon meeting specified conditions in the project cooperation agreement and the internal cooperation agreement.

68. MoEnv, as a government authority, follows national financial rules, and its capacity for procurement will be assessed by UNEP to ensure alignment with United Nations standards. This assessment will influence the procurement monitoring plan, a critical part of the project cooperation agreement shared with GCF as mandated. The project's equipment investments will also adhere to United Nations procurement procedures, with MoEnv conducting an annual compliance audit of all project expenditures.

4.5 Results monitoring and reporting

69. The project aims to deliver benefits under Adaptation Results Areas 1, 2 and 4. It seeks to reach 246,942 direct beneficiaries and 756,000 indirect beneficiaries (accounting for 2.2 and 7.2 per cent respectively of the national population). The project has an implementation period of 8 years and a lifespan of 25 years. The project was reviewed over five rounds, where detailed comments were provided to the AE for improving results management under the intervention.

70. The project theory of change adequately captures the results that are expected to be generated at different levels, including outcomes, co-benefits, outputs and activities. These are linked to underlying barriers and risks, and the theory of change also identifies assumptions that are being made for the project results to materialize.

71. The logical framework provides a clear account of the results, associated indicators and targets under the project. This includes relevant GCF results areas and associated core and supplementary indicators. Importantly, project-level outcome and co-benefit indicators are also identified along with clear mid-term and final targets.

72. The supporting annexes related to monitoring and evaluation were found to be clear and detailed. Relatively minor observations relating to annex 5 (Implementation Timeline) have been noted and shared with the AE during the latest round of review.

4.6 Legal assessment

73. The Accreditation Master Agreement was signed with the Accredited Entity on 15 December 2016, and became effective on 20 February 2017, which was amended and restated pursuant to a first amendment and restatement agreement dated 13 July 2023, and which became effective on 2 August 2023 (the “**AMA**”).

74. The Accredited Entity has provided a certificate confirming that it has obtained all internal approvals, and it has the capacity and authority to implement the project.

75. The proposed project will be implemented in the Hashemite Kingdom of Jordan, a country in which GCF is not provided with privileges and immunities. This means that, amongst other things, GCF is not protected against litigation or expropriation in this country, which risks need to be further assessed. The GCF Secretariat provided a draft agreement on privileges and immunities and a background note to the Government of the Hashemite Kingdom of Jordan most recently on 19 November 2022.

76. The Heads of the Independent Redress Mechanism (IRM) and Independent Integrity Unit (IIU) have both expressed that it would not be legally feasible to undertake their redress activities and/or investigations, as appropriate, in countries where the GCF is not provided with relevant privileges and immunities. Therefore, it is recommended that disbursements by the GCF are made only after the GCF has obtained satisfactory protection against litigation and expropriation in the country or has been provided with appropriate privileges and immunities.

77. To address the matters raised in this section, it is recommended that any approval by the Board is made subject to the following conditions:

- (a) Signature of the funded activity agreement in a form and substance satisfactory to the GCF Secretariat within 180 days from the date of Board approval; and
- (b) Completion of the legal due diligence to the satisfaction of the GCF Secretariat.

Independent Technical Advisory Panel's assessment of FP232

Proposal name:	Jordan Integrated Landscape Management Initiative (JILMI)
Accredited entity:	United Nations Environment Programme
Country:	Jordan
Project/programme size:	Medium

I. Assessment of the independent Technical Advisory Panel

1.1 Overview

1. This funding proposal (FP)¹ for a medium-sized adaptation project in environmental and social safeguards category B is submitted by the Government of Jordan through the United Nations Environment Programme (UNEP) functioning as the proposed project's accredited entity (AE). The executing entities (EEs) are the Government of Jordan, represented by the Ministry of Environment (MOENV), and the International Union for Conservation of Nature (IUCN) through its Regional Office for West Asia (IUCN-ROWA) based in Amman, with a consortium of local non-governmental organizations (NGOs) providing co-finance and technical support.

2. The total cost of the funding proposal is USD 60.5 million, of which USD 44.9 million is proposed as GCF finance and USD 15.6 million as co-finance from the Government of Jordan (GOJ) and the consortium of NGOs – see section 1.6 below for more detail. The project implementation period is eight years, while benefits are calculated over a lifespan of 25 years.

3. The proposed project's objective is to reduce the negative impacts of climate change on water resources in Jordan by improving aquifer recharge using a landscape-level approach. The project focuses on three target watersheds of the northern Jordan Valley – the Yarmouk, Amman Zarqa and Jordan Rift Valley Basins – where agriculture and human well-being are threatened by water scarcity worsened through climate change. The proposed integrated land and water resources management (ILWRM) approach includes: i) improved ecosystem management through land restoration, improved agricultural practices and managed aquifer recharge; ii) rainwater harvesting; iii) evaporation reduction from the King Talal Dam; and iv) strengthened governance, with long-term finance through a Water Fund for upscaling ILWRM across the Jordan Valley.

4. The project involves implementation of four interlinked technical components (as well as monitoring and evaluation, and project management):

Component	Outputs	GCF funding	Co-finance
1. Climate change adaptation governance in Jordan	- Strengthened governance and capacity for climate-resilient ILWRM - Economic assessments developed quantifying the net benefits of ILWRM as an adaptation approach	USD 1.62 million	USD 0.72 million

¹ Funding proposal package for the Jordan Integrated Landscape Management Initiative was received from UNEP on 3 May 2024.

2. ILWRM under conditions of climate change	- Climate-resilient ILWRM practices and ecosystem restoration implemented in three watersheds in the Jordan Valley - A Water Fund established in the Jordan Valley to maintain landscape investment for enhanced water resources	USD 23.1 million	USD 10.53 million
3. Improved protection, supply and use of water resources under climate change	Improved protection, supply and use of water for domestic purposes under climate change conditions	USD 15.21 million	USD 3.13 million
4. Knowledge management and awareness on climate change adaptation	Improved knowledge and awareness of climate change adaptation for landscape and water resources management	USD 3.23 million	USD 0.65 million

1.2 Impact potential

Scale: High

5. The project targets three of the GCF results areas for adaptation, aiming to enhance the resilience of i) the most vulnerable people and communities; ii) health and well-being, and food and water security; and iii) ecosystems and ecosystem services.

6. **Adaptation impact.** The funding proposal highlights the hot and arid baseline climate of Jordan, in which over 90 per cent of its limited annual rainfall is lost through evaporation, and shows how water stress² is worsening with climate change. Severe droughts have become more common in the past 50 years, with the worst recorded drought in 900 years occurring between 1998 and 2012, when rainfall decreased by up to 70 per cent compared with long-term averages, resulting in severe agricultural, socioeconomic and environmental impacts. Climate projections under Representative Concentration Pathway (RCP) 4.5 and RCP 8.5 scenarios for the period 2070–2100 anticipate an increase in average annual temperature by 2–4°C, and a reduction in annual average rainfall, with extended dry periods, a shorter rainfall season, and increased frequency of high-intensity rainfall events that can also cause flash flooding. Staple barley production is expected to decline by up to 50 per cent under future climate conditions, with pressure also on other crops and animal fodder, severely affecting local food supplies.

7. Natural ecosystems and agroecosystems in the country are already being degraded by non-climatic drivers such as overgrazing, unregulated fuelwood harvesting, and cultivation on steep slopes, accelerating water run-off and preventing infiltration. With climate change, it is anticipated that shifts in rainfall will reduce river baseflow and aquifer recharge, and higher temperatures will increase evaporation rates, reducing available water supplies. At the same time, rising temperatures will increase demand for fresh water for irrigation and domestic consumption, encouraging further extraction of groundwater. These combined effects are expected to worsen the existing water deficit of Jordan (i.e. the gap between supply (from all water resources, including non-renewable) and demand in the country), which was assessed at 265 million m³ per year in 2022, and is expected to climb steeply to 835 million m³ per year by 2050.³

8. Project work on the ground in the Yarmouk, King Talal Dam and Kafraïn areas of the Jordan Valley in components 2 and 3 (under IUCN), will engage communities in implementing participatory climate-resilient integrated watershed management plans, developed in

² 2023 data from the [Aqueduct Water Risk Atlas of the World Resources Institute](#) shows that Jordan is one of 25 countries facing extremely high water stress each year.

³ Figures from [Jordan's fourth national communication to the United Nations Framework Convention on Climate Change \(2022\)](#). (Note: the FP quotes the third national communication submitted in 2014.) The water deficit when calculated as the gap between supply from **renewable** water resources and demand is significantly higher at 422 million m³ per year.

component 1 (under the MOENV). Implementation will involve restoring 8,000 ha of forest, rangeland, riparian and wadi ecosystems across the three sites. Protocols will be developed for both passive restoration, via the traditional Hima system,⁴ and active restoration through revegetating degraded areas and afforestation with saplings raised in community nurseries. Active restoration will be complemented by small-scale, in-field rainwater harvesting interventions (e.g. bunds, planting pits, Negarim and Vallerani basins, and run-off strips) and irrigation (e.g. porous hose and deep pipe), improving soil moisture and plant survival in the sites. The independent Technical Advisory Panel (iTAP) welcomes section 8 of the feasibility study on “Best practices and lessons learnt”, which illustrates and details best practice restoration techniques relevant to Jordan, and encourages the AE to consider producing this excellent resource in Arabic as a technical manual.

9. Component 2 also includes support to women and men on new climate-resilient natural resource-based livelihoods – including vegetable gardening (hydroponics and permaculture) and beekeeping – and introduction of water-efficient practices, including selection of climate-resilient crop varieties and permaculture. Learning exchanges will be promoted between communities; training will be provided on equipment maintenance; and private-sector champions, including women, will be supported to mentor farmers to develop microenterprises in post-harvest processing, and to access markets and finance. In component 3, water resources will be enhanced through installing 3,000 rooftop rainwater harvesting tanks on community buildings and homes, with community monitoring and management plans through multi-stakeholder Water Management Committees (established under component 1), including women. Floating solar panels will be installed on the King Abdullah Canal, which will reduce water loss through evaporation, while supplying 1 MW of renewable energy into the local grid for sustainable water pumping.⁵

10. In addition to restoring ecosystem productivity and enhancing soil moisture, ecosystem-based adaptation (EbA) techniques will also be used to replenish shallow aquifers to help adapt to water scarcity. Permeable rockfill dams will be established in wadis⁶ at 12 sites, allowing for managed aquifer recharge, with small check dams established in upstream gullies to trap sediment and further enhance infiltration. As groundwater is replenished, 10 freshwater springs will be rehabilitated, with refurbishment of irrigation canals. The project’s EbA approach represents one element of a multi-pronged strategy by government and partners that includes increased agricultural water efficiency,⁷ desalination, water treatment for reuse, as well as more controversial mega-projects that pump up irreplaceable fossil water from deep aquifers.⁸ It is hoped that the project’s combined interventions in aquifer recharge, ecosystem restoration, rainwater harvesting and evaporation prevention will lead to an increase in the country’s annual water supply of approximately 8.9 million m³ by project end, with a potential cumulative increase of approximately 154.5 million m³ by the end of the project’s 25-year lifespan.

11. The project is expected to directly benefit⁹ a total of 246,942 people (45 per cent of them female) in the Jordan Valley by reducing their climate vulnerability. The FP shows the spread of beneficiaries across i) the Yarmouk site – Khaled Bin Al Waleed and Al Seru

⁴ Originating from an ancient system of sacred sites, a “hima” is a reserved pasture, where a community’s trees and grazing lands are protected from indiscriminate harvesting on a seasonal basis, allowing for natural regeneration of productivity.

⁵ The AE has confirmed that no new boreholes will be drilled, and no new solar photovoltaic pumps will be installed, with GCF funding (in written responses to iTAP questions, received from UNEP on 22 May 2024).

⁶ A “wadi” refers to a river channel or valley that is dry except in the rainy season.

⁷ Tackled in the GCF project, Building resilience to cope with climate change in Jordan through improving water use efficiency in the agriculture sector project (FP155).

⁸ <https://doi.org/10.3390/w15213729>

⁹ Direct beneficiaries are all those who will benefit directly from participating in a project activity or the result of an activity.

municipalities; ii) the King Talal Dam site – New Ardah, Deir Alla and M’addi municipalities; and iii) the Kafraïn site – Shouna Alwasta and Mahes municipalities. Annex 33 to the FP shows the detailed calculations for anticipated beneficiary numbers. For example, under output 3.2, 1,200 smallholder farming households (approximately 6,000 people, i.e. an average of 5 people per household) across the three sites will benefit from on-farm interventions, while 2,800 households across the sites (approximately 14,000 people) will benefit directly from rainwater harvesting infrastructure installed on rooftops. Also included are direct benefits from improved protection, supply and use of water supplies for domestic and/or agricultural usage, and benefits from enhancements in the quality and quantity of ground and surface water through managed aquifer recharge and spring rehabilitation.

12. As ILWRM interventions are scaled up via the Water Fund, to be created during implementation, it is anticipated that an additional 4,000 ha of land could be placed under improved management for water-resource benefits, with at least one additional (adjacent) municipality per watershed site benefiting directly. Water storage potential would thus be increased across these 4,000 ha in addition to the 8,000 ha to be restored directly by the project. Water storage potential¹⁰ is derived from modelling the effects of slowing overland water flows on water infiltration into the groundwater. As the flow slows, so the water spreads out over a larger area and the time available for the water to seep into the ground increases (see section 11.6 of the feasibility study). If the Water Fund is successful in leveraging resources, it could scale up the set of restoration and land management interventions in component 2 to target the entire 12,000 ha, and the additional 74,763 beneficiaries in this area are thus included in the number of direct beneficiaries of the project.

13. A total of 756,000 people (48 per cent of them female) across the broader Jordan Valley are expected to benefit indirectly from: i) enhanced water resources in the northern Jordan Valley; ii) strengthened local governance to implement climate-resilient ILWRM; iii) reduced water loss through evaporation from the King Abdullah Canal; and iv) improved awareness of the impacts of climate change and the adaptation options to address them. Although not included in the indirect beneficiary calculations, the project proponents note that interventions at the regulatory and institutional levels, and the example set by the establishment of a Water Fund, could indirectly benefit the Jordanian population as a whole, amounting to an estimated 10 million people.

14. Monitoring of the project results in terms of success in groundwater replenishment is crucial, both for reporting to GCF, and for generating evidence for the effectiveness and cost-effectiveness of EbA measures. This evidence will be crucial in making the case for corporate and utility water users downstream to contribute to the Water Fund for ongoing catchment management to secure quality and quantity of water supply. The iTAP therefore welcomes the inclusion of explicit monitoring of groundwater levels by water management committees (WMCs), who will take regular piezometer readings at 12 sites downstream of the rockfill dams. The iTAP notes that readings should be undertaken at the start and end of the dry season each year, and combined with data on the amount of water extracted over the last period (which must be continuously monitored by WMCs or other water-user associations), in order to determine the level of the groundwater and the extent to which upstream landscape based interventions have been able to replenish it.

15. The iTAP assesses the impact potential of the project as high.

1.3 Paradigm shift potential

Scale: Medium

16. The proposed project intends to catalyse an institutional paradigm shift within the water sector in Jordan from a compartmentalized, sector-specific approach to land and water

¹⁰ Referred to as “attenuation potential” in the funding proposal.

resources management, to one that is integrated, participatory and adaptation focused, contributing to climate-resilient sustainable development in the country.

17. **Enabling environment.** The project includes legal work to support the Ministry of Water and Irrigation (MWI) to expand the remit of the Jordan Valley Authority (JVA) to include ILWRM matters explicitly, and to establish multisectoral WMCs in the three target watersheds. The project will support the establishment and capacity development of the new WMC structures, aiming to achieve dialogue and cooperation between upstream and downstream stakeholders and undertake collective management actions. In response to a question from iTAP about the government's openness to these new structures and willingness to provide budget support beyond the project, the AE indicated in general terms that "Government buy-in has been shown in multiple consultations, as well as the letters of approval received for the project from MWI and other ministries, through the climate change national committee".¹¹

18. At national scale, the project design includes activities to strengthen the mandate and capacity of major government agencies, including the ministries of environment, agriculture, and water and irrigation. An analysis of relevant national policy and legislation will be undertaken, to develop policy briefs, and propose amendments to relevant legislation, though there is no guarantee that these will be made. Other interventions to remove barriers to integrated climate-risk informed resource management will include the development of land-use guidelines, standards and protocols for climate-resilient watershed management; training of officials to use the results of climate analyses in planning and decision-making, including disaster risk management; and generating evidence of the economic benefits of ILWRM to project beneficiaries for use by policymakers and planners.

19. **Knowledge-sharing and learning.** The proposed project includes specific interventions to improve the current knowledge management system for climate-change adaptation in Jordan, extending the project's influence. This will include the establishment of a national knowledge repository of lessons learned, likely based in one of the country's universities, and gender-responsive best practices for climate-change adaptation, focusing on the ILWRM approach; the training of WMCs and other user groups in the use of climate data and knowledge tools; and the implementation of an outreach programme, with local knowledge hubs supported by trained local officials and community members. Additionally, relevant educational practitioners (50 per cent of whom will be women), will be trained on integrating climate-change adaptation into the syllabuses and curricula of Jordanian universities.

20. **Scaling up of results.** At a technical level, the project interventions are shown to have strong potential for upscaling. The FP outlines four possible scenarios for upscaling the key project interventions (i.e. ecosystem restoration, managed aquifer recharge (rockfill dams), rooftop rainwater harvesting, and evaporation reduction (floating solar panels)). Table 11 in the FP shows the potential impact on water supply under each of the scenarios – conservative, mildly conservative, optimistic and best case scenarios. The AE calculates that, if the ecosystem restoration interventions alone were to be upscaled across the entire northern Jordan Valley (1,147,600 ha), the potential cumulative increase in water resources over 25 years could range from 34 million m³ under a conservative scenario, to 344 million m³ under the best-case scenario. The combined effect of all four interventions at maximum potential over 25 years could make a cumulative total available of between 297 and 3,059 million m³ (or an average of 11.88 to 122.36 million m³ per year¹²).

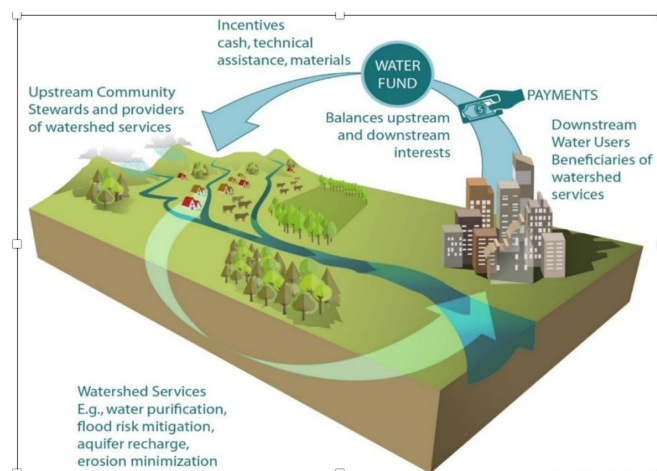
21. **Sustainability of results.** The project's main strategy for resource mobilization is to sustain the gains of the interventions in the three target sites, replicate these in other sites in the target basins, and scale them up across the northern Jordan Valley (and potentially beyond), is

¹¹ Written responses to iTAP questions, received from UNEP on 22 May 2024.

¹² By the calculations of the iTAP, this maximum potential contribution would represent a significant 29 per cent of Jordan's current annual water deficit (when defined as the gap between supply and demand based on renewable water resources – see footnote 3).

the establishment and capitalization of a Water Fund (activity 2.1.4). The fund is planned as a mechanism for leveraging financial resources for watershed management in the long term, from water-dependent stakeholders in the private sector (industry and commercial agriculture) and public sector (municipalities/utilities). Such a fund could ensure the ongoing availability of funding for the maintenance and restoration of ecosystems in the three target watersheds and beyond. The project includes activities to engage multiple stakeholders in the design of the fund, and to establish the necessary legal mechanisms and operational capacity. Anchored in the JVA, the fund will be governed by a multi-stakeholder public-private steering committee that includes representatives of WMCs from the target watersheds, NGO partners, the three key ministries, water utilities, and private-sector partners.

Model for a water fund (fig. 2 from the FP)



22. Activity 2.1.4 also makes USD 7.5 million available for community-level ILWRM subprojects through a GCF project on-granting facility, which “will function alongside the Water Fund’s grant mechanism”. The iTAP asked the AE if the option of using the GCF on-granting funds to kick-start the actual Water Fund had been considered, thereby leveraging immediate private-sector contributions by using the momentum created by GCF investment. The AE explained that establishing the Water Fund will be a lengthy and complex process, requiring that legal mechanisms be put in place in the project’s third year, with the fund unlikely to become operational before the fourth year. The project’s on-granting facility would need to start earlier than this, but can nevertheless help to “establish and embed the on-granting practice that will be upscaled by the Water Fund”.¹³

23. Key potential investors in the Water Fund are listed in the FP – large agri-companies, the Alfaqih Group and Haddadin Farms; large industry, including the Manaseer Group, Arab Potash Company and Jordan Phosphate Mines Company; and water utilities such as the Yarmouk Water Company. As confirmed by the AE,¹⁴ no discussions have yet been held with any of these private-sector players, who will be engaged only during implementation. Given the centrality of the proposed Water Fund to the project’s exit strategy, and the historical significance of powerful vested interests in skewed water delivery in Jordan, the failure to engage these key players during the design phase represents a serious risk to the project’s sustainability.

24. On one small aspect of sustainability, it is noted that the project will contribute the full input costs, worth USD 4.4 million in GCF funds and co-finance, for the community-run tree nurseries that will produce plant material for the restoration interventions (see annex 4 to the FP, budget note B13–B17). The iTAP asked how the shift would be made to a commercial set-up post-project, in which nurseries would cover their own costs and find their own paying customers, and avoid the frequent fate of such enterprises folding after project completion. In response, the AE stated that the community-based nurseries will “be run as income generating activities and sell the produced seedlings to the market – to farmers, restoration projects, etc., ensuring sustainability”.¹⁵ It is unclear, however, what model is envisaged to achieve a successful transition from fully subsidized project nursery to self-sustaining business.

25. Based on all the above, the paradigm shift potential of the project is assessed as medium.

¹³ Written responses to iTAP questions, received from UNEP on 22 May 2024.

¹⁴ Written responses to iTAP questions, received from UNEP on 22 May 2024.

¹⁵ Written responses to iTAP questions, received from UNEP on 22 May 2024.

1.4 Sustainable development potential

Scale: High

26. The proposed project is aligned with the United Nations 2030 Agenda for Sustainable Development, including contributions towards the efforts of Jordan to meet the Sustainable Development Goals (SDGs), namely: SDG 2 (Zero hunger); SDG 3 (Good health and well-being); SDG 5 (Gender equality); SDG 6 (Clean water and sanitation); SDG 8 (Decent work and economic growth); SDG 10 (Reducing inequalities); SDG 12 (Responsible consumption and production); SDG 15 (Life on land); and SDG 16 (Peace, justice and strong institutions). Specific co-benefits anticipated from project interventions are discussed below.

27. **Socioeconomic benefits:** Improved access to clean water will improve farming households' health and sanitation, and reduce healthcare costs accordingly. Social cohesion should also improve due to decrease in conflict over scarce water resources. Project interventions through activity 2.1.3, which will introduce new climate-resilient additional livelihoods and agriculture practices that reduce dependency on water-intensive crops, as well as support for microenterprises for post-harvest processing, should lead to diversified and improved incomes and better nutrition. The re-establishment of drought-tolerant forage shrubs in degraded rangelands through the Hima passive restoration approach should improve livestock condition and productivity. This was demonstrated in a project in the Zarqa River basin, where after two years of practising the Hima system, the dry yield of cereals for fodder in a rain-fed system increased from an average of 40 kg/ha/yr to 113 kg/ha/yr.¹⁶ The floating solar power installation will create employment and income-generating opportunities in solar-panel installation and maintenance.

28. **Environmental benefits:** The project's interventions in the landscape will have numerous environmental co-benefits, beyond the core adaptation results in relation to reduced vulnerability to climate change related water scarcity. The restoration and recharge measures should also lead to decreased pollution of water resources, decreased soil erosion and decreased sedimentation of dams. Promoting climate-resilient agricultural practices will reduce the need for fertilizers that contribute to pollution levels. By improving infiltration through restoration, freshwater and riparian habitats will be less affected by pollutants such as fertilizers which cause eutrophication.¹⁷ Soil quality will improve, and health risks posed to humans by nitrates will be reduced.

29. Some flood reduction co-benefits are expected from the project as well. Improving vegetative cover will allow greater rainfall interception, and reduce the volume of run-off and speed of overland water flow. This will decrease the potential for flash flooding of downstream areas and the associated loss of life and destruction of infrastructure and crops. In response to a question from the iTAP about why the project is primarily focused on drought and not flood, the AE confirmed that the topography is unsuitable for large-scale landscape engineering to capture and divert floodwaters to prevent downstream flooding and maximize water for recharge, and that the limited anticipated flood reduction from revegetation and some capturing of floodwaters in the rockfill dams was more accurately reflected as a project co-benefit.¹⁸

30. **Mitigation co-benefit.** The floating solar panel energy system to be installed over the King Abdullah Canal, with the adaptation benefit of saving up to 0.8 million m³ of water annually through reduced evaporation, will also produce 1 MW of clean energy, contributing to the country's aim of increasing its production of renewable energy, and reducing the reliance of the MWI on fossil-fuel generated energy to supply pumping stations in the area.

¹⁶ [http://nwrn.eu/sites/default/files/nwrn_ressources/f3 - afforestation of reservoir catchments 0.pdf](http://nwrn.eu/sites/default/files/nwrn_ressources/f3_-_afforestation_of_reservoir_catchments_0.pdf)

¹⁷ Eutrophication is over-enrichment of water with nutrients, causing excessive growth of algae that may deplete the water of oxygen.

¹⁸ Written responses to iTAP questions, received from UNEP on 22 May 2024.

31. **Gender and social inclusion:** The proposed project helps to take forward the national strategy for women in Jordan (2020–2025), with the following goals: i) women and girls are able to exercise their economic, political and human rights and freely lead and participate in society free of gender-based discrimination; ii) women and girls enjoy a life free of all forms of gender-based violence; iii) positive gender norms, roles and attitudes support gender equality and women empowerment; and iv) institutions are executing and sustaining policies, structures and services that support the empowerment of girls and women in alignment with the national and international commitments of Jordan.

32. The project will include women in the decision-making for and management of water resources at the community level, including monitoring of water-resource use and maintenance of community-level water infrastructure. This will allow for the specific water-related climate-change adaptation needs of women and girls to be addressed. The project will promote gender-inclusive climate-resilient livelihoods that will benefit female-headed households, and can be implemented at home, and tailored awareness-raising and outreach programmes that will allow women to become leaders in the promotion of climate-change adaptation in their communities. The project's gender action plan (annex 8 to the FP) outlines gender-disaggregated impacts, outcomes and output statements, as well as performance indicators and targets, and includes gender-responsive activities. In response to a question from the iTAP about the use in the FP of both "gender-transformative" and "gender-responsive" to characterize the project's approach, the AE agreed to standardize to "gender-responsive" which was felt to be most accurate term.¹⁹

33. The iTAP assesses the sustainable development potential of the project as high.

1.5 Needs of the recipient

Scale: High

34. The climate of Jordan has historically been arid, with a high degree of rainfall variability and temperatures in summer that often exceed 40°C. Approximately 94 per cent of the annual rainfall of Jordan is lost through evaporation, limiting freshwater availability. These dry conditions resulted in the country's ranking in 2019 as the fifth most water-stressed country in the world, with only 125 m³ of water available per capita per year.²⁰ This baseline water stress is being exacerbated by climate change, with observed trends and future projections indicating decreases in annual rainfall, higher annual temperatures, and more frequent, extreme and prolonged droughts. This water stress will particularly impact upon agricultural productivity, resulting in the need for increased food imports²¹ to ensure food availability.

35. Rising costs of food imports are compounded by the increasing cost of extracting usable water from declining water resources. In providing motivation for the need for the proposed project and the 100 per cent concessionality of GCF funding, the FP includes figures on the country's indebtedness that are now five years out of date. The iTAP notes that Jordan was reported by the International Monetary Fund as having a debt-to-gross domestic product ratio

¹⁹ The following definitions from the United Nations Fund for Population Activities may be useful:

- **Gender sensitive:** "programming that recognizes different needs of women, men, boys and girls and acknowledges gender power dynamics but does not necessarily address these";
- **Gender responsive:** "programming which includes specific action to try to reduce gender inequalities within communities"; and
- **Gender transformative:** "programming which is designed around a fundamental aim of addressing root causes of gender inequality within society".

https://www.unfpa.org/sites/default/files/admin-resource/thematic%20note%201_gender_final.pdf

²⁰ World Resources Institute. 2019. Aqueduct baseline water stress rankings. Available at:

<https://www.wri.org/aqueduct/>

²¹ Jordan imports three times more agricultural products than it exports – a deficit that is compounded in times of drought.

of 89.5 per cent, in December 2023, with total sovereign debt standing at USD 45.5 billion.²² This level of indebtedness mitigates against the country meeting its climate adaptation needs through debt finance for a paradigm shift towards climate-resilient integrated land and water-resource management.

36. The project targets small-scale farming communities in the three target sites (Yarmouk, King Talal Dam and Kafraïn) in the northern Jordan Valley that are shown to be highly vulnerable to climate-change impacts. This is compounded by overextraction of groundwater and poor land-management practices that reduce the recharge rates of groundwater basins. With lowering water tables in many areas, farmers abandon existing wells and seek to invest more money and resources into drilling deeper wells, which may further exacerbate overextraction and inequality in access to water. The Jordan Valley is the primary agricultural region of the country, where most local communities depend on agriculture for their livelihoods, and are directly affected by the intensifying water shortage. Under future climate conditions, decreased agricultural yields will reduce the income potential for local communities which already face high poverty rates. Future climate change will additionally present multiple health risks to Jordan's most vulnerable inhabitants, including children, women and refugees, including an increase in the incidence of heat stress related illnesses, exposure to diseases from reduced water quality, and a reduction in the diversity of food that households can afford, causing malnutrition.

37. Overall, based on the above, the needs of the recipients are rated as high.

1.6 Country ownership

Scale: High

38. The proposed project has been designed following extensive consultation with government, and a range of stakeholders, and is aligned with the country's policies. The roles of the AE and EE, as well as technical partners to support the project upon implementation, are clearly set out in the FP package.

39. **Policy and strategy alignment:** The proposed project is well aligned with the national priorities of Jordan as outlined in various policy and strategy documents, including the country's updated nationally determined contribution (2021) which acknowledges gender-responsive management of watersheds, agriculture and rangelands as central to the country's adaptation strategy, and highlights the need for interventions in managed aquifer recharge, rainwater harvesting, and watershed restoration for surface and groundwater provision. The project is aligned with the priority sectors for climate-change adaptation that were identified in the country's national adaptation plan, the national climate change policy 2013–2020, the aligned national action plan to combat desertification in Jordan 2015–2020, and the national strategy for women in Jordan (2020–2025).

40. In the water resources sector, the project aligns well with the climate change Policy for a resilient water sector (2016) which sets out the integrated water resources management approach upon which the national water strategy (2023) for Jordan is based – including the following four goals:

- (a) "Sustainably manage groundwater resources to restore safe yield levels and protect Jordan's aquifers";
- (b) "Sustainably manage and protect surface water resources and infrastructure";

²² <https://www.ceicdata.com/en/indicator/jordan/national-government-debt#:~:text=Jordan%20National%20Government%20Debt%20reached,bn%20in%20the%20previous%20month>

- (c) “Restructure water institutions to ensure that they have the legal mandates, staff capacity, resources, and incentives to modernize the sector in order to enhance accountability and increase public trust”; and
- (d) “Private sector participation increased to improve operational efficiency and sustainability, introduce innovation and technology... and access alternative funding”²³.
41. **Roles of key entities.** The two main EEs are the Jordanian Ministry of Environment for component 1, and IUCN for components 2–4. The suitability of these organizations for these roles is demonstrated in the FP and its annex 23, and can be summarized as follows:
42. International Union for Conservation of Nature Regional Office for West Asia: Through its member organizations, IUCN supports scientific research and national conservation legislation, policies and practices; and funds, operates or manages thousands of field projects worldwide. The organization has a presence and footprint in Jordan via its regional office (IUCN-ROWA) which is based in Amman, with locally based Jordanian staff who are familiar with the national context and have experience in working with stakeholders in government, communities and the consortium of NGOs which will be technical partners (see below). According to the AE, “the selection of IUCN as EE, as opposed to UNEP, for example, is strategic in that it provides a strong operational foundation for the implementation of the project” (FP, para. 236).
43. Ministry of Environment: The MOENV leads the country’s environmental governance system – including other relevant governmental organizations – in initiating and driving projects, policies, legislation, strategies, and monitoring, and is the national designated authority for GCF. The Climate Change Directorate coordinates high-level climate-change policy formulation and implementation, facilitating the participation of all relevant stakeholders. MOENV is well equipped to fulfil the role of EE for component 1, possessing the necessary technical and financial management capacity. The ministry leads the development of the country’s nationally determined contribution, national communications to the United Nations Framework Convention on Climate Change and national adaptation plan process, and has been involved in a number of related projects in recent years, focusing on, *inter alia*, climate-change adaptation, landscape and water-sector management, and sustainable development.
44. **Technical partners for implementation:** The required expertise to implement project technical activities (components 2 to 4) will be sourced primarily from members of the consortium of NGOs, whose track record and capacity are set out in detail in annex 23 to the FP. In addition to IUCN-ROWA, these organizations include the Jordanian Royal Society for the Conservation of Nature, Royal Marine Conservation Society of Jordan, Princess Alia Foundation, Horizons for Green Development, Jordan Hashemite Fund for Human Development, Royal Botanical Garden, and Future Pioneers Empowering Communities. The selected technical partners will be contractually bound through legal agreements signed with IUCN, and will receive cash advances to carry out activities, dependent on cleared expenditure and progress reports.
45. **Engagement with stakeholders, including civil society.** The proposed project was designed following an extensive consultative process with MOENV, which is responsible for the overall management of the country’s GCF portfolio, and the consortium of NGOs, as well as all other relevant project stakeholders. Key government stakeholders involved were representatives from the Ministry of Agriculture, MWI, JVA, Water Authority of Jordan, Ministry of Planning and International Cooperation, and National Agricultural Research Centre, as well as participating NGOs and local communities. All stakeholder consultations were held in Arabic (with translation provided where necessary) and gender sensitivities were considered. Stakeholders were mapped and consulted at the national level (24 stakeholder organizations)

²³ Jordan’s [National Water Strategy 2023-2040](#)

and local level across the three target areas (South Shounah – 38, Upper King Talal Dam – 21, and Yarmouk – 33 stakeholder organizations).

46. Details of the stakeholder engagement process are summarized in annex 7, covering consultations that took place during four in-country missions from 2018 to 2020. No mention is made of any engagement between November 2020 and the present day, though it can be assumed that progress was delayed by the coronavirus disease 2019 pandemic. Annex 7 to the FP sets out the consultations in detail, including discussions with local communities, NGOs, sectoral experts (ecologists, botanists, hydrologists and social scientists), state and non-state institutions, and national-level stakeholders. A focused round of community consultations was held in late 2020 to explore and document local community views, concerns and priorities on the scope of the proposed project, as part of the validation process. The group of respondents interviewed had strong representation of upstream and downstream farmers, near equal distribution of men and women, and nearly half were young people (aged 18–35 years). The iTAP trusts that the project proponents will refresh the stakeholder engagement during project inception, so as to manage any changes that may have occurred over the last three years.

47. Country ownership of the project is thus seen to be high.

1.7 Efficiency and effectiveness

Scale: Medium to High

48. The GOJ is requesting USD 44,904,203 in grant finance from GCF to implement climate-resilient ILWRM as part of a national response to climate change. This will make a contribution towards the ability of Jordan to balance its water demand with its water supply over the long term, despite evolving climate-change impacts. The GCF investment will be supported by co-finance of USD 1,000,000 in-kind from the GOJ and USD 14,605,748 as a combination of grant and in-kind co-finance from a consortium of NGOs. The combined co-finance forms 26 per cent of the total project costs, while the GCF funds form 74 per cent.

49. The co-finance is described as covering “all baseline costs – such as business-as-usual repairs or maintenance to water supply infrastructure necessary to ensure the sustainability of climate-proofing interventions, [and] GCF funding will solely be used to address the additional impacts of climate change”. In addition to official co-finance reflected in letters of commitment, parallel finance of USD 13,133,000 has been identified for the project, consisting of resources included under aligned initiatives and projects supported by the consortium of NGOs (detailed in annex 30 to the FP).

50. **Cost-effectiveness.** Jordan experiences interconnected climate-change impacts that increase the scarcity of water resources in an already water-stressed country, requiring effective and cost-effective responses. In order to inform decision-making on adaptation interventions, a full economic analysis across all four project outputs was undertaken using a cost-benefit analysis approach – see the economic feasibility analysis in annex 3a and 3b to the FP. This builds on previous studies indicating that the benefits of ecosystem restoration to Jordanian society far outweigh the costs,^{24,25} including improvement of the country’s water supply.

51. The results of this analysis, assuming a 25-year lifespan for the project and a discount rate of 5 per cent, (a social discount rate, on the grounds of the project’s multiple social and environmental public benefits, and not intending to provide a benchmark for competing with

²⁴ Becker N & Katz DL. 2009. An economic assessment of Dead Sea preservation and restoration. In: *The Jordan River and Dead Sea Basin*. Dordrecht: Springer.

²⁵ Myint MM & Westerberg V. 2014. *An Economic Valuation of a Large-scale Rangeland Restoration Project through the Hima System in Jordan. Report for the ELD Initiative by International Union for Conservation of Nature, Nairobi, Kenya.* Available at www.eld-initiative.org

market-based rates of return²⁶) are summarized below. The results shows that the total investment of USD 60,509,951 across all four project outputs will generate a net present value of USD 105 million. The overall benefit to cost ratio is 3.00, meaning that the economic benefits of implementing the project are expected to be three times greater than the costs (to all parties).

Net present value	USD 105 279 874
Benefit/cost ratio	3.00
Economic rate of return	25%
Payback period	9 years
Total project cost (undiscounted costs incurred in years 1–8)	USD 60 509 951

52. According to the sensitivity analysis, the economic summary statistics above are robust to changes in input values. For example, when the value of groundwater to agriculture is reduced by 30 per cent, the economic rate of return (ERR) for the project is still favourable at 22 per cent (and the value of USD 0.5/m³ of water is supported by other studies). A second key parameter is the additional groundwater recharge from the rockfill dams. Given that there is more uncertainty around this quantity, it has been reduced by 50 per cent for the sensitivity analysis. However, this still results in a favourable ERR for the project — with a rate of 20 per cent overall. It is also shown that if no scale-up of activities is achieved via components 1 and 4, the ERR remains favourable at 17 per cent. Overall, this project therefore represents a cost-effective spend for GCF.

53. The economic feasibility analysis also makes an assessment of financial sustainability assessment post-project (in section 6 of annex 3a). For project interventions to be sustainable, there is a need for an “ongoing financial flow to support the operations and maintenance of key infrastructure, equipment, and livelihood and household practices established via project activities”. These operations and maintenance costs are further elaborated in the operations manual framework (annex 21 to the FP), and are assessed against the results of the benefits analysis in order to demonstrate that the project’s exit strategy is financially viable. The floating solar panels will generate revenue through selling electricity, easily covering costs. The value of rainwater harvesting benefits exceed maintenance costs by a factor of five, which means that building owners and households can be expected to cover the maintenance costs using savings achieved from decreased purchase of water from other suppliers, or income generated using the water. The water fund is intended to deliver the financial resources required for ongoing maintenance of the rockfill dams and the ecosystem restoration sites. The analysis shows that the annual benefit in additional groundwater recharge from ecosystem restoration exceeds the associated annual maintenance costs, indicating that sufficient funds could, at least in theory, be leveraged via the fund to cover these costs.

54. The extent to which the Water Fund will be successful in practice is unknown, and the absence of engagement of the key potential private-sector funding partners increases the risk that large-scale capitalization or ongoing replenishment via a payments for ecosystem services model will turn out not to be realistic. There is also no discussion in the FP of specific long-term public revenue sources which might be utilized to support the fund. Section 8.5 of the feasibility study explores lessons learned from water funds globally, and there is a section on previous studies on Jordanian farming communities’ willingness to pay for the preservation and provisioning of environmental resources, but this is not extended to larger corporate water users. A detailed application of global models and lessons to Jordan’s circumstances is not attempted, and the FP indicates that only during the Water Fund’s maturation phase (Years 7 and 8), will a long-term financing strategy be developed, detailing how financial resources for the ongoing operation of the Water Fund post-GCF involvement will be leveraged from public and private sources (see subactivity 2.1.4.9).

²⁶ This is also the middle discount rate of three used in the Economics of Land Degradation initiative (which used 2.5, 5 and 8 per cent).

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55. The project's overall efficiency and effectiveness is rated as medium to high.

II. Overall remarks from the independent Technical Advisory Panel

56. The iTAP believes that the Jordan Integrated Landscape Management Initiative project applies a best practice EbA approach to enhance the resilience of small-scale farmers in the northern Jordan Valley, whose livelihoods, and the ecosystems on which these depend, are highly vulnerable to the negative impacts of climate change. The project also develops national capacity for integrated land and water resource management, and represents an important contribution to the efforts of Jordan to adapt to climate change intensified water scarcity.
57. The iTAP recommends that the Board approve this funding proposal.

Response from the accredited entity to the independent Technical Advisory Panel's assessment (FP232)

Proposal name:	Jordan Integrated Landscape Management Initiative (JILMI)
Accredited entity:	United Nations Environment Programme
Country:	Jordan
Project/programme size:	Medium

Impact potential

iTAP's recommendation to translate the best practice restoration techniques presented in Annex 2 into Arabic for sharing is noted. The information will be translated for sharing by the PMU, while ToRs for the relevant work packages detailed in the procurement plan will annex the information and make requirements for the selection of the partner contingent on experience in implementing Jordan-relevant techniques.

Groundwater monitoring at rockfill dams will be prioritised at both the start and end of the dry season each year. This will be combined with data on the amount of water extracted to determine overall impact on groundwater.

Paradigm shift potential

The AE confirms government's openness to the establishment of WMCs and will be fostered further as the relevant activities under Component 1 (executed by the Ministry of Environment) are implemented (including consultations with relevant partners).

Discussions with potential private sector investors into the Water Fund will be prioritised from the design phase (FP, Activity 2.4.1.1) to ensure both their involvement and support (including investment) are encouraged from an early stage.

Shifting community-based nurseries to commercial units post-project will be informed by similar successful initiatives led by project partners such as RBC.

Sustainable development potential

We agree with iTAP that the potential flood reduction co-benefits of the project are important, despite the project focus being on water resource attenuation (to address a drying trend — drought). While the primary focus is on strengthening water security, there will be flood attenuation co-benefits as report in the FP as water flow across the landscape is slowed and run-off is reduced. Section 11.6 of the FS provides an analysis of the Impacts of restoration interventions on the extent of coverage and speed of flow of rainwater over project target areas to support water harvesting and aquifer recharge. Managed Aquifer Recharge under Activity 3.1.1, will contribute to flood attenuation through the capturing of flood waters in rockfill dams, which will then slowly recharge aquifers with this water.

Mitigation co-benefit: in addition to the mitigation benefit of the suspended solar installation, mitigation co-benefits will also accrue via the project's landscape management interventions (restoration and improved ecosystem management). It is expected that the project will result in 537,390 tCO₂eq GHG emissions (tCO₂eq) reduced as a result of improved land use

management practices (by project end). This is presented as a measurable co-benefit with indicator and target in the project's logical framework.

Needs of the recipient

iTAP's feedback is much appreciated.

Country ownership

The AE thanks iTAP for the positive feedback. Stakeholder engagements will be refreshed during inception to sensitive stakeholders to the project's design, as per the Stakeholder Engagement Plan (Annex 7).

Efficiency and effectiveness

iTAP's feedback is well noted. During the design phase of the Water Fund (Activity 2.4.1.1) in the first two years of the project, the options for the long-term financing of the fund will be identified. Relevant options will continue to be identified and assessed, with partnerships being built throughout the project period, contributing to the development of the long-term financing strategy during years 7 and 8. Much exploratory work would have already taken place by the time the development of the strategy commences. Additionally, the project design phase will also include the detailed application of global models and lessons to Jordan's circumstances, in addition to work already carried out during project design. During Feasibility Study development (Annex 2), best practices and lessons related to the design, establishment and operationalisation of Water Funds around the world were assessed and captured (see Section 8.5 of Annex 2). Very few examples of Water Funds in a similar context to Jordan (none in the MENA region) were found. Hence, details on global models and lessons relevant to Jordan's context are sparse. However, where possible, relevant lessons and modelled were applied to the Jordanian context.

iTAP confirms the robustness of the economic feasibility. We would also like to note here that there are many benefits values associated with ecosystem restoration that were not captured in the economic appraisal, as follows:

Ecosystem restoration will reduce surface water runoff, thereby attenuating downstream flood risks. However, whilst the supply of this service is acknowledged as a co-benefit of ecosystem restoration it has not been modelled due to the uncertainties and complexities surrounding the associated hydrological modelling. Similarly, additional benefits such as an increase in surface albedo, reductions in methane emissions from livestock, capture of methane in the soil and improved biodiversity are not quantified and monetized meaning that the ecosystem restoration benefits can only be seen as partial at best.

With regards to albedo, the restoration is likely to lead to an increase in the cover of indigenous grasses. This in turn likely to increase albedo, which mitigates global warming¹, because grasses, particularly in the dry season, have a lighter coloration than the shrubs that prevail in degraded ecosystems in Jordan. In terms of methane emissions, it is well documented that restoration of ecosystems increases the quality of fodder available for

¹ Haslter et al. 2023. Accounting for albedo change to identify climate-positive tree cover restoration. Nature Communications. <https://doi.org/10.1038/s41467-024-46577-1>

livestock, which in turn greatly reduces the methane production from livestock². The amount of methane produced per head of cattle can, for example, potentially be reduced by tens of kilograms per year³. This effect will contribute to climate change mitigation as methane is a powerful greenhouse gas. In addition to reducing methane emissions, recent studies show that it is likely that restoration of rangelands will also capture methane in topsoils (in the order of 10 kgC/ha/year)⁴. Lastly, with regards to biodiversity, the restoration will in time increase the diversity of indigenous plants, which in turn improves the quality of habitat for a wide range of indigenous animals, including insects, birds, reptiles and mammals.

The details can be found in Annex 4: Financial and Economic appraisal.

Overall remarks from the independent Technical Advisory Panel:

The AE welcomes the positive feedback from the iTAP, as well as its recommendation for the FP to be approved by the GCF Board.

² Herrero et al. 2016. Greenhouse gas mitigation potentials in the livestock sector. Nature Climate Change. <https://doi.org/10.1038/nclimate2925>.

³ Hristov et al. 2013. Mitigation of methane and nitrous oxide emissions from animal operations: A review of enteric methane mitigation options. Journal of Animal Science, Volume 91, Issue 11, November 2013, Pages 5045–5069, <https://doi.org/10.2527/jas.2013-6583>

⁴ He et al. 2024. Meta-analysis shows the impacts of ecological restoration on greenhouse gas emissions. Nature Communications. <https://doi.org/10.1038/s41467-024-46991-5>

JORDAN INTEGRATED LANDSCAPE MANAGEMENT INITIATIVE (JILMI)

ANNEX 8

GENDER ASSESSMENT AND GENDER ACTION PLAN



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1 Background

The overall objective of the Jordan Integrated Landscape Management Initiative (JILMI) is to contribute to enhanced resilience of representative Jordanian ecosystems and vulnerable communities in the Jordan Rift Valley, Zarqa River Basin and the Gulf of Aqaba through integrated landscape and water resources management (ILWRM). This will include the promotion of climate-smart land- and sea-use planning, small-scale sustainable energy opportunities and the diversification of local livelihoods and economy options.

Experience and evidence indicate that women, youth and children are among the most vulnerable to the impacts of climate change, and that a large proportion of vulnerable people who are highly dependent on local natural resources for their survival are women and children. The Government of Jordan has therefore established that gender-sensitive planning, resource-allocation and implementation in the context of the Jordan Integrated Landscape Management Initiative will be a priority. Supporting transformational change both in the lives of female beneficiaries and in prompting gender mainstreaming in national climate change policymaking and implementation was therefore considered a priority in stakeholder consultations.

To fulfill the project's design requirements, this assessment was undertaken in accordance with GCF guidelines to inform the project design on gender barriers, constraints and opportunities for women to benefit from the project's activities. In addition, the Gender Action Plan will ensure women and other marginalised groups are included in the project's activities. The Gender Action Plan is based on the proposed project logical framework and focused on avoiding and minimising the negative impacts of the project activities on the local communities, and maximising the positive impacts and social benefits focusing on women and marginalised groups.

1.1 Desk review and Field Research

To design this gender analysis as part of preparation for a Gender Action Plan, an accelerated review was undertaken of supporting documents related to gender mainstreaming in climate change, as well as previous gender analysis conducted by other donors in Jordan — in addition to DOS available data.

This gender assessment identifies gender considerations that are relevant to the project and examines potential gender mainstreaming opportunities. Therefore, stakeholder consultations were undertaken and focus group discussions (FGDs) were carried out in three sample provinces selected to benefit from the project (Al yarmouk, Al kafreen and tal al Romman) with 60 total participants, of which more than 50% were women. This was done to highlight gender and social norms and to identify key entry points for women's participation in the project activity within the study area.

The consultations helped to identify constraints/barriers to women's participation, their opportunities and prioritize the areas for capacity building to mainstream gender in the future activities of the project.

2 Jordan context on gender equality

Jordan is ranked 138th out of 153 countries in the World Economic Forum 2020 Global Gender Gap Report — it is part of the five countries with the highest absolute decrease in its overall score in recent years, with the main sub-index that led the decrease being economic participation. Challenges facing women's economic participation include: i) low wage levels and low minimum wages; ii) high unemployment rates; iii) failure to regulate foreign (migrant) labor; iv) a large

unregulated sector; and v) non-alignment between (academic and vocational) education outputs and the labor market needs¹.

Inequalities remain in participation in economic and political spheres, as well as in legal rights related to marriage, inheritance, divorce and employment. There are indications that, within society, boys are preferred to girls and decision-making within the home is largely in the hands of husbands and male family members. Structural and systemic gender inequality is a major obstacle impeding women's empowerment in Jordan, preventing women from exercising their right to lead a full life in both public and private spheres. Gender inequality affects the level to which women have control over resources such as wealth, land, property, water, fertile soil and transportation, which in turn reduces their ability to enter public life, establish a business, find employment, travel to markets, stand for office and speak out in public.

2.1 *Women's economic empowerment*

The socio-economic situation in Jordan is categorized by high unemployment rates, economic inequality, political marginalization of youth and women, and a mismatch between educational skills and skills sought by the labor market².

Population size: 8,686,607million (2019)
Overall population sex ratio (male/female): 1.06 male/female (2019)
GDP (US \$ Billions): \$ 42.29 (2018)
Religions: Muslim 97.2% (official: predominantly Sunni), Christian 2.2% (majority: Greek Orthodox)
Life expectancy at birth: 78.7 /male, 81.5/ female (2019)
Ethnic Groups: 98% Arab 1% Circassian (Adyghe) 1% Armenian
Fertility rate: 3.17 children born/woman
Live births average per day (26.93 in an hour) 2019
Infant Mortality 12.8 (infant deaths per 1,000 live births)2019
Gender Gap Index 2020: 138 (out of 153 countries)
The total dependency ratio of population in Jordan is 66.9 % (2019)
Literacy rate 2019: 96.66%

In the second quarter of 2019, the Labor Force Participation Rate in Jordan decreased to 34.60%, from 35.10% in the first quarter of 2019. From 2017 until 2019, the Labor Force Participation Rate averaged 37.13%, reaching an all-time high of 40.60% in the first quarter of 2017 and a record low of 34.60% in the second quarter of 2019³. The Jordanian government has notably succeeded in closing the gender gap in education. However, it has been slow to recognize how this translates into opportunities for women's contribution to the economy.

¹WEF. 2020. Global Gender Gap Report 2020. Available at: http://www3.weforum.org/docs/WEF_GGGR_2020.pdf

² WEF. 2020. Global Gender Gap Report 2020. Available at: http://www3.weforum.org/docs/WEF_GGGR_2020.pdf

³ Trading economics, 2019. Jordan Employment Rate. Available at: <https://tradingeconomics.com/jordan/employment-rate>

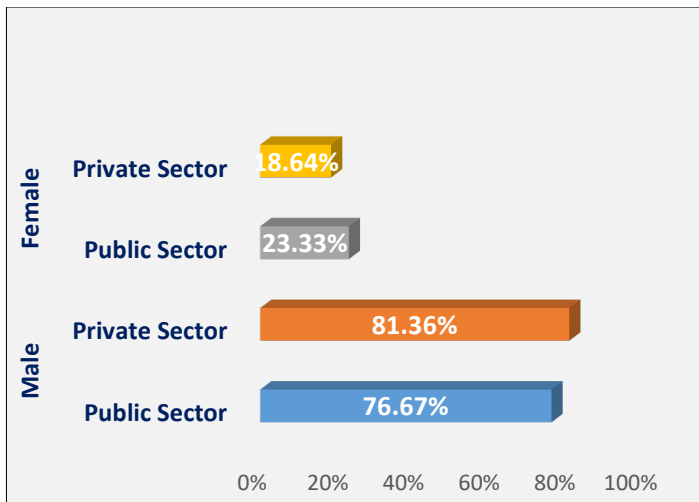


Figure 1. The number of economically active Jordanians (aged 15+ years) by sex and sector.

The Jordanian labor market is characterized by large gender differences. Labor force participation among women is below 14%, compared to ~64% for men⁴. As a result, unemployment rates among women are almost twice as high as for men. These differences are strongly related to the traditional social structures that place a higher value on women's reproductive roles. Women's roles and responsibilities have traditionally been defined within the household context. As a result, any economic benefits women can gain tend to be considered for family and household, rather than individual, causing invisibility of women's economic contributions.

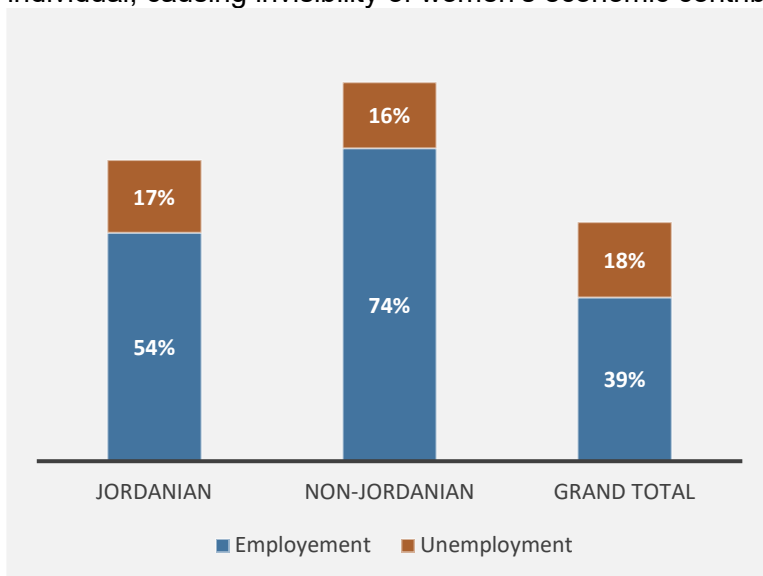


Figure 2. Population Age 15+ Years by Economic Activity Status, Sex, Educational Level & Nationality (% age Distribution) 2019

Women in Jordan are expected to take on a disproportionate amount of family-related responsibilities, which creates pressure on them because it is challenging to balance professional and family responsibilities. Therefore, women's participation decreases significantly after marriage because of the perceived difficulties in balancing professional and household responsibilities. The working environment, whether there are many men or not, influence women's decisions. Generally, workplaces that do not require long commutes and which are "female friendly" are considered desirable⁵.

⁴ The World Bank, 2019. Gender Data Portal. Available at: <http://datatopics.worldbank.org/gender/country/jordan>

⁵ Wilson Center, 2019. Available at: <https://www.wilsoncenter.org>

Unemployment in Jordan increased to 19.2% in 2019, although this was not completely out of line with other countries in the region. DOS statistics for 2019 place the rate for men at an average of 17.1% and women's unemployment at 27.2% (Figure 2)⁶.

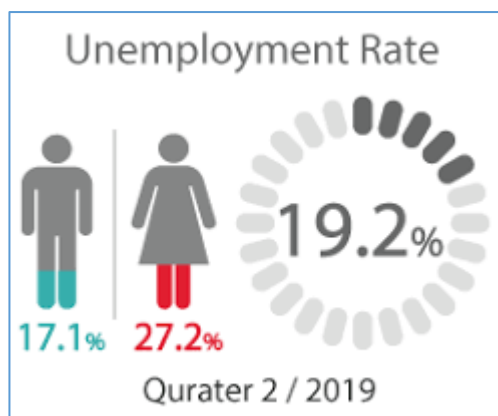


Figure 3. Unemployment disaggregated by gender

Although educational opportunities for young women and men in Jordan are equal, their ease of entry into the labor market is not. This occurs because of several reasons, including the limited range of jobs considered to be appropriate and acceptable for women and employers. There are also preferences for hiring men, who are considered to be more in need of jobs as they must provide for families. For university graduates (holding a bachelor's degree), male unemployment is 18.5% in 2018 and for women 73.7% in 2018, underlining the difficulties faced even by educated women in entering the Jordanian job market.

In Jordan, women's participation in the labour market is limited by gender-specific social constraints, such as cultural and class-based expectations regarding women's household roles and unofficial 'occupational segregation', where various jobs are seen as appropriate for men or for women. Some legal regulations reinforce notions of a women's work as being secondary to that of her husband and diminish incentives for married women to work. For example, women employees in the public sector are not entitled to receive family allowance if the fathers of their children are alive and fit for work. Moreover, a deceased wife's pension cannot be transferred to her husband unless he is unfit for work and does not have any other source of income⁷.

There are extraordinary differences for women in the work force by sector, with the public sector employing half of all employed women in Jordan. There are many social taboos around women working in male-dominated environment, such as construction or engineering. Many poor women work out of economic necessity, although a substantial number of working women come from middleclass families. Jordanian women are more likely to work as professionals (48.8% of all employed women), and technicians and associate professionals (24.0%). Women constitute 54.6% of all employees in the education sector, 45.5% in health and social work, and 28.9% in banking⁸.

The 1996 Labor Code applies to all employees in the formal sector, with the exception of domestic, agricultural and unpaid workers. It stipulates normal working hours, allows for temporary contracts and outlines terms of leave and dismissal as well as giving the Minister of Labor the authority to either temporarily or permanently shut down any enterprise in which an incident of sexual

⁶ DOS, 2019

⁷ ETF, 2009. Women and work in Jordan: Case study of tourism and ICT sectors. Available at: <http://www.silviacambie.com/wp-content/uploads/2010/01/womenwork-in-jordan.pdf>

⁸ National Center for Human Resources Development. Available at: <http://www.almanar.jo/en/documents/Mayssa%20Gender%20report.pdf>

harassment (as defined in the Penal Code) has taken place. The employment of civil servants is governed by the Civil Service Statute amended in 2007⁹.

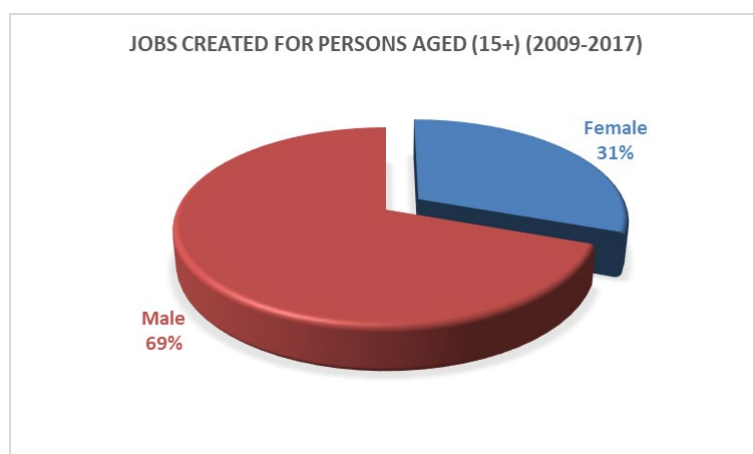


Figure 4. Jobs created for persons aged 15+ (2009–2017).

Cultural constraints also clearly militate against women working in areas of low prestige, in jobs requiring travel and unsocial working hours or contact/proximity with strangers, these often in areas with the most critical labor shortages, such as tourism¹⁰.

2.2 Women's political participation

A gender quota system for political participation in the Parliament was adopted in 2003 and revised in 2015 to increase the number of seats reserved for women parliamentary candidates from 6 to 15 (of 130 total available seats for elected representatives). However, women's political participation in Jordan remains low. The 2020 Global Gender Gap Index ranks Jordan at 113^h under the political empowerment section¹¹. Moreover, the 2020 Women in Parliament Index ranks Jordan 113th out of 153 countries. Traditional gender stereotypes are used to justify low levels of women participation in the political sphere. Women are poorly represented in the political life, which hinders them from effectively contributing in the decision-making process by not conveying the actual priorities, policies and practices for women. Therefore, politics in Jordan is considered a male domain and decisions are taken by men and therefore generally serve male interests.

Table 1. Politics (Kingdom) (2008-2018)

	Female (%)	Male (%)
The Cabinet	24	76
The Diplomatic Corps at all levels	20.1	79.9
Ambassadors	10.2	89.8
The Upper House of Parliament (Senates)	15.3	84.7
The Lower House of Parliament (Deputies)	15.4	84.6
Local Councils	27.8	72.2
Political Parties	35.3	64.7
Chambers of Industry	8.7	91.3
Chambers of Commerce	0	100

⁹ UNDP, 2018. Jordan Gender justice and the law. Available at: <https://www.undp.org/content/dam/rbas/doc/Gender%20Justice/English/Full%20reports/Jordan%20Country%20Assessment%20-%20English.pdf>

¹⁰ Sectoral assessment for the Jordan Valley Links project.

¹¹ WEF. 2020. Global Gender Gap Report 2020. Available at: http://www3.weforum.org/docs/WEF_GGGR_2020.pdf

2.3 *Peace, and Security*

Jordan is one of 80 countries that has developed a National Action Plan for the implementation of Security Council resolution 1325 on Women, Peace and Security. Resolution 1325 is a landmark achievement of the international community coming together to address the importance of women's participation in sustaining peace and building security. The Jordanian National Action Plan (JONAP) included actions to drive women's engagement in local, regional, and national processes of peacebuilding and social inclusion.

2.4 *Rule of law*

Jordan ratified the International Convention on the Elimination of All Forms of Discrimination against Women (CEDAW) in 1992. However the country maintains reservations on some articles, such as giving equal rights to women with regard to nationality of children, same rights and responsibilities during marriage and divorce¹², rights and responsibilities as parents, as well as personal rights — including the right to choose a family name, a profession and an occupation. However, in 2009, a Cabinet decision lifted its reservations on giving women freedom of mobility and choice of residence without consent of their husbands or other male family members. Some progress has also been achieved with regards to policy, legislation and procedures adopted to promote gender equality and protect women's rights in general, but improvements are still needed. As the Arab Women Organization concludes "The steps did not address the core of the problem; the patriarchal system itself as there is no political will to genuinely promote equality in the public and political sphere."¹³

Legislations in Jordan is civil, with the exception of the Personal Status Law that is based on religion. Therefore, it is highly dependent on the interpretations of the religious precipices. Despite some progressive amendments, significant legal Inequalities persist in issues related to personal status (marriage, child custody, inheritance, divorce), Penal Code, labor laws, and, violence against women is not sufficiently addressed¹⁴. This situation contributes towards the legal perpetuation of the inferior status of women in society, which plays an essential role in the continuation of gender stereotypes.¹⁵

2.5 *Health*

The health situation in Jordan is amongst the best in the Middle East because of the secure and stabile conditions in the Kingdom. It is also a result of a set of effective development plans and projects which included health as an important element and essential part of sustainable development. The health sector in Jordan as such has witnessed remarkable development and reflected positively on the wellbeing of citizens. The general health 2018 indicators have reflected the quality and efficiency of the delivered health services, putting Jordan in an advanced rank globally.

¹² JIF, 2019. Available at: <https://jordaningoforum.org>

¹³ Women's Rights in Jordan, Arab Women Organization, 2013

¹⁴ Gender Profile JORDAN, Institute for women studies in the Arab word,2018

¹⁵ AWO, 2013

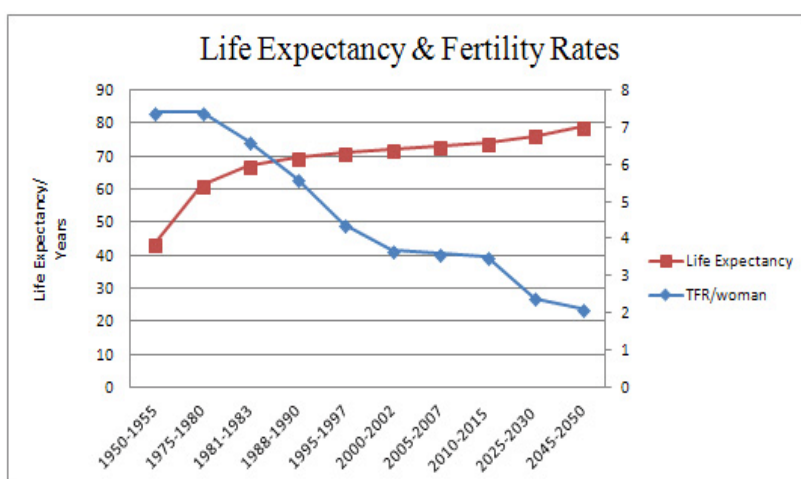


Figure 5. Life expectancy and fertility rates.

There was a comparatively rapid decline in fertility between the 1990 and 2002 — according to JPFHS surveys — from 5.6 children per woman to 3.7 children per woman (Figure 5). The TFR remained relatively stable between 2007 and 2012, ranging from 3.5 to 3.8 children per woman, before declining to 2.7 children per woman in 2017-18¹⁶.

Among women age 30–49, the median age at first birth is approximately 26 years. This is related to educational level, as the most educated women tend to marry later (by approximately five years) than women with middle education levels¹⁷.

Approximately half of married women (49%) want no more children or have been sterilized. However, ~7% of married women are undecided about whether they want more children, but men are much less likely than women to express a desire to limit childbearing where ~31% are undecided about future childbearing. The mean ideal family size among currently married women (age 15–49) is 2.9 children; compared with men, who have a mean ideal family size of 3.8 children.

The proportion of married women age 15–49 currently using any contraceptive method decreased from 61% in 2012 to 52% in 2017-18. In this regard, it was found that ~ 26% men in some governorates tend to decide about use of family planning for their wives which can impact women health both physically and psychologically.

Mortality rates are generally higher among children born to Syrian women than among children born to Jordanian women or women of other nationalities. Overall, the under-five mortality rate is 25 deaths per 1,000 live births among children whose mothers are Syrian and 16 deaths per 1,000 live births among children whose mothers are Jordanian or of other nationalities. However, women face some constraints in receiving medical and health care services. Many factors can prevent women from getting medical advice or treatment for themselves when they are sick, particularly during pregnancy and at the time of delivery. 42% of married women age 15–49 in Jordan report having at least one of the specified problems in accessing health care. The most frequently reported problems were: i) having to take transport (25%) and not wanting to go alone (24%); ii) distance to a health facility and Maternal Health Care; and iii) difficulty in getting money for treatment (both 22%). The problem reported least often was getting permission to go for treatment (9% of respondents).

The total fertility rate in rural areas is higher (3.1 children) compared to urban areas (2.7). This result is in line with the average size of the Jordanian family of 4.8 person, as determined by the General

¹⁶ Jordan Population and Family Health Survey, 20017-18. Available at: <https://dhsprogram.com/pubs/pdf/FR346/FR346.pdf>

¹⁷ Jordan Population and Family Health Survey, 20017-18. Available at: <https://dhsprogram.com/pubs/pdf/FR346/FR346.pdf>

Population and Housing Census – 2015. The largest disparity in total fertility rate is among Syrian women (at 4.7), compared with 2.6 children per Jordanian woman and 1.9 per woman from other nationalities.¹⁸ Among women overall, mean ideal number of children is slightly higher in rural (4.0 children) than urban (3.8 children) areas, this number is higher among women of Syrian nationality (4.1 children) than among women of Jordanian nationality.

2.6 Education

Jordan has made great progress in increasing access to education, with nearly universal primary enrollment, gender parity in Jordanian schools, and a 98% adult literacy rate (Figure 6). Despite widespread access to education, challenges remain.

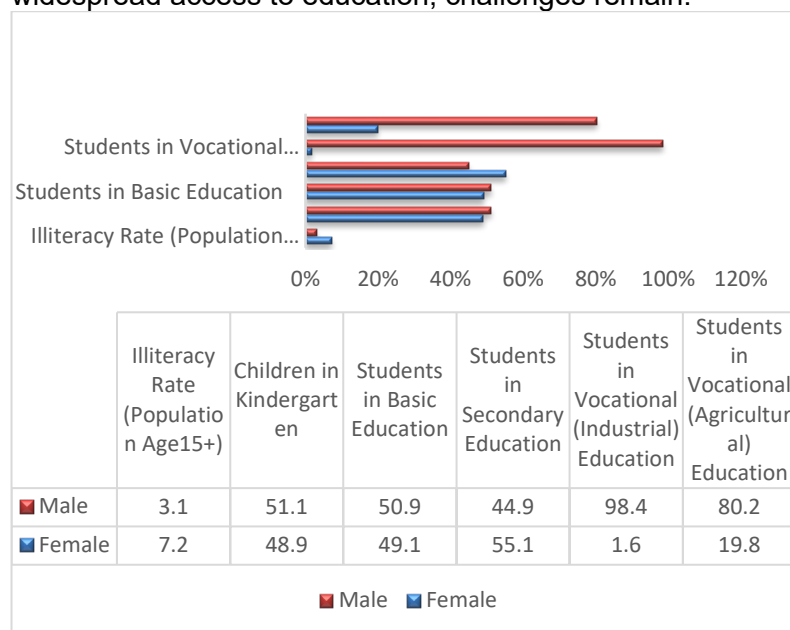


Figure 6. Indicators of Education (2009–2018).

Schools are overcrowded and in varying states of disrepair, teachers receive limited training and support, and learning outcomes remain low, with only a fifth of second and third graders reading at grade level. The growing population of Jordanians and the influx of refugees have placed additional pressure on Jordan's public education system, and children and youth who fall out of the system lack opportunities to get back on track. Ensuring that all children and youth have the opportunity to gain skills and knowledge to succeed is essential to their economic productivity.

Jordan has almost achieved its target of universal access to primary education — with 97% of children in school — and has maintained gender parity in education since 1999. However, progress in school enrolment has not benefited all children equally, including children with disabilities and refugee children, and improving the quality of education requires greater focus and investment. Children from poor socio-economic backgrounds, children involved in child labor and those with disabilities are at a higher risk of being out of school. Children with multiple and overlapping vulnerabilities are at greater risk of either dropping out or being out of education altogether. Early childhood development opportunities occur primarily in homes, with only 13% of children aged 3 to 4 years old accessing Kindergarten 1 (KG1); and 59% of those aged 5 to 6 years old attending Kindergarten 2 (KG2).

Sending a girl to school allows her to build confidence and contribute to the country's economic, social and political development. Although education in Jordan has reached gender parity in 1999,

¹⁸ DOS, 2019. Launching the results of the Key Indicators Report of the Population and Family Health Survey 2017-2018. Available at: <http://dosweb.dos.gov.jo/dhs-2017>

social norms and traditions, along with other factors, block Jordanian girls from fully utilizing their education in the job market¹⁹. More female students are enrolled in secondary school than male students. Boys are more likely to drop out of school due to poor academic achievements, violence, bullying and the need to find employment. Lagging boys' performance may be tied to teacher quality, availability of male teachers, violence in schools, availability of quality school environments and unfavourable student-teacher ratio²⁰.

Meanwhile, girls are more likely to be kept at home because of protection concerns and household responsibilities such as caring for younger siblings or ailing parents. Girls are also far more likely to be married at an early age. There is no specific gender disparity in Jordanian primary schools. Over 94 % of girls attend school compared to 95 % of boys. Girls in rural areas are just as likely as girls in urban areas to attend school. Approximately 10% of girls who are of secondary school age (12–17 years old) are not participating in the education system, compared to 15% of male youth of the same age. Although the percentage is in favor of girls, it is not satisfactory and is primarily the result of child marriage or low school performance followed by dropout.

Since 14% of the country lives below the poverty line, child marriage occurs often among Jordanian girls in order for parents to be relieved of their financial responsibility. Around 13% of girls in Jordan marry before turning 18 years of age. Among men age 20-49, by contrast, only 1% were married before their 18th birthday. Over 86% of girls who marry under the age of 18 have only finished their elementary education. The median age at first marriage among women aged 25–49 is 22.7 years. However, the median age at first marriage cannot be calculated for men for the same age because less than 50% of men had married before age 25. The situation is not better for the Syrian refugee girls coming to Jordan since 33% of them are already married. This significantly decreases their chances of school enrollment²¹.

The Syrian refugee crisis has strongly impacted on education in Jordan. It has created overcrowded classes and increased educational costs for the government. The government strives to improve its educational standards for girls and boys alike, despite the setback caused by the humanitarian crisis. UNICEF is partnering with the Ministry of Education to educate refugees and supply classroom furniture and learning materials. Plan International Jordan creates safe child-friendly spaces for Jordanian and Syrian refugee children under the age of 5, which increases their chances of attend primary school²².

¹⁹ Klein, G. 2018. Top 10 facts about girls' education in Jordan. Available at: <https://borgenproject.org/top-10-facts-about-girls-education-in-jordan/>

²⁰ Jordan National Youth Assessment 2015. Available at: https://pdf.usaid.gov/pdf_docs/PA00KBZD.pdf

²¹ Klein, G. 2018. Top 10 facts about girls' education in Jordan. Available at: <https://borgenproject.org/top-10-facts-about-girls-education-in-jordan/>

²² HRW. 2016. Barriers to education for Syrian refugee children in Jordan. Available at: <https://www.hrw.org/report/2016/08/16/were-afraid-their-future/barriers-education-syrian-refugee-children-jordan>

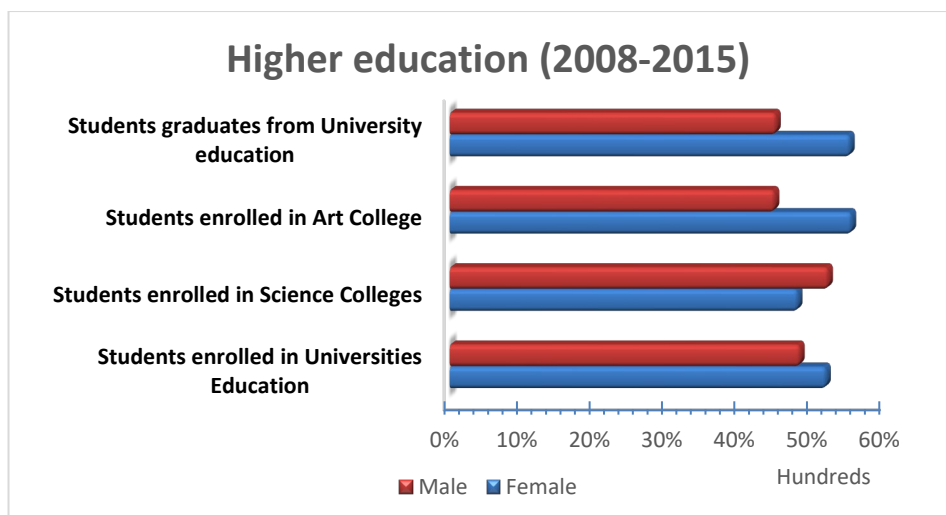


Figure 7. Higher education in Jordan.

Jordan has one of the highest literacy rates for girls in the Middle East, which is estimated to be as high as 97%. However, this educational advancement does not transfer over to the workforce. Jordan has one of the world's lowest rates of women participating in the workforce, at 13.2%. If a gender gap in Jordan's workforce continues to exist, the country will experience a reduction in potential GDP growth of 0.5 to 0.9% per year.

A good education is no guarantee that the girl will find employment. Approximately 30% of Jordanian women with a university degree or above are unemployed. The percentage of woman that believe there are obstacles to women's employment is estimated at 76%. They consider these obstacles to have a cultural and religious background that pressures women to stay at home, as well as a lack of women's job opportunities. There is a large socio-economic gap that exists in Jordan. In 2009, only 16% of girls from underprivileged households excelled above level 2 math, compared to 57% of girls from wealthier households. Costly private schools that usually offer better education are reserved for the upper class of society²³.

Learning outcomes in Jordan are poor throughout basic and secondary education. 70% of students in Grades 2 and 3 are reading without comprehension²⁴. Students consistently perform poorly in international standardized tests, including in mathematics and science²⁵.

Education for women and men as well as vocational training programs in the growing sectors of the country's economy will contribute to macroeconomic growth and also to gender equity²⁶. According to the theory of demographic transition, fertility begins to decline when at least half of the female population is educated. In Jordan, education has progressed much faster than changes in behaviour concerning motherhood. This is due to the low employment rate of women because of social constraints related to the reproductive role for females. Women tend to have one child more than desired. This is especially the case for the poorest women, while the gap narrows in the wealthiest categories who manage to obtain their desired fertility. The recent report of the Population and Family Health Survey 2017–2018, showed a decline in the Total Fertility Rate from 5.6% in 1990 to 3.7% in 2002 and 3.5 % in 2012 to reach 2.7% in 2017-2018, which means that a woman in Jordan has an average of 2.7 children throughout her life.

²³ Klein, G. 2018. Top 10 facts about girls education in Jordan. Available at: <https://borgenproject.org/top-10-facts-about-girls-education-in-jordan/>

²⁴ National Committee for Human Resources Development, 2016

²⁵ UNICEF, 2019. Jordan. Available at: <https://www.unicef.org/jordan>

²⁶ https://pai.org/wp-content/uploads/2012/01/SOTC_PIB.pdf

2.7 Gender norms

Jordan has witnessed changes over the past decades that included the social construction and the prevailing values as a result of major changes in the educational, economic and technological aspects. These changes appeared as family patterns began to shift from the extended family to the nuclear family, as a result of urbanization, education and economic situation. As a result, the perception of women's education and work, reproduction and social relations — particularly in urban and civilized areas — and the change of the standards that control behavior.

However, the tribe still represents the primary social structure (paternal/masculine), and its orientations and representation are masculine as well. Therefore, this system of values, customs and traditions define an individual's awareness and concepts, and form their culture and behaviour especially for women. As a result, gender role allocation focuses on the man as head of household and the main source of income for the family and the woman as wife and mother. According to prevailing cultural norms, interactions between the sexes and mobility for women are limited so that family honour and dignity can be maintained. The norms and values that underscore this pattern place numerous obstacles in the way of women who hope to fulfil and balance their productive and reproductive roles.

Decision-making in the household depends on the issue at hand and is influenced by level of education, residency, age, status in the family and control of assets. With the exception of decision on women's own health care most decisions are made jointly by the couple. Regarding decision-making in the household most decisions regarding health care, major household purchases, and visits to their family or relatives are made jointly. The lowest decision-making ability of women pertain to major household purchases²⁷.

2.8 Women-headed households in Jordan

Women-headed households constitute about 17.5% of Jordan's households. By region of residence, religion, and household size they differ little from households in the country at large. 392,300 families out of 2.242 million families are woman-headed. 94.7 percent of the women who head their families are economically inactive. 105,642 households received monthly assistance from the National Aid Fund in 2020²⁸. Nearly half of women-headed households (48 percent) are living below the poverty line, compared with about 13 percent in the population at large. Their annual spending is less than one-third of the nationwide average. Very few of the women household heads work outside the home: 78 percent of them are not and have never been employed. Their education levels are low: about 54 percent are either illiterate or have only basic schooling, and about 27 percent have secondary education. The problems faced by women household heads include malnutrition (a problem in nearly one in five of these families); social stigma attached to themselves and their children; problems in rearing their sons, many of whom drop out of school and become socially withdrawn; and pressures from their own or their husband's family²⁹.

2.9 Women in the agricultural sector

Although the contribution of the agriculture sector to Jordan's Gross Domestic Product (GDP) and labour force has declined over the last decades, women's participation in the sector remains a critical source of employment for the country's poorest citizens, and also serves as a major source of subsistence and food security in the country³⁰. While only 2% of the total working population in the country — and 0.9% of its total female labour force — was employed in agriculture as of 2014,

²⁷ Gender Analysis report, 2015

²⁸ <https://jordantimes.com/news/local/392300-families-jordan-women-headed-—-sigi#:~:text=AMMAN%20—%20Women%2Dheaded%20households%20in,of%20Statistics%20figures%27%20for%202020.>

²⁹ https://www.worldscientific.com/doi/10.1142/9781783267347_0006

³⁰ UN women, 2018. Women's Participation in the Agricultural Sector, Rural Institutions and Community Life.

approximately 25% of the total number of poor people who live in rural areas continue to depend on agriculture as a primary source of livelihood³¹.

The agricultural sector in Jordan is also known to have the highest proportion of informal workers compared to other economic sectors. Approximately 16% of women who work in the agricultural sector are informally employed, which is higher than the proportion of men (~5%). According to FAO, women in Jordan are also actively involved in home-based agricultural activities, typically managing small homestead gardens and looking after the family plot³².

Despite women's seemingly active role in the agriculture sector, a review of existing literature and secondary data suggests that limited up-to-date information is available on the subject of women's involvement in the sector, possibly due to its informal nature³³. While some research exists on the challenges facing the agricultural sector in general, very limited up-to-date gender-disaggregated data is available on the challenges specific to women working in the sector.

The majority of the rural population depend either directly or indirectly on agriculture for their livelihoods. The rural poor are expected to face the most severe consequences of climate change because of their greater dependence on agriculture, their relatively lower ability to adapt and the high share of income they spend on food. Climate change could undermine the progress that has been made in poverty reduction and negatively impact on food security and economic opportunities for vulnerable rural populations. The negative impacts of climate change will be strongly by women.

The role of women in the rural economy is substantial. Women are traditionally responsible for the household economy and are active in agriculture as well. Women make crucial contributions in agriculture and rural enterprises in drylands as farmers, animal caretakers, workers and entrepreneurs. For rural women and men, land is the most important asset to support production and provide for food, nutrition and income security. However, while having the legal right to own land, social norms diminish women's access to land by favouring male inheritance and providing disincentives for women to purchase land³⁴. Legal and social restrictions on land inheritance have particularly adverse effects on poor women, since often in poorer, rural areas the primary way in which wealth is transferred is through land inheritance. Moreover, the acquisition of land by single or married women is viewed as outside the usual social norms, which is not the case for single or married men, as is the registration of wives as joint owners of property of their husbands. Likewise, given the customary patrilineal inheritance systems, where property is transferred to and through male members of the family, women often have to perform unpaid work on family farms. For instance, women account for only 5% of agriculture holders in Jordan, though the number of large (10 ha and above) agricultural area owned by men is twice as large as that of female holders³⁵. In a recent study, it was found that the majority of female respondents were workers on and not owners of the farms³⁶. Only 4.8% of the respondents were owners of the farm, while only 1.6% of them reported that her farm ownership is of private partnership nature³⁷.

Women empowerment programs may be another source of support for farms as these farms already utilize women labour in agricultural operations. In fact, 59.8% of the farms surveyed have women among their labour. The nature of work performed by women includes cutting and picking (38.2%),

³¹ Food and Agriculture Organization of the United Nations (FAO)

³² UN Women. 2018. Women's participation in the agricultural sector, rural institutions and community life. Available at: https://www.alnap.org/system/files/content/resource/files/main/reach_jor_unw_agriculture_report_final_unw_format.pdf

³³ UN Women, 2018. Women's participation in the agricultural sector, rural institutions and community life. Available at: <https://jordan.unwomen.org/en/digital-library/publications/2018/womens-participation-in-the-agricultural-sector-rural-institutions-and-community-life>

³⁴ Climate Change Profile Jordan, ministry of foreign affairs ,2008

³⁵ Sweidan, M. 2016. Gender Perspective on Measuring Asset Ownership for Sustainable Development in Jordan. Available at: https://unstats.un.org/unsd/gender/Finland_Oct2016/Documents/Jordan_paper.pdf

³⁶ Clean Technology and Environment Intervention in the Jordan Valley, 2019

³⁷ Clean Technology and Environment Intervention in the Jordan Valley, 2019

harvesting (63.6%), packaging (72.7%) as well as other tasks (5.5%).³⁸ Besides, it appears that women employed in farms are asked to practice multi-tasks. None of them were involved in fertilizers, waste composting, or marketing within the region, even though they are willing to do so, and have asked on several occasions to be involved in the mentioned activities.

2.10 *Women and land ownership*

Property ownership is a male domain in Jordan, where women are dependence on men for housing. A patriarchal pattern of power dominates both inheritance and property. While the inheritance rights of women are formally enshrined in the constitution, in Islamic law (Sharia), and in the customary law particularly common in the steppe regions, female heirs continue to face social pressure to renounce their rights in favor of male heirs. Most women either do not receive the share of inheritance that the law entitles them to or they are simply denied their right to housing and land. Jordanian Sharia Court data show that in 2014 a third of heirs fully relinquished their inheritance rights through the process of takha^{ruj}, the voluntary “opting out” of their rights (exclusion). Records do not specify their gender, but survey evidence shows most of those who requested takha^{ruj} were women. A survey conducted in the Irbid governorate in 2010 by the Jordanian National Forum for Women indicated that 20 percent of women had renounced their inheritance rights and that three-quarters of them had done so willingly. Jordanian women still appear to abide by the belief that they will harm the economic and social status of their brothers if they take up their right to inheritance. These social norms and beliefs have led Jordanian women to internalize and condone the privileges of men in matters of inheritance. Besides social pressure, women are deprived of their inheritance in several other ways. One takes the form of a donation to male heirs before a person’s death. Another is when owners and their male heirs choose to leave a property undivided, which prevents female heirs from using or selling their share for years and even decades. This often happens with agricultural land and family-owned buildings. In most cases male heirs give symbolic gifts to women, called badal or takr^{im}, which are worth far less than the value of shares they are legally entitled to receive. In general, women are kept in the dark about the real value of assets such as land or an apartment³⁹.

As a consequence of such cultural norms, only 6.8 percent of ever married women owned their houses and only 7 percent owned land in 2012. A recent trend has seen women managing to register residential apartments in their name upon marriage, mainly in the major cities of Amman and Irbid. Women owned 19.5 percent of registered apartments in Jordan in 2014. But since apartments constitute 42 percent of housing units, mainly built as a form of investment, that translates into women owning a mere 10 percent of all apartments. In comparison, women owned 22.5 percent of securities, whether bonds, shares, or options in 2011. Husbands often register apartments and securities in the name of their wives to protect their assets from creditors, if their business goes bankrupt. Less than 3 percent of Jordanian people are Christian, but historically they have owned large tracts of land and are significant players in the nation. Sharia law has been applied to Christian property inheritance since the Ottoman period in Jordan, and although Christian tradition stipulates that women be granted the same shares as men, Christian deputies have never asked for the inheritance law to be revised⁴⁰.

Women’s rights activists have succeeded in amending the Jordanian Personal Status Law No. 36 of 2010 by working with the Sharia Supreme Court (Qa^{di}-al-Quda^h), the country’s highest religious and legal institution. In 2011 seven amendments were made and two articles added: articles 318 and 319 on property and exclusion. The most important amendment introduced a mandatory three months after the death of the owner before either exclusion (takha^{ruj}) or power of attorney (waka^{la}) could be invoked. Another amendment stated that an attorney dealing with an estate was

³⁸ Clean Technology and Environment Intervention in the Jordan Valley, 2019

³⁹ https://arabstates.gltn.net/wp-content/uploads/2021/02/Enforcing-Womens-Rights-to-Property-in-Jordan_Myriam-Ababsa.pdf

⁴⁰ https://arabstates.gltn.net/wp-content/uploads/2021/02/Enforcing-Womens-Rights-to-Property-in-Jordan_Myriam-Ababsa.pdf

under obligation to inform the heirs of all properties included in an inheritance. Another specified that a judge must explain to the heirs the legal consequences of the exclusion of inheritance; this has resulted in judges asking for better education for women so that they can better defend their rights. The improvement of women's rights to inheritance has been proffered since the 1960s as a solution to poverty and a means of preventing women from falling into destitution. Property rights and security of tenure are basic not only to shelter but also to improvement of livelihoods, economic prosperity, and sustainable development. The United Nations' human settlement program, UN-Habitat, and the Office of the High Commissioner for Human Rights support property rights and the security of tenure under the Right to Adequate Housing Framework. Jordan is working on incentives for better inclusion of women in the economy, and inheritance and property should be a major dimension of this policy⁴¹.

2.11 Women and Water

Jordan suffers from water scarcity — a fact that places severe strain on its social and economic development prospects. Currently almost 60% of Jordan's water is used for irrigated agriculture. Water allocation for domestic use comprises less than 30%. In terms of drinking water allocation, cuts below a certain threshold endanger the very survival of people.

Women play an important role in water management. They are most often the collectors, users and managers of water in the household as well as farmers of irrigated and rainfed crops. Because of these roles, women have considerable knowledge about water resources, including quality and reliability, restrictions and acceptable storage methods, and are key to the success of water resources development⁴². Decisions about water rationing have a major impact on women, since they are responsible for household activities that require water, such as cooking and cleaning. The gender assessment study conducted by USAID in 2012 highlighted that, both women and men mentioned that in neighbourhoods where water is only available during certain days and times, women must be at home during those periods in order to complete their household chores.⁴³

Women are not only responsible for food preparation, but also for keeping the family healthy, which means taking care of sick family members. The combination of these family roles makes them potentially strong advocates for clean water. Therefore, it is important to educate women on the connection between safe drinking water, contamination of the water supply, and the impact on the health of the family (especially children). The environmental movement in Jordan already has strong women advocates who are showing interest in policymaking and political engagement, making them potential role models for other women. Research in Jordan has identified that programmes which involved women in their activities were very successful in achieving their objectives. They had a greater impact on conserving water resources and helping local communities to achieve acceptable living conditions and to adopt the sustainable use of water resources⁴⁴.

2.12 Climate change

In the fight against climate change, the role of women as agents of change is frequently overlooked. Women are often perceived as passive members of households or indirect victims of climate change. However, it is critical to recognise the actual role of women within their communities as educators, practitioners, and influential agents of change. Empowering women to harness their leadership capacities is therefore essential. Women can inform and implement actions and strategies related

⁴¹ https://arabstates.glttn.net/wp-content/uploads/2021/02/Enforcing-Womens-Rights-to-Property-in-Jordan_Myriam-Ababsa.pdf

⁴² Al-Naber, S. and Shatanawi, M., 2004. The roles of women in irrigation management and water resources development in Jordan. *Integration of gender dimension in water management in the Mediterranean region: INGED project*. Bari: CIHEAM, pp.97-113.

⁴³ USAID, 2012. Jordan Gender Analysis and Assessment Report.

⁴⁴ UNDP, 2013. Jordan Poverty reduction strategy. Available at: <https://www.undp.org/content/dam/jordan/docs/Poverty/Jordanpovertyreductionstrategy.pdf>

to water management, energy, agriculture and food security, economic growth and livelihoods, waste management, and policymaking.

In this regard, a study released in 2010 by the International Union for the Conservation of Nature (IUCN) demonstrated the strength of women's ability to enact social change by utilising their knowledge, skills and capacities, therefore informing Jordan's climate change policies. The resulting gender-sensitive Climate Action Plan (ccGAP) outlined a framework for integrating a gender perspective in climate change efforts, acknowledging the fundamental role of women in the sectors of water, agriculture, energy, and waste management and the vast contribution that women could make to climate change adaptation and mitigation⁴⁵.

3 Demographic Information

Jordan, is an Arab kingdom in the Middle East. It is located between the Mediterranean and Red Seas and contains the massive Jordan River. The land itself is mostly flat and covers 34,495 square miles (89,342 square kilometres) of surface area.

95-97% of Jordanians are of Arab decent, but there are many other ethnic groups in the country. There are ~500,000 Iraqis, over 500,000 Syrians, ~30,000 Kurds, ~5,000 Armenians Assyrian Christians make up 0.8% of the population. In 2019, the population of Jordan was estimated at ~10 million, which makes it the 89th most populous nation. Jordan is an extremely urban country, with nearly 90% of the people living there residing in cities or the surrounding metro areas. Jordan has become the destination of several waves of migrants from Palestine and recently from Iraq and Syria. The total number of Syrian refugees in Jordan is an estimated at 1.4 million persons. UNHCR reports that of the refugees of Syrian nationality, % are living under the poverty line. Most refugees registered with UNHCR live in urban areas where competition for resources and services with Jordan communities is most intense.

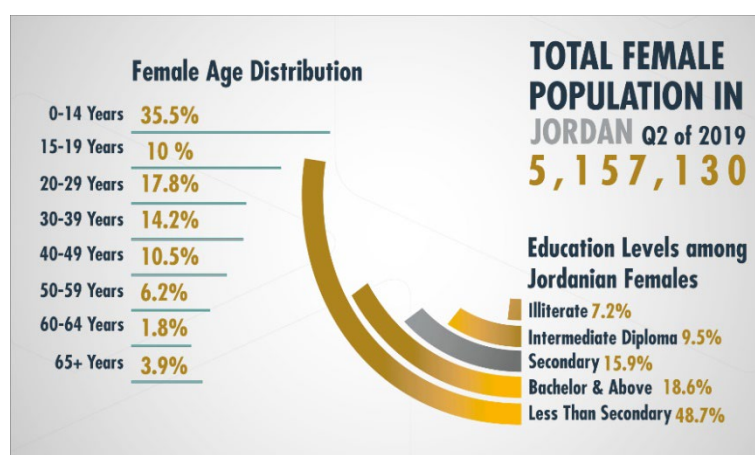


Figure 8. Age distribution and education statistics⁴⁶.

⁴⁵ UN Women, 2018. Rural women and climate change. Available at: <https://data2.unhcr.org/en/documents/download/66494>

⁴⁶ Worldometers. 2019. Jordan Population. Available at: <https://www.worldometers.info/world-population/jordan-population/>

The influx of Syrian refugees has increased the demand for energy in the Jordan and total residential energy consumption has risen significantly in recent years⁴⁷. According to the results of the Population and Housing Census 2015 and the Migration streams, a significant increase in population growth occurred between the periods of the censuses 2004–2015, reaching ~5.3%.

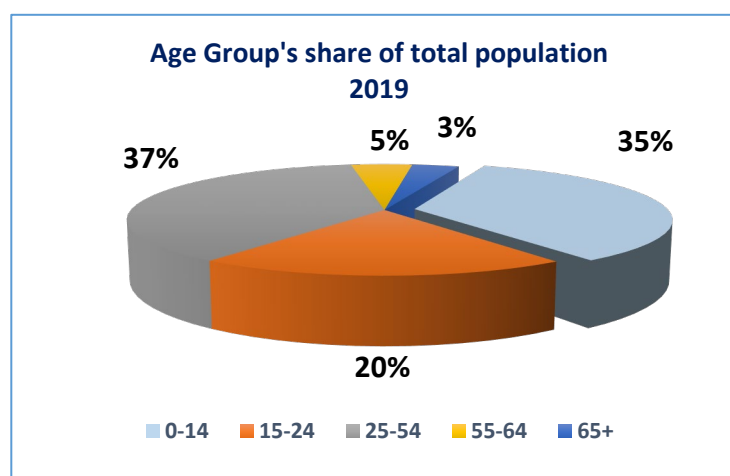


Figure 9. Age group share of total population⁴⁸.

The overwhelming majority of the population of Jordan is Arab, and mostly Sunni Muslim. Jordanian society also includes ethnic minorities, including Circassians, Kurds, Turkomans, Chechens, and Armenians. A substantial number of Jordanians are Christian, the second most numerous religious community, and there is a small Druze community. The Jordanian state as a whole is very supportive of all the cultural, ethnic, and religious components of its society⁴⁹. Traditional social institutions such as the family and tribal system, as well as modern institutions — for example family associations and charitable organizations — are very influential in the preservation of the identity of the smaller groups. They also play an important role in supporting a collective Jordanian cultural identity. Political, social and religious leadership, as well as civil-society organizations, are also very influential in preserving diversity.

The Jordanian Constitution stipulates freedom of religion to all Jordanians, regardless of ethnic or religious origin. Access to political, civil, and welfare rights is guaranteed to all Jordanians by the Constitution. Although in family matters the law is heavily influenced by Islamic jurisprudence, the Christian community has its own judicial councils to resolve disputes between followers of the various Christian denominations.

⁴⁷ UN Women. 2018. Women's participation in the agricultural sector, rural institutions and community life. Available at: https://www.alnap.org/system/files/content/resource/files/main/reach_jor_unw_agriculture_report_final_unw_format.pdf

⁴⁸ Worldometers. 2019. Jordan Population. Available at: <https://www.worldometers.info/world-population/jordan-population/>

⁴⁹ World population review. 2019. Jordan Population. Available at: <http://worldpopulationreview.com/countries/jordan-population/>

Jordan's economy is among the smallest in the Middle East, with insufficient supplies of water, oil, and other natural resources, underlying the government's strong reliance on foreign assistance. Other economic challenges for the government include chronic high rates of unemployment and underemployment, budget and current account deficits, and government debt⁵⁰. Jordan has a very young population. In 2018, about 35% of Jordanians was under the age of 15 (male 1,835,094 and female 1,735,773), whereas, 20% are 15–24 years of age. The estimated Jordan population growth rate is ~2% for the year 2019.

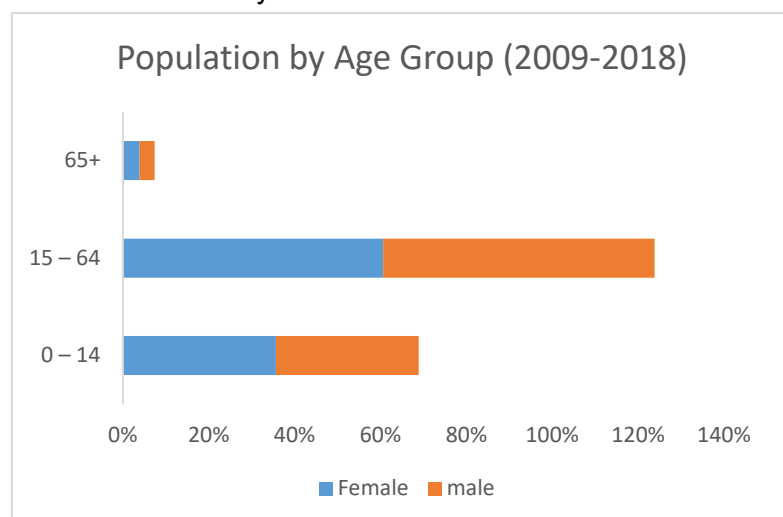


Figure 10. Population by age group.

New research findings highlighted the relationship between demography and age structural factors, showing that demography is an important factor for development. Youth should be one of the main economic assets for a country or a region, therefore, benefiting from this age structure will be through sound policies, which ensure that, young people enter the labor market, they are adequately educated and jobs exist for them. Education, training and a well-functioning labor market are prerequisites to obtain the benefits of progress through this demographic transition⁵¹.

Economic growth will have been downbeat in the third quarter, after slowing slightly in the second quarter. In the first two months of the quarter, bank lending growth in year-on-year terms averaged below Q2's over four-year low, suggesting fixed investment growth was depressed in Q3. Moreover, industrial production decreased in July compared to the same month a year earlier for the third consecutive month. However, spending increased in July from the same month a year earlier, which should have supported the economy somewhat, although the government's fiscal deficit widened in January–July in annual terms. In addition, inflationary pressure was lower in Q3, which should have supported private consumption⁵².

⁵⁰ <https://theodora.com/wfbcurrent/jordan/jordan>

⁵¹ <http://dosweb.dos.gov.jo/>

⁵² <https://www.focus-economics.com/>

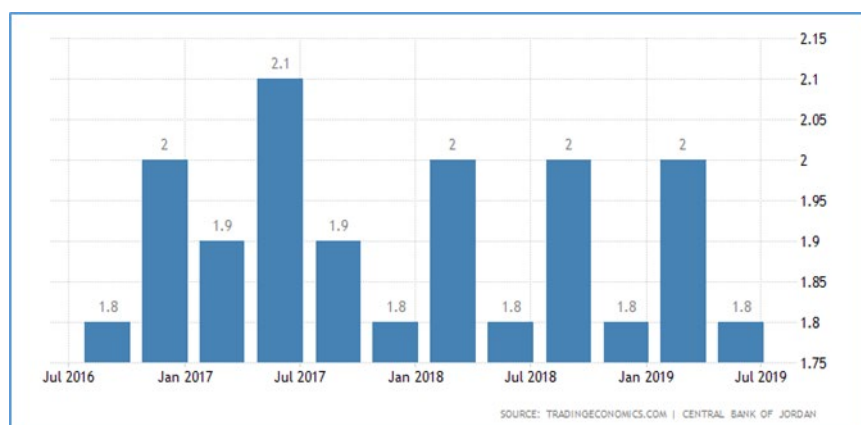


Figure 11. Jordan GDP 2016-2019.

Jordan's real GDP registered a growth of 2% in 2018, a bit lower than growth in 2017, constrained by structural weaknesses and a difficult regional setting. On the supply side, a strong services sector (contributing 1.5% to GDP growth), followed by the industrial sector (0.4% to GDP growth), and provided the dominant sector. On the demand side, growth from a significant improvement in net exports was countered by a major contraction in private demand during the first half of 2018. Investment growth was moderate due to a major reduction in public capital expenditures and low inflows of private investment.⁵³

4 Legal Framework

4.1 Jordan Constitution

The Constitution of the Hashemite Kingdom of Jordan stipulates that all Jordanians shall be equal before the law. There shall be no discrimination between them as regards to their rights and duties on grounds of race, language, or religion. Reference in the Constitution to the rights of Jordanians includes both men and women. However, the Constitution does not specifically prohibit gender discrimination. National reports issued by Jordan arising from international human rights conventions recognize that the Constitution does not distinguish between women and men.

4.2 National laws

Table 1. The main laws of Jordan relevant to gender justice.

National Laws	Year
Constitution of Jordan	1952
Crime Prevention Law, No. 7	1954
Nationality Law, No. 6	1954
Penal Code, No. 16 with amendments	1960
Labour Code, No. 9	1996
Law on Protection from Domestic Violence, No. 6	2008
Human Trafficking Law, No. 9	2009
Temporary Personal Status Law, No. 36	2010
Law on Electronic Crimes, No. 27	2015
Shelters for Vulnerable Persons System, No. 17	2016
Law on Protection from Domestic Violence, No. 15	2017
Law on the Rights of People with Disabilities, No. 20	2017
Regulation of Flexible Work System, Regulation No. 22	2017

⁵³ <https://www.worldbank.org/en/country/jordan/publication>

While women have been appointed to the judiciary since 1995, they are poorly represented in the judicial system — only 19% of judges and 27% of all practising lawyers.

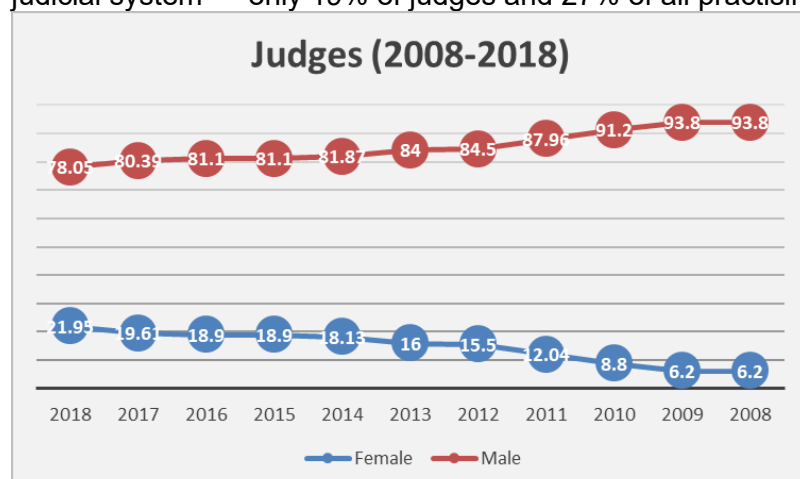


Figure 12. Women's representation in the judiciary (as judges).

In 2018, the first woman was appointed to the Court of Cassation, but no women serve in the Sharia Courts. Although qualified women are permitted to serve in this role, those who oppose the idea of having women in Sharia courts state that women are too emotional to occupy such positions. The presence of female lawyers and judges encourages women to claim their rights through the legal system and increases their knowledge about their legal rights⁵⁴.

4.3 Jordan's International Commitments

Jordan is signatory to all major international instruments pertaining to women's rights and gender equality, including: i) the Universal Declaration of Human Rights; ii) the International Covenant on Economic, Social and Cultural Rights; iii) the International Covenant on Civil and Political Rights; iv) the Convention on the Political Rights of Women; v) the Convention on the Rights of the Child; vi) the United Nations Convention against Transnational Organized Crime and its Protocol to Prevent, Suppress and Punish Trafficking in Persons, Especially Women and Children; vii) the Arab Charter of Human Rights; and the International Labor Organization (ILO) conventions on equal remuneration and discrimination in respect of employment and occupation, among others. Jordan is signatory to the Convention on the Elimination of All Forms of Discrimination against Women (CEDAW, ratified in 1992)⁵⁵.

However the country maintains reservations on the article 9 (2) (which is supposed to give equal rights to women with regard to nationality of children), to article 16 paragraph 1 (c) (same rights and responsibilities during marriage and at its dissolution), (d) (same rights and responsibilities as parents), and (g) (same personal rights, including the right to choose a family name, a profession and an occupation). Jordan maintains its reservations on article 16 (1) (c) and (d) because these are not compatible with Shari'a provisions. In 2009, a Cabinet decision lifted its reservations on paragraph 4 of Article 15, and gave women freedom of mobility and choice of residence without consent of their husbands or other male family members.⁵⁶

Although the government made improvements in the legislations, the provision leaves a gap, Jordan still have to take actions in some of the discriminatory laws against women. For example, Jordan should address the Jordanian Nationality Law, which does not allow Jordanian women married to foreign-born spouses to pass on their nationality to their spouses and children, and stop what is

⁵⁴ <https://www.undp.org/content/dam>

⁵⁵ OECD. 2019 Social Institutions and Gender Index. Country profile: Jordan. Available: <https://www.genderindex.org/wp-content/uploads/files/datasheets/2019/JO.pdf>

⁵⁶ JIP, 2018. Archives for 2018. Available at: <https://jordaningoforum.org/2018>

called “administrative detention”, which is used as a protective measure for women considered to be at risk of being killed by a family member, and allows her indefinite detention under the country’s 1954 Crime Prevention Law. It is worth noting that, despite a minimum age of marriage set at 18, judges are allowed to approve marriage agreements for girls at 15 years old, which contributes to the rise of child marriage in Jordan, including in the Syrian refugee community.⁵⁷

4.4 *National Women’s mechanisms*

4.4.1 The Jordanian National Commission for Women JNCW

The Commission was established in 1992, it was the first national mechanism mandated to achieve gender equity and social justice by improving the status of women, enhancing their economic, social and political participation, safeguard their gains, and defend their rights. The responsibilities assigned to JNCW and translated in its mandate focused on policy, laws and legislations, networks, exchange of knowledge and expertise and the national women strategy.

JNCW lack of government commitment to integrate gender within its various ministries which results in scattering JNCW’s efforts here and there instead of systematic approach in the handling of women issues. This affects JNCW role and transforms their tasks to a more ad hoc basis. JNCW remains an active participant in any request from the Government to ensure gender integration and addressing women issues. JNCW works strategically with Ministries of Planning, Justice, Labour, Social Development and Public sector reform to integrate comprehensive programs on gender equality and women empowerment.⁵⁸

JNCW works strategically with Ministries of Planning, Justice, Labor, Social Development and Public sector reform to integrate comprehensive programs on gender equality and women empowerment. Also, JNCW built strategic partnerships with Working NGOs for effective results on the ground and youth groups to change the current cultural practices and spread new culture of tolerance and respect for women’s rights.⁵⁹

JNCW drafted the first national strategy for women in 2003, it serves as road map and guiding document for government to promote the rights of women under each component. The latest national strategy for women 2013–2017 was endorsed by the cabinet in 2013, covering human security, political empowerment, economic empowerment, media and ICT, integration of gender in policies and legislations, Institutional improvement for organizations working on women issues and the popular culture on women empowerment.

4.5 *Governmental Mechanisms to promote Gender equality and women empowerment*

The governmental gender focal point network was established by Cabinet decree in 1996. The network has representation of most governmental organizations. They are gender units of four Ministries namely, Ministry of Labour, Planning, Social Development and Health.

4.5.1 Directorate of Women, Ministry of Labour⁶⁰

In 2006 the Ministry of Labour had a division for women. It was renamed in 2008 to take on the responsibilities of a directorate according to the new organizational structure. The tasks of the women directorate include economic empowerment, as well as awareness and education. The

⁵⁷ JIP, 2018. Archives for 2018. Available at: <https://jordaninfoforum.org/2018>

⁵⁸ UN Women, 2015. Assessment of National Women Machineries. Available at: <https://www2.unwomen.org>

⁵⁹ UN Women, 2015. Assessment of National Women Machineries. Available at: <https://www2.unwomen.org>

⁶⁰ UN Women, 2015. Assessment of National Women Machineries. Available at: <https://www2.unwomen.org>

directorate worked on amending the labour law to include article on creating nurseries in the workplace, conducting training on pay equality and implementing in cooperation with ILO program on pay equity. However, the directorate needs to be empowered. There is a need to win the support and championship of the senior management to strategically mainstream gender within the departments/directorates of the Ministry and adequate funding to implement certain initiatives.

4.5.2 Gender Unit, Ministry of Planning and International Cooperation

The department develops economic policies through conducting research and studies. It contributes to determining objectives, policies and development indicators with full coordination with JNCW and the gender focal points in the ministries, government institutions and non-governmental organizations. In addition, one of the department's mandates is to build a system to monitor and evaluate policies and programs and their impact on gender equity.

As other gender units in other ministries, the gender unit does not have the proper implementation mechanisms to ensure gender is properly addressed at all levels of the ministry, besides the limited financial and human resources to provide technical assistance on gender mainstreaming.

4.5.3 Gender Unit, Ministry of Social Development

The gender unit falls under the supervision of the directorate of Studies and Policy Development. The directorate's objective is to prepare strategic studies, as well as to follow up on progress and implementation. The directorate formulates policies, analyses data and social norms. However, the unit is operated by only one staff member and lacks sufficient financial and human resources. In addition, leadership in the area of gender analysis and mainstreaming is required. There is also a need to have a customized manual to give guidance to the gender units on how to mainstream gender in policies, plans, programmes and budgets.

4.5.4 Ministry of Health

There is a gender focal point that has received advanced trainings on gender and gender responsive budgeting through the trainings provided by UN Women and other EU programs. The gender focal point is in continuous cooperation with JNCW and UN Women on all matters related to gender within the Ministry. However, the position, authority and resources of the gender units should be strengthened, and the structure of the gender unit revised.⁶¹

4.5.5 National Laws

Since 2007, the Government of Jordan has taken significant steps to address issues raised by civil society regarding the status of women in national law.

4.5.6 Labor Law of 2008

Jordan has ratified the International Labour Organization (ILO) Equal Remuneration Convention and included its provisions in Jordanian law, legally discrimination in employment on the basis of gender or marital status is not permitted. Sexual harassment is an offence and abuse of women and children in the workplace is liable to punishment. However, no monitoring system is in place, which indicates that these laws are not activated applied. Implementation of laws is negatively affected by lack of capacities, prevalent patriarchal norms and gender roles, and the domination of men in the public sphere.

⁶¹ UN Women, 2015. Assessment of National Women Machineries. Available at: <https://www2.unwomen.org>

Therefore, a significant pay gap remains even for the same type of work, the wage gap between genders has reached 13.8 % in the public sector and 14.2% in the private sector. In the health sector, the gap was at 31.8% and in private education it had reached 30.2%⁶². In terms of social benefits, the Civil Service Code remains discriminatory in some of its articles. Family allowances are automatically paid to men, while it is granted to women only in certain circumstances. Within the social security system, pension payments to the family are only paid to a woman's family under limited circumstances despite the fact that social security contributions are equal.

Many firms discriminate against hiring women due to the perceived higher additional costs they occur for maternity leave. Several reforms have been introduced in recent years to address this discrimination, but they still do not go far enough to close the gender gap in labor legislation.

The labor law amendment made in 2008 condemning sexual harassment in work place and the new Civil Services Statute 2007⁶³ that considers sexual harassment in public service was a positive step, however activation of these two provisions is limited and no procedures are put in place to raise women's awareness on the importance of these laws. Also, cultural norms constrains women from reporting on sexual harassment incidents so not to be shamed and humiliated.

4.5.7 Personal Status laws

The Personal Status Law (No. 61 of 1976) which is also known as "Family Law", organizes all matters related to the family including children. The Law is adapted from Islamic jurisprudence.

The Personal Status Law is meant to guide decisions and regulation in matters of marriage, divorce, custody, and inheritance. In 2012, the Government of Jordan introduced some positive major revisions to the Personal Status Law, it address past Inequalities, as in the field of inheritance, in which the amended Law establishes a three-month waiting period before a women can waive her inheritance rights after the death of her husband or father, which prevent her male relatives to approach her to sign documents that would waive her inheritance rights without her knowledge⁶⁴. Another reform that is very beneficial for divorced Jordanian men is the clause that divorced parents are able to see their children at their homes instead of police stations or local organisations.

Under the Nationality Law, women cannot pass citizenship to their children or spouse in the same way as men. Jordanian women married to non-Jordanian men cannot pass their citizenship to their children or husbands, although they may retain their own Jordanian citizenship⁶⁵. In November 2014, the government approved granting Jordanian women married to foreigners and their children a set of privileges. These privileges did not amount to Jordanian women being allowed to pass on their citizenship to their children and spouses (a right Jordanian men enjoy) but substantially eased their lives⁶⁶.

4.5.8 Election Law

Women are poorly represented in the political sphere, which prevents them from successfully contributing in the decision-making process that would shape the political priorities, policies and practices for women. The quota system was introduced in the 2003 elections through a legislative amendment, reserving 6 out of 110 seats for women candidates. The quota was doubled to 12 seats in the Election Law of 2010, resulting in a total number of 13 women in the Lower House of

⁶²Jordan Times, 2018. Gender pay gap in public sector reaches 13.8 percent – committee. Available at: <http://www.jordantimes.com/news/local/gender-pay-gap-public-sector-reaches-138-cent-%E2%80%94-committee>

⁶³ The UPR Report on Women's Rights in Jordan submitted by the AWO Coalition-Feb. 2013

⁶⁴ OECD. 2019 Social Institutions and Gender Index. Country profile: Jordan. Available: <https://www.genderindex.org/wp-content/uploads/files/datasheets/2019/JO.pdf>

⁶⁵ <https://www.undp.org/content/dam/rbas/doc>

⁶⁶ OECD. 2019 Social Institutions and Gender Index. Country profile: Jordan. Available: <https://www.genderindex.org/wp-content/uploads/files/datasheets/2019/JO.pdf>

Parliament; 12 filled by quota and one by direct competition. The new electoral law implemented in the 2013 parliamentary elections increased the number of parliamentary seats to 130 and women's quota from 12 to 15 seats, it kept the level of representation of women to around 11%, far short of the Millennium Development Goals (MDG).

On the other hand, some steps have been made in easing the atmosphere for organizing as part of the government's recent efforts to promote democratic reform in the areas of freedom of assembly, freedom of association and freedom of expression. However, these changes in legislation have not been translated into practice in ways to produce the desired effect of real freedom and equality⁶⁷.

4.5.9 Protection against Domestic Violence

In Jordan, domestic violence is widely regarded as a family matter that affects large numbers of women, children and adults across their lifespan. In cultural and religious matters, Jordanian society has conservative attitudes towards gender roles within society, and the behaviour of men and women is strictly defined and formed according to the social system. The structure of the Jordanian family and the distribution of power within it is one of the reasons behind violence in Jordan. In other words, men according to the cultural norms have the power and exercise complete authority over all the family's affairs due to the financial power they have. On the other hand, women are expected to fulfil their childbearing and homemaking roles, including loyalty to their husbands and preserving the reputation of their families, otherwise they are condemned socially and may be exposed to physical punishment. To avoid such consequences, women have to comply with social and cultural rules and should obey all instructions and orders given by men to avoid punishment⁶⁸. In this regard, women in Jordan still widely accept that violence against them by men could sometimes be justifiable. As a result, women do not seek help from governmental agencies.

Article 98 of the Penal Code was applied to reduce penalties for men who perpetrated violent crimes against women, such as murder or assault, after the woman had committed a dishonourable act. Article 98 was amended in 2017 to prevent it from being used to reduce penalties for so-called 'honour' crimes against women⁶⁹. Jordan has undertaken a number of procedural, administrative, judicial and legislative measures to combat and eliminate domestic violence. It has created many public welfare institutions and has given them a general mandate to provide protection programmes, promoted in all regions of the kingdom. Departments for family protection have been established at both the Public Security Directorate and the Ministry of Social Development, and a National Council for Family Affairs has also been created.

4.5.10 National government strategies and policies for climate change and gender

The Ministry of Environment (MOE) is primarily responsible for overseeing the policy and legal frameworks that guide climate change mitigation and adaptation efforts of the country, including the development of the Third National Communication (TNC) to the United Nations Framework Convention on Climate Change (UNFCCC) and the Intended Nationally Determined Contributions (INDC). The MOE works closely with the Ministry of Agriculture, the Ministry of Health and the Ministry of Water and Irrigation, which is responsible for managing water resources by implementing irrigation policy, determining water allocation, constricting water infrastructure and establishing water conservation programs.

Jordan has prepared climate strategies, communications to the UNFCCC, and action plans including:

- Initial National Communication (1999)

⁶⁷ Women's Rights in Jordan, Arab Women Organization, 2013

⁶⁸ Protection against Domestic Violence in Jordanian Law and International Conventions, Laith K. Nasrawin, Arab Law Quarterly 31 (2017)

⁶⁹ Available at: <https://www.undp.org>

- Second National Communication (2009)
- Third National Communication (2014)
- National Strategy and Action Plan to Combat Desertification (2015–2020)
- National Climate Change Policy and Sector Strategic Guidance Framework (2013–2020)
- Climate Change Adaptation and Low Emission Development Strategy (2013)
- Water for Life: Jordan’s Water Strategy (2008–2022)
- Climate Change Policy for a Resilient Water Sector (2016)
- Intended Nationally Determined Contribution (INDC) 2015, submitted as its First NDC in November 2016.
- A National Green Growth Plan for Jordan (2017)

4.5.10.1 The National Climate Change Policy

The National Climate Change Policy (2013–2020) provides an overview of Jordan’s strategies to combat climate change across various sectors and actions that the Jordan will implement in the coming years. Special attention is given to “vulnerable groups” that stand to be severely impacted by the negative effects of climate change, as well as strategies to address gender imbalances between men and women (Jordan was the first Middle Eastern country to include gender considerations in its national climate change policies).⁷⁰

4.5.10.2 Third National Communication

Third National Communication (TNC) (2014) builds on the objectives and proposed actions of the Climate Policy. The TNC includes projected impacts of climate change on Jordan, as well as a comprehensive mitigation assessment and a detailed inventory of GHG emissions. Using downscaling projection models, it reports that Jordan will be highly vulnerable to climate change impacts, especially the projected decrease in precipitation and increase in temperature and dry spells. It notes that this is an extreme concern for a country that is among the poorest in availability of water resources in the world and facing a huge challenge of accommodating the massive influx of refugees from neighbouring countries. The TNC contains comprehensive vulnerability assessments for major sectors in Jordan, including assessments of four rural communities, and identifies opportunities for mitigation and adaptations that Jordan will pursue with support of the international community⁷¹.

5 Situational Analysis — Jordan Valley Targeted Areas

5.1 Socio economic profile

The Jordan Valley forms part of the larger Jordan Rift Valley. The internationally recognized World Heritage values of the Jordan Valley are strongly related to its unique historic, religious, cultural, economic, and environmental values, due to its typical rift valley topography. The lower part of the Jordan River (LJR) originates at the Sea of Galilee and meanders along 200 km down to the Dead Sea through the Jordan Valley.

Climatically, the Jordan Valley is characterized by hot dry summers and mild wet winters, becoming progressively drier moving southward through the valley towards the Dead Sea. Climate change impacts are likely to intensify the water supply-related problems in the Jordan Valley⁷². Impacts of climate change includes a foreseen reduction in local annual water resources with a maximum of 20 % by 2050 and increasing temperatures and related surface water evaporation rates based on 2008 climate change impact assessment.

⁷⁰ The National Climate Change Policy of the Hashemite Kingdom of Jordan 2013–2020

⁷¹ Third National Communication, 2014

⁷² https://link.springer.com/chapter/10.1007/978-3-319-30036-8_2

The northern part of the East Bank of the Jordan Valley in Jordan will be impacted most negatively by climate change, with a foreseen substantial reduction of annual and winter rainfall, although summer rainfall will increase slightly.

The study area houses a population of about (561,510)⁷³. The information with regard to the population numbers in the study area have been obtained through the Jordanian Department of Statistics (DOS). The population growth rate for the Jordanian section of the study is estimated at 2.2 %⁷⁴.

The Jordan Valley houses large number of informal foreign workers originating mainly from Egypt and Iraq, and lately including some refugees from Syria. It is estimated that ~108,964 households live in the Jordan Valley today, some of them employed in the agricultural sector⁷⁵.

Table 2. Estimated population in the study area (2018) ⁷⁶

	Male	Female	Total
Al Arda sub district	9,100	7,790	16,890
Ain Al Basha	100,390	90,450	190,840
Al Showneh al Janoubiyeh	31,600	25,330	79,350
Al aghwar al shamaleyeh	70,720	61,380	132,100
Bani Kinana	72,160	70,170	142,330
Total	283,970	255,120	561,510

Due to unavailable updated data sources for directorates and sub directorates sex disaggregated data, and for the purpose of this assessment, DOS available data for the year 2015⁷⁷ were used to describe the socio-economic profile for the selected project areas as follows.

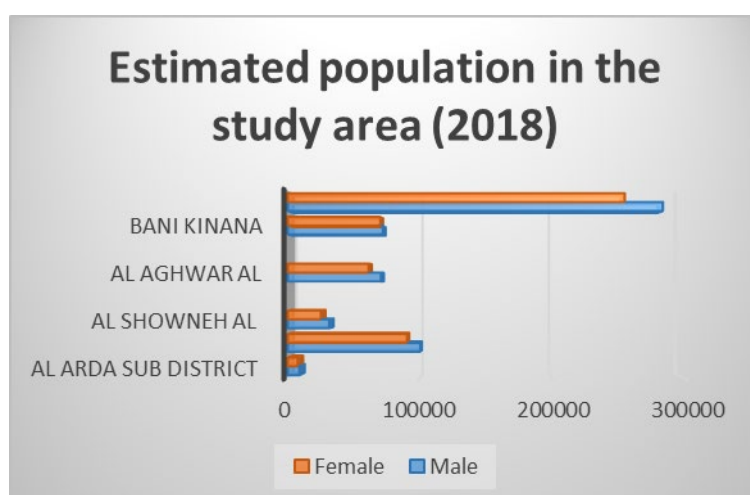


Figure 13. Estimated population in the study area (2018).

⁷³ DOS statistics, 2018

⁷⁴ DOS statistics, 2015

⁷⁵ https://link.springer.com/chapter/10.1007/978-3-319-30036-8_2

⁷⁶ DOS statistics, 2018

⁷⁷ Citypopulation, 2017. Jordan: Administrative Division. Available at: <https://www.citypopulation.de/en/jordan/admin>

Name	Population	
Ain El Bash	176,726	
Males	92,949	52.59%
Females	83,777	47.41%
Urbanization		
Rural	14,306	8.10%
Urban	162,420	91.90%
Nationality		
Jordanian	142,621	80.70%
Foreigners	34,105	19.30%
Poverty rate	20.9 ⁷⁸	
Unemployment rate +15⁷⁹		
Male	16.9%	
Female	24%	

Name	Population	
Al showneh al Janobeyeh	52,714	
Males	29,260	55.51%
Females	23,454	44.49%
Urbanization		
Rural	39,040	81.82%
Urban	8,674	18.18%
Nationality		
Jordanian	40,218	84.29%
Foreigners	12,504	26.21%
Poverty	20.9	
Unemployment rate +15		
Male	16.9%	
Female	24%	

Name	Population	
Al Aghwar al shamaleyeh	122,330	
Males	65,491	53.54%
Females	56,839	46.46%
Urbanization		
Rural	17,751	14.51%
Urban	104,579	85.49%
Nationality		
Jordanian	107,197	87.63%
Foreigners	15,133	12.37%

⁷⁸ DOS, 2008

⁷⁹ DOS, 2018

Poverty	15%
Unemployment rate +15	
Male	12.6%
Female	30.7 %

Name	Population	
Bani Kinana	131,797	
Males	66,856	50.71%
Females	64,971	49.29%
Urbanization		
Rural	37,317	28.31%
Urban	94,480	71.69%
Nationality		
Jordanian	112,735	85.54%
Foreigners	19,062	14.46%
Poverty	15%	
Unemployment rate +15		
Male	12.6%	
Female	30.7 %	

13% of the total population lives in rural areas, where poverty is more prevalent than in urban areas and living conditions are difficult. As an arid country with less than 5% of land available for farming and little to no rainfall, prospects of self-sustainability through farming are limited. The most vulnerable groups include large rural households with low literacy and education levels, women-headed households, households 12% with sick or elderly people, and households that do not own land or have very little land.

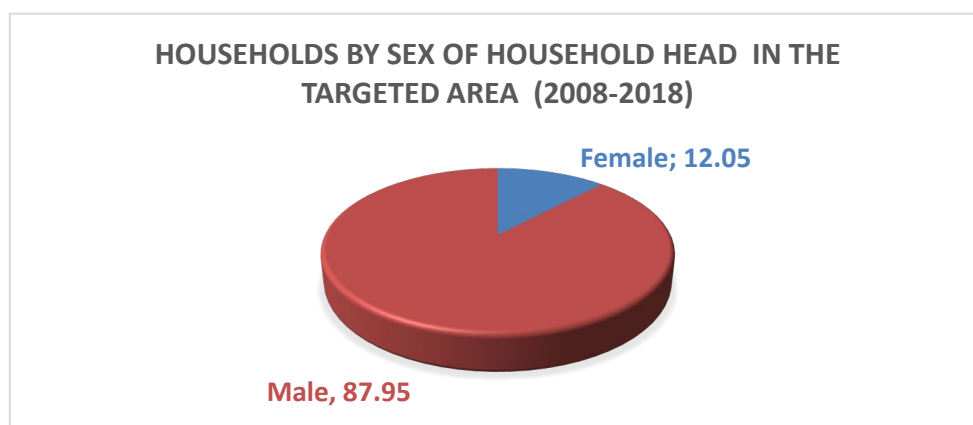


Figure 14. Households by sex of household head in the target area (2008–2018).

Local citizens of the Jordan Valley are involved in mixed farms that covered crop and livestock production systems. In addition, some Bedouins also live in the Jordan Valley and use the lands as grazing ground for their sheep and goats during the winter months because of its warm climate and available fodder for their animals. The livestock will be impacted by climate change, whereas self-sufficiency degree (SSD) for most food items would decline by 2050. Of great concern, production

of red meat would be highly affected by climate change, as rangelands would not provide enough forage for sheep and goats.⁸⁰

In the Jordan Valley there is a small group of wealthy agricultural entrepreneurs, but also to a large group of labourers who live close to the poverty line of JD 32.6 per person per month. The negative impacts of climate change will be strongly felt by women. Women make crucial contributions in agriculture and rural enterprises as farmers, animal caretakers, workers and entrepreneurs. Land is the most important asset to support production and provide for food, nutrition and income security, however, while having the legal right to own land, social norms diminish women's access to land by favoring male inheritance and providing disincentives for women to purchase land.⁸¹

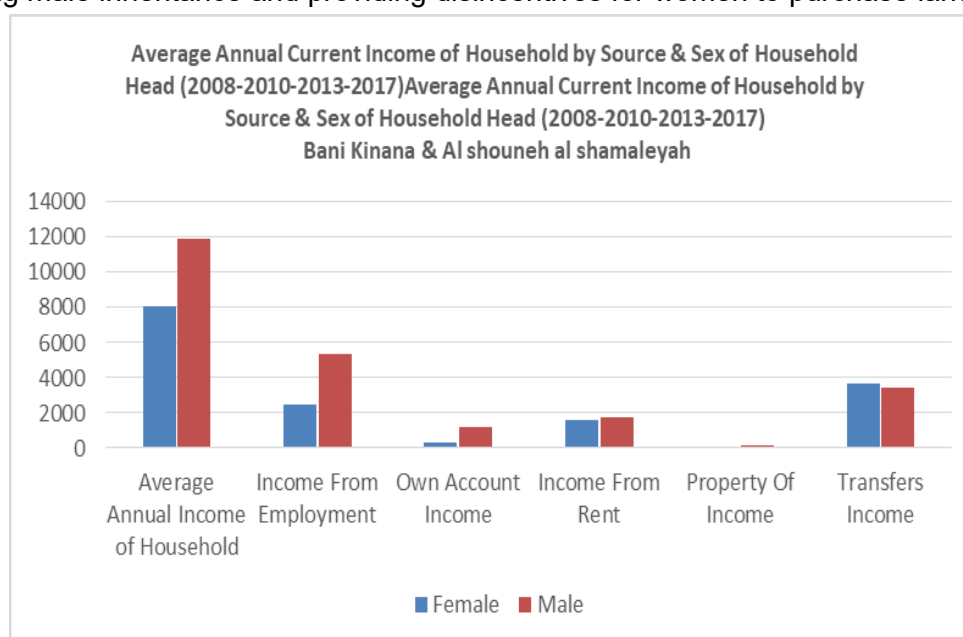


Figure 15. Average annual income of household by source and sex of household hold in Bani Kinana and Al shouneh al shamaleyah.

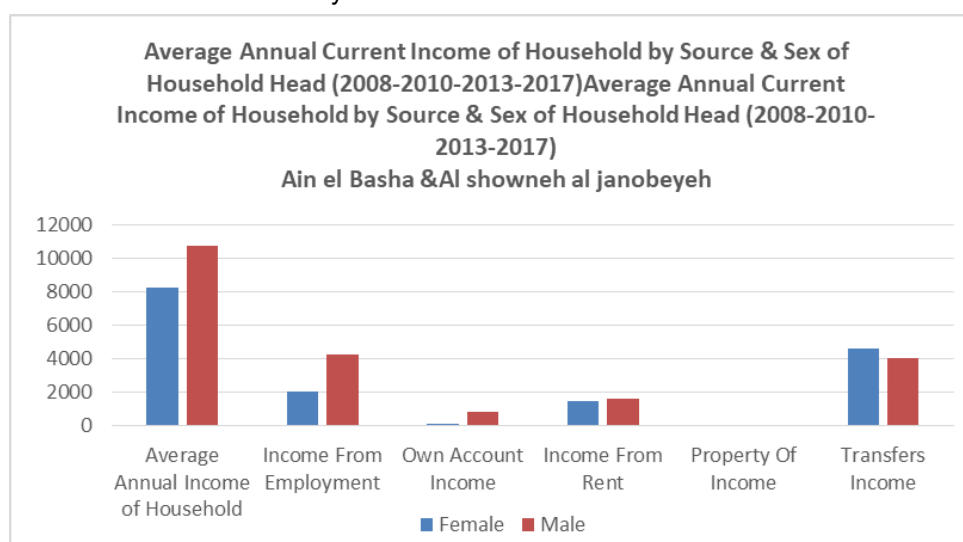


Figure 16. Average annual income of household by source and sex of household hold in Ain el Basha and Al showneh al janobeyeh.

⁸⁰ Impact of Climate and Land Use Changes on Water and Food Security in Jordan: Implications for Transcending "The Tragedy of the Commons"

⁸¹ Climate change profile, 2018

Industries in the Jordan valley are mostly small industries for the construction sector and package industry. There are several quarries that produce materials for the construction of buildings and infrastructures in the north-eastern governorates of Jordan. Some quarries export marble. There are also several metal processing plants in the central and southern part of the Jordan valley that produce metal frameworks for greenhouses and install these for commercial farmers. Small plants are located across the region, that produce wooden and plastic crates and boxes for the commercial farmers, for packing their produce in accordance with the demands of the export markets.

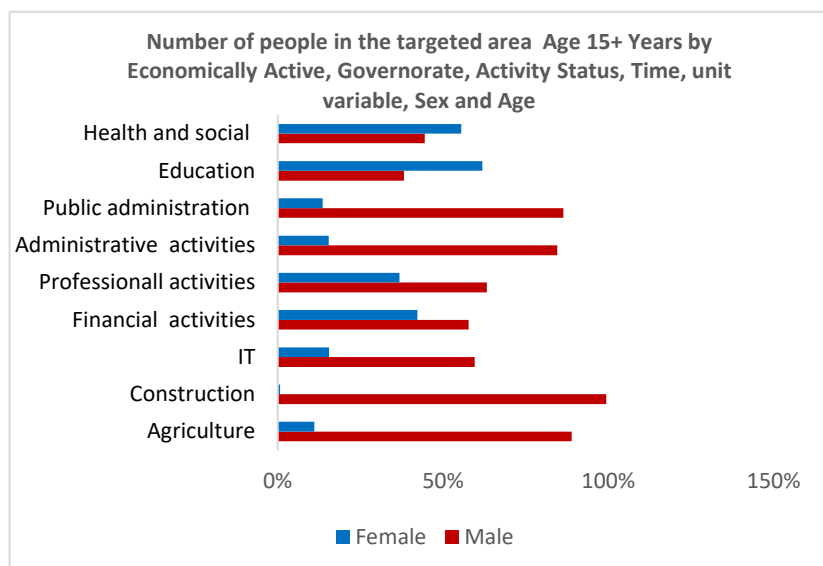


Figure 17. Percentage of people in the target area, aged 15+ employed in main economic sectors.

In terms of women's engagement in the agricultural sector specifically, FAO statistics show that as of 2014, 0.9% of the country's total female population was employed in agriculture, and this could be related to the high prevalence of informal employment in this sector. It's worth noting that home-based agriculture was found to be the most common type of agricultural activity undertaken by women.

Most of women in the targeted areas are in health and education sectors 56% and 62% respectively, while the majority of men are working in agriculture (89%) compared with only 11% of women⁸².

Meanwhile, women were found to be engaging in home-based agriculture to produce food for household consumption. Women's engagement in agricultural activities has economic importance for their household. The majority of FGD women participants confirmed that women engaged in both paid agricultural labour and home-based agriculture contribute to their household's income that enhanced their ability to independently take decisions within the household.

The FGD participants stated that, the majority of women who engaged in home-based agriculture in are Jordanian, compared to Syrian women which is linked to their or their family's ability to own land, unlike Syrians. These motivating factors and variations within them are important to take into account when identifying avenues for programming.

5.2 Education

⁸² DOS, 2018

Illiteracy rate in the targeted area for women is relatively high for women aged 15+ (9.7%), with 9.2% in Ain El Basha, comparing with 3.8% among men. However, the school enrolment rates are similar for both boys and girls, however the enrolment ratio is higher among girls enrolled in the secondary education 53% comparing with 46% for boys, where they enrol in the vocational education 98%.

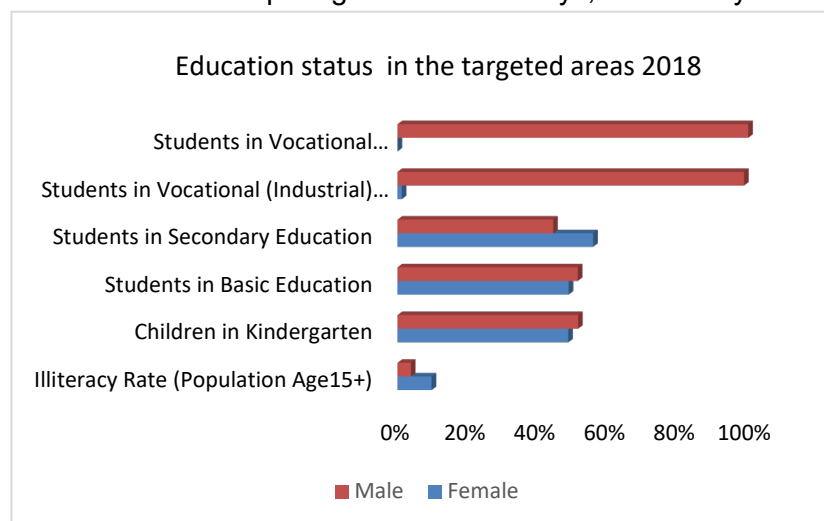


Figure 18. Education status in the target area (2018).

However, the unemployment rate among women is approximately 27%, which is higher than for men (15%). This rate is even higher for single (never married) women (41.5%) and widows (19.5%).

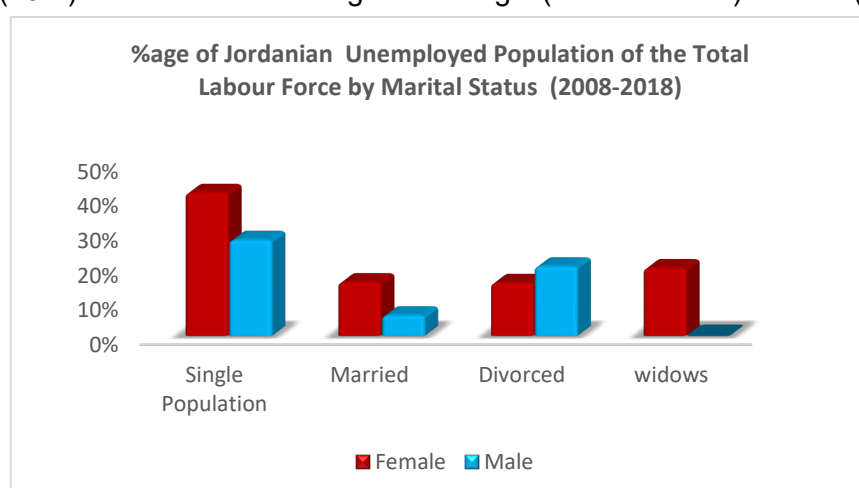


Figure 19. Percentage of unemployed of the population by marital status.

5.3 Access to finance and resources

The majority of women are not engaging in any financial loans, FGD said that they used to get revolving funds. However, these funds were not well controlled by the donors and become a financial burden for women. Most of these funds were for the purpose of family issues such as schools and universities. On the other hand, women have no control of these funds, since husbands usually take these funds to be used for other purposes. Men usually refuse to guarantee the fund for their wives, as a result, women will not be able to pay back the money which leads to social and legal distresses.

Many research assessments on the revolving funds impact indicated that, women entrepreneurs in Jordan face considerable gender-specific barriers to accessing loans, such as having to provide a guarantee from their husbands and being subject to larger collateral amounts. Studies has indicated that men are usually interested in acting as guarantor if they can have control over the money; few women have the means to guarantee loans.

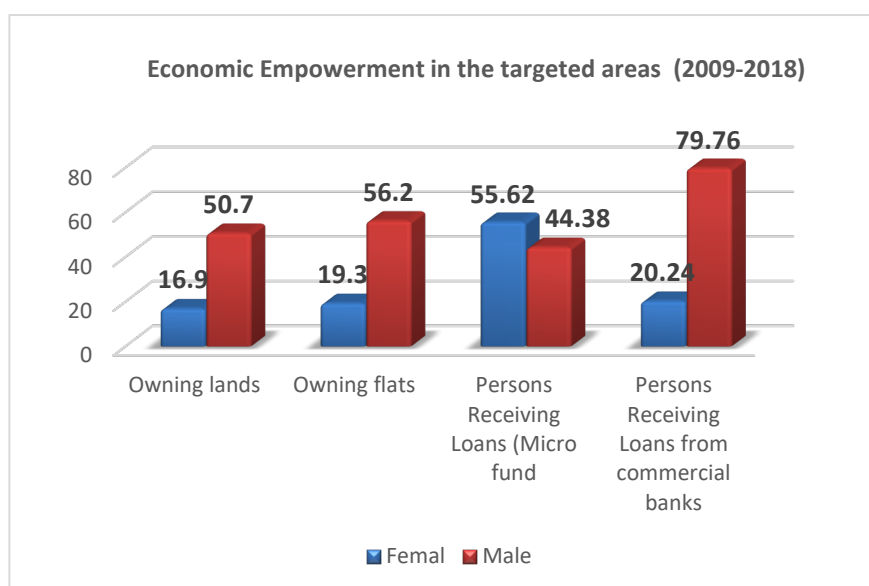


Figure 20. Economic empowerment in the target area (2009–2018).

However, FGD participants mentioned that few women owned lands and farms (10%), most of them are as an inheritance from their families managed by male family members such as husbands, brothers and sons. DOS figures show that only 16.5% of women owning lands comparing with 51% of men, also show that women tend to access micro finance 56% comparing with 20.2% access loans from commercial Banks .

5.4 *Women participation in the public sphere and the women CSOs involvement in the decision-making process in their communities*

The Jordan Valley Solidarity Network is supporting community-based organizations in the valley that provide social services to poor and vulnerable individuals, households and Syrians. Some of the CBOs in the targeted areas participated in the FGD and contribute to the discussion of social impact on proposed interventions. One of the key CBOs is the Jordan Hashemite Fund for Human Development (JOHUD). JOHUD supports national initiatives to improve the management of scarce water resources, through a wide range of water related projects. To build sustainable awareness and to spur behavioural change on the local level in water resource management.

The other key CBO is the Jordan Environment Society (JES). The objective of JES includes, but is not limited to, protecting the environment and its basic elements of water, air, soil and wildlife, based the belief that the environment is both the endowment and responsibility of all people. In addition of the Royal Society for Conservation of Nature to create, manage and advocate for a national network of protected areas to conserve Jordan's biodiversity and support local community development, while also promoting wider public support and action for the protection of the natural environment within Jordan and neighboring countries.

Finally, IUCN, Jordan to influence, encourage and assist society to conserve the integrity and diversity of nature and to ensure that any use of natural resources is equitable and ecologically sustainable. A range of women CSOs operate in their areas, there is some focus on empowerment, such as training, they also targeting boys and men, especially in awareness activities and revolving funds. On the other hand, CBOs may enjoy a level of public trust and be considered to be safe or acceptable spaces for women to gain training and receive information.

However, these organisations are not seen as a powerful tool to communicate with women and other vulnerable groups in the local communities, it was explained that they don't have a network relation

with each other nor the other official or private entities in the community. Those CSOs are approached by donors, providing them with trainings such as food processing and handmade crafts, however these trainings are not reflected or support for women in enhancing their social situations because of poor marketing. In addition, some CSOs are active in delivering awareness sessions for the local communities such as JOUHD community centre in Al Yarmouk on water management, environment and drugs for youth.

5.5 *Social norms and cultural constraints*

There was much evidence to indicate a gendered division of labor within households in the three areas, with women (and to some extent girls) responsible for cooking, cleaning, childcare and 'indoor' domestic work and men responsible for working outside of the home. The family size reported by focus group participants generally ranged from 6 to 10, though a family size of more was reported where the man had more than one wife. Women participants indicated that there is a high rate of divorce cases for girls aged 18–23 which lead them to be more vulnerable because they are unemployed and don't have financial power.

Where feasible, participants were asked to discuss the decision-making dynamic around child spacing (as a proxy indicator for women's access to decision making within the family). Nearly all said that reproductive decisions are the domain of the husband, with the possibility of significant involvement by either mother in law.

In terms of financial decision making, participants varied. There was a fair degree of consensus that women make decisions about small, daily expenditures such as food, though while women said they were given a spending allowance by their husbands, the latter mostly denied providing an allowance for their wife's autonomous use. Some financial decisions, such as schooling and health, were said to be taken by both men and women together. The general picture that emerged was one where, though the wife may participate in and even control small spending decisions, financial decision making ultimately lies in the hands of the husband.

Norms around mobility, women's mobility was not entirely restricted by all husbands, it is dependent upon the husband's preference. Most men and women reported that if the husband was not at home and, for example, shopping needed to be done, then the woman could go out for that purpose. Similarly, however, most participants reported that if the husband or an older son was available, it would be preferable that he would be the one to leave the home and go out. Participants mentioned that a significant percentage of youth are still unemployed and still actively looking for suitable jobs. Employed individuals mainly work in different unsustainable jobs, in the military sector, or in daily-paid fluctuating jobs (per day jobs).

5.6 *Specific findings in line with Gender Analysis Framework*

1. It was observed that, only 30% of the FGD women participants know about the climate change and how it may affect women and children more, saying that women are responsible for taking care of children and the doctors' visits. In addition they are more affected by risks associated with rain floods and poor sanitation facilities. While the majority of men (80%) participants said that the climate change conditions will affect both men and women the same way. Given that there is a different understanding between men and women related to the impacts of climate change between genders there is an opportunity to address this through the project's awareness-raising interventions under Output 4.1 — see Activity 4.1.1 in the GAP below. This will also be addressed through the livelihoods- (Output 2.1) and water supply and management-based activities (Output 3.1), where the specific needs of female beneficiaries related to vulnerabilities and climate change adaptation are emphasised during implementation — see Activities 2.1.3 and 3.1.3 in the GAP.

2. Men control access to resources in which men own the key productive assets such as land, livestock and medium to large businesses. For climate change interventions especially in dry lands, this means that activities which include the use of productive resources like land or livestock should include men and women in decision-making. Secondly, any proposed land-use change relating to dry land conservation should include both local men and women as key stakeholders. This provides an opportunity for the representation of women in decision-making bodies to be enhanced, as well as focusing on improved productivity and income generation that promote the participation of women and their access to resources. Under Activity 1.1.4, the establishment of Watershed Management Committees (WMCs) will be supported, local decision-making bodies focused on the climate-resilient management of watersheds and their resources. This is an opportunity to improve the representation and participation of women in the management of their local watersheds and associated resources to support climate change adaptation in their communities — see Activity 1.1.4 in the GAP.
3. Access to credit is a key constraint in improving women's economic productivity in the selected areas. Hence it is important that the project focuses on improving women's access to credit through the Water Fund in the Jordan Valley to pay for ongoing watershed restoration in the upper-catchment areas of the project's three target sites. The on-granting facility and the Water Fund's grant mechanism (Activity 2.1.4) provides an opportunity for women to access grant finance to support the implementation of gender-responsive ILWRM initiatives, empowering them to contribute to climate change adaptation in their own communities that directly benefits local women. All proposals submitted for consideration for funding by the on-granting facility will need to be gender responsive, promoting the inclusion of women, considering their roles in society and have activities specifically tailored for women and align with the GAP (see below) — proposals will be reviewed by JILMI's gender expert to confirm this.
4. The project can provide support to the Ministry of Environment, to activate gender mainstreaming actions stated in the climate change policy and involve women in the decision-making process. Output 1.1 includes the drafting of amendments to Jordan's planning, legal and policy framework for agriculture, land, water and natural resources, to mainstream climate-resilient ILWRM (Activity 1.1.2). This provides an opportunity to further mainstream gender into policy- and decision-making processes, contributing to a gender-responsive cross-sectoral response to the impacts of climate change in Jordan. Participants in this activity will include at least 50% female representation — see Activity 1.1.2 of the GAP.
5. Women's illiteracy, poverty and unemployment are high in the targeted areas, which make them more vulnerable to climate change impacts. In addition, more investment is needed to improve women's participation in alternative livelihoods, to ensure that their rights and interests are recognized and encouraging home-based farming. Gender-sensitive schemes can support vulnerable groups in the targeted areas and particularly women-headed households. Under Activity 2.1.3 (Introduce gender-responsive climate-resilient additional livelihoods and agricultural practices in the project's three intervention areas), up to 75% of beneficiary households will be female headed, depending on the livelihood. Livelihoods have been selected and will be further tailored during implementation to be gender-responsive — see Activity 2.1.3 in the GAP. Activity 3.1.4 will see local communities, including women, empowered to monitor and manage local water supplies more effectively. This will include developing gender-responsive community-based water use monitoring and managing plans for project beneficiary communities; implementing a gender-inclusive outreach campaign to train community members to monitor and manage the use of clean water (100% female representation); and promoting additional livelihoods related to the monitoring and management of clean water (including at least 50% female beneficiaries) — see Activity 3.1.4 in the GAP.
6. Increased collaboration and partnership with CSOs, NGOs and the entities working on climate change issues can help increase awareness of gender, and environmental sustainability issues

in the targeted areas. This awareness raising should seek to increase women's knowledge and equal rights in particular. In addition, CSOs could support gender mainstreaming by involving whole communities, men, and local leaders, to overcome sensitivity to the traditional divisions of labor that may help to design the project better to be more socially acceptable. Activity 4.1.1 will see the implementation of a gender-responsive climate change adaptation outreach programme across the Jordan Valley, bringing together local CSOS, NGOs and other entities. The outreach programme will create awareness around climate change issues, including those which are gender specific, through a specific model prioritizing gender and youth — see Activity 4.1.1 in the GAP.

6 Gender Action Plan (GAP)

Jordan Integrated Landscape Management Initiative (JILMI)

Impact Statement

Research and anecdotal evidence indicate that women, children and youth are among the groups most vulnerable to the impacts of climate change. In addition, a large number of vulnerable people in communities that are highly dependent on local natural resources for their survival are women and children. Therefore, the Government of Jordan has established that gender-sensitive planning, resource-allocation and implementation in the context of the Jordan Integrated Landscape Management Initiative (JILMI) will be a priority. Supporting transformational change both in the lives of female beneficiaries and in prompting gender mainstreaming in national climate change policy-making and implementation was therefore seen as a priority in stakeholder consultations.

Expected total numbers of direct and indirect beneficiaries (reduced vulnerability or increased resilience), are:

- 246,942 people directly benefit from project interventions (135,736 men and 111,206 women)
- 756,000 people indirectly benefit from the project interventions (355,320 women and 400,680 men)

The project will also:

- ensure gender inclusion in all the policies reviewed to be more gender sensitive;
- ensure sustainability and long-term impact of gender-affirmative actions;
- build capacity for government institutions in acknowledging gender differentials in responding to climate risks;
- ensure that women should take leadership with regards to project components related to them, however, without the marginalization of men with the intent to avoid their resistance and negative influence; and
- increase social accountability.

Outcome statement

Increased resilience to climate change of at least 111,206 women in project area through direct improvement of their capacities to resources for agricultural/agroforestry practices and water management, climate smart agriculture and agroforestry practices and/or to grant facility.

Project Activity	Gender Activity	Indicators	Targets	Indicative Timeline ⁸³	Responsibilities	Budget
Output 1.1: Strengthened governance and capacity for climate-resilient integrated landscape and water resources management (ILWRM) and Output 1.2: Economic assessments developed quantifying net benefits of ILWRM as an adaptation approach						

⁸³ To be confirmed at project inception.

Activity 1.1.1 Strengthen the mandate and capacity of main government institutions — including MoEnv, MWI and their subsidiaries⁸⁴ — for climate-resilient ILWRM in Jordan	1. Ensure that gender considerations related are integrated into the mandates of main government institutions for climate-resilient ILWRM in Jordan. The Gender Officer will engage with the relevant government institutions to ensure that strengthened mandates: • Consider the needs, roles and knowledge of women and men in ILWRM; and • empower women to play a role in ILWRM activities. 2. Ensure that at least 30% of the national-level government staff trained on the interpretation and application of climate data and results of climate analyses and risk assessments are female. 3. Ensure that at least 30% of the national-level government staff trained on the interpretation and application of climate data and results of climate analyses and risk assessments are female.	1. Integration of gender considerations into descriptions of mandates. 2. No. of national-level female staff attending trainings — baseline is 0. 3. No. of local-level female staff attending trainings — baseline is 0. 4. Sections on gender considerations included in the guidelines, standards and protocols.	1. Gender considerations integrated into mandate descriptions by project end. 2. At least 15 (30%) female staff per training — out of a total of 50 trainees. 3. At least 17 (30%) female staff per training — out of a total of 57 trainees. 4. Gender considerations 100% integrated into documents by project closure.	1. By Q4 Y1 2. By project closure 3. By project closure 4. By project closure	Gender officer and Policy and Institutional Officer	<i>The cost of implementing the GAP for Activity 1.1.1 will be covered by the Gender Officer (USD 60,000 salary per year). Additional costs are outlined below.</i> <i>Transport allocation for female stakeholders: \$ 5,000</i> <i>Other costs are embedded in the activity costs, particularly workshops meeting and training (\$169,500)</i>
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⁸⁴ Such as the Jordan Valley Authority (JVA).

	4. Draft guidelines, standards and protocols for ILWRM in the Jordan Valley to be gender-responsive in their content, design and distribution, by taking the following actions: • Gender Officer will support the design of the review methodology for developing the guidelines standards and protocols to ensure women's adaptation needs and interests are fully considered. • Women and women's groups are represented in stakeholder consultations to inform the development of the outputs, including women-specific focus groups and/or workshops. • Ensure that role of women in ILWRM in the Jordan Valley is clearly detailed and considerate of their responsibilities at home and in society.					
Activity 1.1.2: Draft amendments to Jordan's policy, planning and	1. Draft policy briefs and recommendations to be gender-responsive in their content, design and distribution, by taking the following actions:	1.a. Inclusion of gender considerations sections in the draft policy briefs and recommendations.	1.a. Gender considerations 100% mainstreamed into the policy briefs and recommendations by project end.	1.a. By project closure 1.b. By Q4 Y4	Gender officer and Policy and Institutional Officer	The cost of implementing the GAP for Activity 1.1.2 will be covered by the Gender

legal frameworks for agriculture, land, water and natural resources, to mainstream climate-resilient ILWRM as a climate change adaptation approach.	• The Gender Officer, with support from the Policy and Institutional Officer, will support the design of the review methodology for reviewing key plans, policies and legislature to ensure women's adaptation needs and interests are fully considered when policy briefs and recommendations for the mainstreaming of climate-resilient ILWRM are drafted. • Women and women's groups are represented in stakeholder consultations for conducting the review of key plans, policies and legislature, including women-specific focus groups and/or workshops. • The project's Policy and Institutional Officer, in collaboration with the Gender Officer, will lead the development of evidence-based advocacy messages that showcase the effects of the project's gender responsiveness and women's participation and how it is contributing	1.b. Development of evidence-based advocacy messages on gender-responsiveness under the project. 2. No. of female policy- and decision-makers trained — baseline is 0. 3. No. of female national- and sub-national staff trained — baseline is 0.	1.b. Advocacy messages developed and integrated into policy briefs and disseminated to key governmental stakeholders. 2. At least 13 (50% of 26 trainees) female staff per training. 3. At least 10 (50% of 20 trainees) and 20 (50% of 40 trainees) female staff per national- and sub-national training, respectively.	2. By project closure 3. By Q3 Y2		Officer (USD 60,000 salary per year). Additional costs are outlined below. Transport allocation for female stakeholders: \$ 5,000 Other costs are embedded in the activity costs, particularly workshops meeting and training (\$62,000)
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	to the protection and realization of women's rights. 2. Ensure that at least 50% of the policy- and decision-makers trained to integrate climate-resilient ILWRM into plans, policies and legislature are female. Actions include: • Targeting female staff to invite to sessions. • Providing transport to female staff. 3. Ensure that at least 50% of the national- and sub-national-level development and land use planners trained on climate-resilient ILWRM planning are female. • Targeting female staff to invite to sessions. • Providing transport to female staff.					
Activity 1.1.3. Generate evidence of the economic benefits of ILWRM to project beneficiaries for use by policymakers and planners.	1. Include gender consideration, emphasising women's empowerment and gender equality in reports and recommendations through: • Ensuring women are represented in working sessions for report distribution, by encouraging participation	1.a. Full and summary reports include sections on 'Gender Considerations' with gender-sensitive reporting, and recommendations for enhancing women's empowerment and gender equality in the application of findings.	1.a. Reports finalized by project end. 1.b. All project reports have an IASC GAM score of 3 or higher	1.a. By project closure 1.b. By Q4 Y7	Gender officer	The cost of implementing the GAP for Activity 1.1.3 will be covered by the Gender Officer (USD 60,000 salary per year). Additional costs

	and scheduling and locating meetings that are sensitive to women's duties and child-minding responsibilities. - Ensuring women are represented in providing inputs to the modelling methodology to value ecosystems, by encouraging participation and scheduling and locating stakeholder engagements that are sensitive to women's duties. - The Gender Officer will screen all full and summary project reports for the IASC GAM score (using the IASC online tool) and will provide inputs and advise to ensure they reach the agreed rating	1.b. IASC Gender with Age marker score for all project reports ⁸⁵ .				are outlined below. Transport allocation for female stakeholders: \$ 2,500 per site year for years 2 and 7 (\$15,000) Provision of childcare services (years 2 and 7): \$2,500 per site per year (\$15,000) Other costs are embedded in the activity costs, particularly workshops meeting and training (\$44,000)
Activity 1.1.4: Strengthen the mandate and capacity of non-state actors for climate-	1. Ensure that women are adequately represented in WMCs and are able to participate in their activities. Actions will include:	1. % of female WMC members. 2. No. of female WMC members trained — baseline is 0.	1. At least 50% female membership. 2. At least 45 (50% of 90 trainees) female WMC	1. By Q4 Y1 2. By Q3 Y8 3.a. By Q1 Y2	Gender officer	The cost of implementing the GAP for Activity 1.1.4 will be covered by the Gender Officer (USD

⁸⁵ <https://www.iascgenderwithagemarker.com/en/home/>

resilient integrated land and water resources management in the Jordan Valley.	- WMC capacity development activities specifically targeting women to ensure that all participants have the same knowledge to work together more effectively. Include training modules for women and men, training participants on GBV and access to community feedback mechanisms, complaints boxes, and referral pathways to obtain support for survivors of GBV. - Schedule training/meetings at times when women can attend. This may include organizing training outside of lunch hours to avoid conflicts with women's household responsibilities and limiting the length of training so that women could attend and still see to other activities.	3.a. Women's needs have been considered in the planning process (No. of needs assessments). 3.b. Number of plans that reflect the needs and interests of local women. 4.a. Number of gender-responsive NGO capacity-building programmes. 4.b. IASC Gender with Age marker score for the capacity-building programme⁸⁶.	members trained. 3.a. One women's needs assessment report to inform the planning process. 3.b. Three gender-responsive plans (1 for each target area) 4.a. One gender-responsive capacity-building programme developed and implemented.	3.b. By Q4 Y2 4.a. By Q2 Y8		60,000 salary per year). Additional costs are outlined below. Transport allocation for female stakeholders: \$ 1,500 per site year from years 1 to 8 (\$36,000) Provision of childcare services (years 1 to 8): \$1,500 per site per year (\$36,000) Needs assessment of women in project target communities to inform project implementation: \$ 22,500 (\$ 7,500 per target site) Other costs are embedded in the activity
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⁸⁶ <https://www.iascgenderwithagemarker.com/en/home/>

	• Provide transportation if travel distances would hamper the participation of women. • Organize childcare services, or in spaces where children may be accommodated during training or meetings.						costs, particularly workshops meeting and training (\$134,000)
	2. Ensure that at least 50% of the WMC members trained on climate-resilient ILWRM planning are female. • Schedule meetings at times when women can attend. This may include organizing training outside of lunch hours to avoid conflicts with women's household responsibilities and limiting the length of training so that women could attend and still see to other activities. • Provide transportation if travel distances would hamper the participation of women. • Organize childcare services, or in spaces where children may be						

	accommodated during training or meetings. 3. Ensure the full and equal participation of women in the development of gender-sensitive participatory climate-resilient integrated watershed management plans, including: • Carrying out an in-depth needs assessment of women in each target watershed to inform the planning process; • Hosting separate meetings to capture women's views during the planning process and have representatives of women's groups publicly represent their voices during decision-making processes; • Organize meetings at a time of day and location conducive to their schedules and sensitive to women's domestic responsibilities; and • Promoting awareness among women and men on incidence of					
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	and avoiding gender-based violence as well as access to community feedback mechanisms, complaints boxes, and referral pathways to obtain support. 4. Ensure that climate-resilient ILWRM capacity-building programme for local NGOs will include strengthening the gender responsiveness of the NGOs work in the Jordan Valley. The Gender Officer will screen the capacity-building programme for the IASC GAM score (using the IASC online tool) and will provide inputs and advise to ensure they reach the agreed rating of 3.					
Output 2.1: Climate-resilient ILWRM practices and ecosystem restoration implemented in three watersheds in the Jordan Valley						
Activity 2.1.1: Support the implementation of participatory climate-resilient integrated watershed management	1. Ensure the full and equal participation of women in the implementation of gender-sensitive participatory climate-resilient integrated watershed management plans, including:	1. % of participants involved in the implementation of plans that are women 2.a. % of women represented in participatory M&E systems	1. At least 50% by project completion 2.a. At least 50% by project completion 2.b. At least one group/CSO per	1. By project closure 2.a. By project closure	Gender Officer	<i>The cost of implementing the GAP for Activity 2.1.1 will be covered by the Gender Officer (USD 60,000 salary per year).</i>

plans (developed under Activity 1.1.4) in the Yarmouk, King Talal Dam and Kafraïn areas of the Jordan Valley.	- The Gender Officer will support the WMCs and local partners in the development of inclusive work plans to ensure women's participation in implementation; - Hosting separate meetings to capture women's views during the planning process and have representatives of women's groups publicly represent their voices during decision-making processes; - Organize meetings at a time of day and location conducive to their schedules and sensitive to women's domestic responsibilities; and - Providing transport for women who live far distance from the restoration sites. 2. Ensure the representation of women in community-based participatory mechanisms for monitoring and evaluating change in community climate	2.b. No. of women's groups/CSOs from the 3 target areas represented in the participatory M&E systems	target area (3 in total)	2.b. By project closure		<i>Additional costs are outlined below.</i> Transport allocation for female stakeholders: \$ 2,500 per year from years 3 to 8 (\$15,000) Provision of childcare services for female restoration workers (years 3 to 8): USD 180,000 (USD 10,000 per year per site)
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	resilience and behavioural change, and that monitoring is gender disaggregated. Actions will include: • Target the members of women's groups/CSOs in the three target areas for membership. • Schedule meetings at times when women can attend. This may include organizing training outside of lunch hours to avoid conflicts with women's household responsibilities and limiting the length of training so that women could attend and still see to other activities. • Provide transportation if travel distances would hamper the participation of women. • Organize childcare services, or in spaces where children may be accommodated during training or meetings. • Ensure that training is provided to the individuals involved in					
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	the monitoring mechanisms to ensure that monitoring is gender sensitive, and data is collected in a gender-disaggregated manner.					
Activity 2.1.2: Restore ecosystems in the watersheds of the Yarmouk, King Talal Dam and Kafraïn areas of the Jordan Valley.	1. Ensure that female stakeholders have equal opportunities to be hired as labour for the implementation of restoration opportunities and are paid the same as men. Actions include: • Organise women-only training sessions and allow participants to feel comfortable to express their ideas and needs. Include training modules for women and men training participants on GBV and access to community feedback mechanisms, complaints boxes, and referral pathways to obtain support for survivors of GBV. • Implement capacity development activities by ensuring that all participants have the same knowledge to work together more	1.a. % of restoration labourers that are female.	1.a. At least 30%	1.a. By project closure	Gender Officer	The cost of implementing the GAP for Activity 2.1.2 will be covered by the Gender Officer (USD 60,000 salary per year). Additional costs are outlined below. Transport allocation for female stakeholders: \$20,000 per year from year 3 to 8 (\$120,000) Provision of childcare services: \$20,000 per year from year 3 to 8 (\$120,000)

	effectively. Activities can also be targeted to the needs of the women participants. - Engage women stakeholders by planning activities and meetings at a time of day and in a location conducive to their schedules and sensitive to their domestic responsibilities. - Provide childcare services for women during trainings and when working on restoration activities. - Provide transport for women to restoration sites so that work is not inhibited. - Develop workplans for women that are considerate of their obligations at home.					Other costs are embedded in the activity costs, particularly workshops meeting and training (\$540,000)
Activity 2.1.3. Introduce climate-resilient additional livelihoods and agricultural practices in the project's three intervention areas.	1. Ensure that female-headed households benefit from permaculture/home gardening and beekeeping, which allow them to work from home, but do not negatively impact on their responsibilities to their families. Actions include: Working with WMCs and Technical Partners to develop selection	1.a. % of women benefitting from beekeeping as an alternative livelihood 1.b. % of women benefitting from permaculture/hydroponics as an alternative livelihood	1.a. 50% of total are female-headed households 1.b. 75% of total are female-headed households 2. At least 360 women farmers or 30% of the total (1,200)	1.a. By Q3 Y8 1.b. Q4 Y4 3. Q4 Y2 4. Q4 Y2	Gender officer	The cost of implementing the GAP for Activity 2.1.3 will be covered by the Gender Officer (USD 60,000 salary per year). Additional costs are outlined below.

	criteria that are targeted at vulnerable and marginalised groups, including women. Female headed households will be specifically targeted, particularly as the alternative livelihoods activities are considered to have a relative low workload. • Provide targeted training for women at their homes or provide transport to training sessions. • Schedule training at times when women can attend. This may include organizing training outside of lunch hours to avoid conflicts with women's household responsibilities and limiting the length of training so that women could attend and still see to other activities. • Organize childcare services, or in spaces where children may be accommodated during training. • The implementation of livelihood activities will consider the needs,	2. No. and % of women farming practicing climate-resilient agriculture 3. % of direct beneficiaries receiving agricultural inputs that are female	3. At least 30% (based on the % of farmers in the target region that are female)		Transport allocation for female stakeholders: \$5,000 per year from year 2 to 8 (\$ 35,000) Provision of childcare services: \$5,000 per year from year 2 to 8 (\$35,000) Other costs are embedded in the activity costs, particularly workshops meeting and training (\$442,000)
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	aspirations, capacities and limitations of women. 2. Ensure that female farmers are benefiting from climate-resilient agriculture support and training. Actions include: • Ensure that women farmers and their role in the agricultural sector are better understood before devising an engagement strategy which aims towards empowering them to co-lead the transformative approach for the farming sector in the target areas; • Take a flexible approach in forming the groups for coordination and capacity building, considering both mixed-sex groups and single – sex groups; • If deemed beneficial, organise women-only training sessions and where possible invite female extension workers to allow participants to feel comfortable to express their ideas and needs;					
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	• Schedule training at times when women can attend. This may include organizing training outside of lunch hours to avoid conflicts with women's household responsibilities and limiting the length of training so that women could attend and still see to other activities; • Include training modules for women and men training participants on GBV and access to community feedback mechanisms, complaints boxes, and referral pathways to obtain support for survivors of GBV; • Provide transportation if travel distances would hamper their participation; • Organize childcare services, or in spaces where children may be accommodated during training; and • Ensure the location of the demo plots allows women to easily access these without spending too much time travelling					
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	also considering any protection/GBV risks that may arise from travelling to and from the demo plots. 3. Ensure women are receiving agricultural inputs and are participating in climate-resilient agriculture activities, including through the following actions: • Ensuring women are fully able to participate in the saving associations; and • Ensuring women are equally benefitting from the distribution of agriculture inputs, including vulnerable women who may not be part of a farmers' cooperative.					
Activity 2.1.4: Establish a Water Fund in the Jordan Valley to administer a payment for ecosystem services system to maintain landscape investment for enhanced	1. Ensure the full and equal participation of women in the design of a gender-sensitive Water Fund, including: • Hosting separate meetings to capture women's views during the planning process and have representatives of women's groups publicly represent their voices during	1. Water Fund that reflects the needs and interests of local women. 2. RFPs submitted under the on-granting facility that are gender sensitive and inclusive of the needs of women.	1. One Water Fund designed in consideration of the needs and interests of local women. 2. Gender sensitive and inclusive RFPs.	1. Q4 Y2 2. Q4 from Y3-Y7 3. By Q4 Y7	Gender officer	<i>The cost of implementing the GAP for Activity 2.1.4 will be covered by the Gender Officer (USD 60,000 salary per year)</i> Other costs are embedded in the activity

water resources.	decision-making processes; and Organize meetings at a time of day and location conducive to their schedules and sensitive to women's domestic responsibilities. 2. Ensure that all proposals submitted under the on-granting facility are gender sensitive, promoting the inclusion of women, considering their roles in society and have activities specifically tailored for women and align with the GAP as a whole.	3. Amount of funds allocated to women-targeted investments through the on-granting facility.	3. At least USD 3,000,000 by project closure			costs (\$310,000)
Output 3.1 Improved protection, supply and use of water for domestic purposes under climate change conditions						
Activity 3.1.1. Implement measures to improve the quantity and quality of ground and surface water supplies within the project sites.	NA	NA	NA		NA	NA
Activity 3.1.2. Install 1 MW of floating solar	NA	NA	NA		NA	NA

panels over 1,200 small-scale irrigation dams/tanks on smallholder farms across the three project sites (400 per site) to reduce water loss from evaporation						
Activity 3.1.3: Introduce small-scale rainwater harvesting interventions in the project's three target areas to improve access to clean water under climate change conditions in a gender-responsive manner.	1. Ensure that female headed households benefit from rainwater harvesting interventions so that access to clean water is improved, reducing the burden on women to collect water from off-site sources. Actions include: An exercise to determine the beneficiary profile and select the technologies to ensure these are appropriate to address specific needs of women. This may include having separate meetings with women to determine their needs and discuss various technology options and how they would benefit them.	1. % of women targeted as beneficiaries of water harvesting interventions, and trained on their operation and maintenance, that have improved access to clean water. 2. % of female-headed households (of total) benefitting from the installation of rainwater harvesting tanks	1. 100% at project completion 2. 50% of total households are female headed	1. By Q4 Y4 2. By Q4 Y4	Gender officer	<i>The cost of implementing the GAP for Activity 3.1.3 will be covered by the Gender Officer (USD 60,000 salary per year). Additional costs are outlined below.</i> Beneficiary profiling exercise: \$ 10,000 (year 2) Assessment on impact of water harvesting: \$ 25,000 (year 8)

	Develop and support the implementation of an assessment (e.g. through focus group discussions or interviews with key informants) to determine the impact of water harvesting and on water availability and women's access to clean water.					
Activity 3.1.4: Empower communities, including women, to monitor and manage clean water supplies more effectively	- Ensure the full and equal participation of women in the development of gender-sensitive community-based water use monitoring and managing plans, including: - Hosting separate meetings to capture women's views during the planning process and have representatives of women's groups publicly represent their voices during decision-making processes; - Organize meetings at a time of day and location conducive to their schedules and sensitive to women's domestic responsibilities; and - Promoting awareness among women and men on incidence of and avoiding gender-based	- Number of plans that reflect the needs and interests of local women. - Gender sensitive awareness-raising campaign a. % of beneficiaries trained to monitor and manage the use of clean water that are women. Ratio of women in stakeholder consultations b. % of women that are selected as waterwise ambassadors in their communities - No. and % of beneficiaries trained on and practicing livelihoods related to water monitoring and	a. 3 plans (one for each of the three target watersheds) b. Equal participation of women (at least 50% representation) in the development of plans - Developed and implemented across the 3 target areas a. 100% by project end b. At least 50% of women members (including farmers) selected as waterwise ambassadors	a. By Q4 Y5 b. By Q4 Y5 - By project closure a. By Q4 Y6 b. By Q4 Y6 - By Q4 Y6	Gender officer	The cost of implementing the GAP for Activity 3.1.4 will be covered by the Gender Officer (USD 60,000 salary per year). Additional costs are outlined below. Transport allocation for female stakeholders: \$5,000 per year from year 5 to 8 (\$ 20,000) Supporting women's groups to become waterwise

	violence as well as access to community feedback mechanisms, complaints boxes, and referral pathways to obtain support. 2. Ensure that women are targeted and reached to raise awareness on the water monitoring and managing plans. Actions include: • Designing awareness raising activities that are gender-sensitive and speak to the needs of women, • Organize awareness-raising activities at a time of day and location conducive to their schedules and sensitive to women's domestic responsibilities, including home visits; and • Arrange transport for women to attend awareness-raising activities if necessary. 3. Ensure that women are targeted and take part in water monitoring and managing training. Actions will include:	management that are women	4. At least 57% or 57 women (out 100 beneficiaries).		ambassadors (one group supported per project site from years 5 to 8): USD 20,000 Other costs are embedded in the activity costs, particularly workshops meeting and training (\$373,500)
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	• Training activities specifically targeting women to ensure that all participants have the same knowledge to work together more effectively. Include training modules for women, training participants on GBV and access to community feedback mechanisms, complaints boxes, and referral pathways to obtain support for survivors of GBV. • Schedule training/meetings at times when women can attend. This may include organizing training outside of lunch hours to avoid conflicts with women's household responsibilities and limiting the length of training so that women could attend and still see to other activities. • Provide transportation if travel distances would hamper the participation of women.					
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	• Organize childcare services, or in spaces where children may be accommodated during training or meetings. • Select women (including female farmers) to be waterwise ambassadors for the monitoring mechanisms. 4. Ensure that women are targeted for training on additional livelihoods related to the monitoring and management of clean water in target communities. Actions include: • Working with WMCs and Technical Partners to develop selection criteria that are targeted at vulnerable and marginalised groups, including women. Women will be specifically targeted. • Provide targeted training for women at their homes or provide transport to training sessions. • Schedule training at times when women can					
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	attend. This may include organizing training outside of lunch hours to avoid conflicts with women's household responsibilities and limiting the length of training so that women could attend and still see to other activities. Organize childcare services, or in spaces where children may be accommodated during training. The implementation of livelihood activities will consider the needs, aspirations, capacities and limitations of women.					
Output 4.1: Improved awareness of climate change adaptation for land and water resources management						
Activity 4.1.1 Implement a climate change adaptation outreach programme across the Jordan Valley	1. Ensure that climate change adaptation outreach programme is gender-sensitive, in its content design and distribution and is designed to target both men and women. Actions include: Needs of men and women considered in the design of the programme, and that outreach activities are gender-responsive.	1. Gender-responsive outreach programme design including activities targeting women and men	1. Gender integrated 100% into outreach programme design	1. Y1	Q4 Gender Officer	<i>The cost of implementing the GAP for Activity 4.1.1 will be covered by the Gender Officer (USD 60,000 salary per year).</i> Childcare services: \$7,500 per year from years 2 to 8 (\$52,500)

	• Awareness-raising on the impacts of climate change focuses on how men and women are affected differently and what their vulnerabilities and adaptation needs are. • Women, women's groups and institutions working with women are targeted with specific awareness-raising activities. • Schedule outreach activities at times when women can attend. This may include organizing activities outside of lunch hours to avoid conflicts with women's household responsibilities and limiting the length of activities so that women could attend and still see to other responsibilities. • Organize childcare services, or in spaces where children may be accommodated during outreach activities. • Identification of communication channels and media that are appropriate for women.					Transport for women: \$7,500 per year from years 2 to 8 (\$52,500) Other costs are embedded in the activity costs, particularly workshops meeting and training (\$525,000)
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	Providing transport for women to outreach activities as necessary. Inclusion of key outreach activities for women at the household level — considerate of women's daily schedules and availability, and preferred engagement approaches.					
Activity 4.1.2. Improve knowledge management for climate change adaptation in Jordan to support national and sub-national decision-making, and build communities of practice.	- Ensure that knowledge management approaches are gender sensitive and include a focus on collecting lessons learned on climate-resilient ILWRM that are relevant to the adaptation of both men and women. Actions include: - Designing the long-term monitoring programme for the generation of lessons learned on climate-resilient ILWRM to focused on the collection of gender disaggregated information and data related to adaptation benefits; and - Presenting best-practices and lessons learned in a gender-sensitive manner	- Integration of gender sensitivities into the knowledge management process % of user groups trained that are female	100% by project end - At least 30%	- By project closure - By Q4 Y7	Gender officer	<i>The cost of implementing the GAP for Activity 4.1.2 will be covered by the Gender Officer (USD 60,000 salary per year).</i> Childcare services: \$5,000 per year from years 2 to 8 (\$35,000) Transport for women: \$5,000 per year from years 2 to 8 (\$35,000) Other costs are embedded in the activity costs,

	(indicating how approaches benefit men and women differently, and how implementation amongst genders differs). 2. Ensure that women are included in the training of user groups on the use knowledge management tools to increase their access to knowledge/information of climate-resilient ILWRM approaches. Actions include: • Targeting women's groups for training; • Ensuring that the knowledge/information is presented in formats that is relevant to both men and women; • Schedule training activities at times when women can attend. This may include organizing activities outside of lunch hours to avoid conflicts with women's household responsibilities and limiting the length of activities so that women could attend and still					particularly professional and contractual services (\$420,000)
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	see to other responsibilities; • Organize childcare services, or in spaces where children may be accommodated during training; and • Providing transport for women to outreach activities as necessary.					
Sub-total of dedicated costs Gender Officer and Gender mainstreaming costs		The Gender Officer will oversee all activities related to gender mainstreaming, including the implementation of action plans. The officer will ensure that the project-level grievance redress mechanism (GRM) (See Annex 6: ESMP) will be accessible to women and men, including to women who are subject to GBV, SEAH and sensitive to their special confidentiality and protection needs and refer cases to existing protection mechanisms accordingly.				\$ 1,369,500
Sub-total of project activity costs						\$2,660,000
Total amount related to implementation of gender action plan						\$4,029,500
